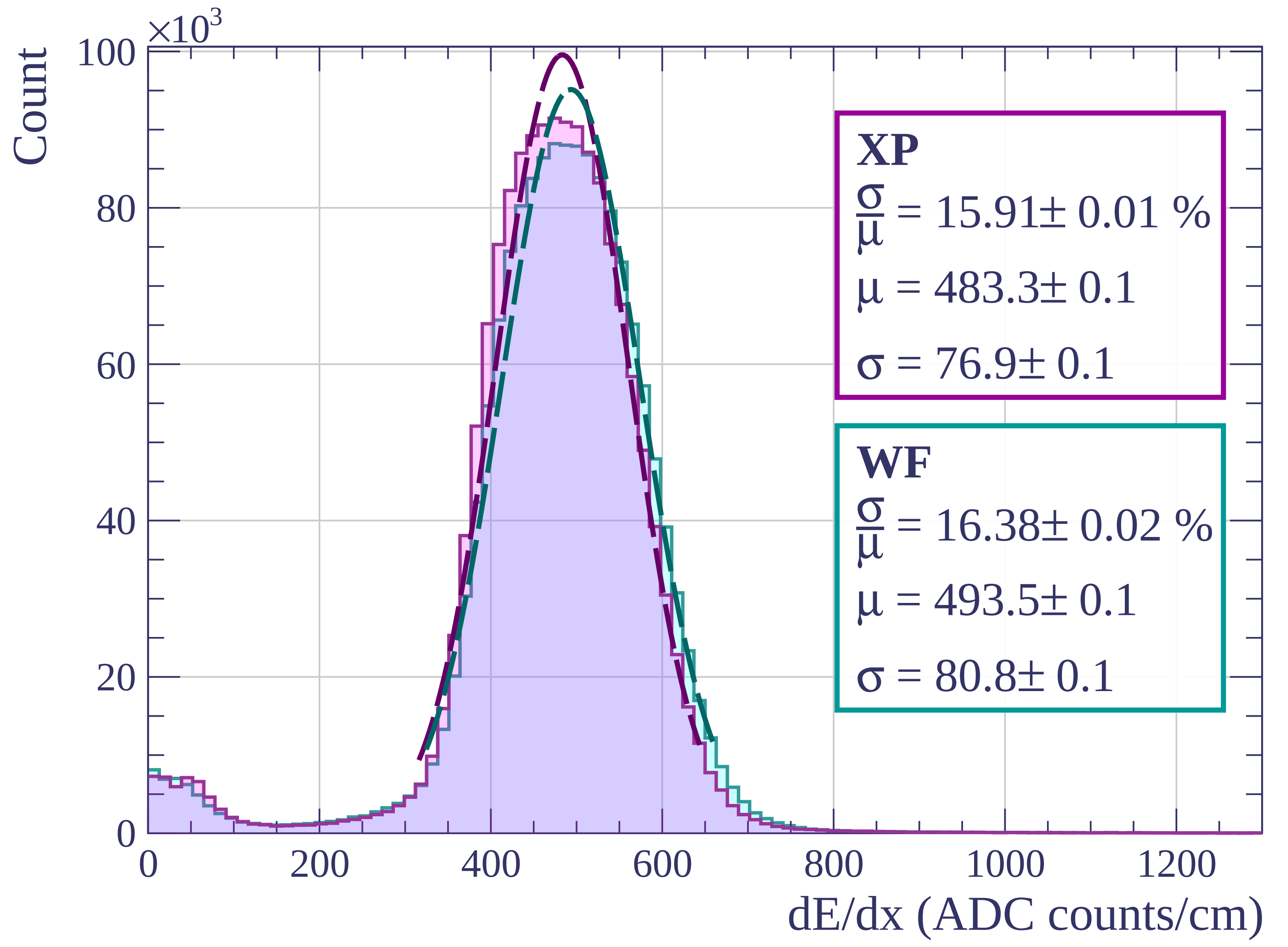
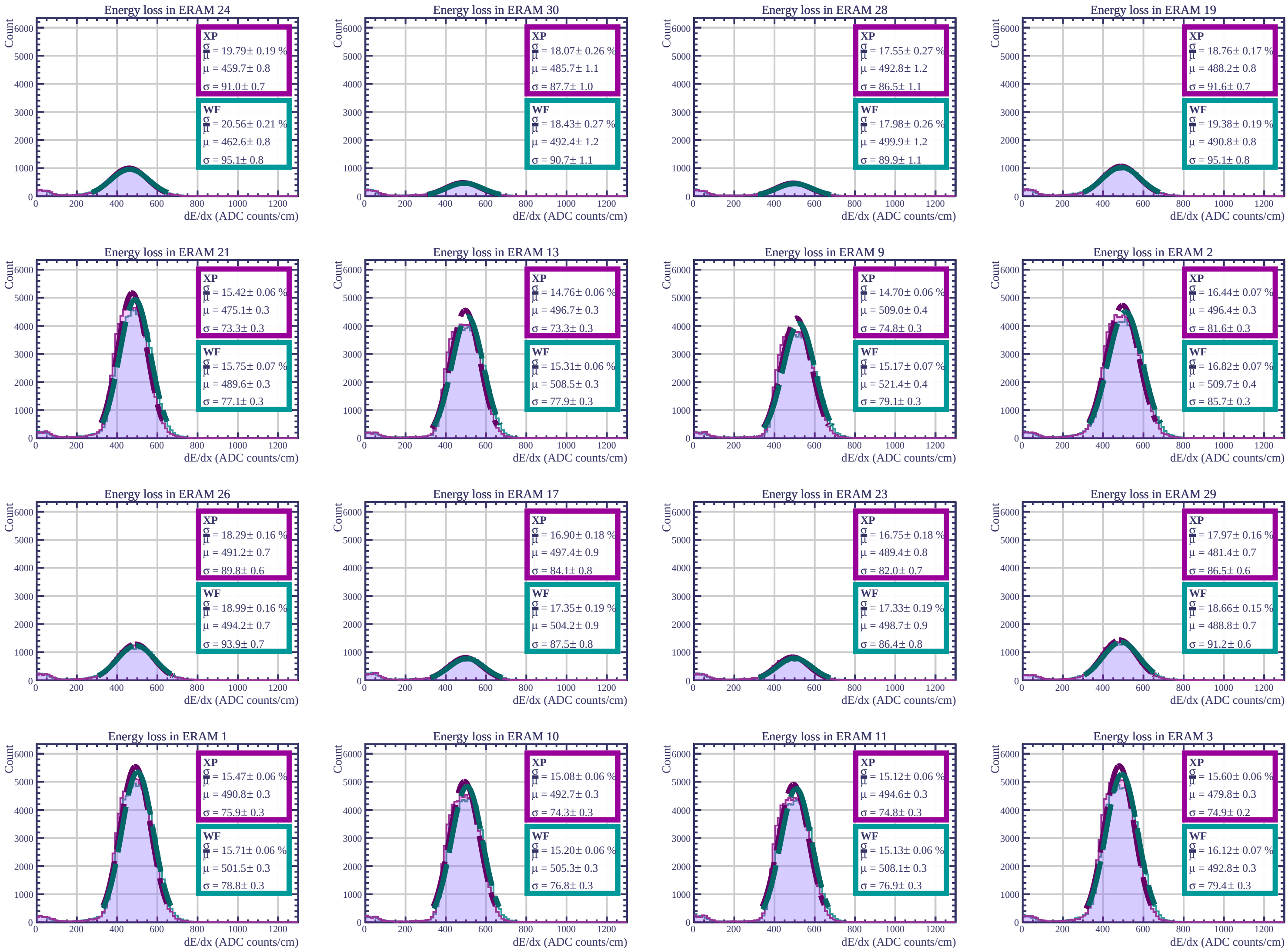
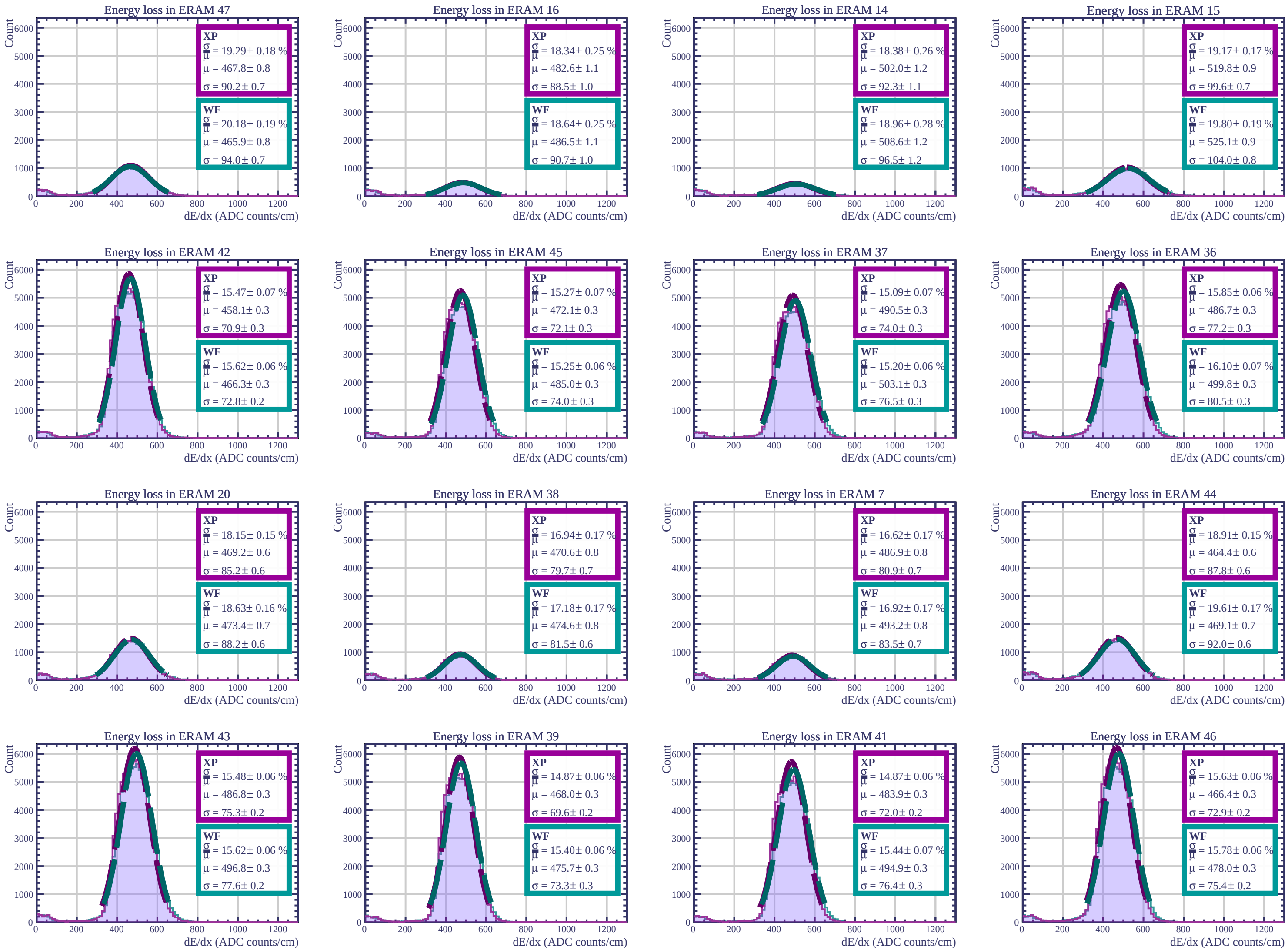
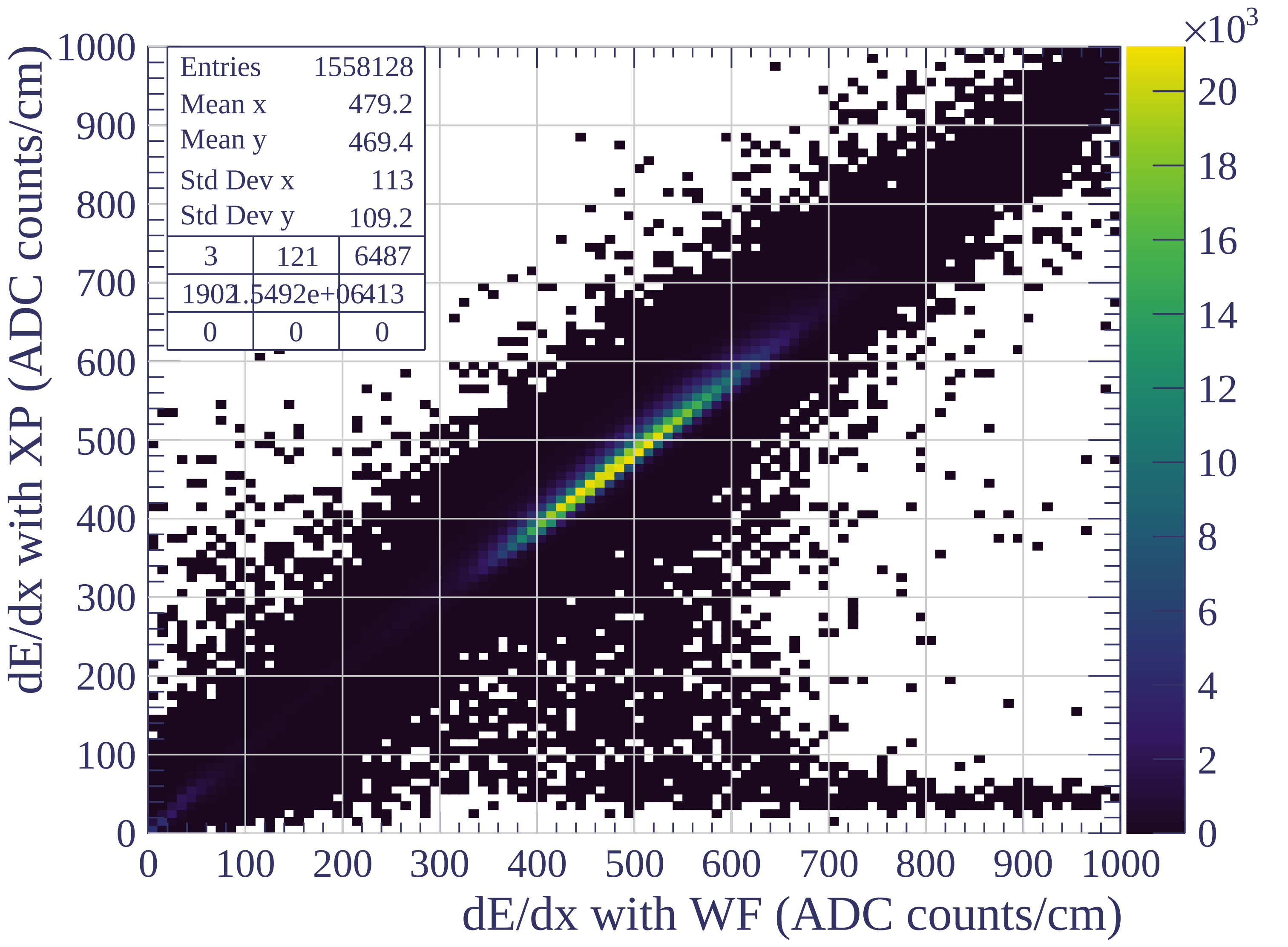


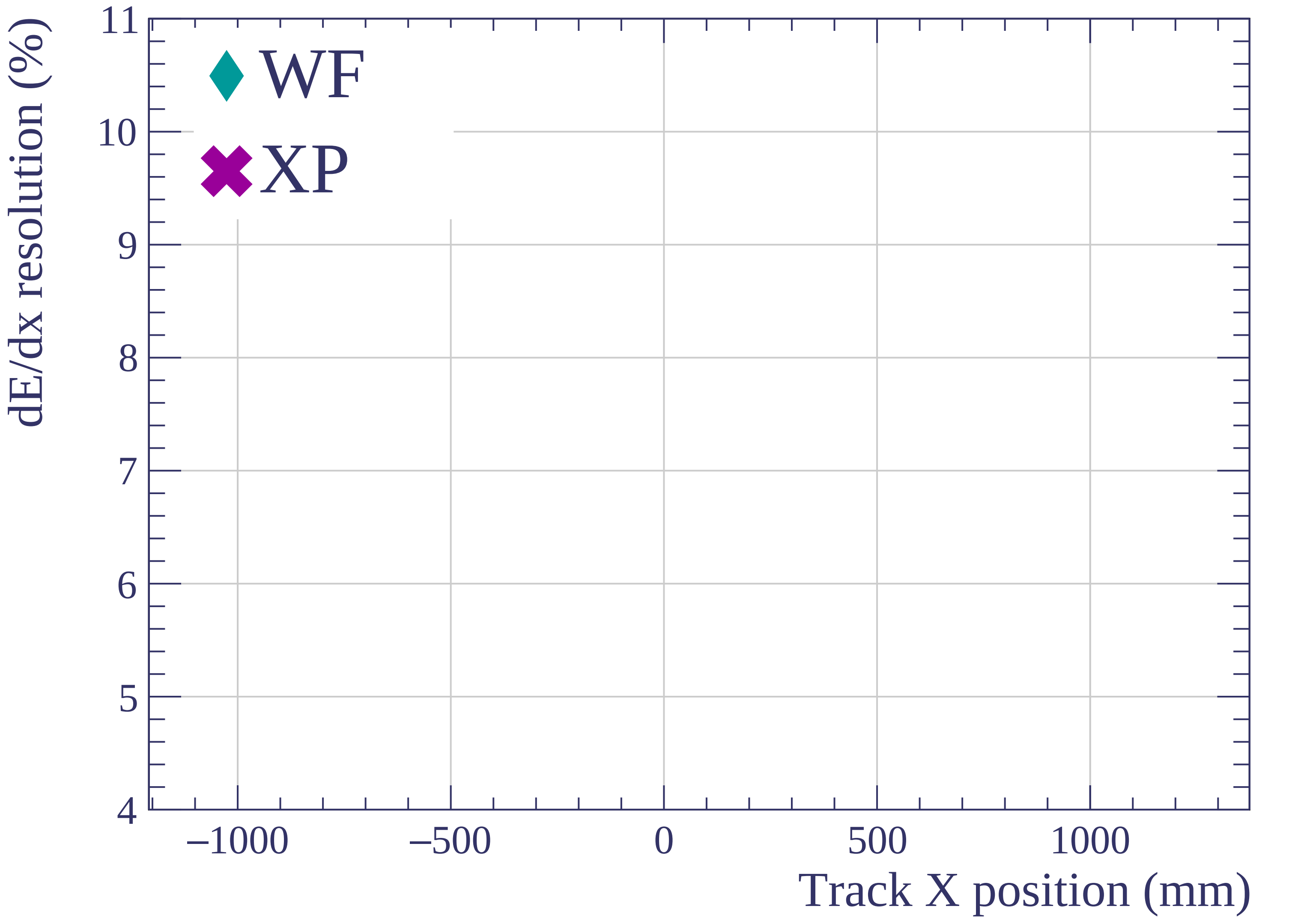
×10^3
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 15.91± 0.01 %
μ = 483.3± 0.1
σ = 76.9± 0.1
WF
σ/μ = 16.38± 0.02 %
μ = 493.5± 0.1
σ = 80.8± 0.1



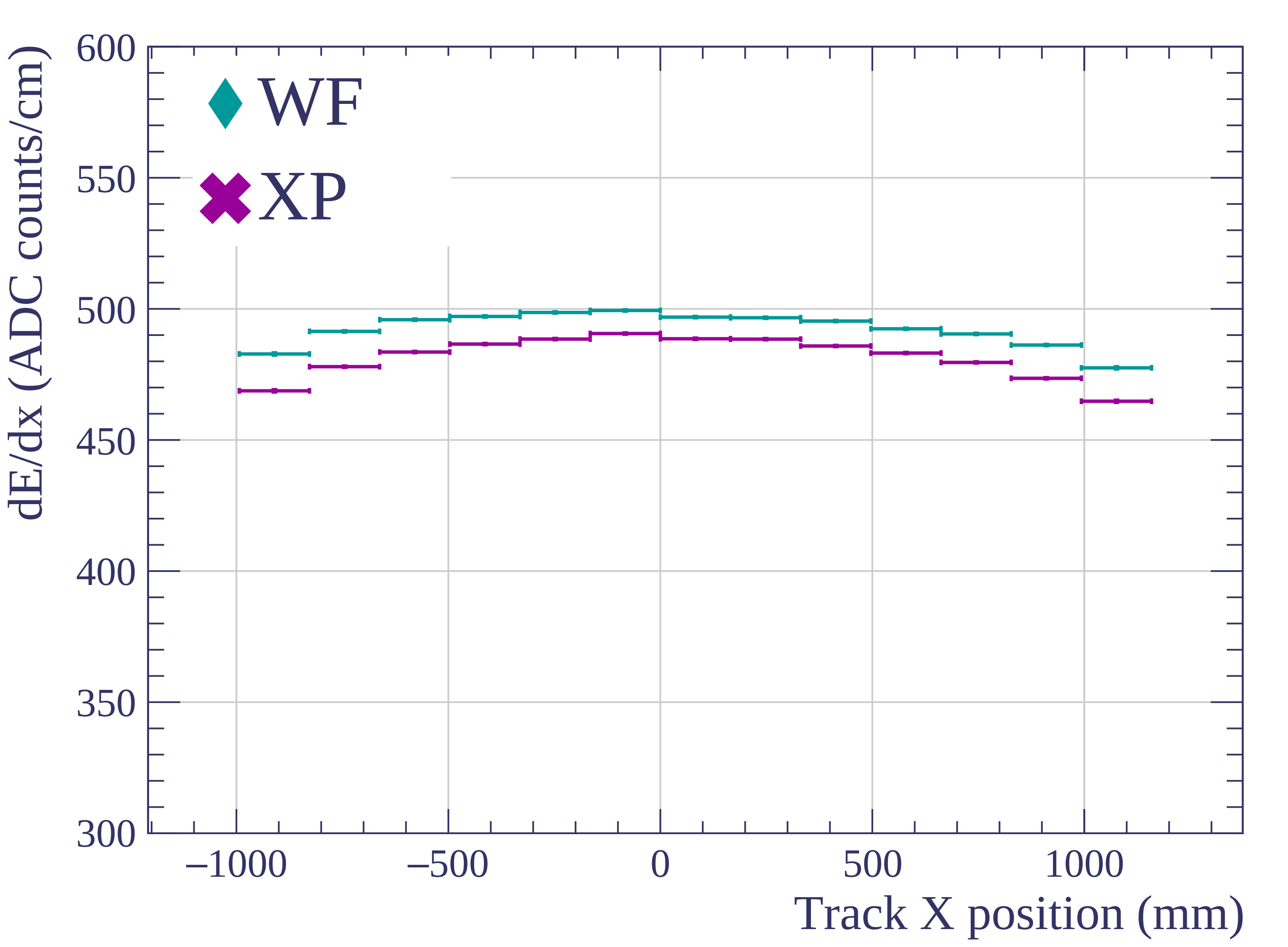
Energy loss in ERAM 24
XP
σ/μ = 19.79± 0.19 %
μ = 459.7± 0.8
σ = 91.0± 0.7
WF
σ/μ = 20.56± 0.21 %
μ = 462.6± 0.8
σ = 95.1± 0.8
Energy loss in ERAM 30
XP
σ/μ = 18.07± 0.26 %
μ = 485.7± 1.1
σ = 87.7± 1.0
WF
σ/μ = 18.43± 0.27 %
μ = 492.4± 1.2
σ = 90.7± 1.1
Energy loss in ERAM 28
XP
σ/μ = 17.55± 0.27 %
μ = 492.8± 1.2
σ = 86.5± 1.1
WF
σ/μ = 17.98± 0.26 %
μ = 499.9± 1.2
σ = 89.9± 1.1
Energy loss in ERAM 19
XP
σ/μ = 18.76± 0.17 %
μ = 488.2± 0.8
σ = 91.6± 0.7
WF
σ/μ = 19.38± 0.19 %
μ = 490.8± 0.8
σ = 95.1± 0.8
Energy loss in ERAM 21
XP
σ/μ = 15.42± 0.06 %
μ = 475.1± 0.3
σ = 73.3± 0.3
WF
σ/μ = 15.75± 0.07 %
μ = 489.6± 0.3
σ = 77.1± 0.3
Energy loss in ERAM 13
XP
σ/μ = 14.76± 0.06 %
μ = 496.7± 0.3
σ = 73.3± 0.3
WF
σ/μ = 15.31± 0.06 %
μ = 508.5± 0.3
σ = 77.9± 0.3
Energy loss in ERAM 9
XP
σ/μ = 14.70± 0.06 %
μ = 509.0± 0.4
σ = 74.8± 0.3
WF
σ/μ = 15.17± 0.07 %
μ = 521.4± 0.4
σ = 79.1± 0.3
Energy loss in ERAM 2
XP
σ/μ = 16.44± 0.07 %
μ = 496.4± 0.3
σ = 81.6± 0.3
WF
σ/μ = 16.82± 0.07 %
μ = 509.7± 0.4
σ = 85.7± 0.3
Energy loss in ERAM 26
XP
σ/μ = 18.29± 0.16 %
μ = 491.2± 0.7
σ = 89.8± 0.6
WF
σ/μ = 18.99± 0.16 %
μ = 494.2± 0.7
σ = 93.9± 0.7
Energy loss in ERAM 17
XP
σ/μ = 16.90± 0.18 %
μ = 497.4± 0.9
σ = 84.1± 0.8
WF
σ/μ = 17.35± 0.19 %
μ = 504.2± 0.9
σ = 87.5± 0.8
Energy loss in ERAM 23
XP
σ/μ = 16.75± 0.18 %
μ = 489.4± 0.8
σ = 82.0± 0.7
WF
σ/μ = 17.33± 0.19 %
μ = 498.7± 0.9
σ = 86.4± 0.8
Energy loss in ERAM 29
XP
σ/μ = 17.97± 0.16 %
μ = 481.4± 0.7
σ = 86.5± 0.6
WF
σ/μ = 18.66± 0.15 %
μ = 488.8± 0.7
σ = 91.2± 0.6
Energy loss in ERAM 1
XP
σ/μ = 15.47± 0.06 %
μ = 490.8± 0.3
σ = 75.9± 0.3
WF
σ/μ = 15.71± 0.06 %
μ = 501.5± 0.3
σ = 78.8± 0.3
Energy loss in ERAM 10
XP
σ/μ = 15.08± 0.06 %
μ = 492.7± 0.3
σ = 74.3± 0.3
WF
σ/μ = 15.20± 0.06 %
μ = 505.3± 0.3
σ = 76.8± 0.3
Energy loss in ERAM 11
XP
σ/μ = 15.12± 0.06 %
μ = 494.6± 0.3
σ = 74.8± 0.3
WF
σ/μ = 15.13± 0.06 %
μ = 508.1± 0.3
σ = 76.9± 0.3
Energy loss in ERAM 3
XP
σ/μ = 15.60± 0.06 %
μ = 479.8± 0.3
σ = 74.9± 0.2
WF
σ/μ = 16.12± 0.07 %
μ = 492.8± 0.3
σ = 79.4± 0.3
Count
dE/dx (ADC counts/cm)

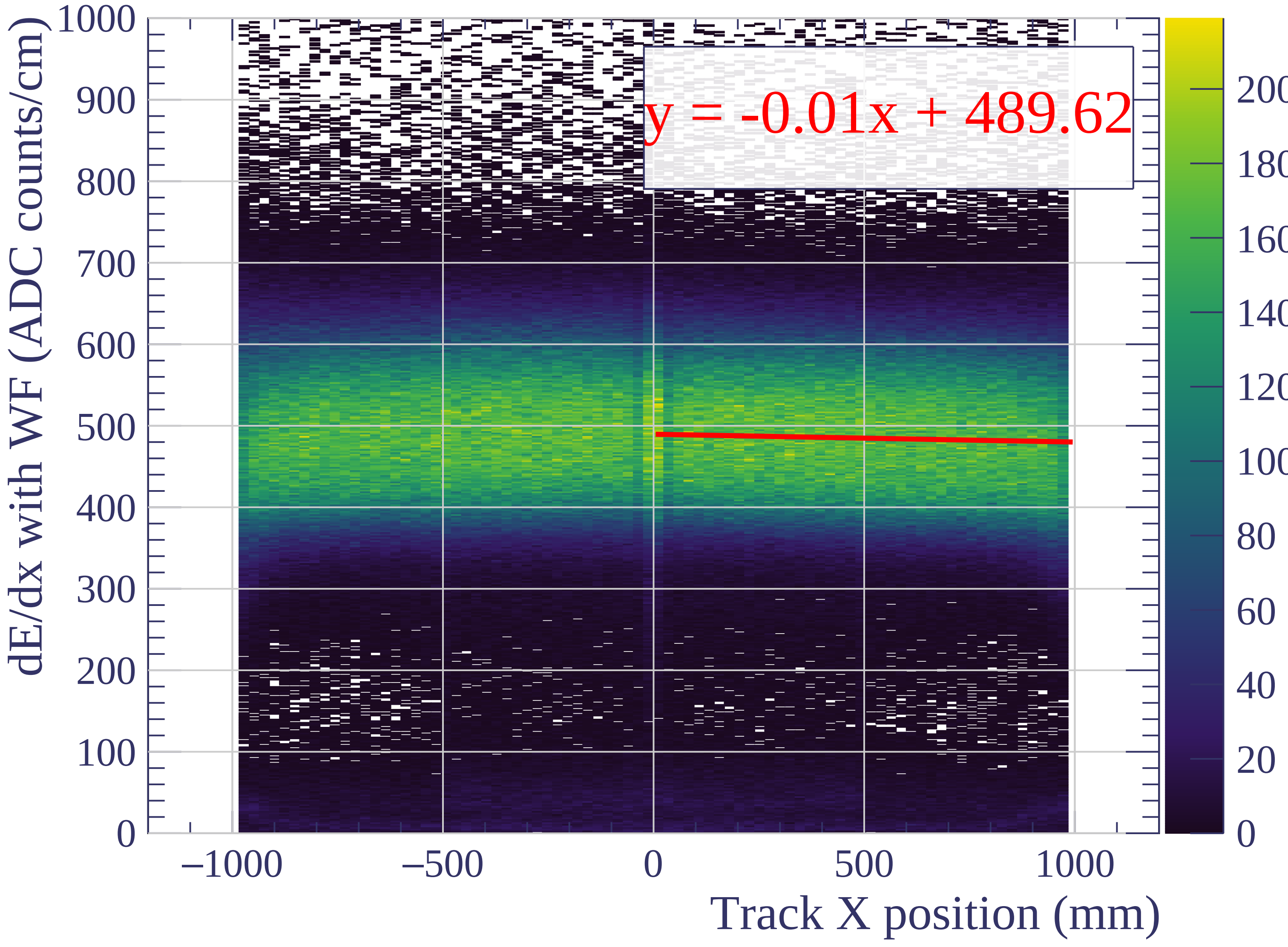

Energy loss in ERAM 47
XP σ/μ = 19.29± 0.18 % μ = 467.8± 0.8 σ = 90.2± 0.7
WF σ/μ = 20.18± 0.19 % μ = 465.9± 0.8 σ = 94.0± 0.7
Energy loss in ERAM 16
XP σ/μ = 18.34± 0.25 % μ = 482.6± 1.1 σ = 88.5± 1.0
WF σ/μ = 18.64± 0.25 % μ = 486.5± 1.1 σ = 90.7± 1.0
Energy loss in ERAM 14
XP σ/μ = 18.38± 0.26 % μ = 502.0± 1.2 σ = 92.3± 1.1
WF σ/μ = 18.96± 0.28 % μ = 508.6± 1.2 σ = 96.5± 1.2
Energy loss in ERAM 15
XP σ/μ = 19.17± 0.17 % μ = 519.8± 0.9 σ = 99.6± 0.7
WF σ/μ = 19.80± 0.19 % μ = 525.1± 0.9 σ = 104.0± 0.8
Energy loss in ERAM 42
XP σ/μ = 15.47± 0.07 % μ = 458.1± 0.3 σ = 70.9± 0.3
WF σ/μ = 15.62± 0.06 % μ = 466.3± 0.3 σ = 72.8± 0.2
Energy loss in ERAM 45
XP σ/μ = 15.27± 0.07 % μ = 472.1± 0.3 σ = 72.1± 0.3
WF σ/μ = 15.25± 0.06 % μ = 485.0± 0.3 σ = 74.0± 0.3
Energy loss in ERAM 37
XP σ/μ = 15.09± 0.07 % μ = 490.5± 0.3 σ = 74.0± 0.3
WF σ/μ = 15.20± 0.06 % μ = 503.1± 0.3 σ = 76.5± 0.3
Energy loss in ERAM 36
XP σ/μ = 15.85± 0.06 % μ = 486.7± 0.3 σ = 77.2± 0.3
WF σ/μ = 16.10± 0.07 % μ = 499.8± 0.3 σ = 80.5± 0.3
Energy loss in ERAM 20
XP σ/μ = 18.15± 0.15 % μ = 469.2± 0.6 σ = 85.2± 0.6
WF σ/μ = 18.63± 0.16 % μ = 473.4± 0.7 σ = 88.2± 0.6
Energy loss in ERAM 38
XP σ/μ = 16.94± 0.17 % μ = 470.6± 0.8 σ = 79.7± 0.7
WF σ/μ = 17.18± 0.17 % μ = 474.6± 0.8 σ = 81.5± 0.6
Energy loss in ERAM 7
XP σ/μ = 16.62± 0.17 % μ = 486.9± 0.8 σ = 80.9± 0.7
WF σ/μ = 16.92± 0.17 % μ = 493.2± 0.8 σ = 83.5± 0.7
Energy loss in ERAM 44
XP σ/μ = 18.91± 0.15 % μ = 464.4± 0.6 σ = 87.8± 0.6
WF σ/μ = 19.61± 0.17 % μ = 469.1± 0.7 σ = 92.0± 0.6
Energy loss in ERAM 43
XP σ/μ = 15.48± 0.06 % μ = 486.8± 0.3 σ = 75.3± 0.2
WF σ/μ = 15.62± 0.06 % μ = 496.8± 0.3 σ = 77.6± 0.2
Energy loss in ERAM 39
XP σ/μ = 14.87± 0.06 % μ = 468.0± 0.3 σ = 69.6± 0.2
WF σ/μ = 15.40± 0.06 % μ = 475.7± 0.3 σ = 73.3± 0.3
Energy loss in ERAM 41
XP σ/μ = 14.87± 0.06 % μ = 483.9± 0.3 σ = 72.0± 0.2
WF σ/μ = 15.44± 0.07 % μ = 494.9± 0.3 σ = 76.4± 0.3
Energy loss in ERAM 46
XP σ/μ = 15.63± 0.06 % μ = 466.4± 0.3 σ = 72.9± 0.2
WF σ/μ = 15.78± 0.06 % μ = 478.0± 0.3 σ = 75.4± 0.2
Count
dE/dx (ADC counts/cm)


Entries 1558128
Mean x 479.2
Mean y 469.4
Std Dev x 113
Std Dev y 109.2
3 121 6487
1902 1.5492e+06 6413
0 0 0
dE/dx with XP (ADC counts/cm)
dE/dx with WF (ADC counts/cm)
×10^3

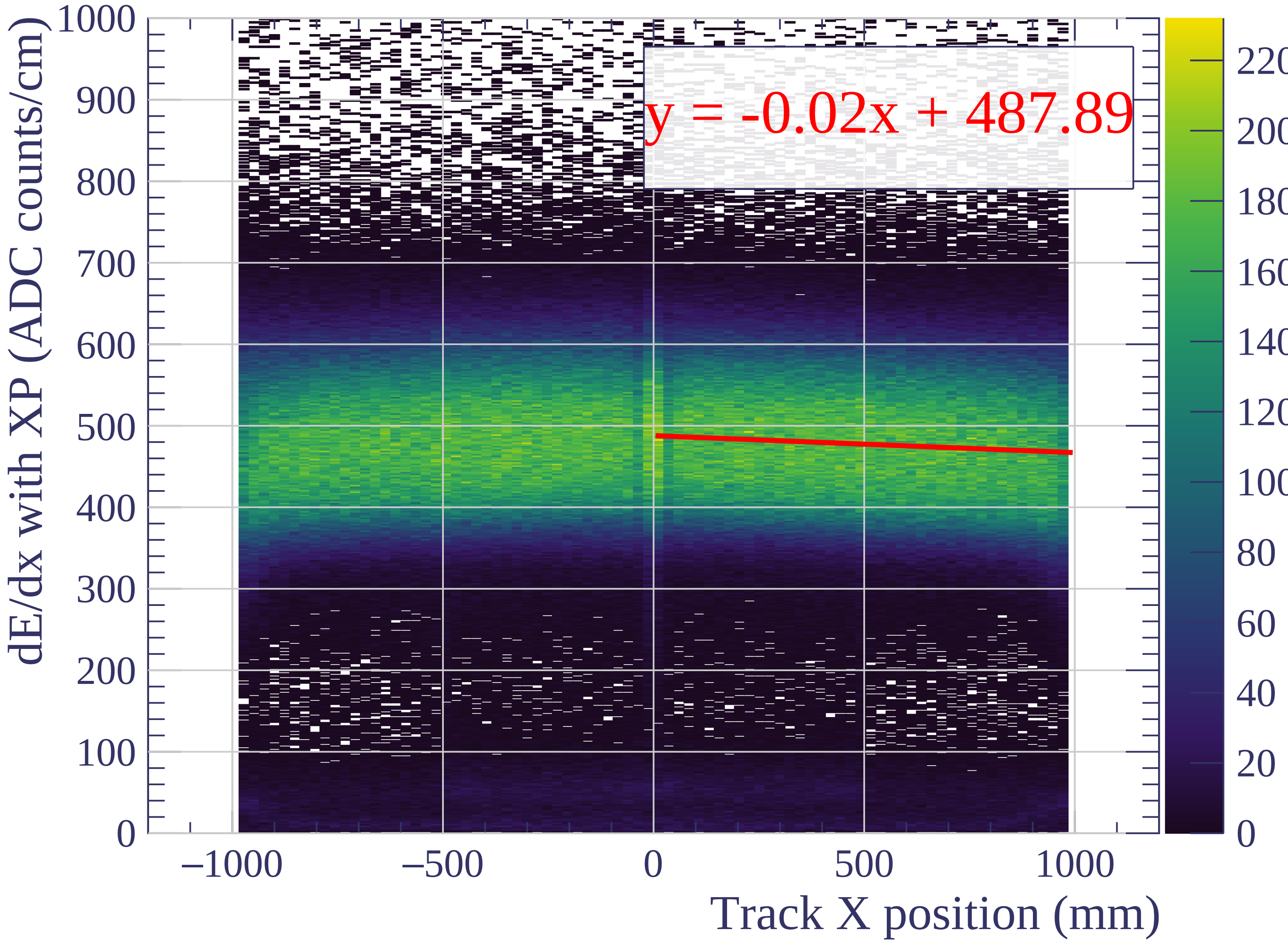


WF
XP
dE/dx resolution (%)
11
10
9
8
7
6
5
4
−1000
−500
0
500
1000
Track X position (mm)

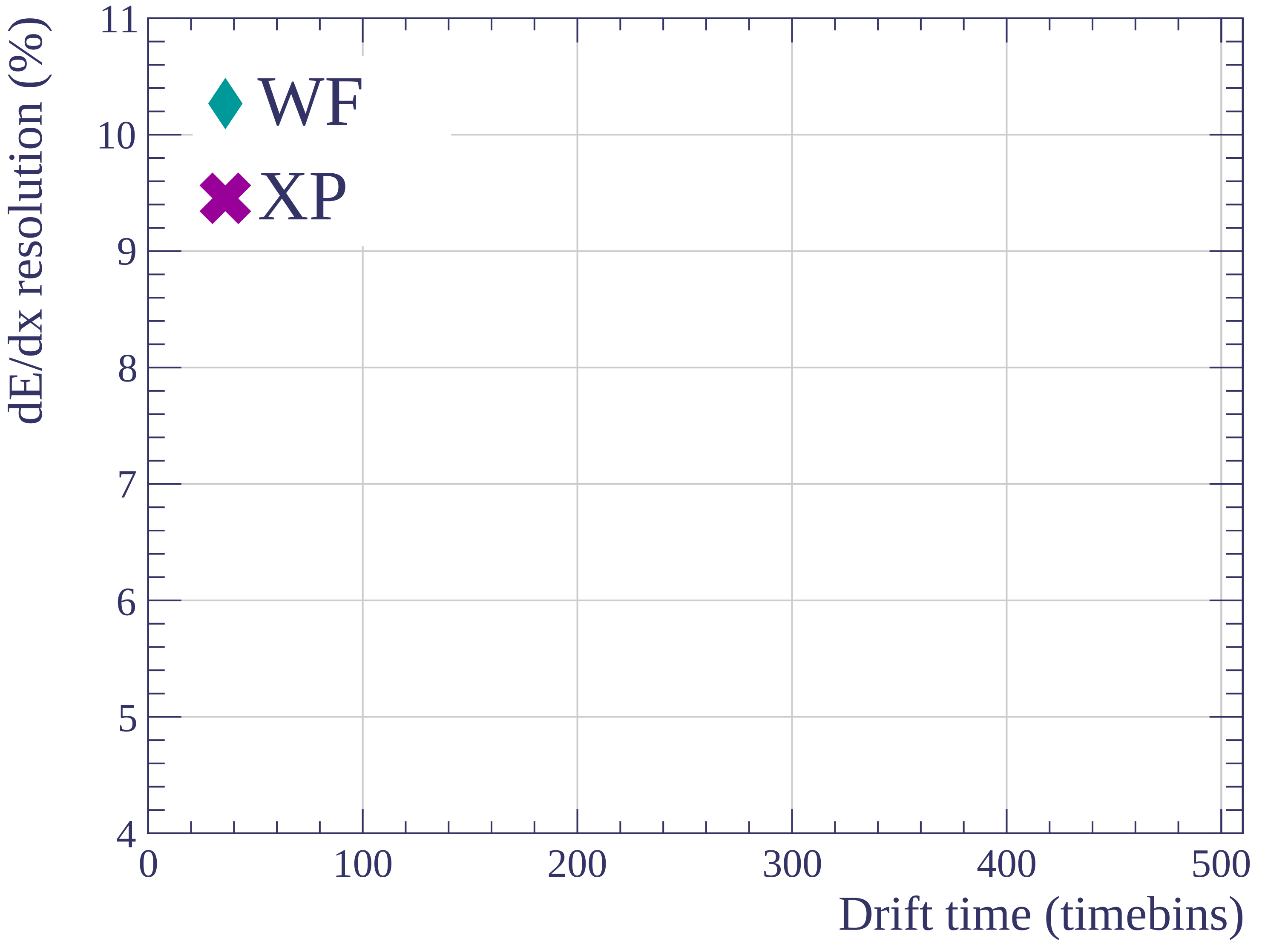

WF
XP
dE/dx (ADC counts/cm)
Track X position (mm)
300
350
400
450
500
550
600
−1000
−500
0
500
1000



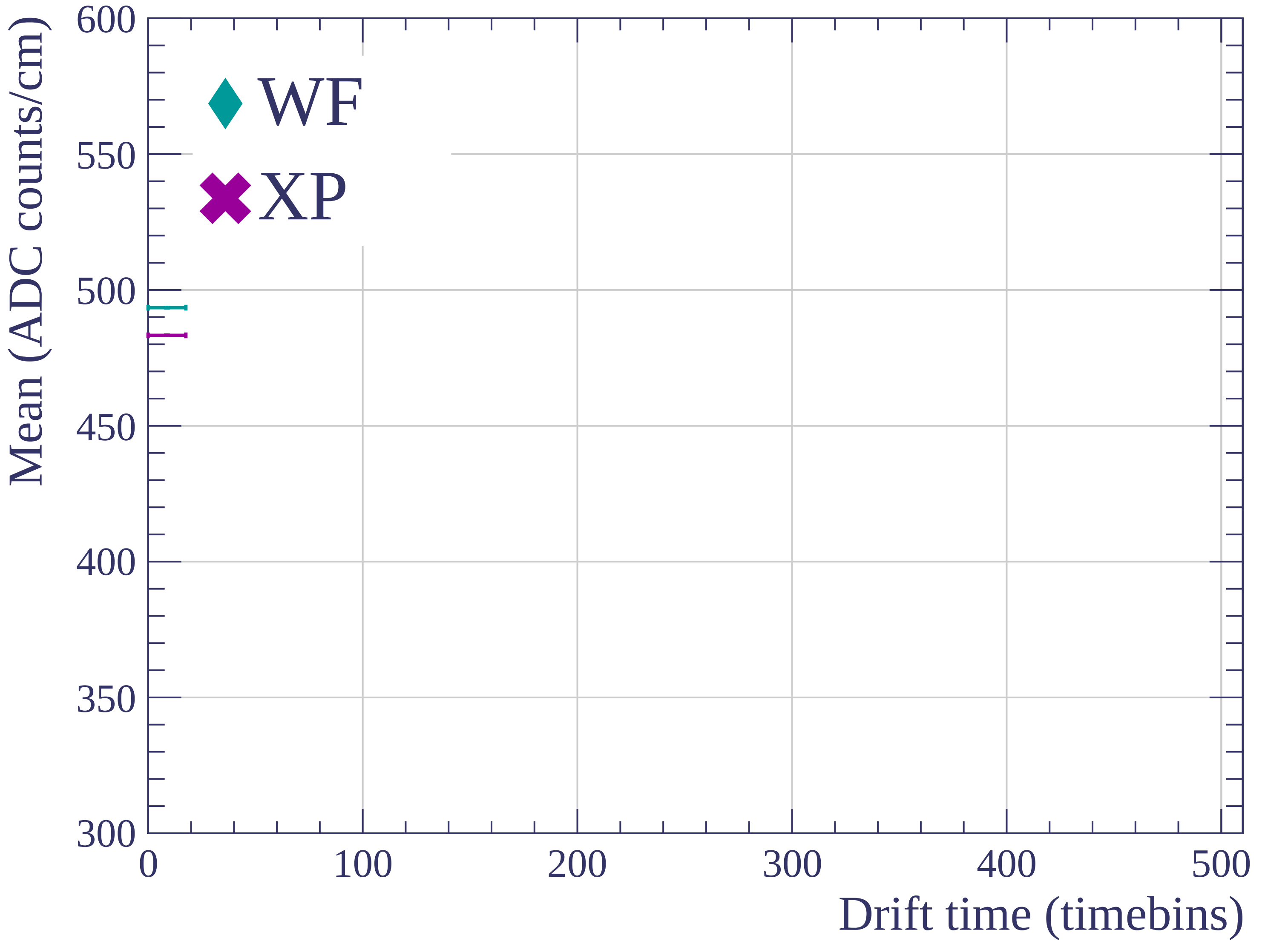
y = -0.01x + 489.62
dE/dx with WF (ADC counts/cm)
Track X position (mm)
0
100
200
300
400
500
600
700
800
900
1000
−1000
−500
0
500
1000
20
40
60
80

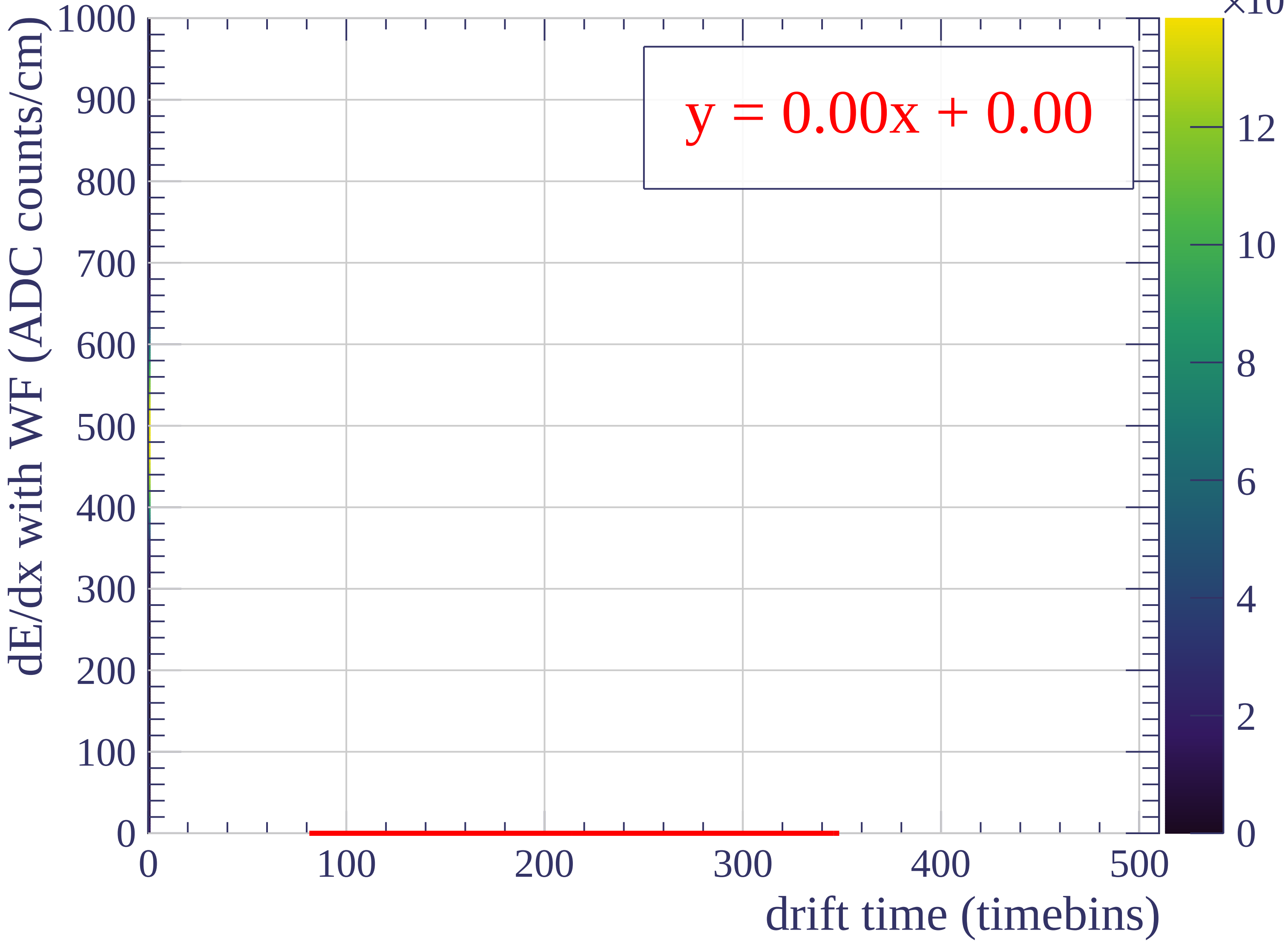


y = -0.02x + 487.89
dE/dx with XP (ADC counts/cm)
Track X position (mm)
0
100
200
300
400
500
600
700
800
900
1000
−1000
−500
0
500
1000

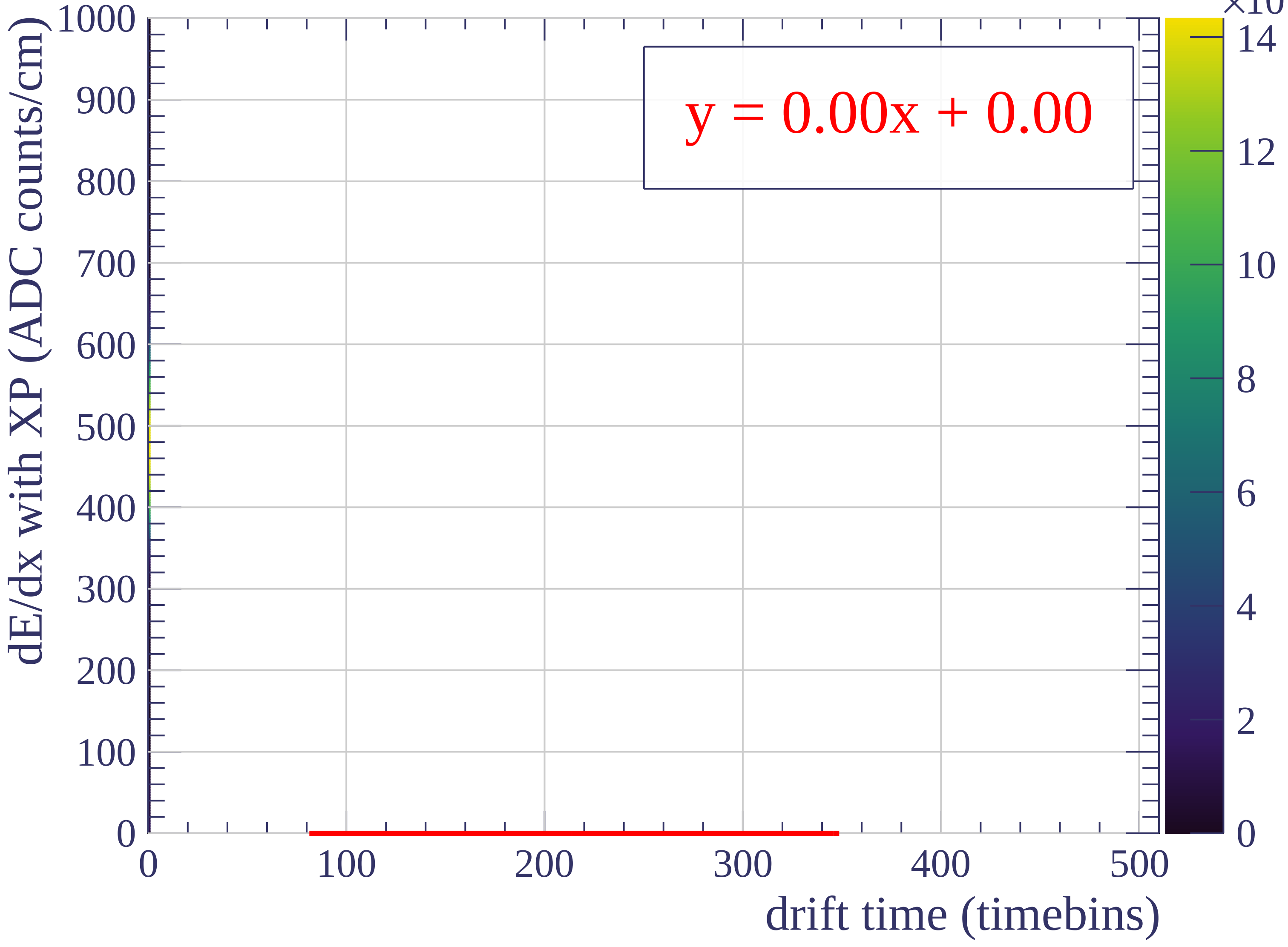


WF
XP
dE/dx resolution (%)
Drift time (timebins)
4
5
6
7
8
9
10
11
0
100
200
300
400
500

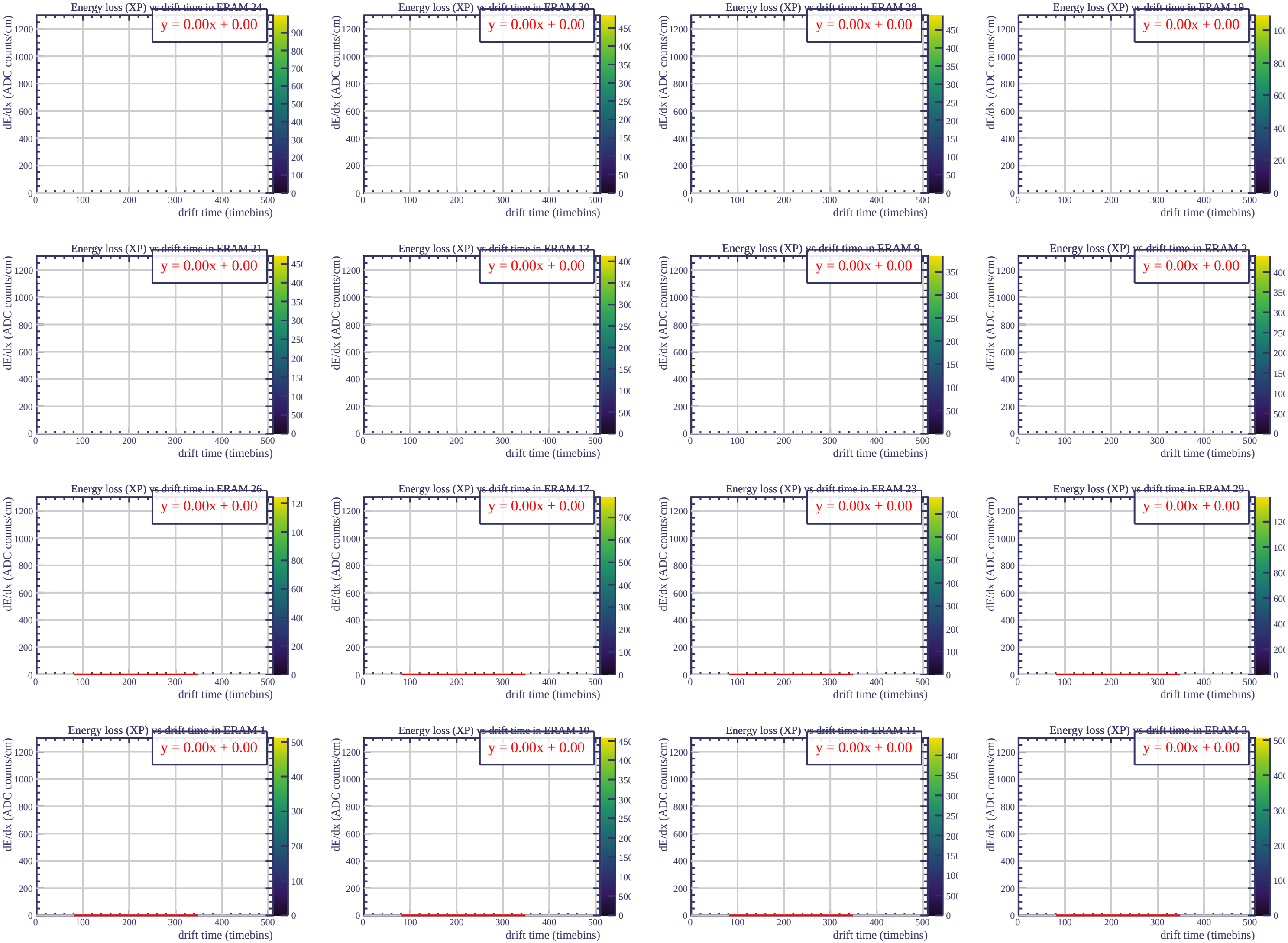

WF
XP
Mean (ADC counts/cm)
600
550
500
450
400
350
300
0
100
200
300
400
500
Drift time (timebins)



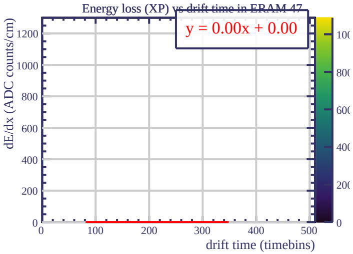
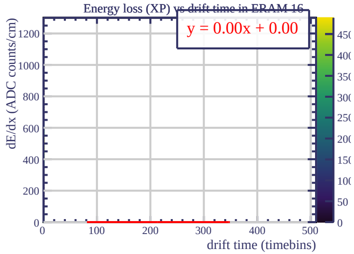
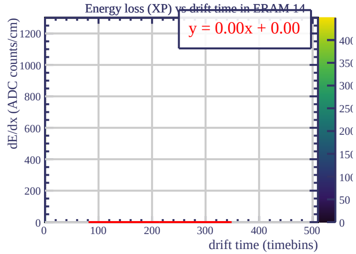
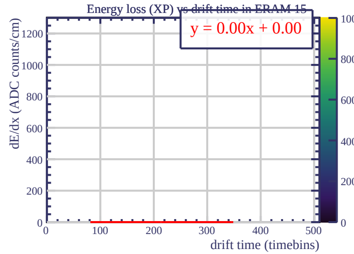
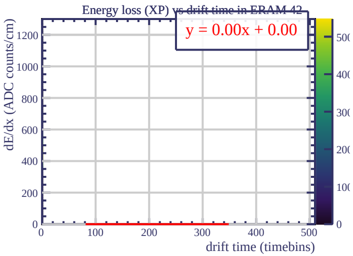
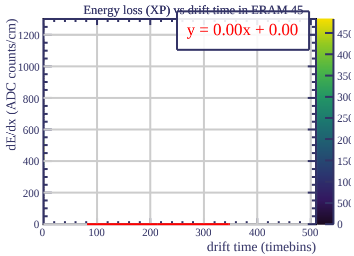
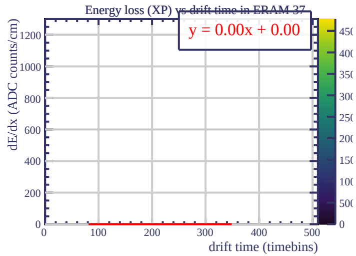
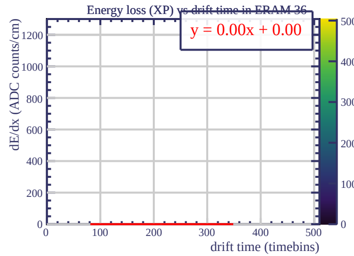
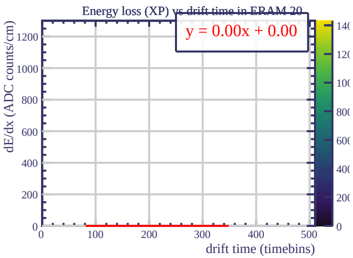
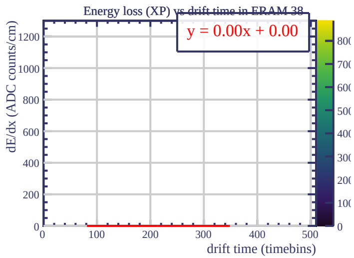
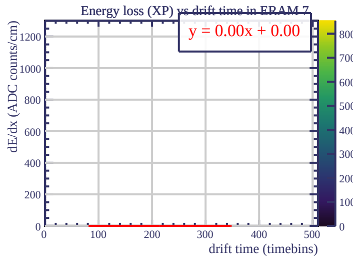
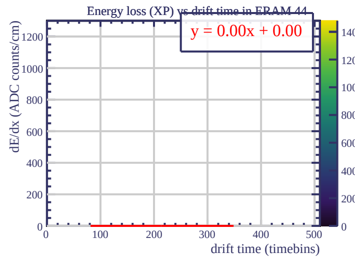
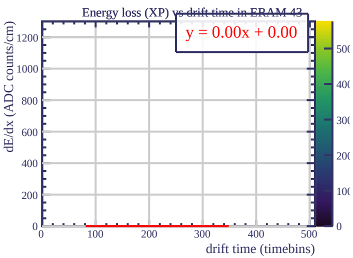
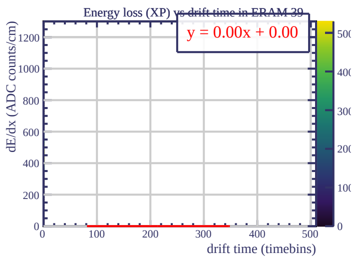
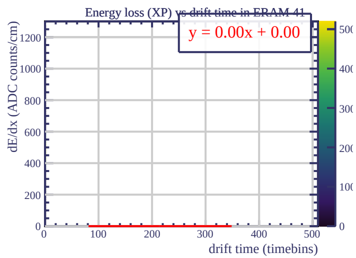
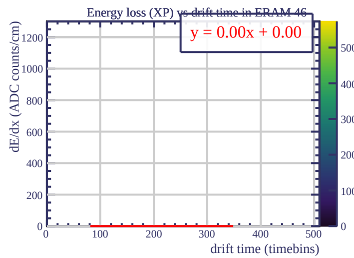
y = 0.00x + 0.00
dE/dx with WF (ADC counts/cm)
drift time (timebins)
1000
900
800
700
600
500
400
300
200
100
0
0
100
200
300
400
500
12
10
8
6
4
2
0

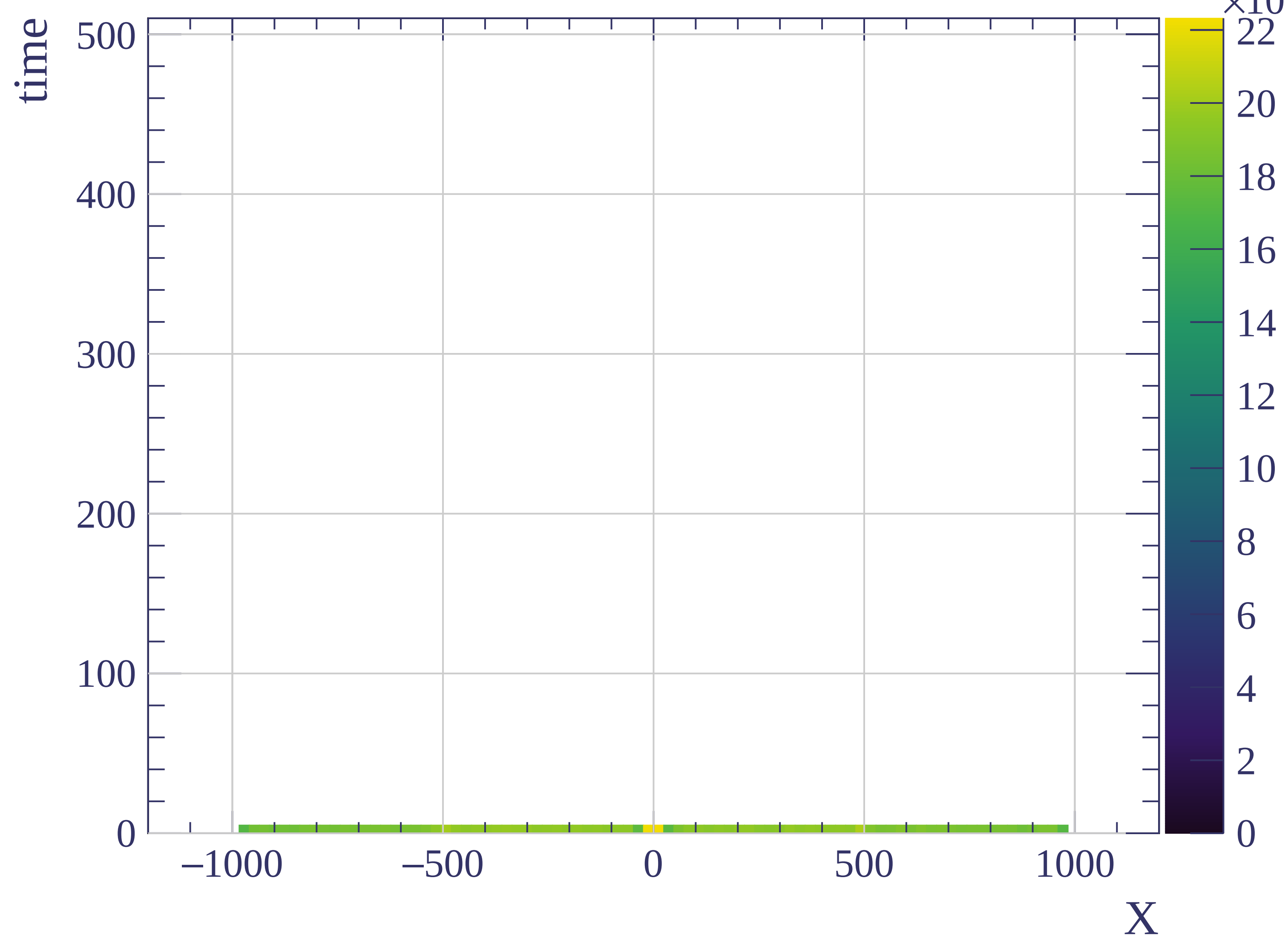


y = 0.00x + 0.00
dE/dx with XP (ADC counts/cm)
drift time (timebins)
0
100
200
300
400
500
600
700
800
900
1000
0
100
200
300
400
500
0
2
4
6
8
10
12
14

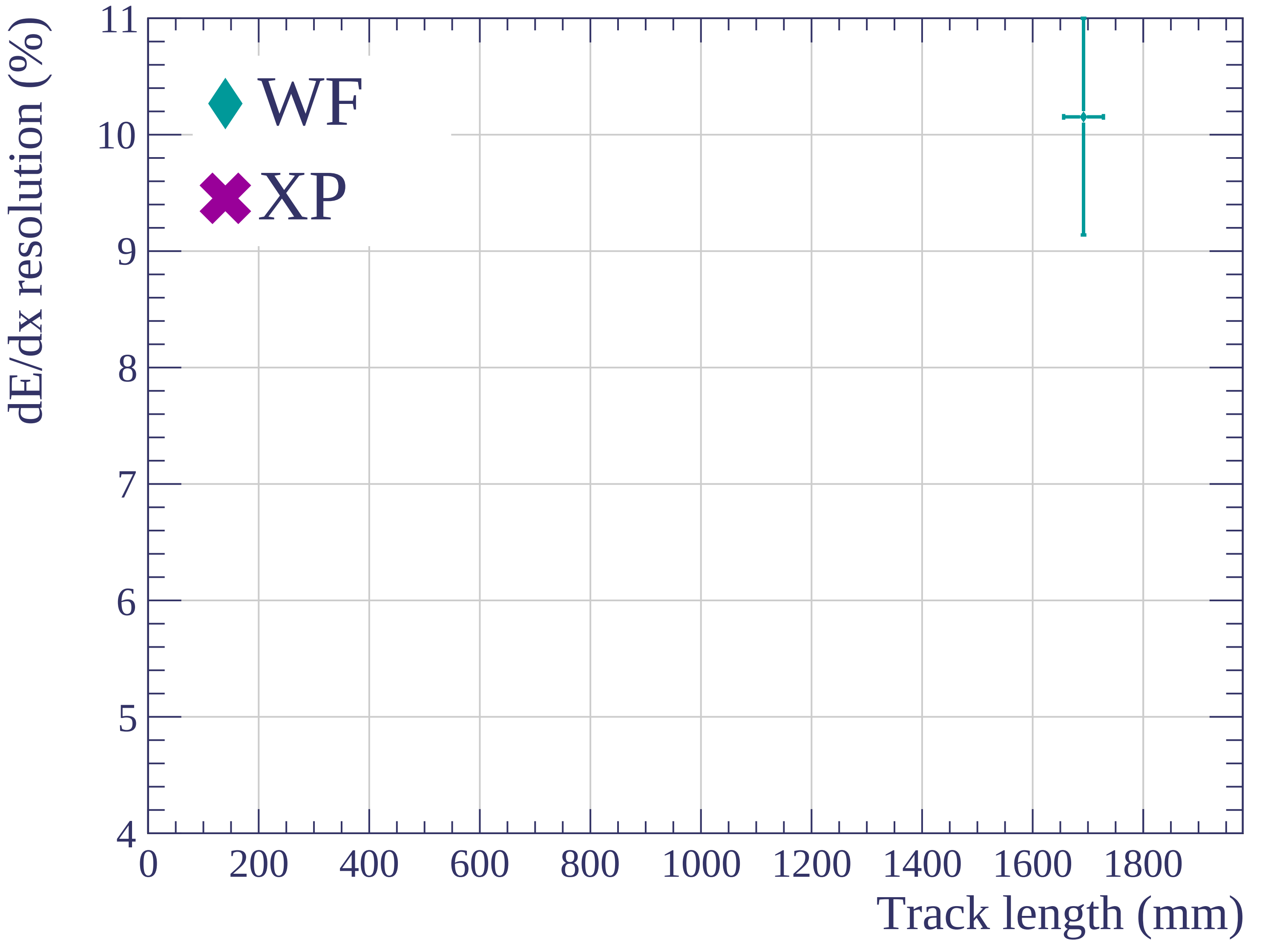


Energy loss (XP) vs drift time in ERAM 24
Energy loss (XP) vs drift time in ERAM 30
Energy loss (XP) vs drift time in ERAM 28
Energy loss (XP) vs drift time in ERAM 19
Energy loss (XP) vs drift time in ERAM 21
Energy loss (XP) vs drift time in ERAM 13
Energy loss (XP) vs drift time in ERAM 9
Energy loss (XP) vs drift time in ERAM 2
Energy loss (XP) vs drift time in ERAM 26
Energy loss (XP) vs drift time in ERAM 17
Energy loss (XP) vs drift time in ERAM 23
Energy loss (XP) vs drift time in ERAM 29
Energy loss (XP) vs drift time in ERAM 1
Energy loss (XP) vs drift time in ERAM 10
Energy loss (XP) vs drift time in ERAM 11
Energy loss (XP) vs drift time in ERAM 3
y = 0.00x + 0.00
dE/dx (ADC counts/cm)
drift time (timebins)

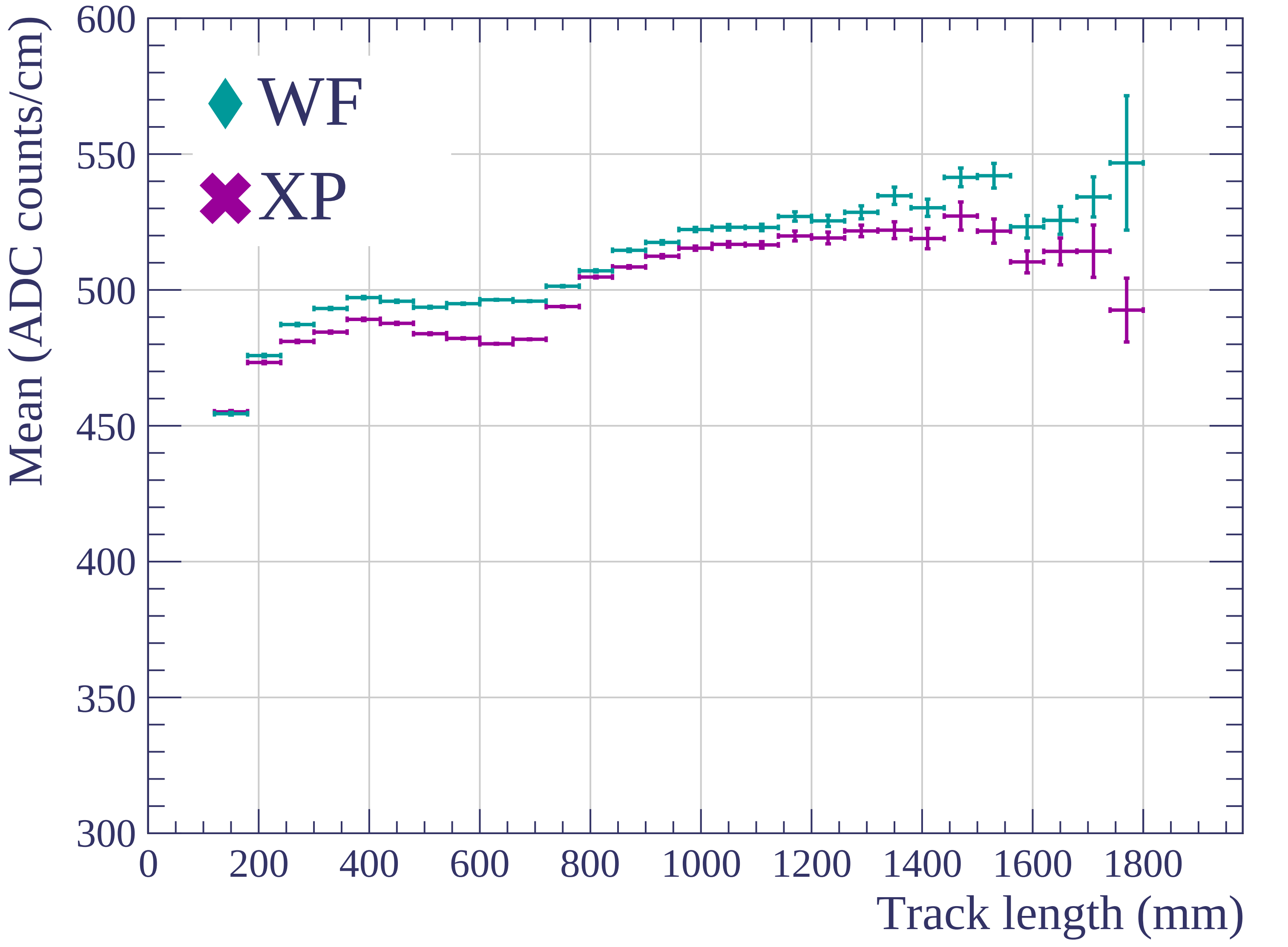

Energy loss (XP) vs drift time in ERAM 47
y = 0.00x + 0.00
dE/dx (ADC counts/cm)
drift time (timebins)

Energy loss (XP) vs drift time in ERAM 16
y = 0.00x + 0.00
dE/dx (ADC counts/cm)
drift time (timebins)

Energy loss (XP) vs drift time in ERAM 14
y = 0.00x + 0.00
dE/dx (ADC counts/cm)
drift time (timebins)

Energy loss (XP) vs drift time in ERAM 15
y = 0.00x + 0.00
dE/dx (ADC counts/cm)
drift time (timebins)

Energy loss (XP) vs drift time in ERAM 42
y = 0.00x + 0.00
dE/dx (ADC counts/cm)
drift time (timebins)

Energy loss (XP) vs drift time in ERAM 45
y = 0.00x + 0.00
dE/dx (ADC counts/cm)
drift time (timebins)

Energy loss (XP) vs drift time in ERAM 37
y = 0.00x + 0.00
dE/dx (ADC counts/cm)
drift time (timebins)

Energy loss (XP) vs drift time in ERAM 36
y = 0.00x + 0.00
dE/dx (ADC counts/cm)
drift time (timebins)

Energy loss (XP) vs drift time in ERAM 20
y = 0.00x + 0.00
dE/dx (ADC counts/cm)
drift time (timebins)

Energy loss (XP) vs drift time in ERAM 38
y = 0.00x + 0.00
dE/dx (ADC counts/cm)
drift time (timebins)

Energy loss (XP) vs drift time in ERAM 7
y = 0.00x + 0.00
dE/dx (ADC counts/cm)
drift time (timebins)

Energy loss (XP) vs drift time in ERAM 44
y = 0.00x + 0.00
dE/dx (ADC counts/cm)
drift time (timebins)

Energy loss (XP) vs drift time in ERAM 43
y = 0.00x + 0.00
dE/dx (ADC counts/cm)
drift time (timebins)

Energy loss (XP) vs drift time in ERAM 39
y = 0.00x + 0.00
dE/dx (ADC counts/cm)
drift time (timebins)

Energy loss (XP) vs drift time in ERAM 41
y = 0.00x + 0.00
dE/dx (ADC counts/cm)
drift time (timebins)

Energy loss (XP) vs drift time in ERAM 46
y = 0.00x + 0.00
dE/dx (ADC counts/cm)
drift time (timebins)



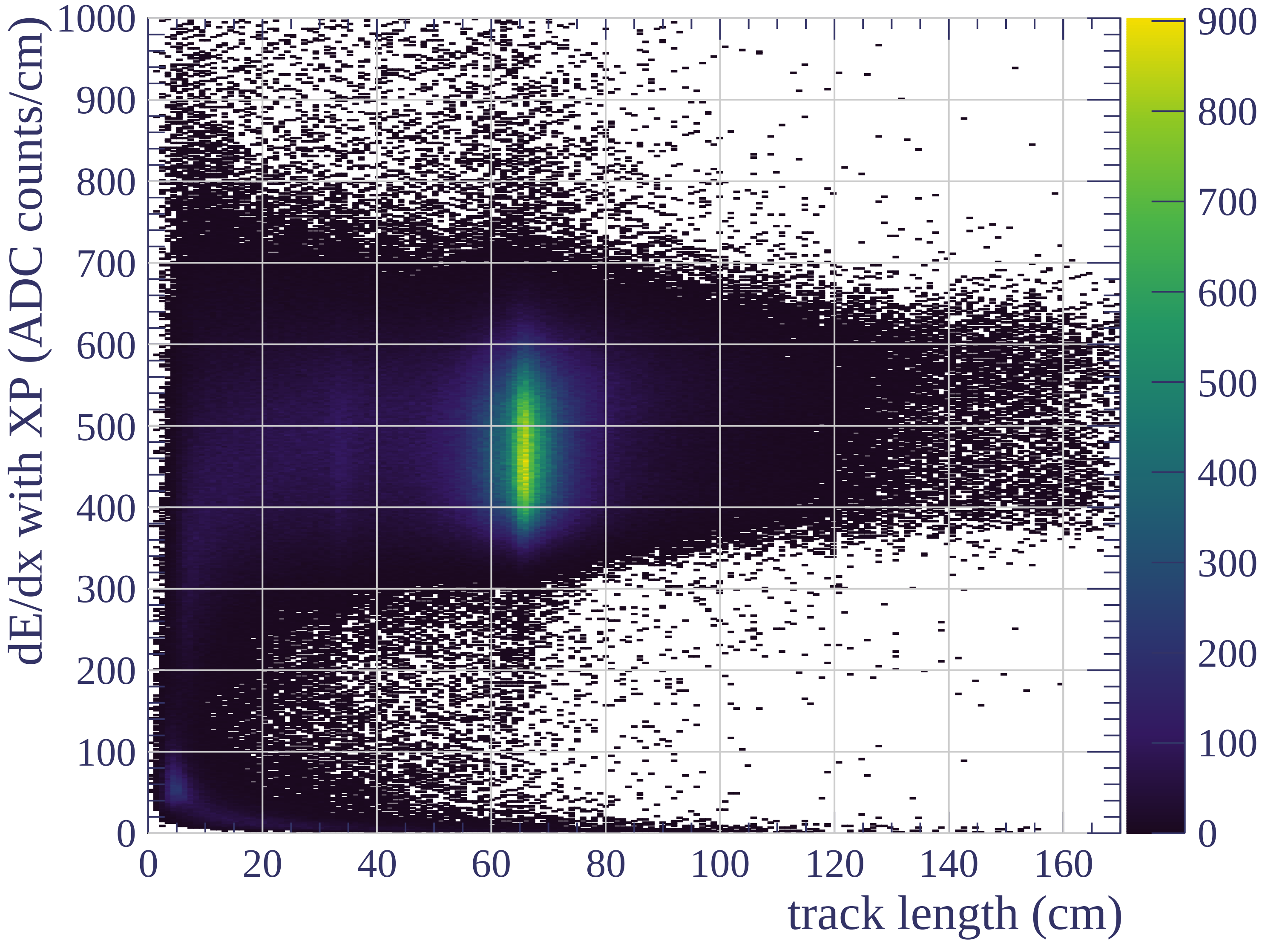
time
500
400
300
200
100
0
−1000
−500
0
500
1000
X
×10
22
20
18
16
14
12
10
8
6
4
2
0

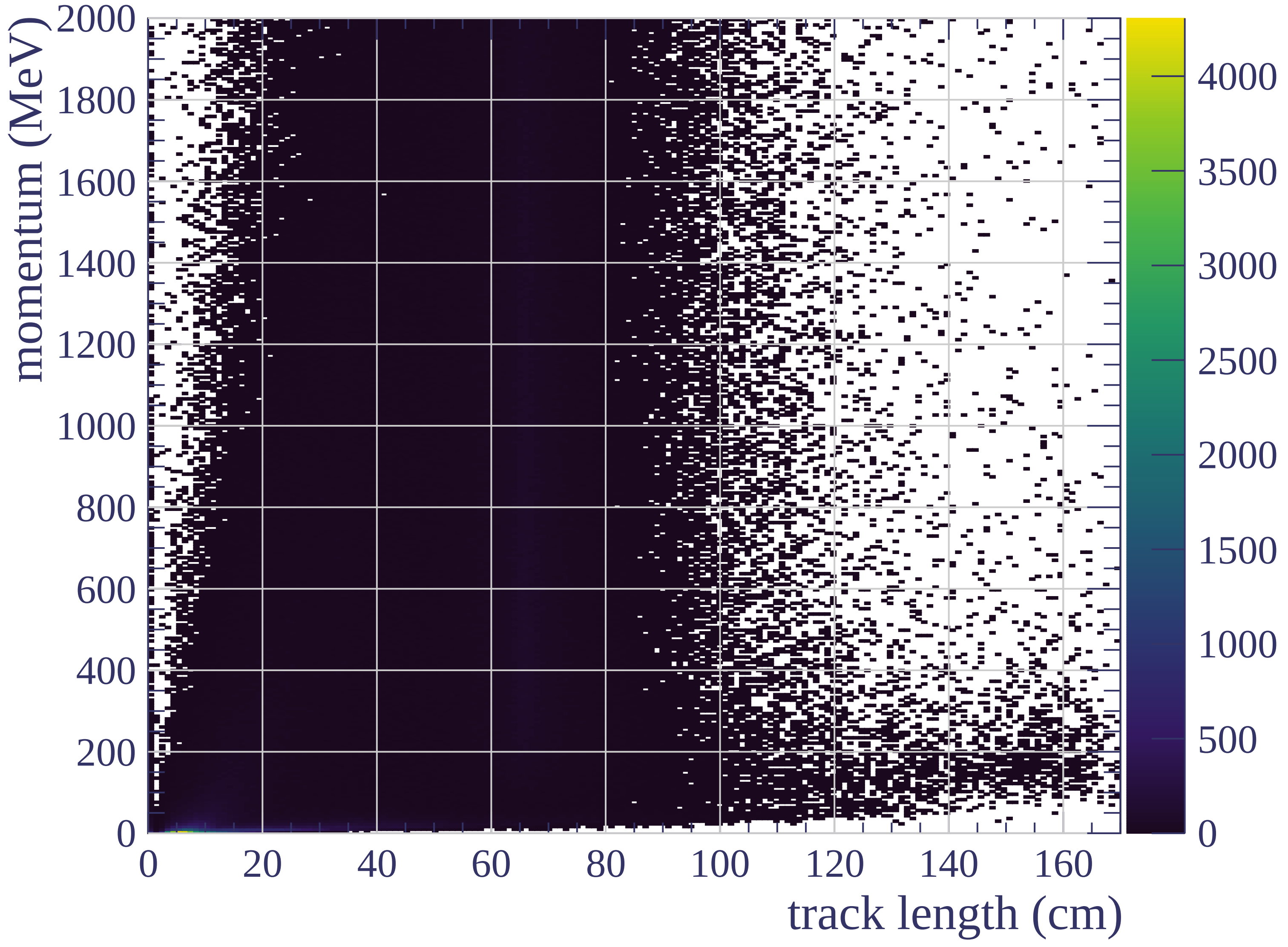


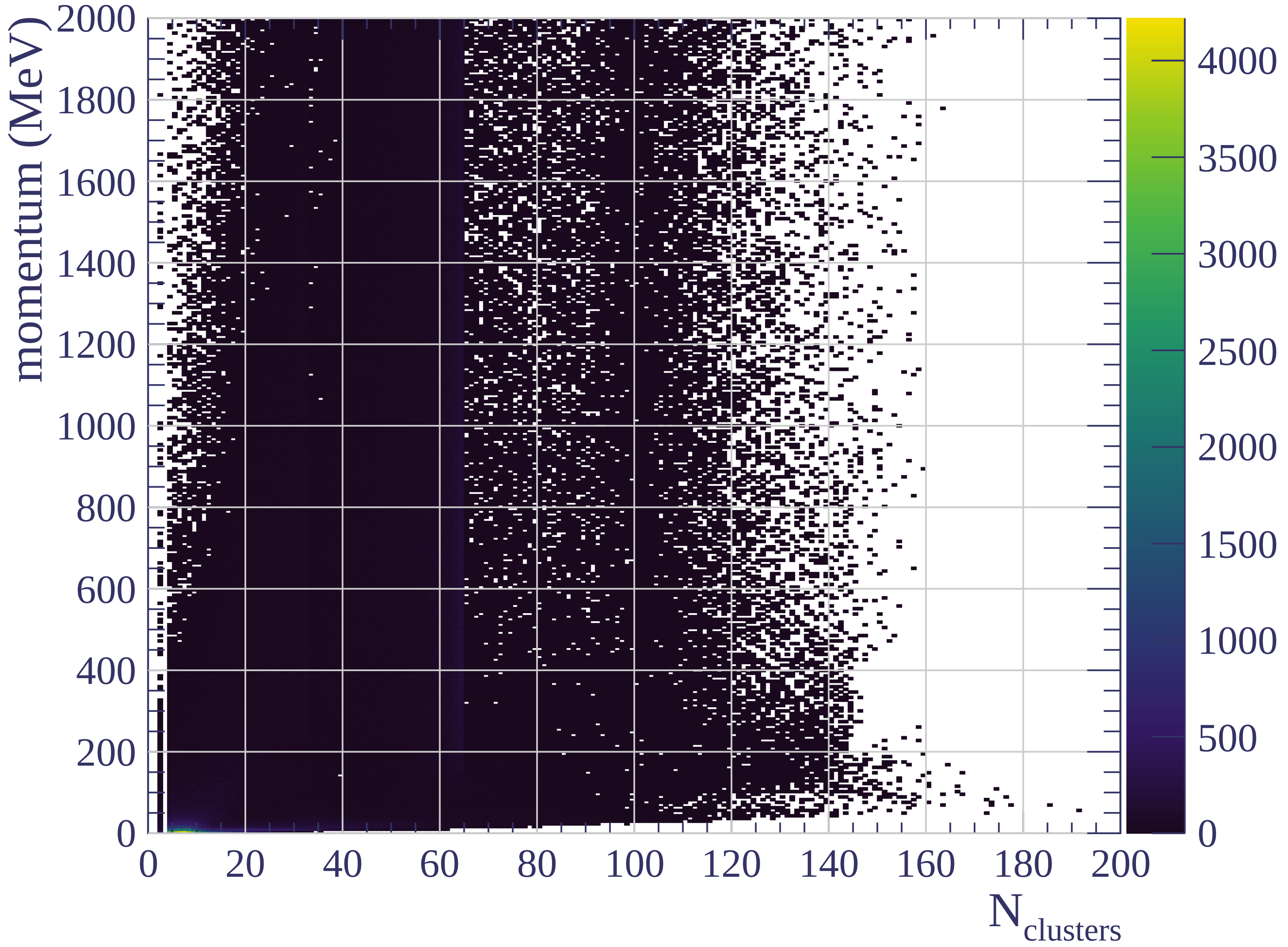
WF
XP
dE/dx resolution (%)
Track length (mm)
4
5
6
7
8
9
10
11
0
200
400
600
800
1000
1200
1400
1600
1800

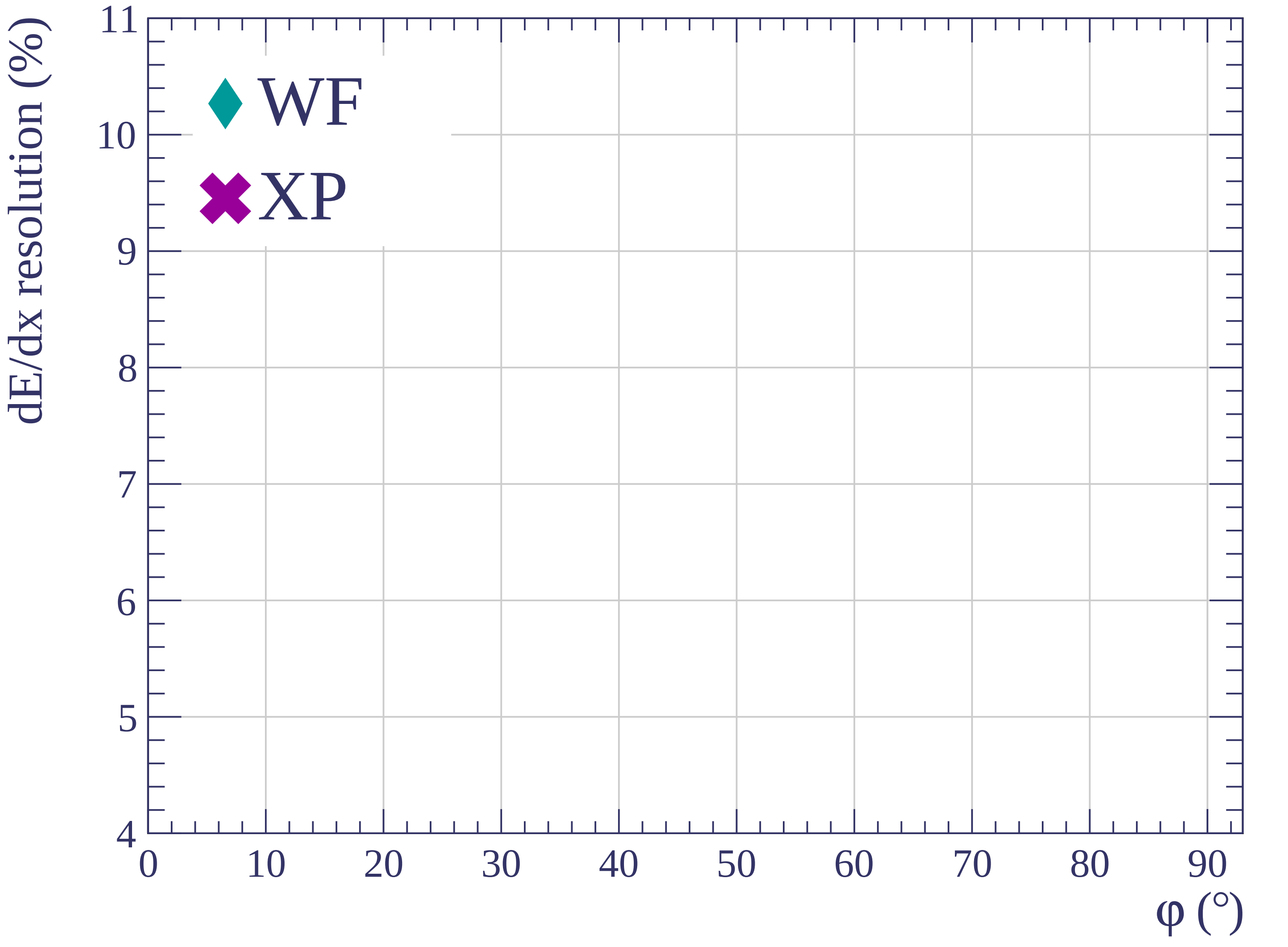


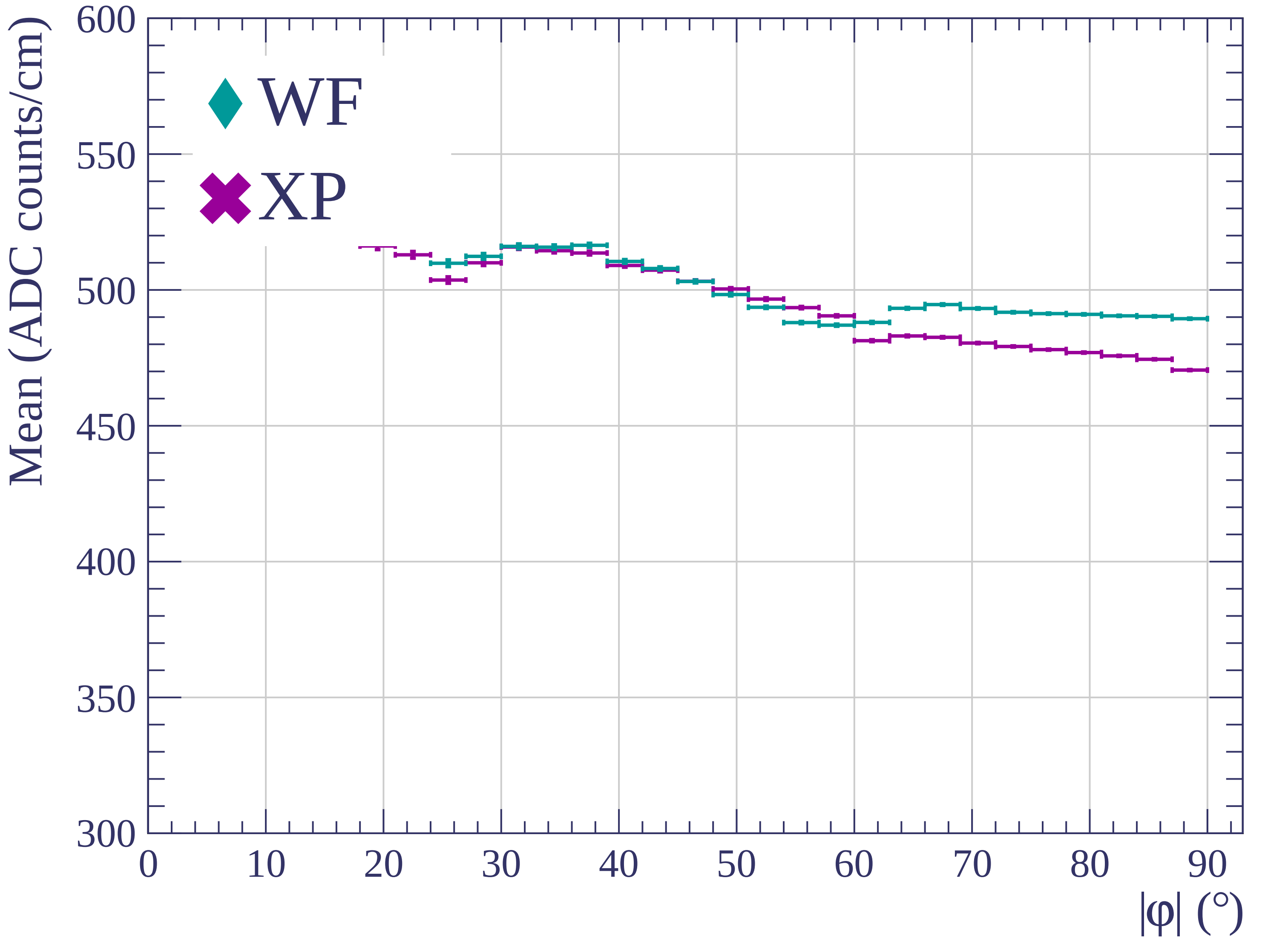
WF
XP
Mean (ADC counts/cm)
Track length (mm)
300
350
400
450
500
550
600
0
200
400
600
800
1000
1200
1400
1600
1800


dE/dx with XP (ADC counts/cm)
1000
900
800
700
600
500
400
300
200
100
0
0
20
40
60
80
100
120
140
160
track length (cm)
900
800
700
600
500
400
300
200
100
0

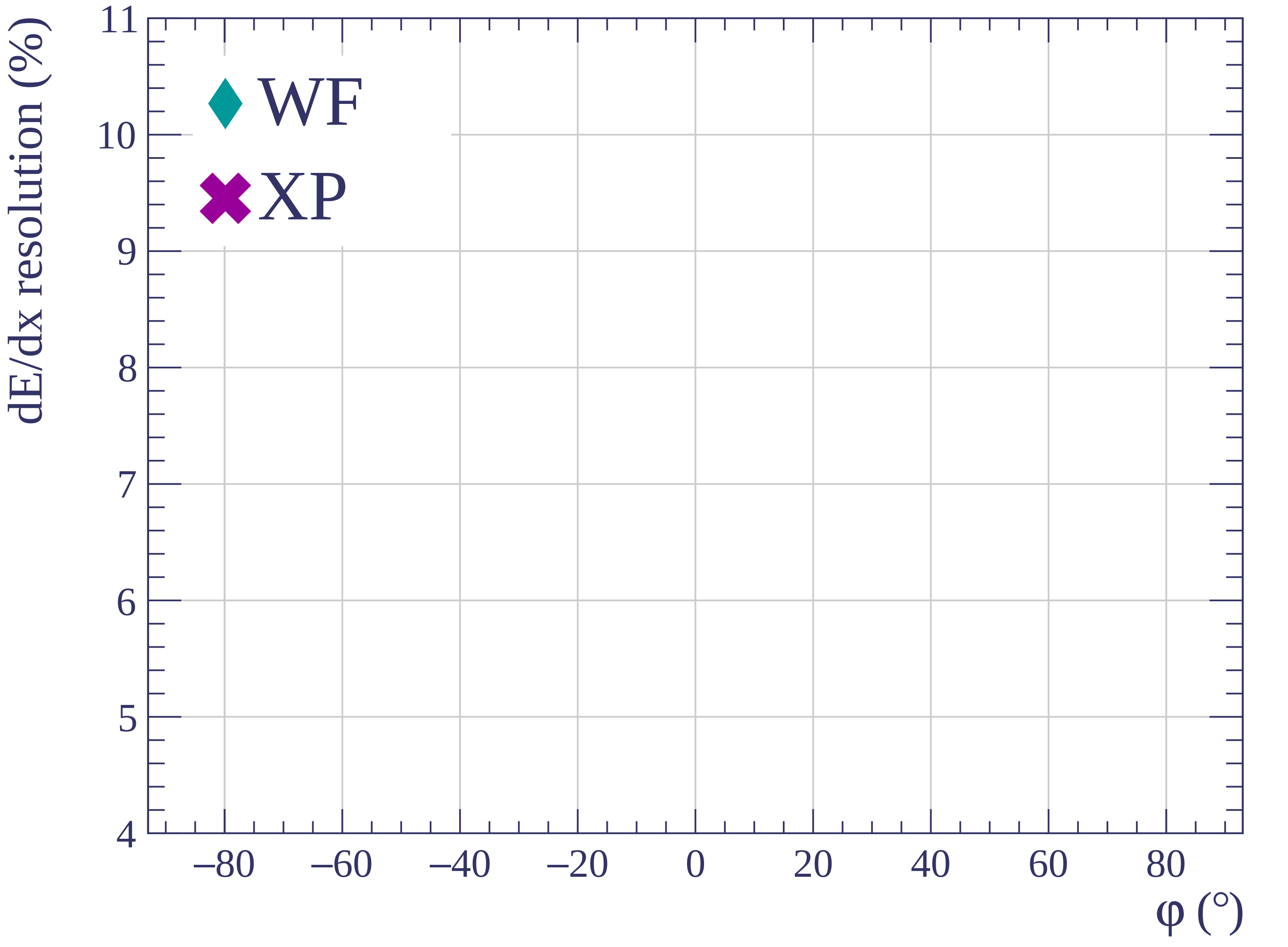

momentum (MeV)
track length (cm)
0
200
400
600
800
1000
1200
1400
1600
1800
2000
0
20
40
60
80
100
120
140
160
0
500
1000
1500
2000
2500
3000
3500
4000


momentum (MeV)
2000
1800
1600
1400
1200
1000
800
600
400
200
0
0
20
40
60
80
100
120
140
160
180
200
$N_{clusters}$
4000
3500
3000
2500
2000
1500
1000
500
0

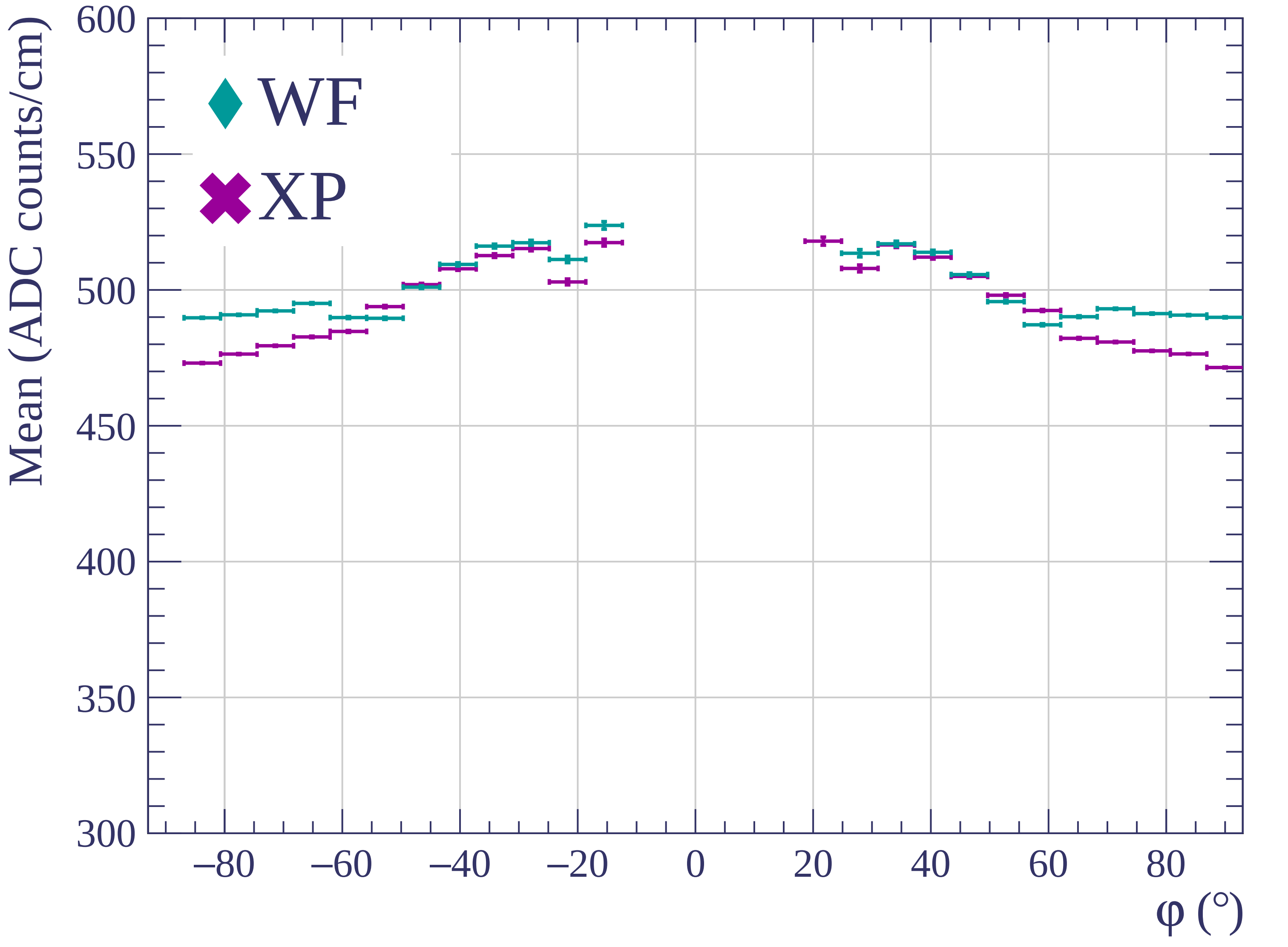

WF
XP
dE/dx resolution (%)
φ (°)
4
5
6
7
8
9
10
11
0
10
20
30
40
50
60
70
80
90



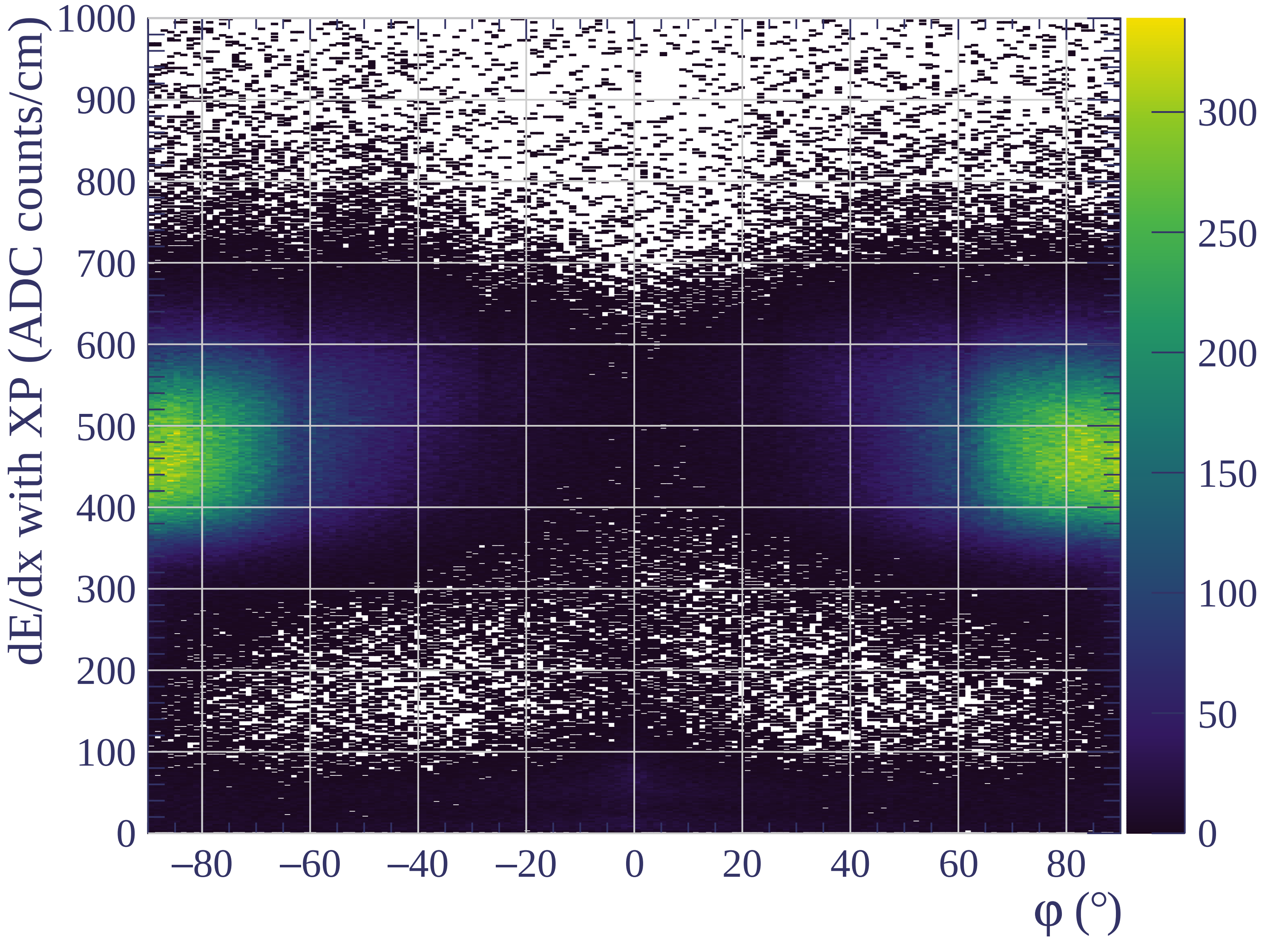
Mean (ADC counts/cm)
600
550
500
450
400
350
300
WF
XP
0
10
20
30
40
50
60
70
80
90
|φ| (°)

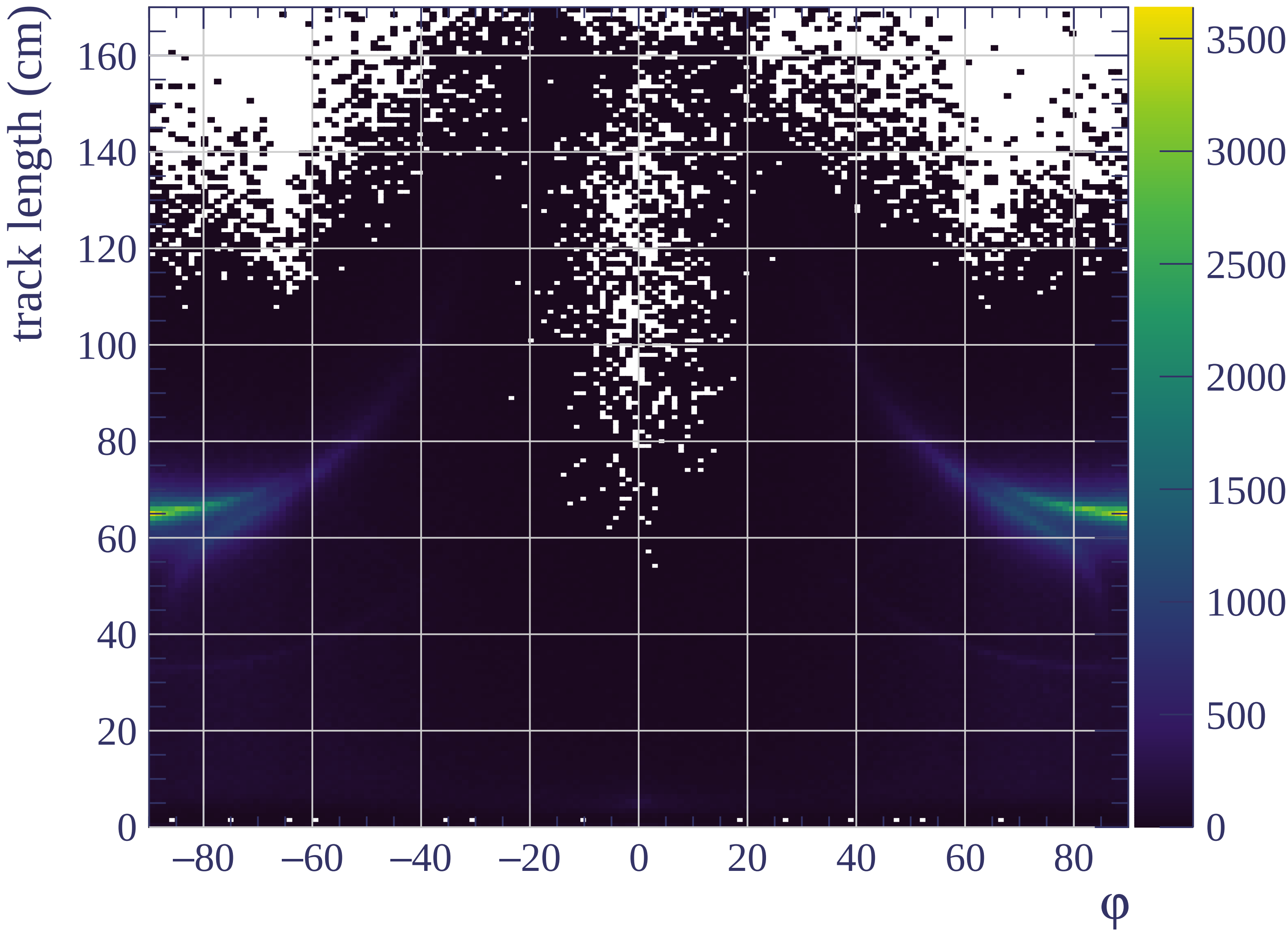


dE/dx resolution (%)
φ (°)
WF
XP
11
10
9
8
7
6
5
4
−80
−60
−40
−20
0
20
40
60
80

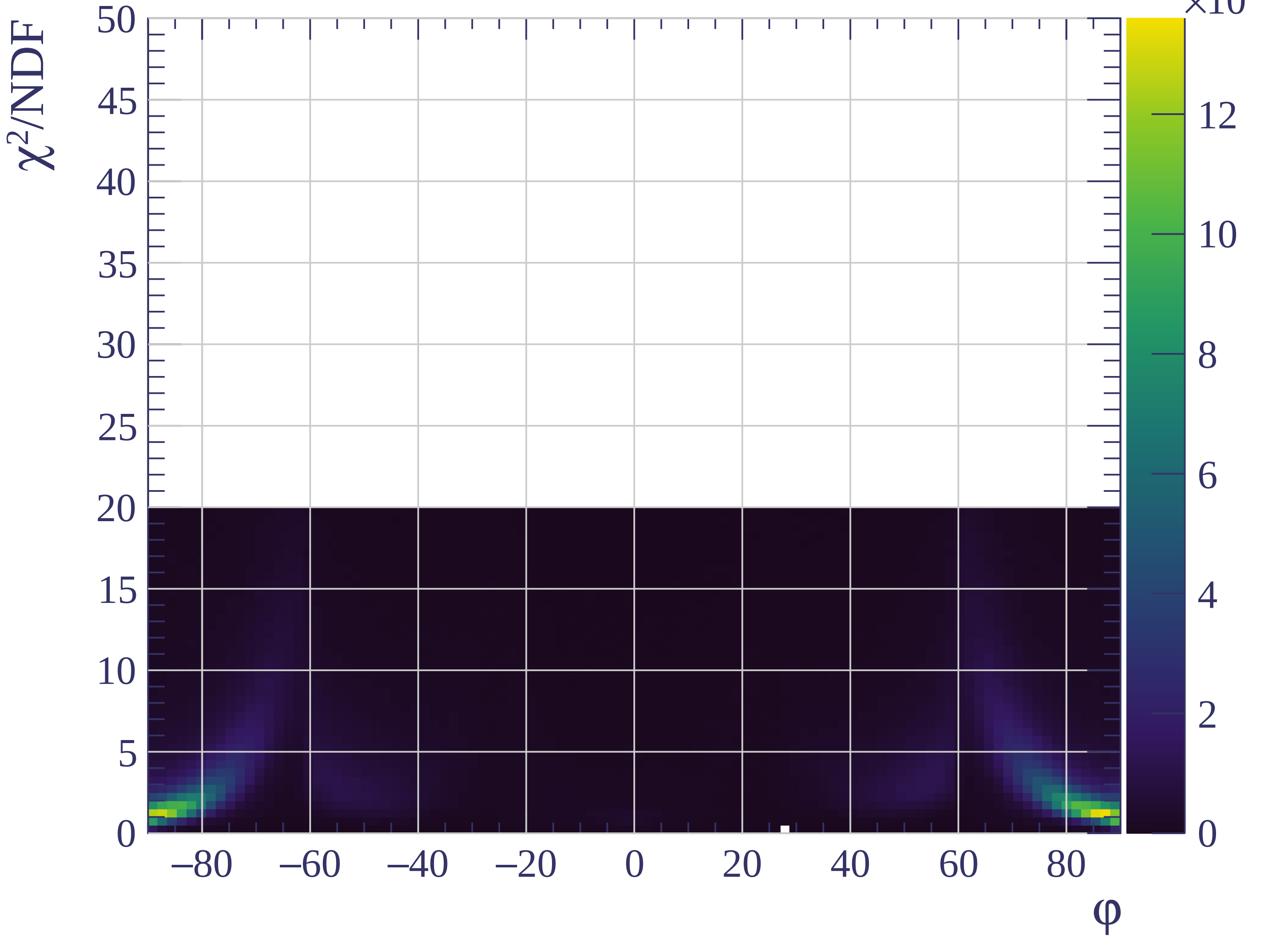


WF
XP
Mean (ADC counts/cm)
600
550
500
450
400
350
300
−80
−60
−40
−20
0
20
40
60
80
φ (°)

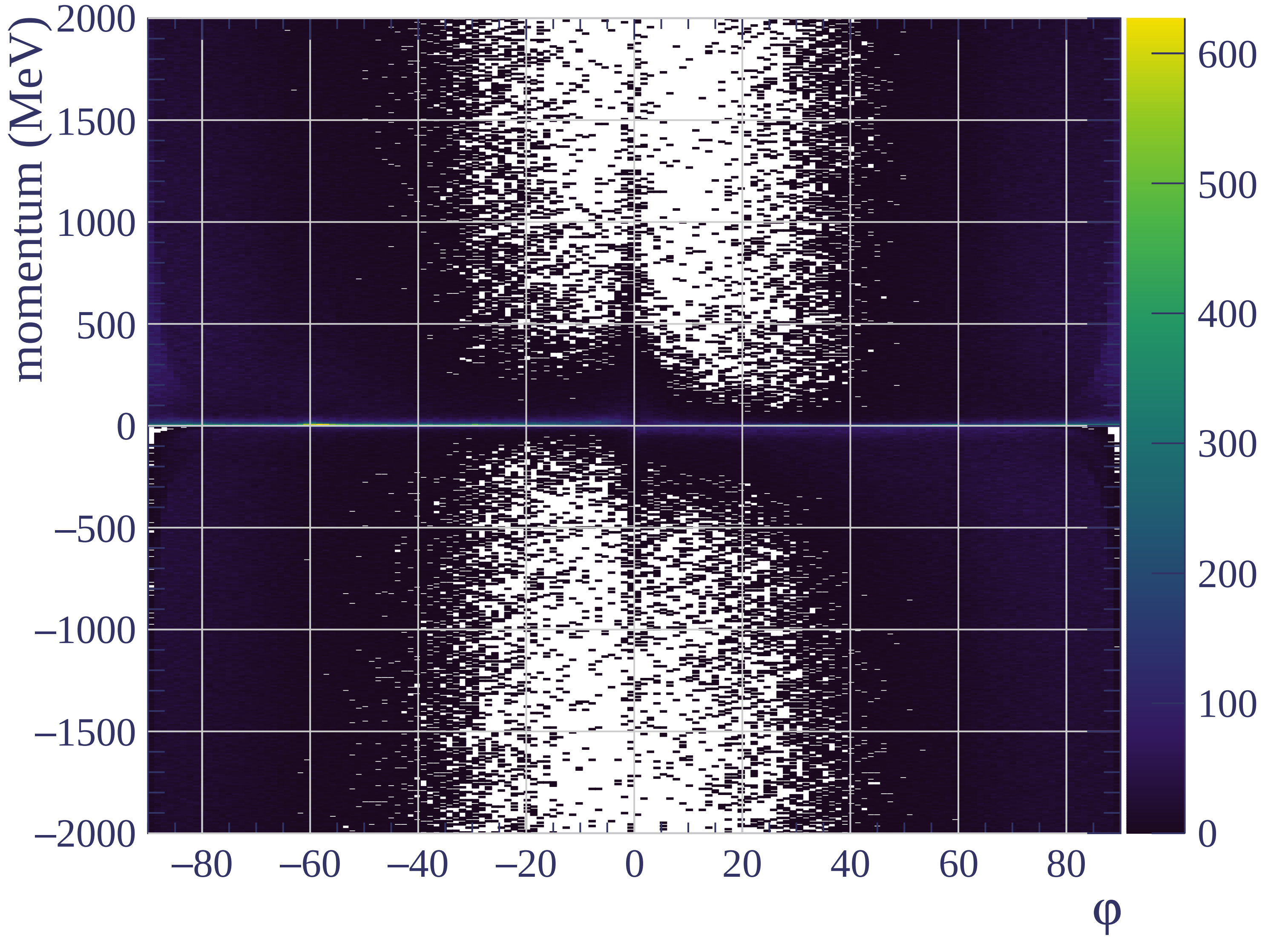

dE/dx with XP (ADC counts/cm)
1000
900
800
700
600
500
400
300
200
100
0
−80
−60
−40
−20
0
20
40
60
80
φ (°)
300
250
200
150
100
50
0



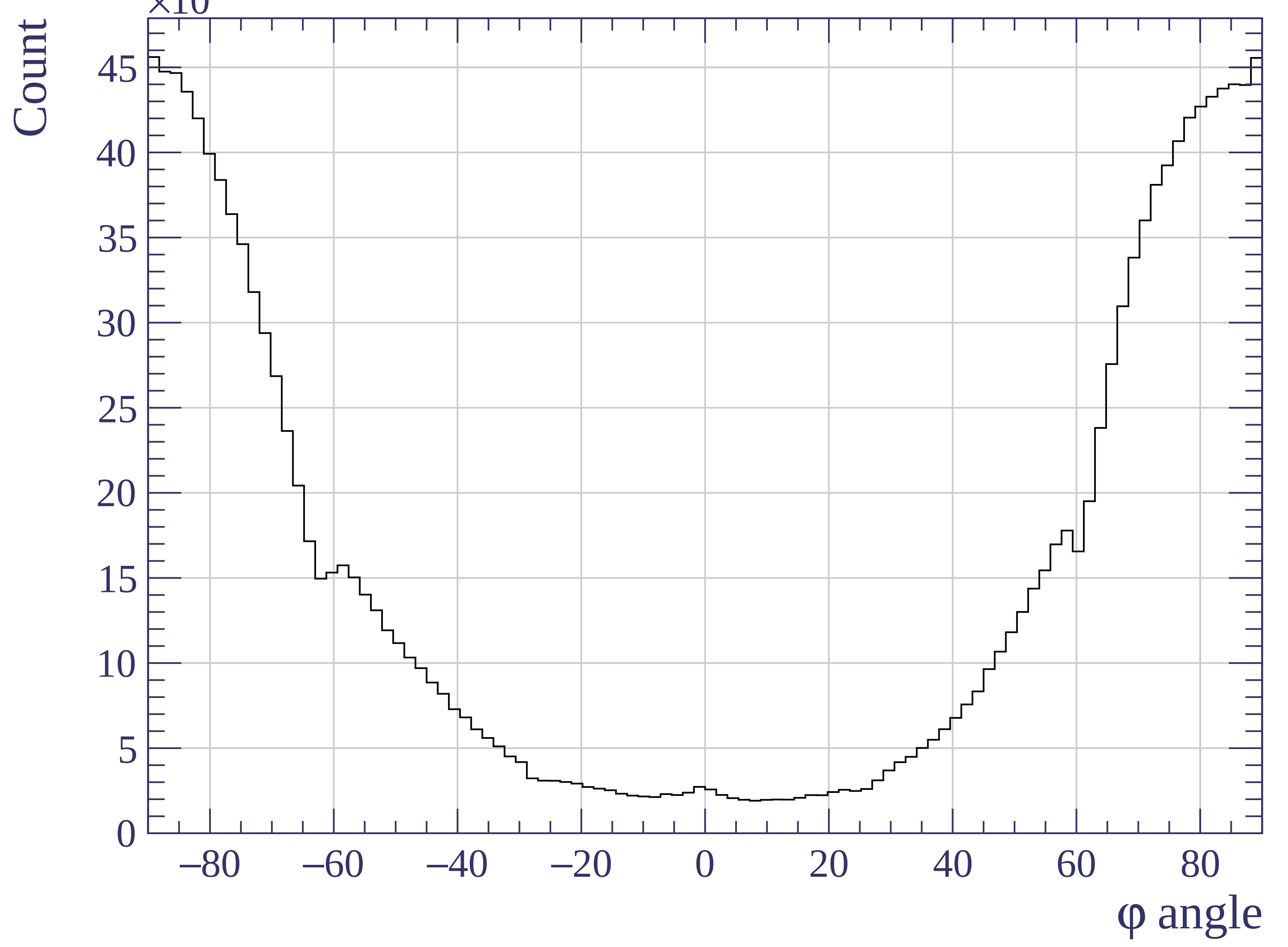
track length (cm)
160
140
120
100
80
60
40
20
0
−80
−60
−40
−20
0
20
40
60
80
φ
3500
3000
2500
2000
1500
1000
500
0



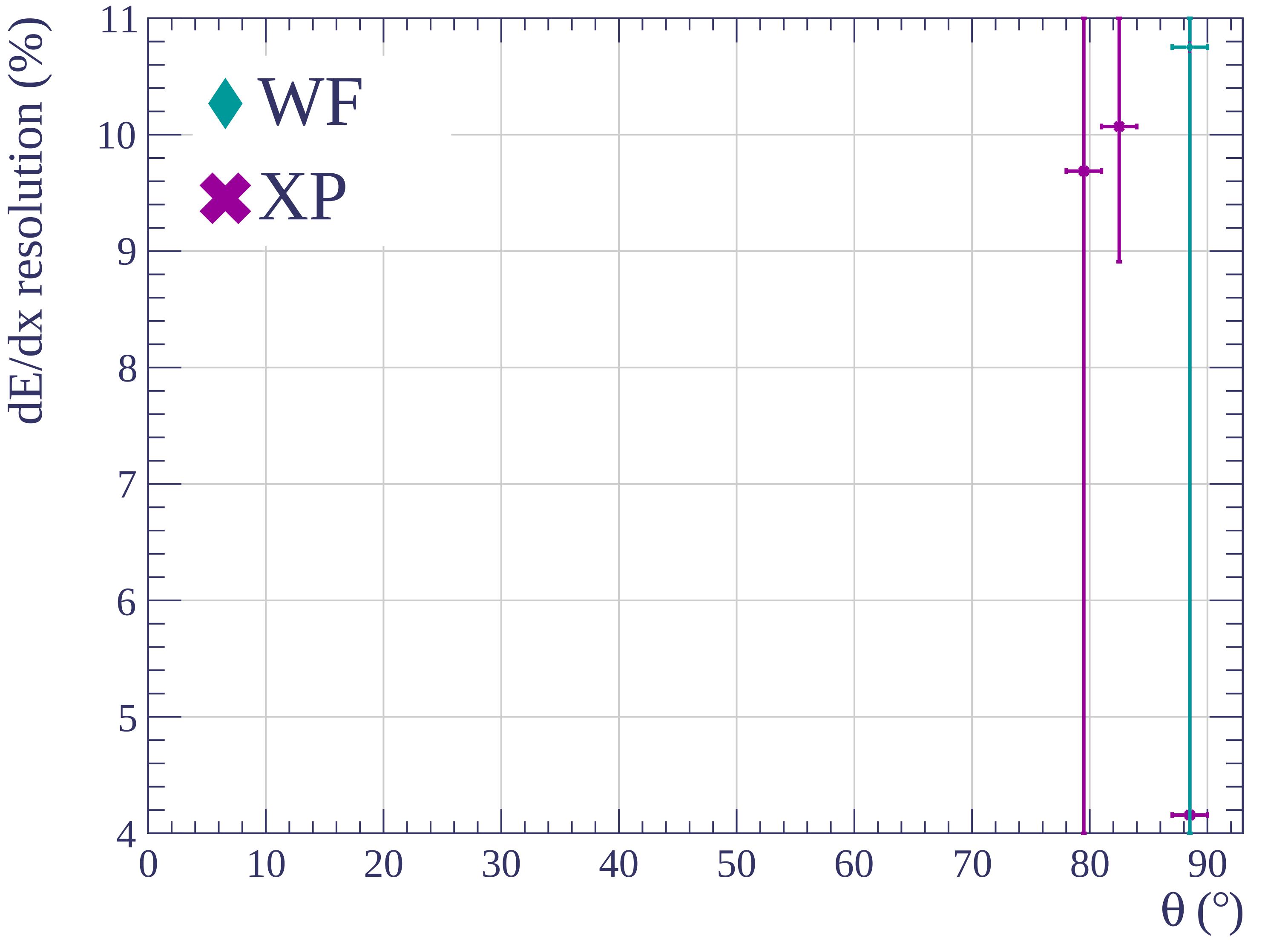
χ^2/NDF
50
45
40
35
30
25
20
15
10
5
0
−80
−60
−40
−20
0
20
40
60
80
φ
×10
12
10
8
6
4
2
0



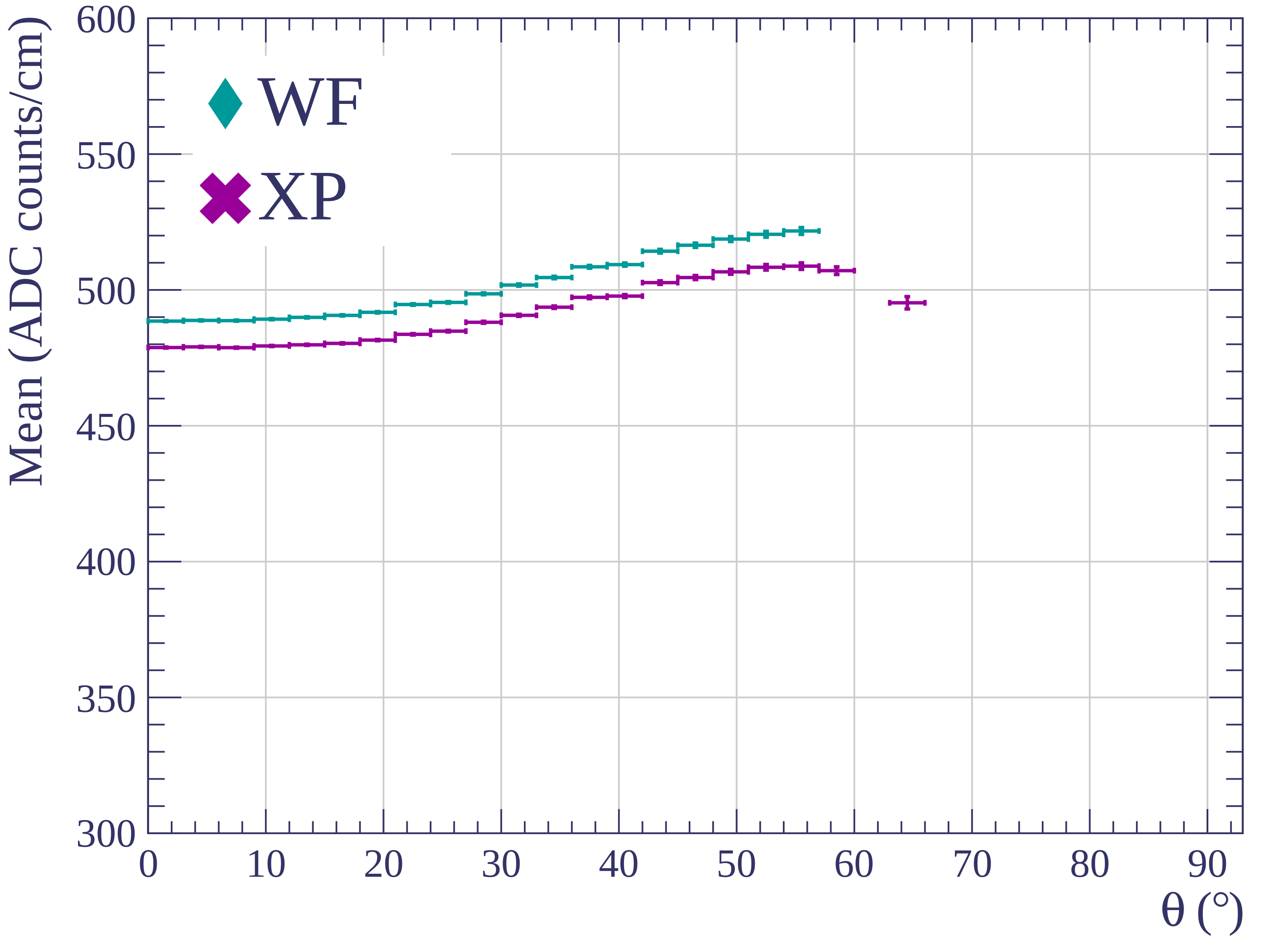
momentum (MeV)
2000
1500
1000
500
0
−500
−1000
−1500
−2000
−80
−60
−40
−20
0
20
40
60
80
φ
600
500
400
300
200
100
0



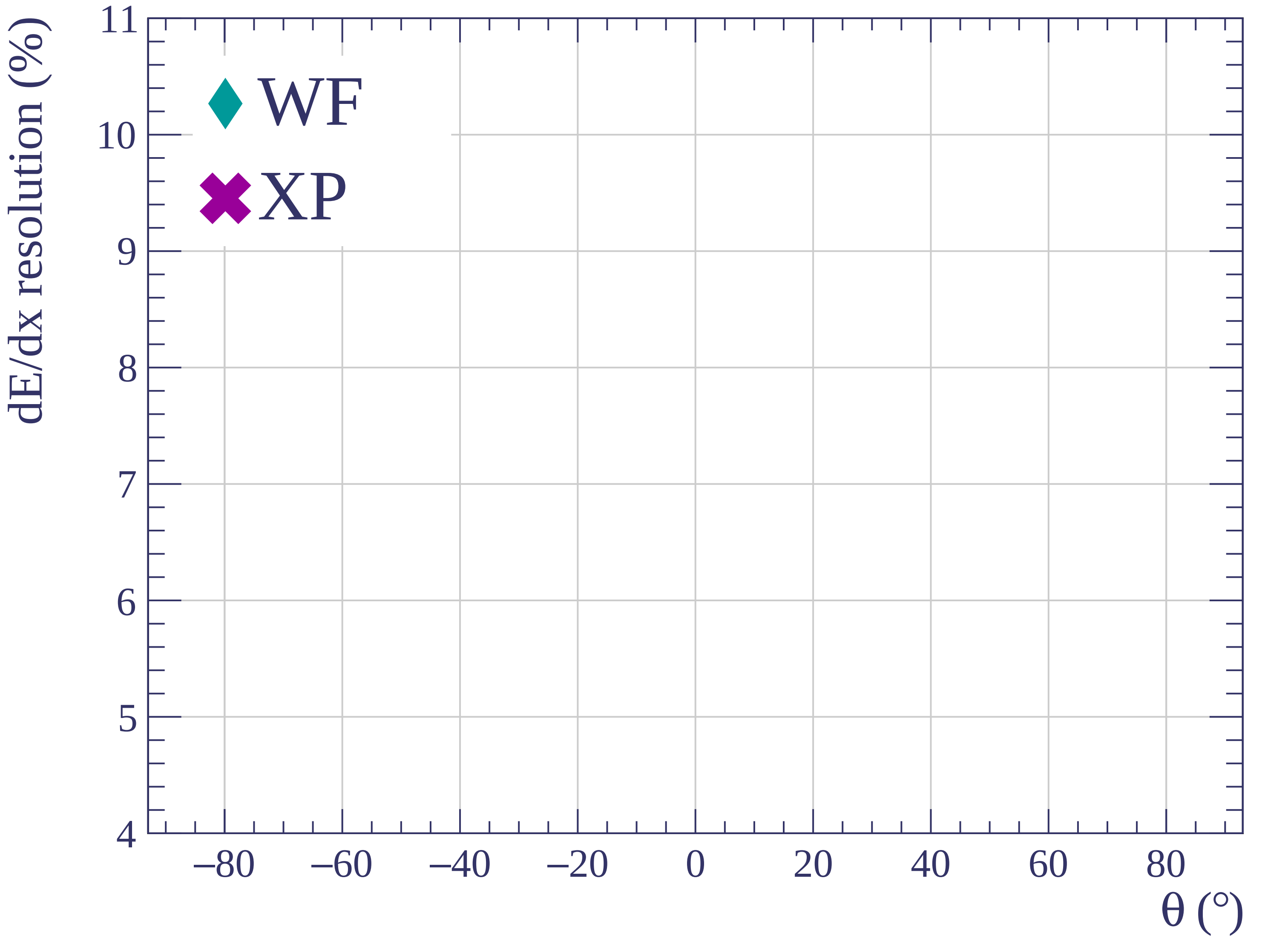
×10
Count
45
40
35
30
25
20
15
10
5
0
−80
−60
−40
−20
0
20
40
60
80
φ angle



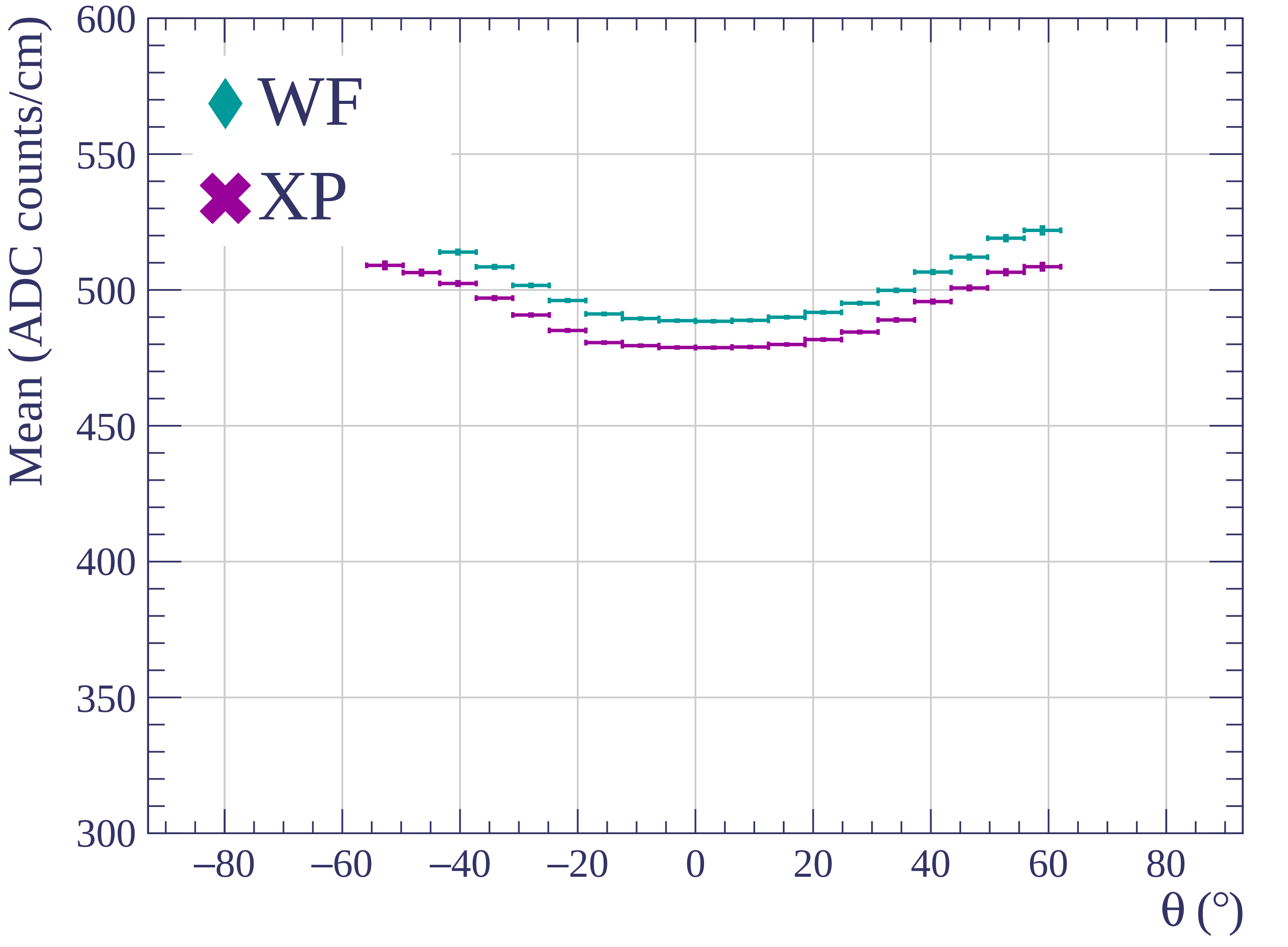
WF
XP
dE/dx resolution (%)
θ (°)
4
5
6
7
8
9
10
11
0
10
20
30
40
50
60
70
80
90

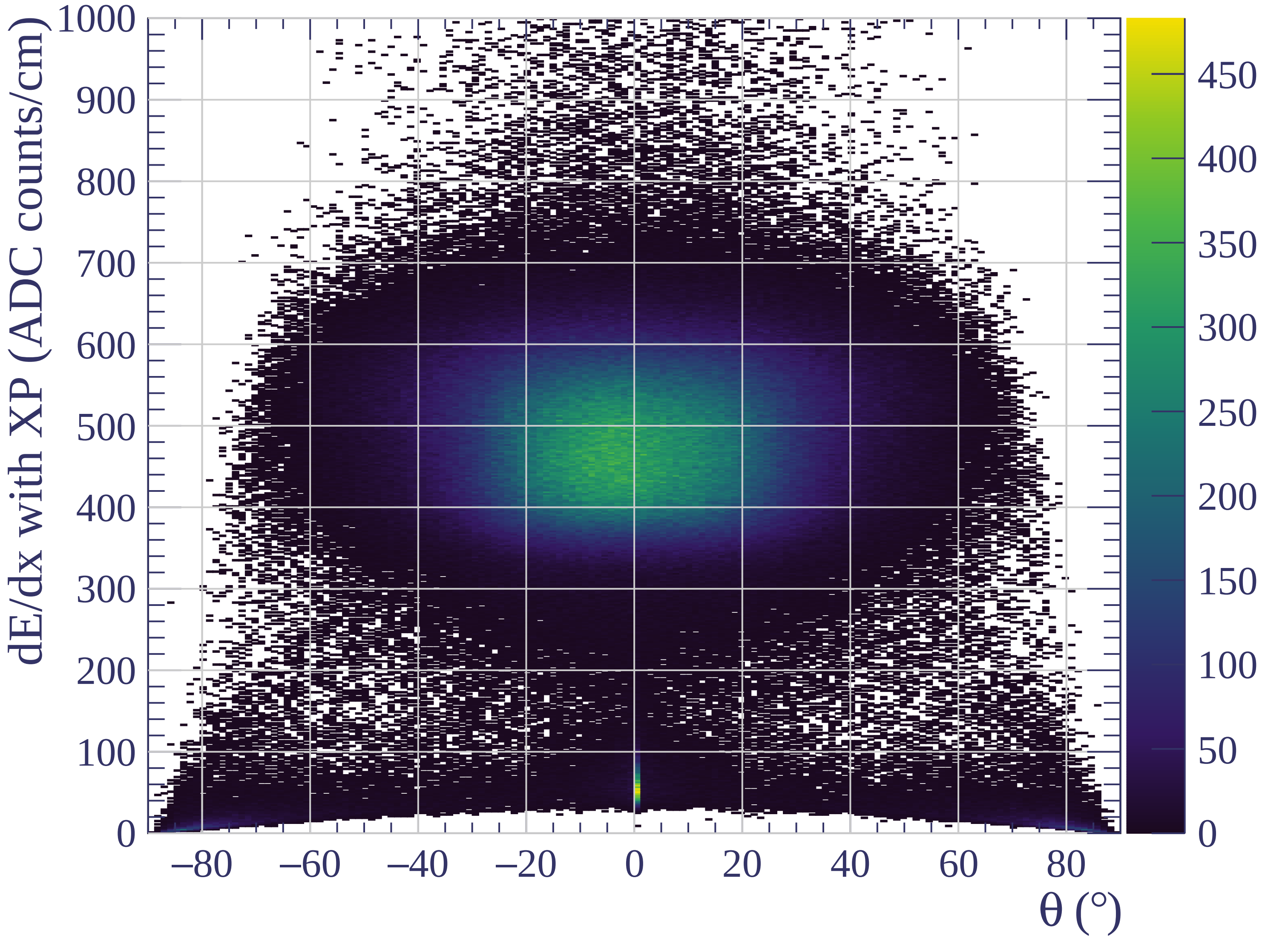


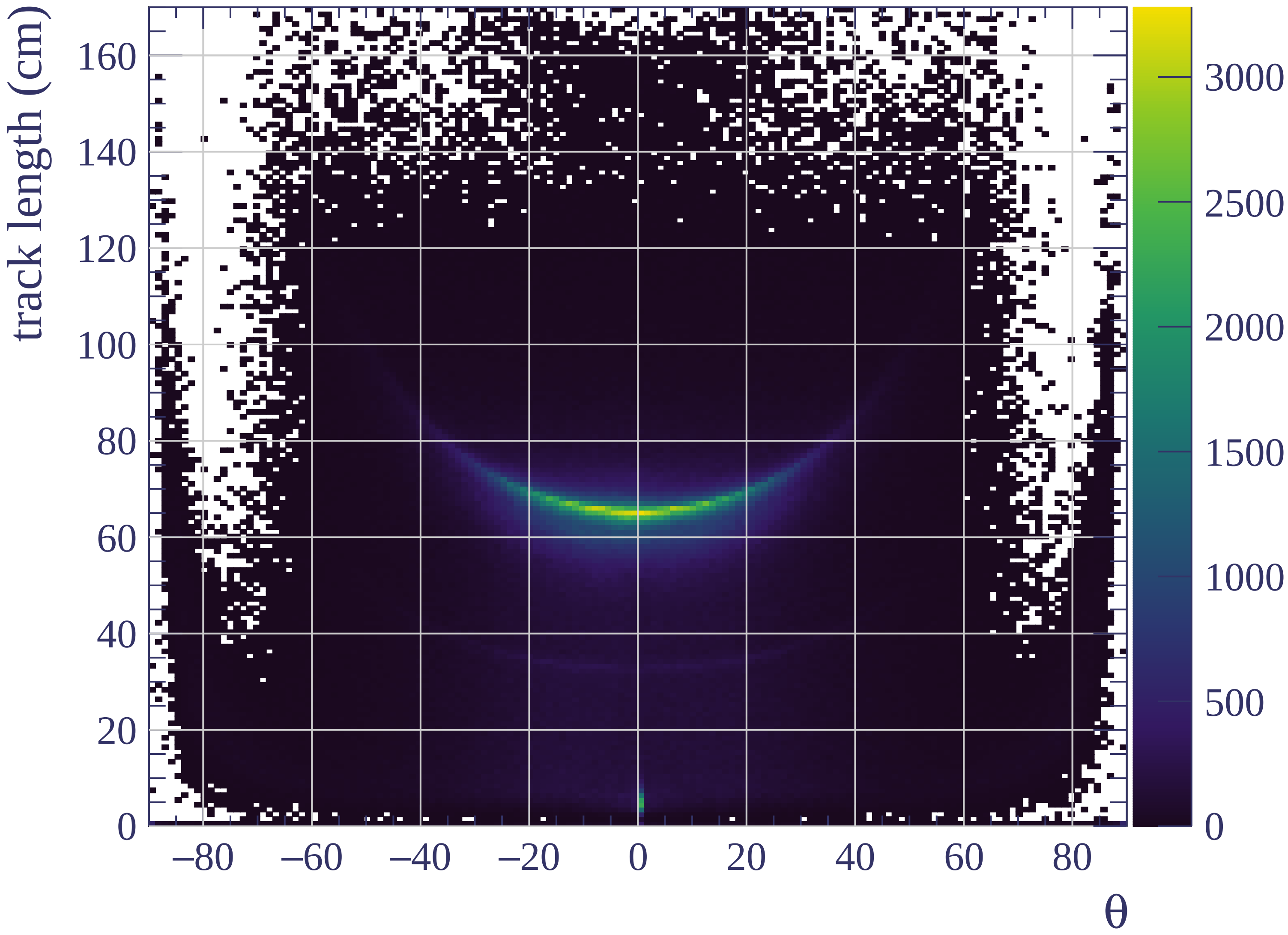
WF
XP
Mean (ADC counts/cm)
600
550
500
450
400
350
300
0
10
20
30
40
50
60
70
80
90
θ (°)



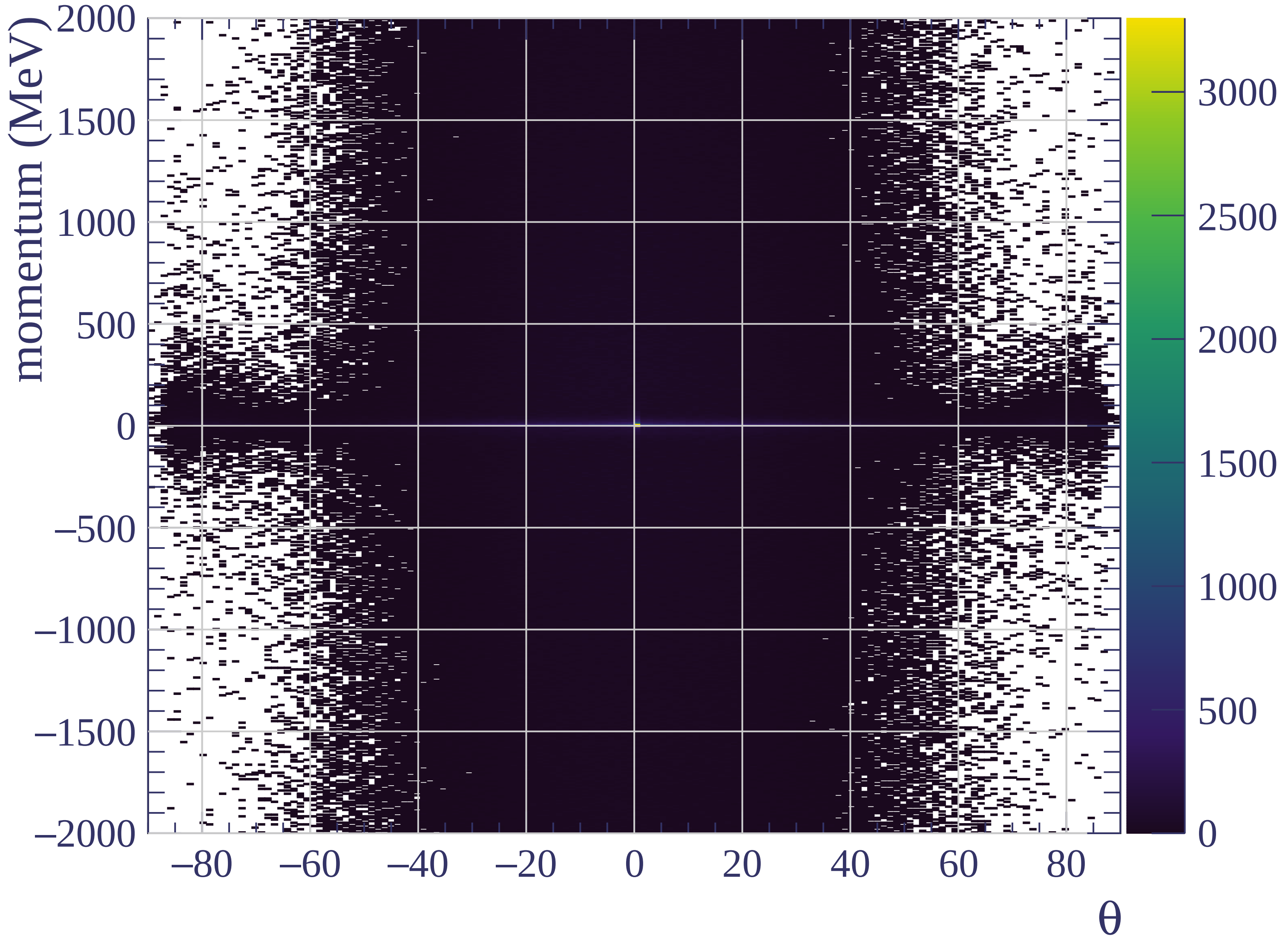
WF
XP
dE/dx resolution (%)
11
10
9
8
7
6
5
4
−80
−60
−40
−20
0
20
40
60
80
θ (°)

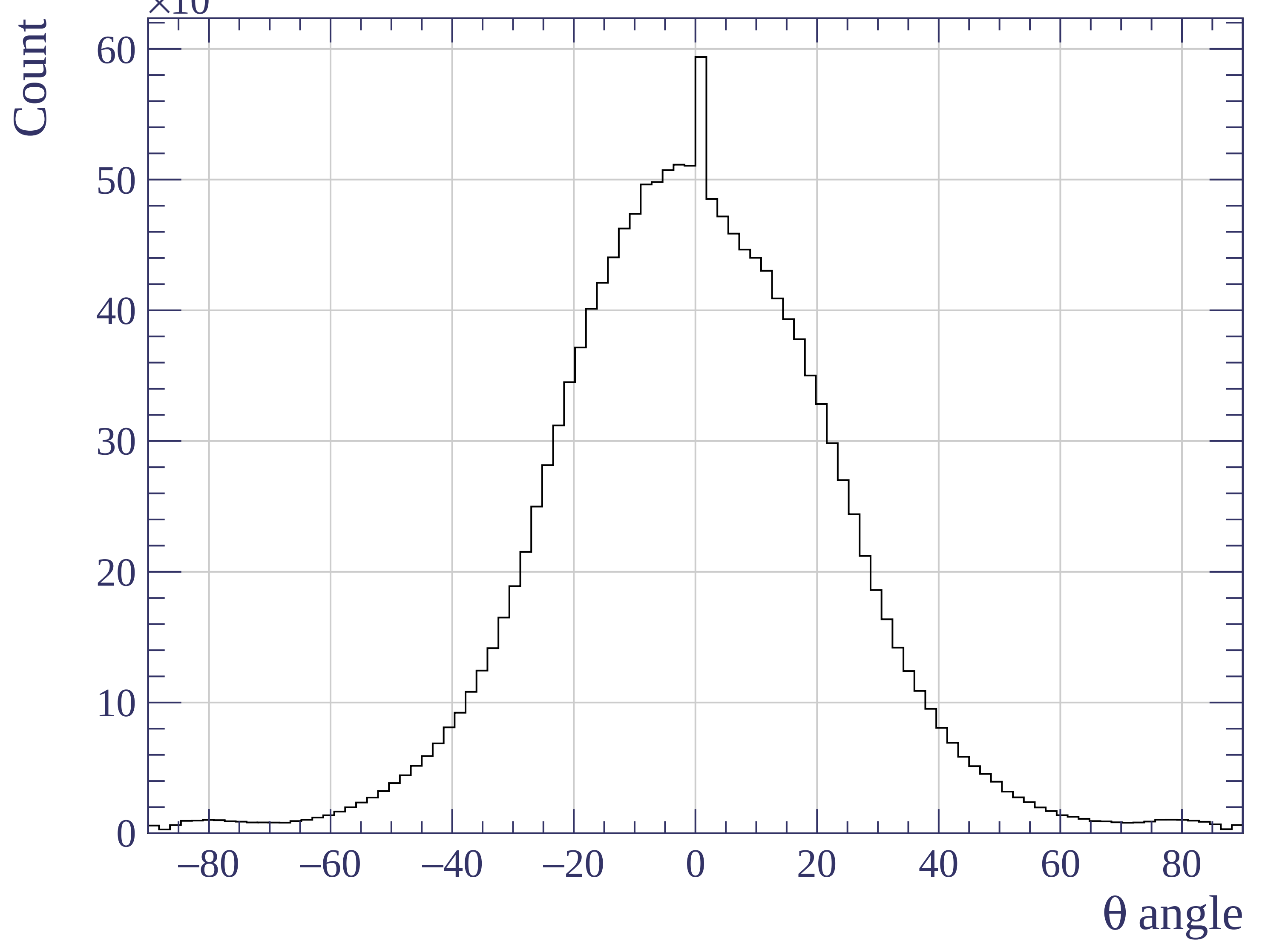

WF
XP
Mean (ADC counts/cm)
600
550
500
450
400
350
300
−80
−60
−40
−20
0
20
40
60
80
θ (°)

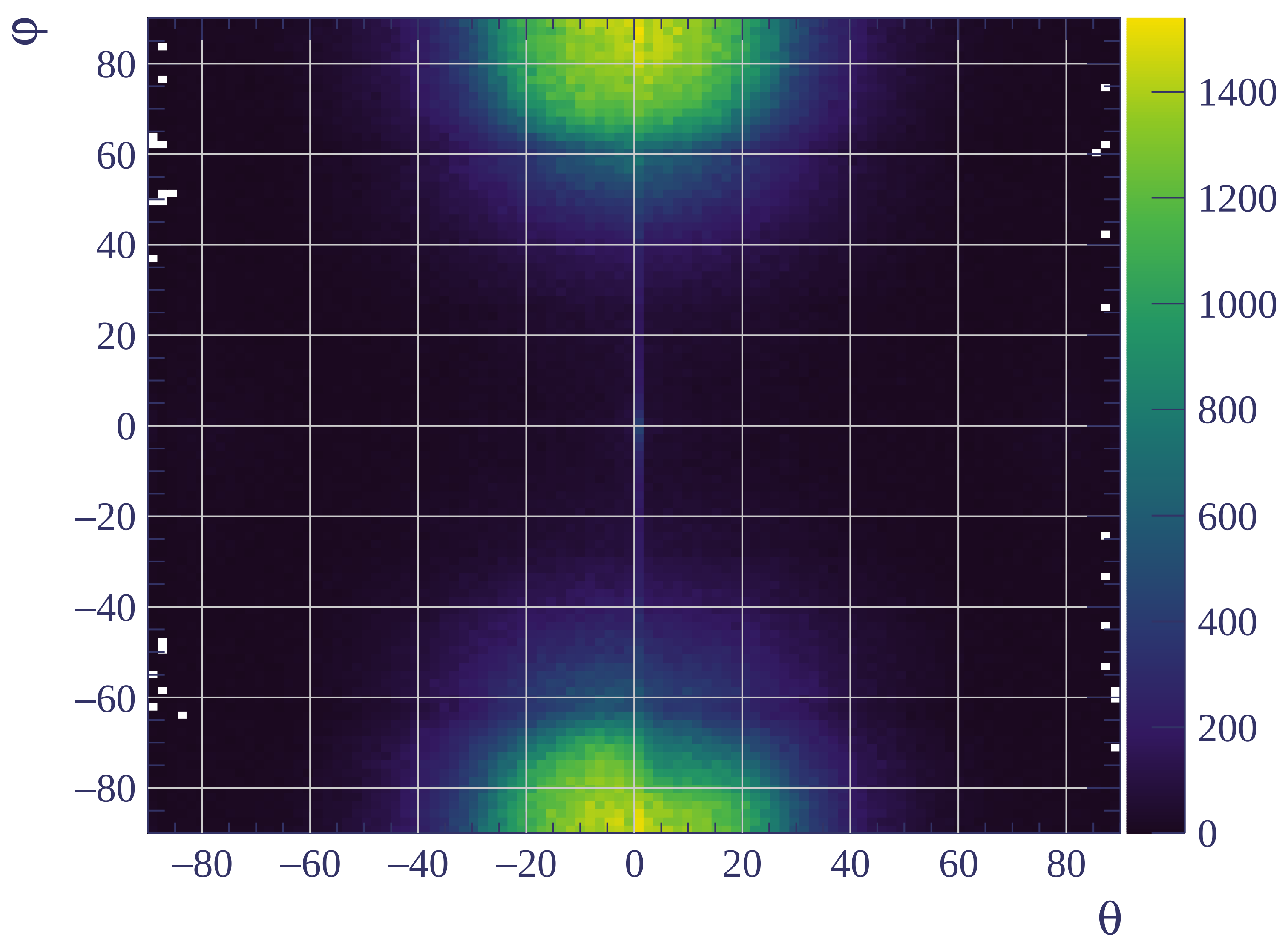

dE/dx with XP (ADC counts/cm)
θ (°)
1000
900
800
700
600
500
400
300
200
100
0
−80
−60
−40
−20
0
20
40
60
80
450
400
350
300
250
200
150
100
50
0

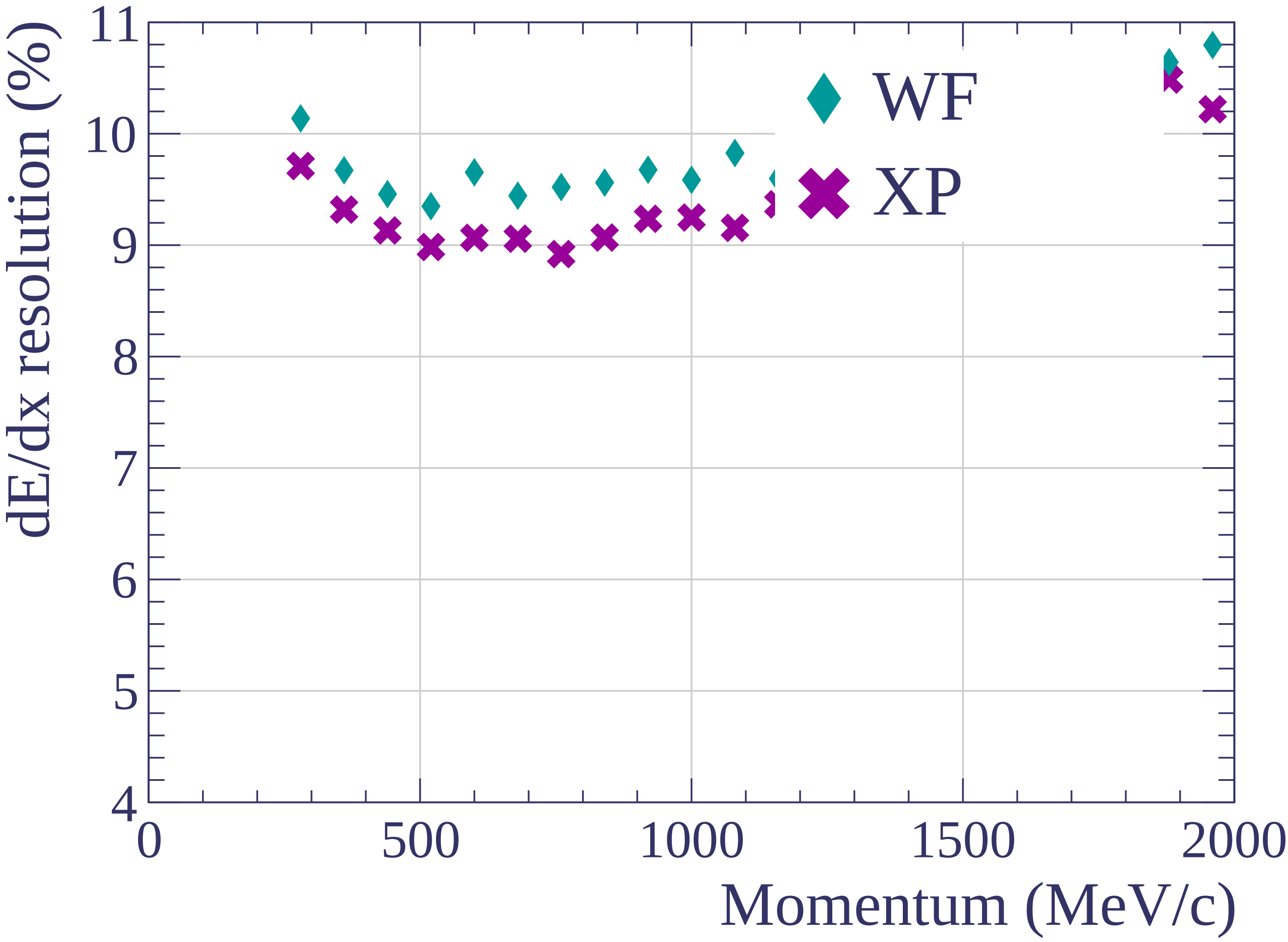


track length (cm)
160
140
120
100
80
60
40
20
0
−80
−60
−40
−20
0
20
40
60
80
θ
3000
2500
2000
1500
1000
500
0

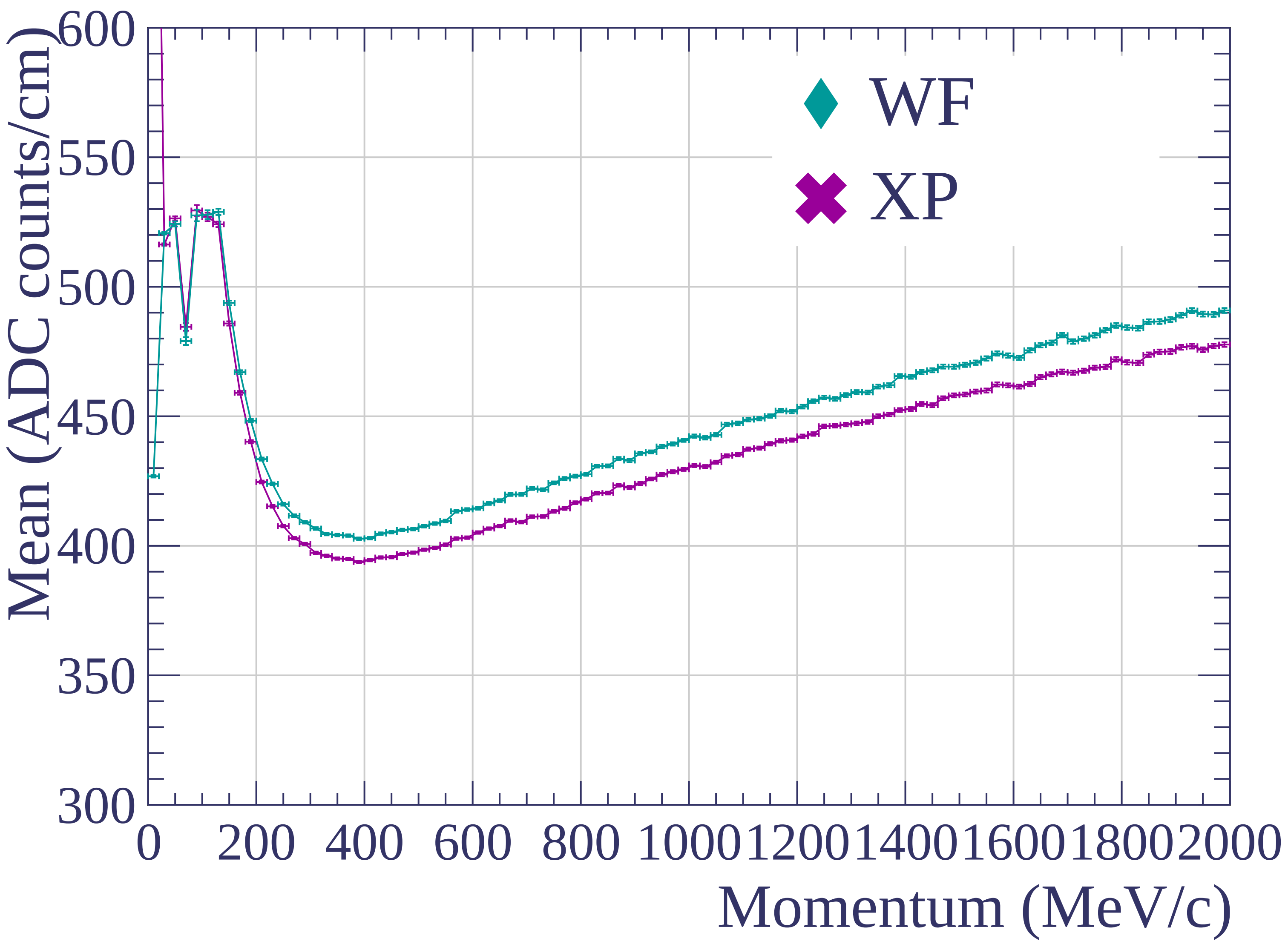

momentum (MeV)
2000
1500
1000
500
0
−500
−1000
−1500
−2000
−80
−60
−40
−20
0
20
40
60
80
θ
3000
2500
2000
1500
1000
500
0

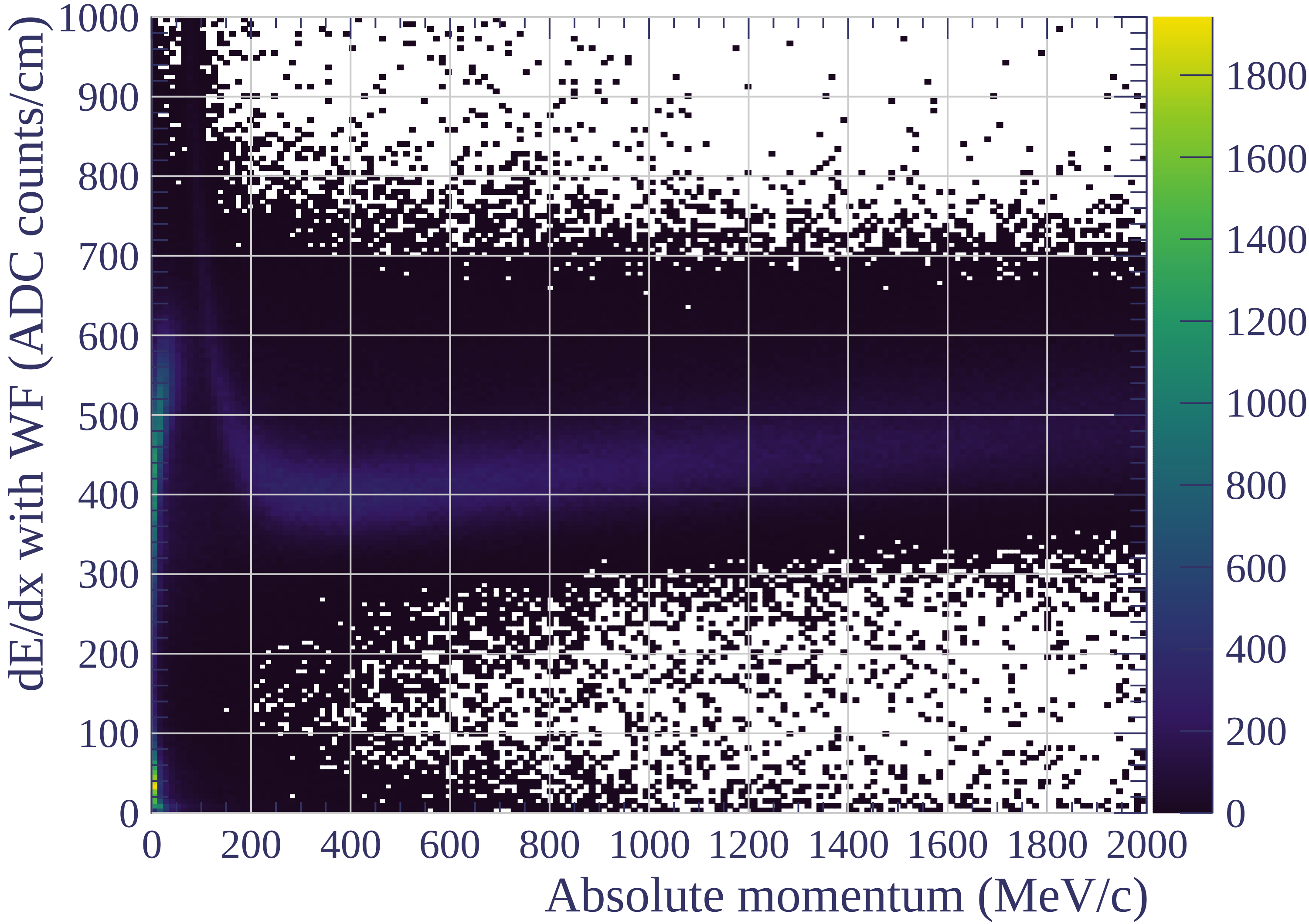

×10
Count
60
50
40
30
20
10
0
−80
−60
−40
−20
0
20
40
60
80
θ angle

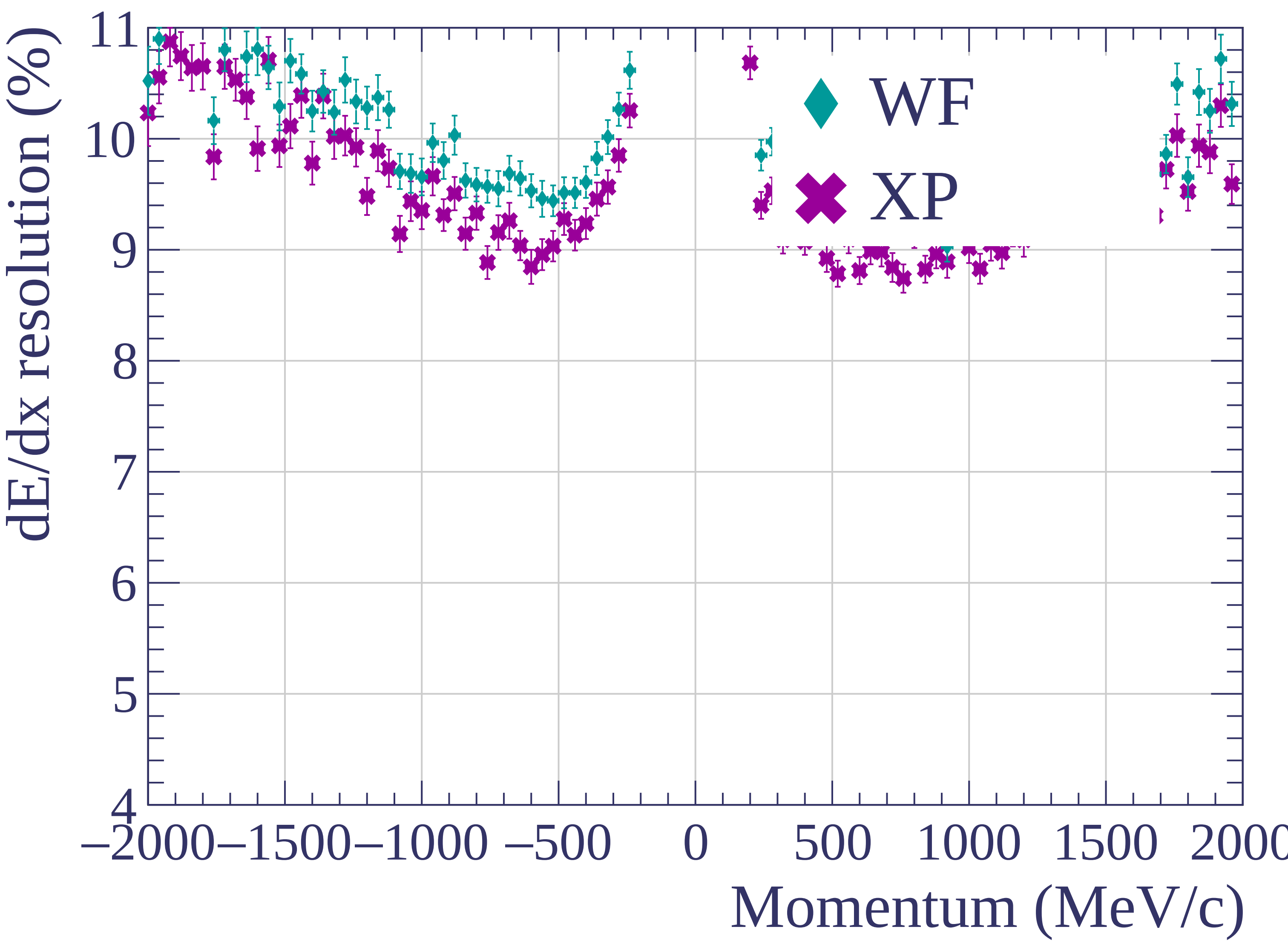

φ
80
60
40
20
0
−20
−40
−60
−80
−80
−60
−40
−20
0
20
40
60
80
θ
1400
1200
1000
800
600
400
200
0



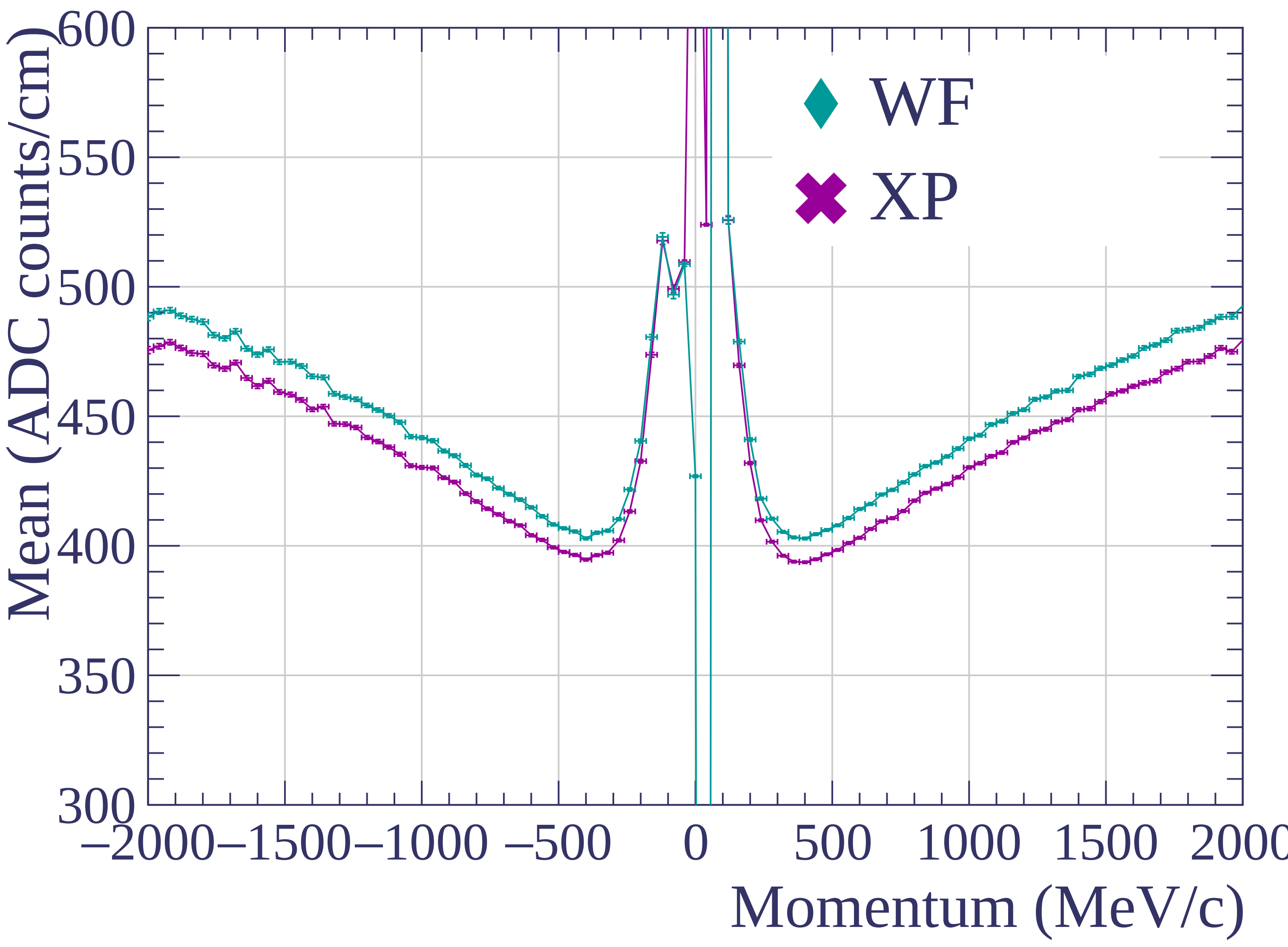
dE/dx resolution (%)
11
10
9
8
7
6
5
4
0
500
1000
1500
2000
Momentum (MeV/c)
WF
XP

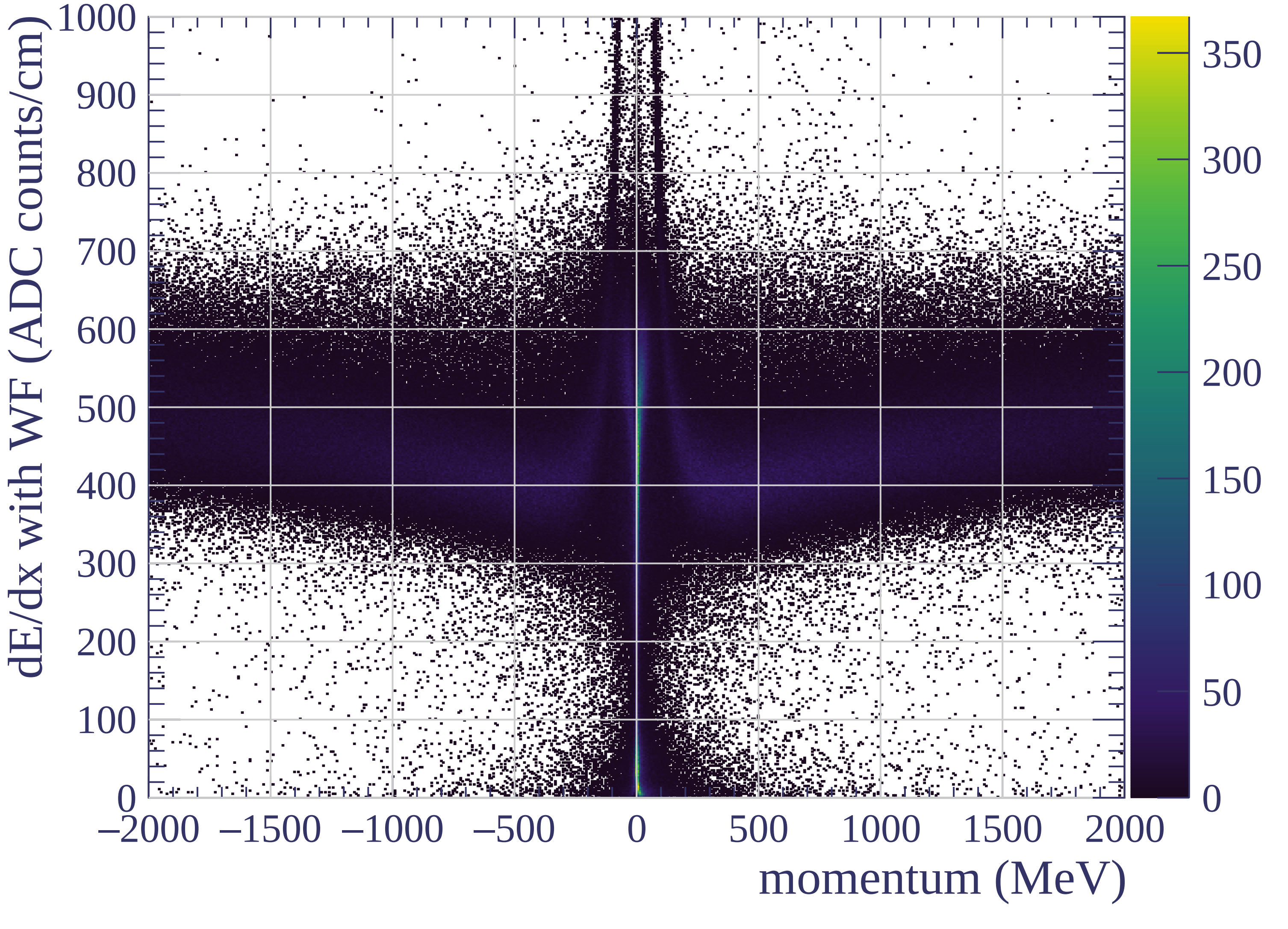

Mean (ADC counts/cm)
600
550
500
450
400
350
300
0
200
400
600
800
1000
1200
1400
1600
1800
2000
Momentum (MeV/c)
WF
XP


dE/dx with WF (ADC counts/cm)
Absolute momentum (MeV/c)
0
100
200
300
400
500
600
700
800
900
1000
0
200
400
600
800
1000
1200
1400
1600
1800
2000
0
200
400
600
800
1000
1200
1400
1600
1800

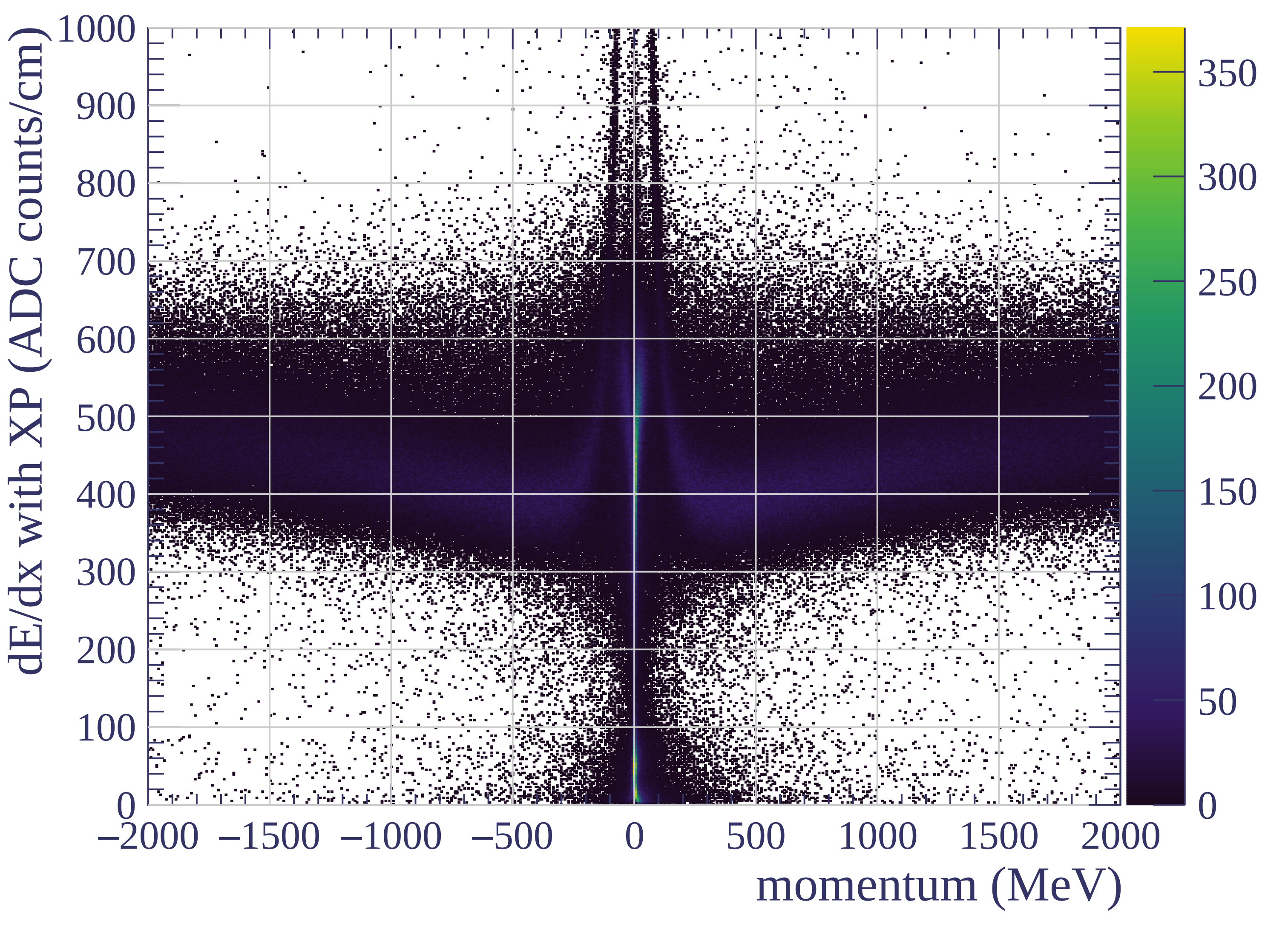


dE/dx resolution (%)
Momentum (MeV/c)
WF
XP
11
10
9
8
7
6
5
4
−2000
−1500
−1000
−500
0
500
1000
1500
2000

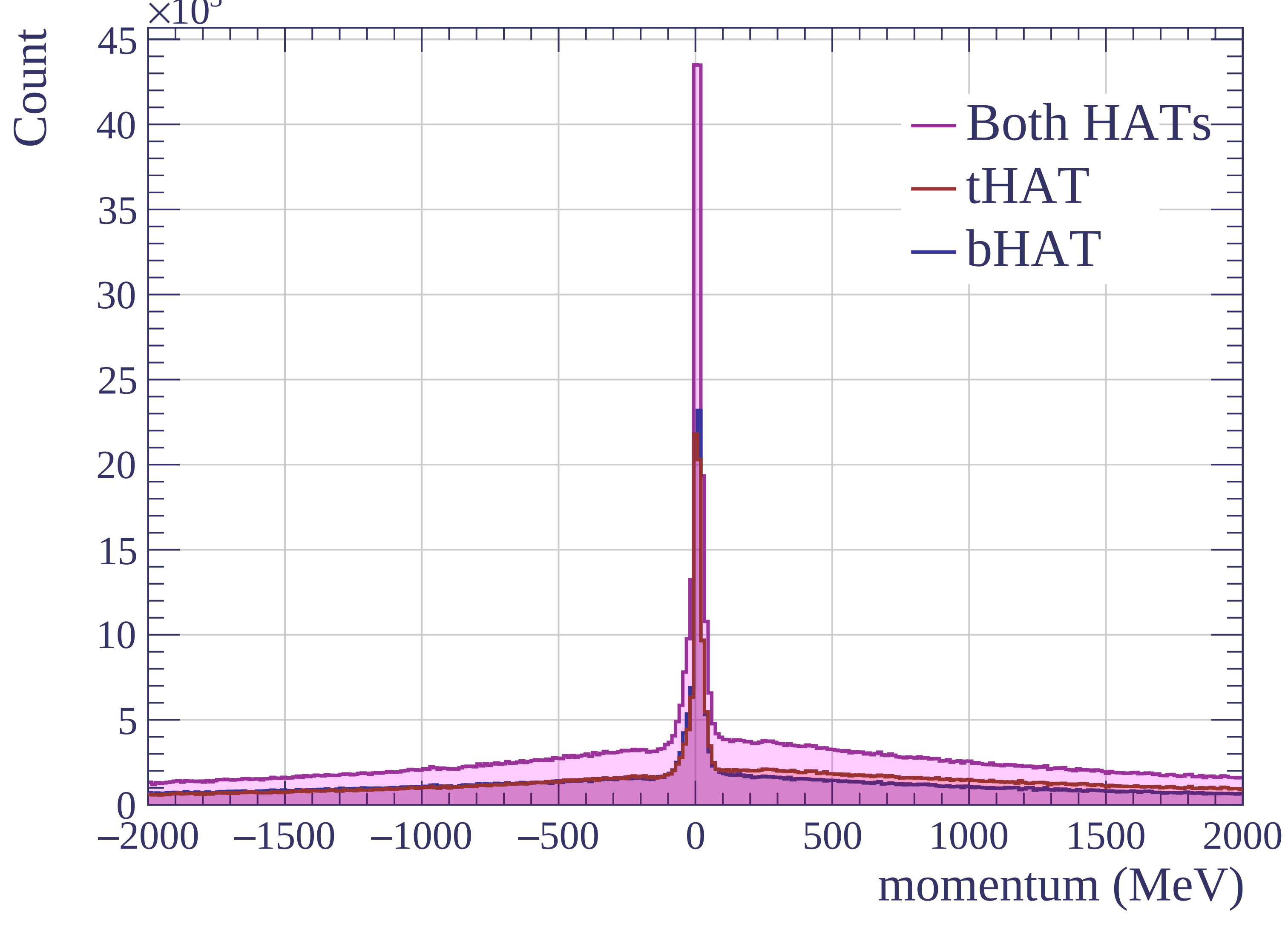

WF
XP
Mean (ADC counts/cm)
600
550
500
450
400
350
300
−2000
−1500
−1000
−500
0
500
1000
1500
2000
Momentum (MeV/c)



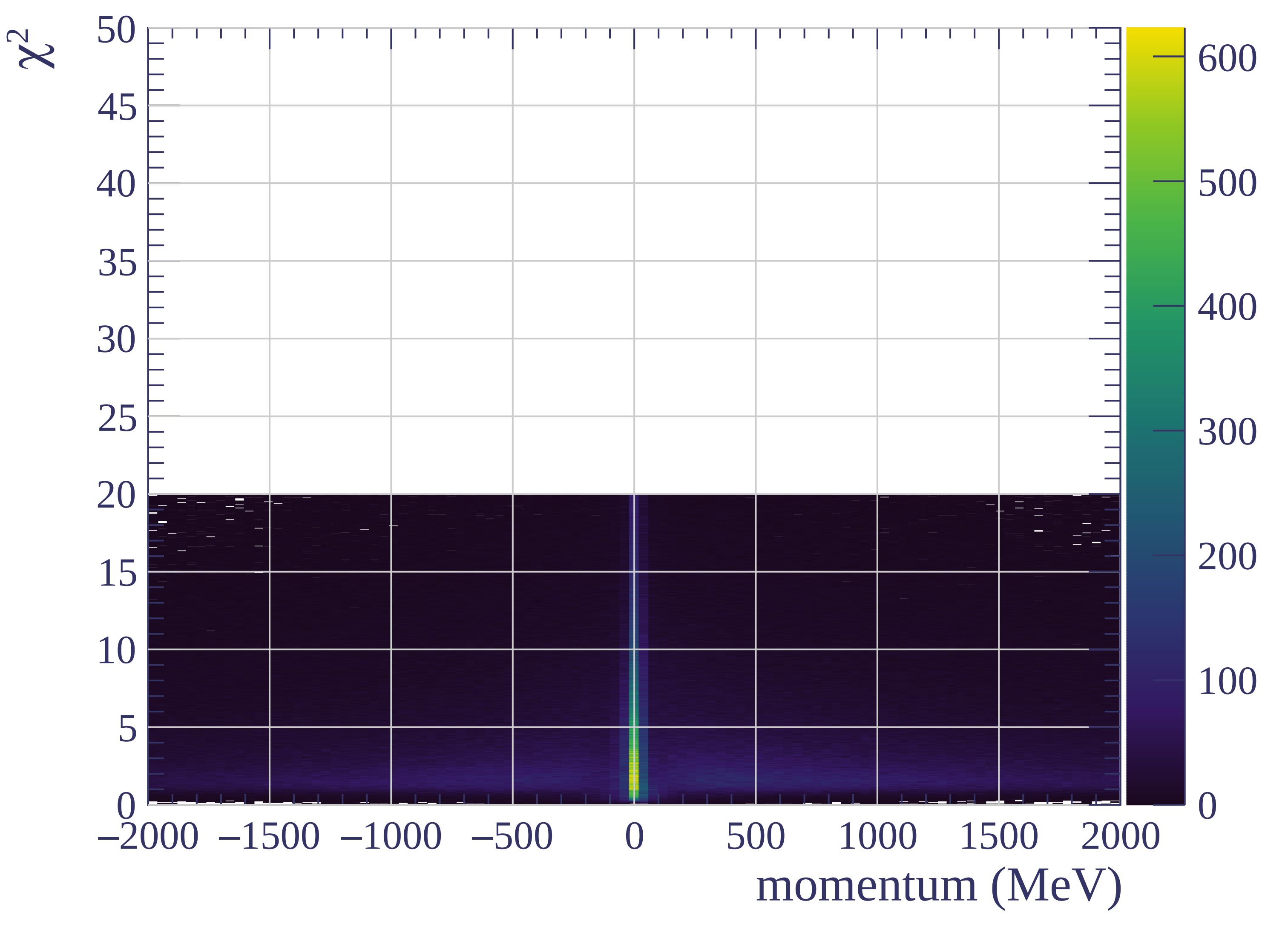
dE/dx with WF (ADC counts/cm)
1000
900
800
700
600
500
400
300
200
100
0
−2000
−1500
−1000
−500
0
500
1000
1500
2000
momentum (MeV)
350
300
250
200
150
100
50
0



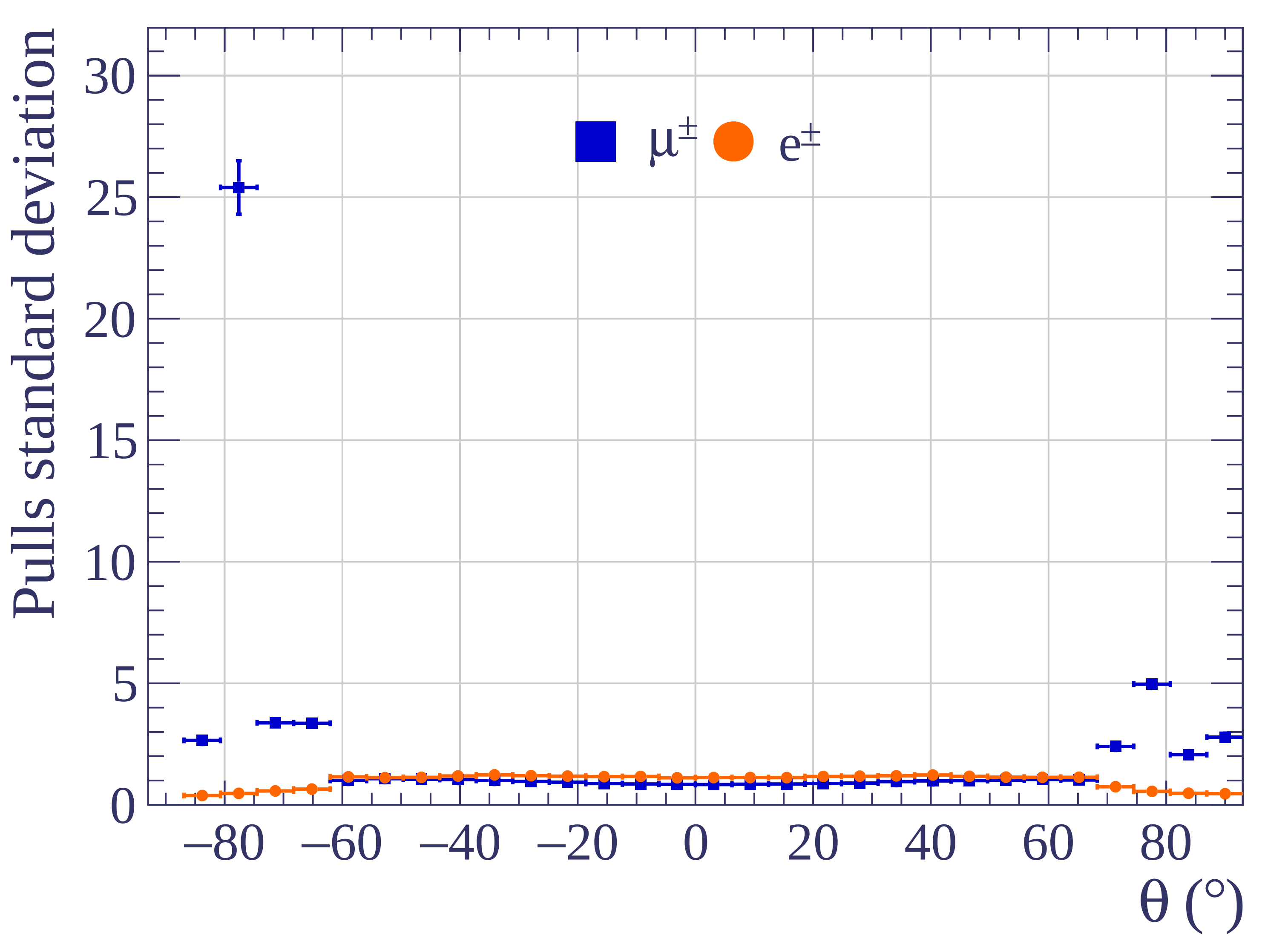
dE/dx with XP (ADC counts/cm)
1000
900
800
700
600
500
400
300
200
100
0
−2000
−1500
−1000
−500
0
500
1000
1500
2000
momentum (MeV)
350
300
250
200
150
100
50
0



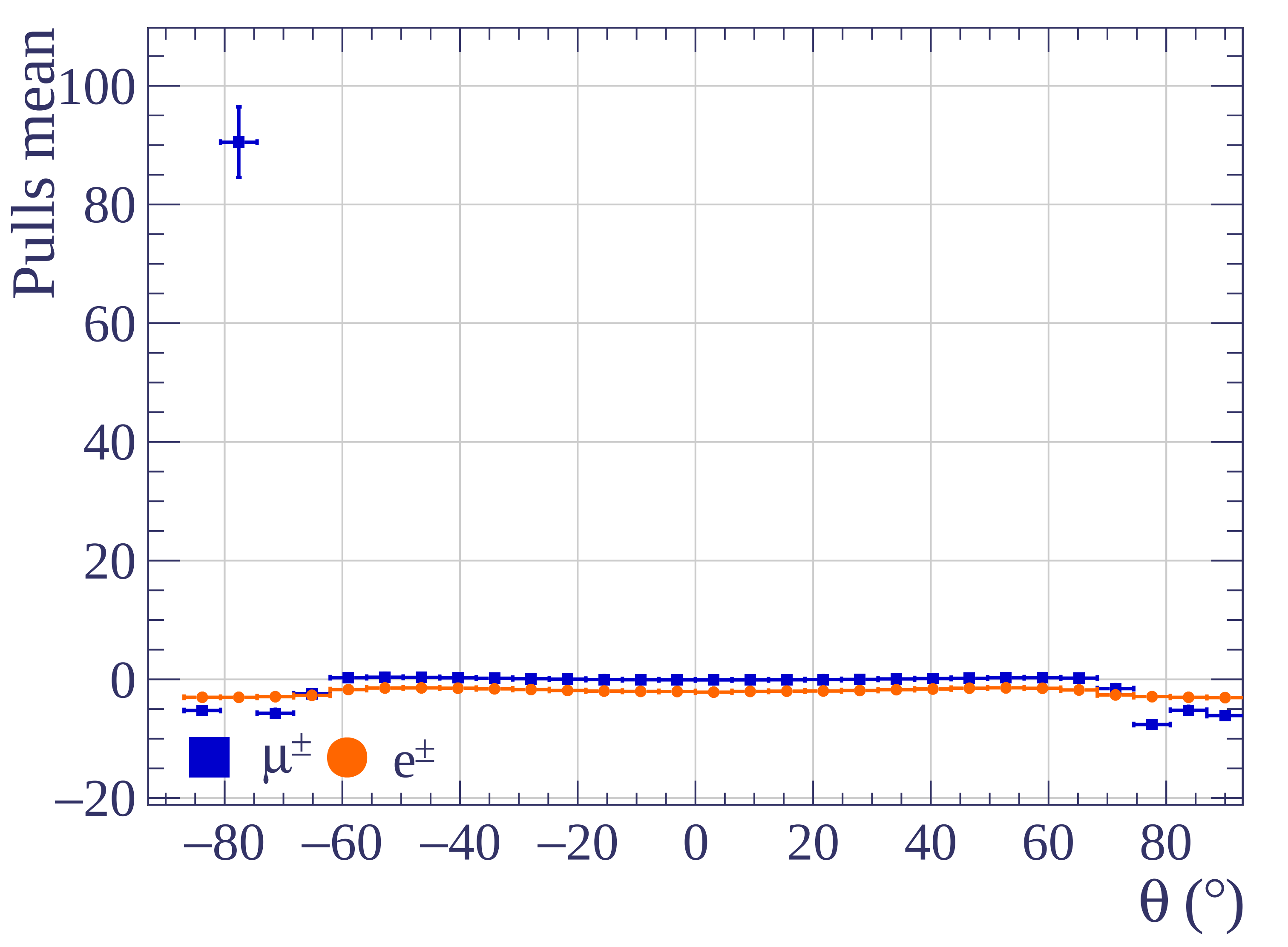
×10³
Count
45
40
35
30
25
20
15
10
5
0
−2000
−1500
−1000
−500
0
500
1000
1500
2000
momentum (MeV)
Both HATs
tHAT
bHAT



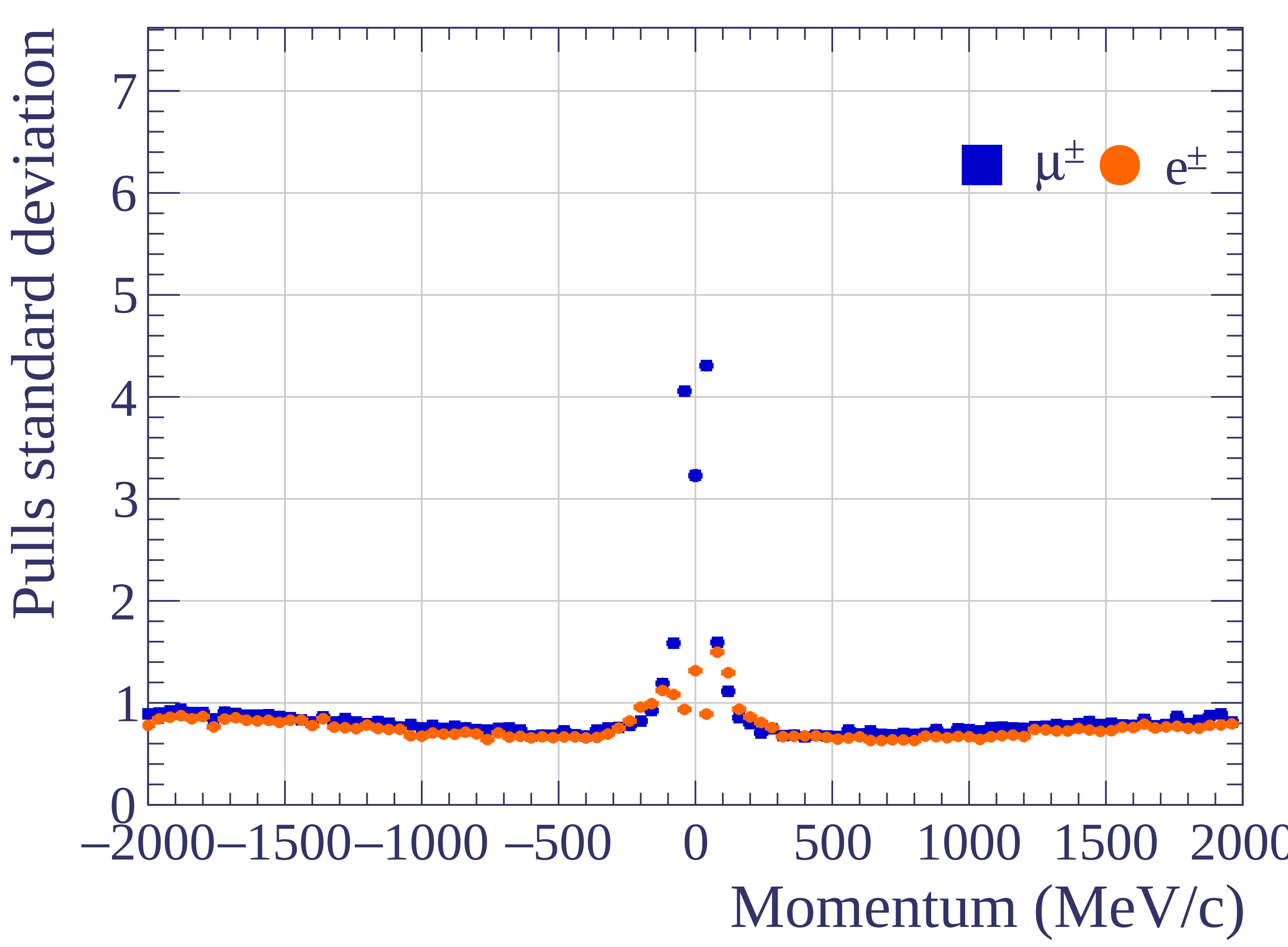
χ^2
50
45
40
35
30
25
20
15
10
5
0
−2000
−1500
−1000
−500
0
500
1000
1500
2000
momentum (MeV)
600
500
400
300
200
100
0



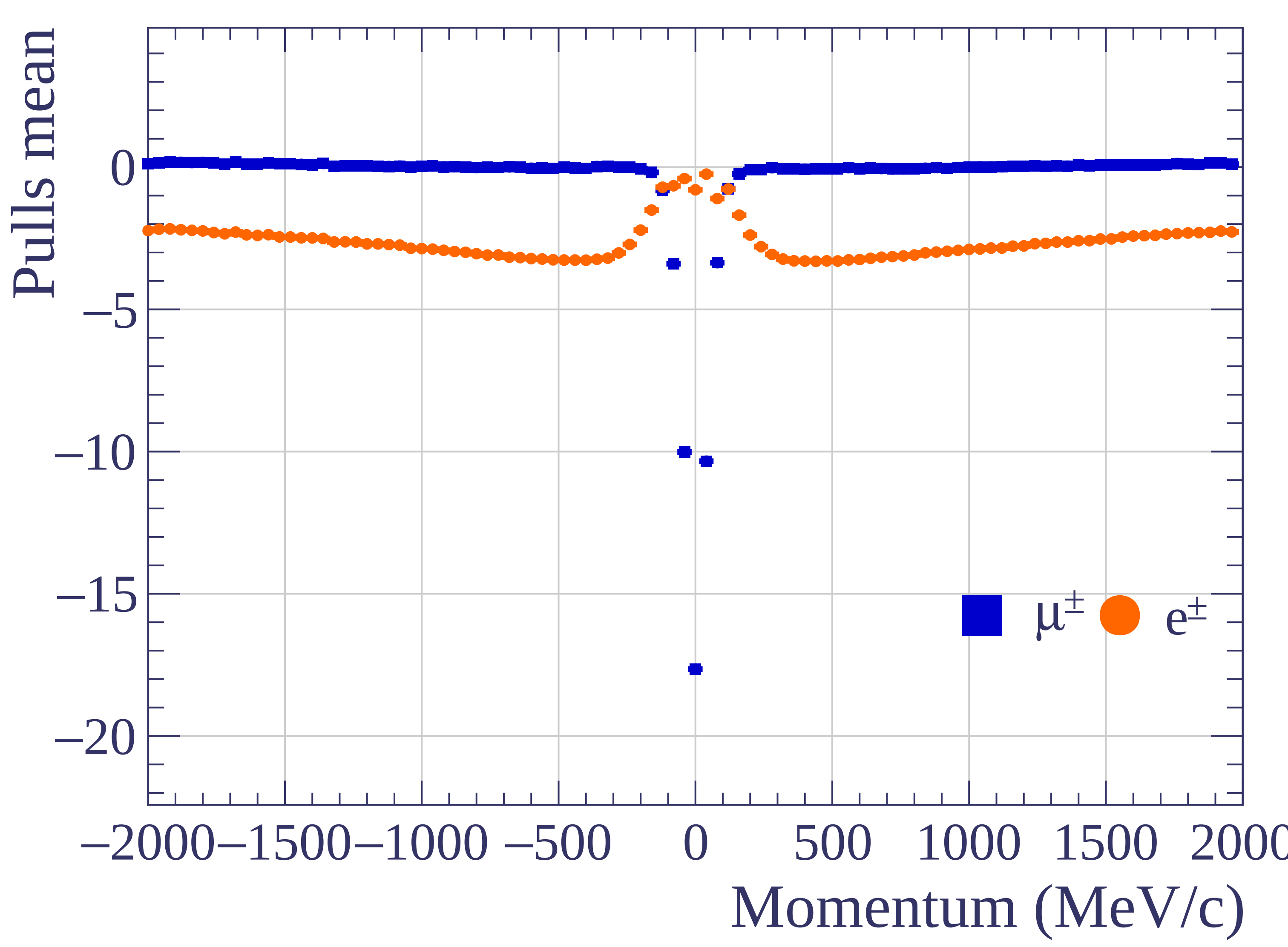
Pulls standard deviation
μ±
e±
θ (°)
0
5
10
15
20
25
30
−80
−60
−40
−20
0
20
40
60
80

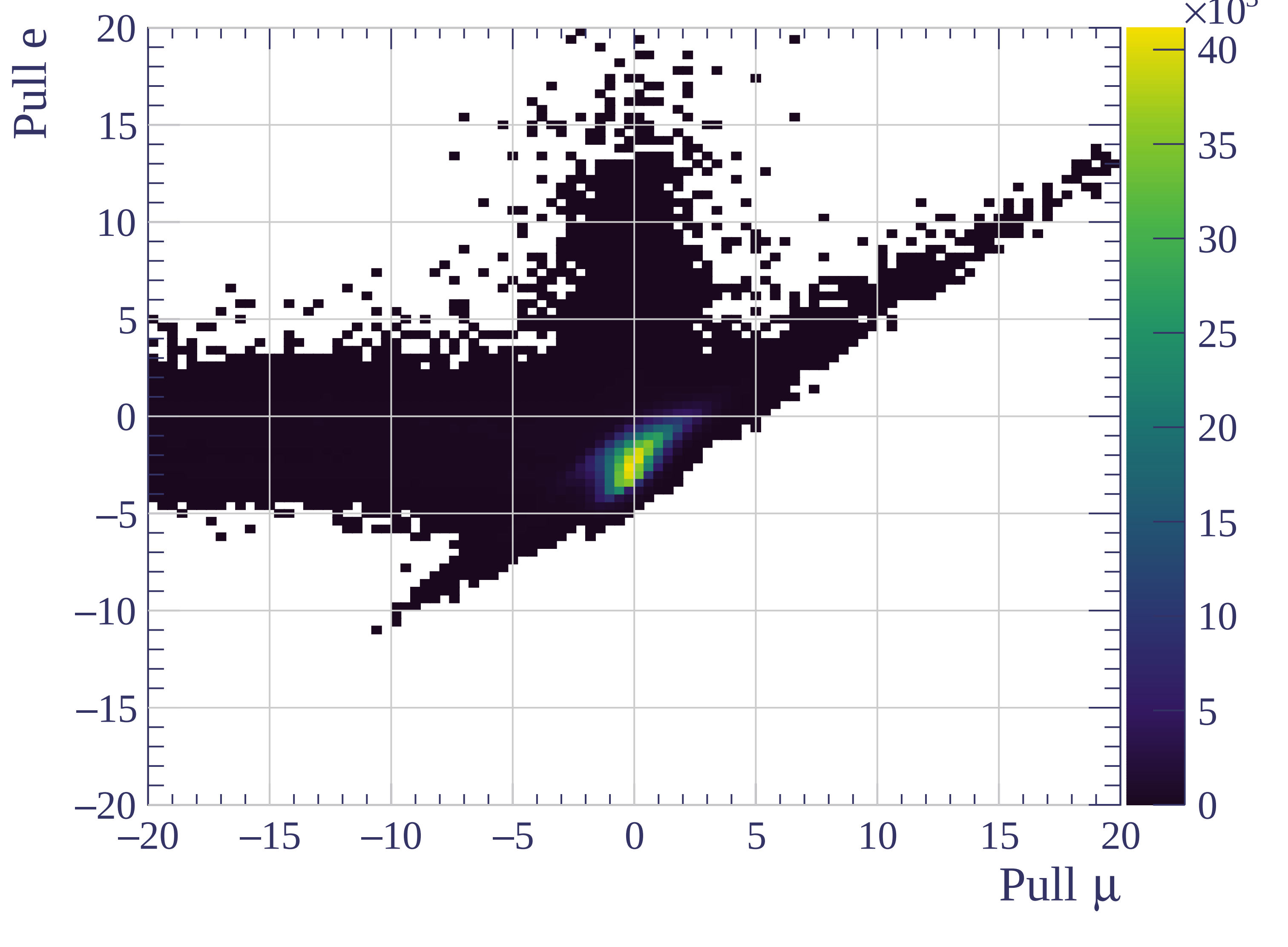


Pulls mean
100
80
60
40
20
0
−20
−80
−60
−40
−20
0
20
40
60
80
θ (°)
μ±
e±

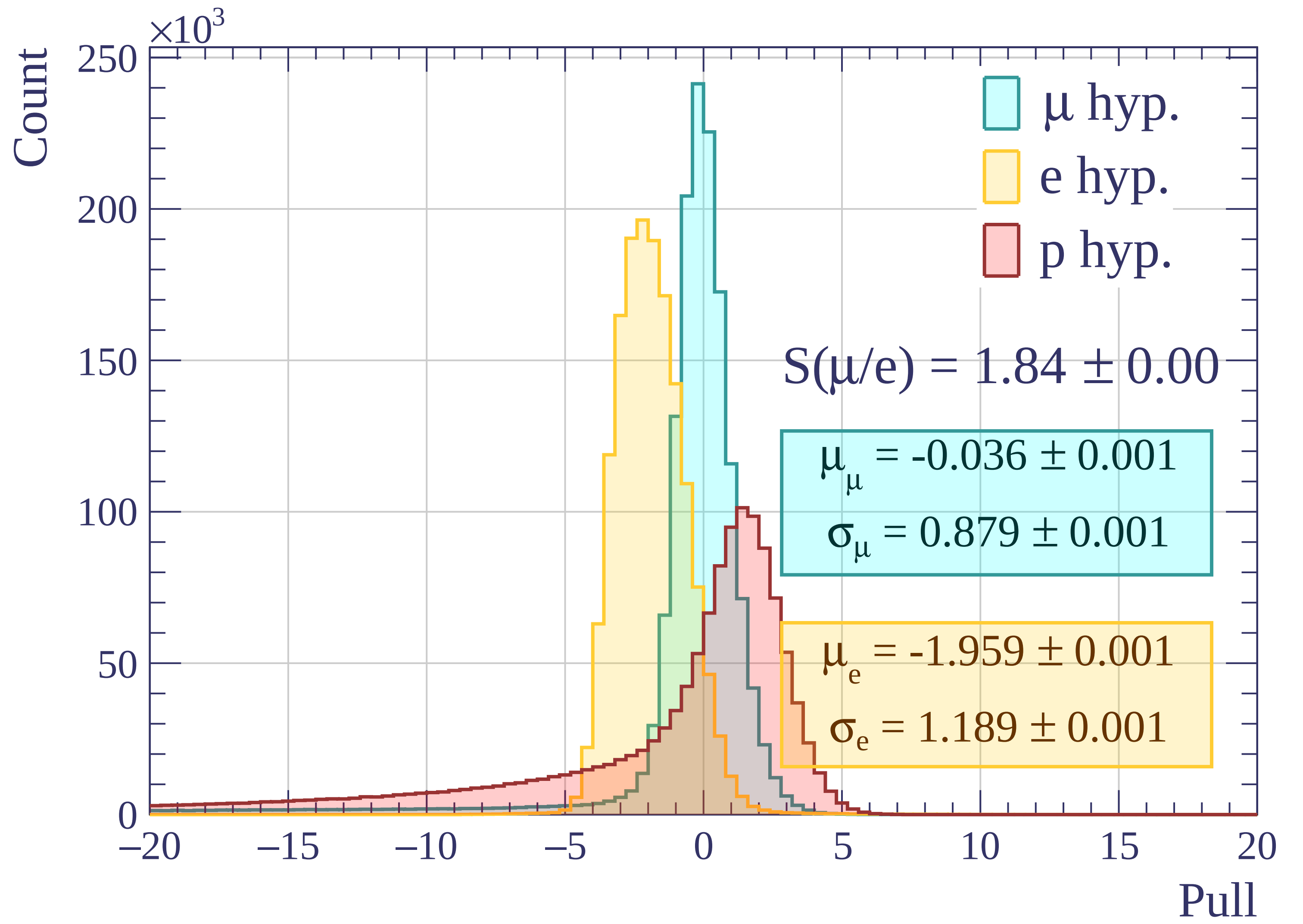


Pulls standard deviation
0
1
2
3
4
5
6
7
−2000
−1500
−1000
−500
0
500
1000
1500
2000
Momentum (MeV/c)
$\mu^{\pm}$
$e^{\pm}$

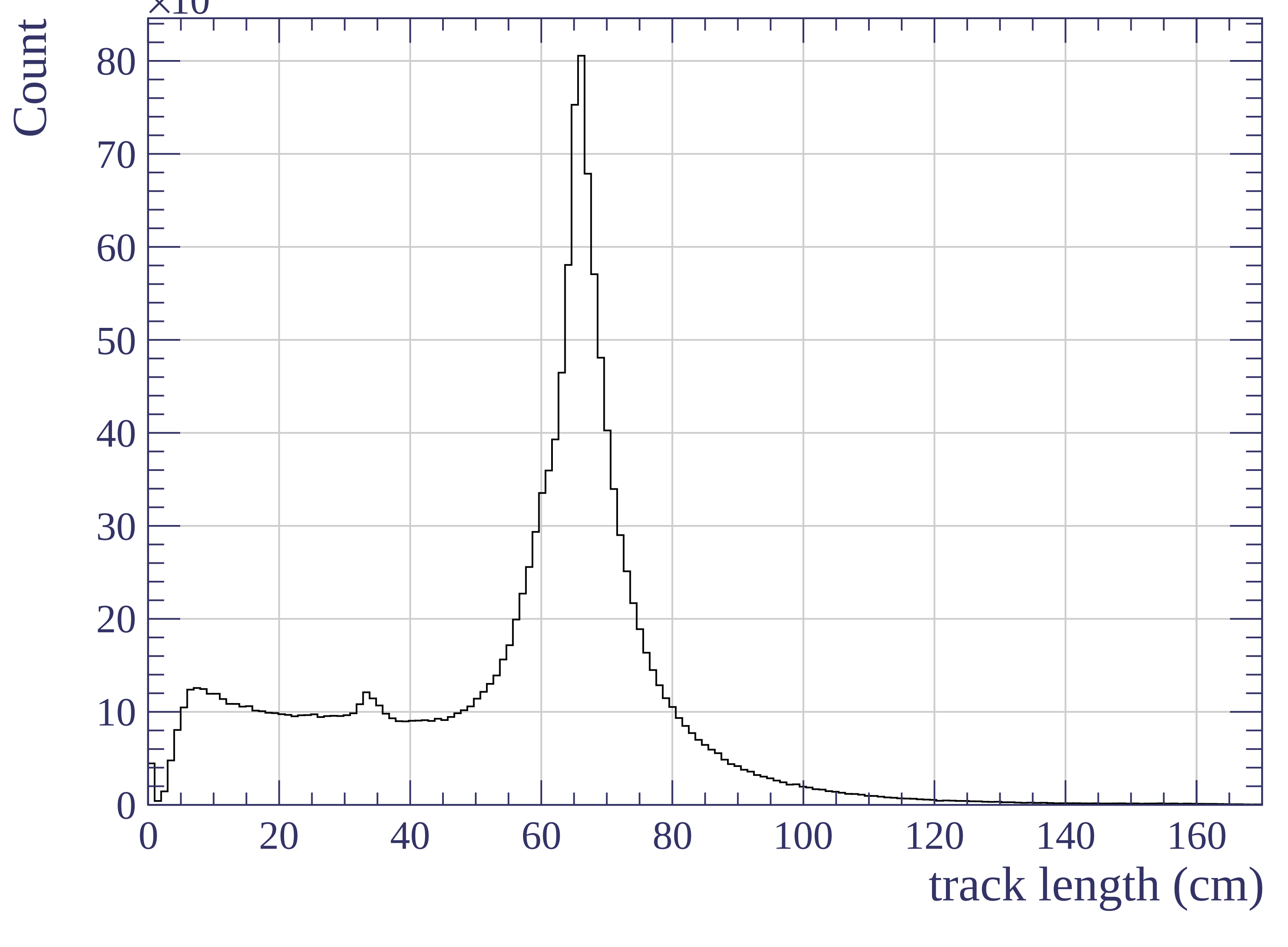

Pulls mean
0
−5
−10
−15
−20
−2000
−1500
−1000
−500
0
500
1000
1500
2000
Momentum (MeV/c)
μ±
e±



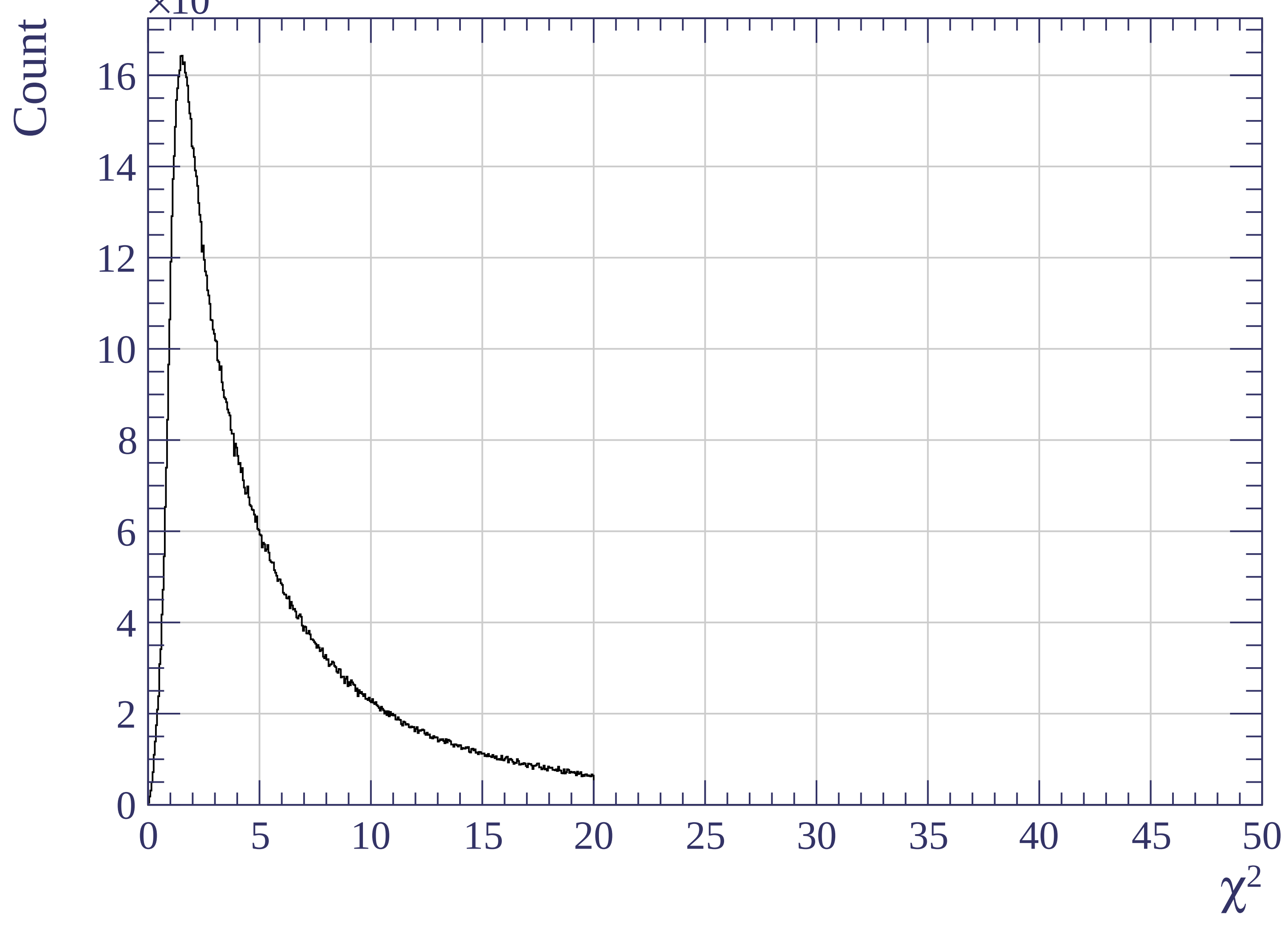
Pull e
Pull μ
×10³

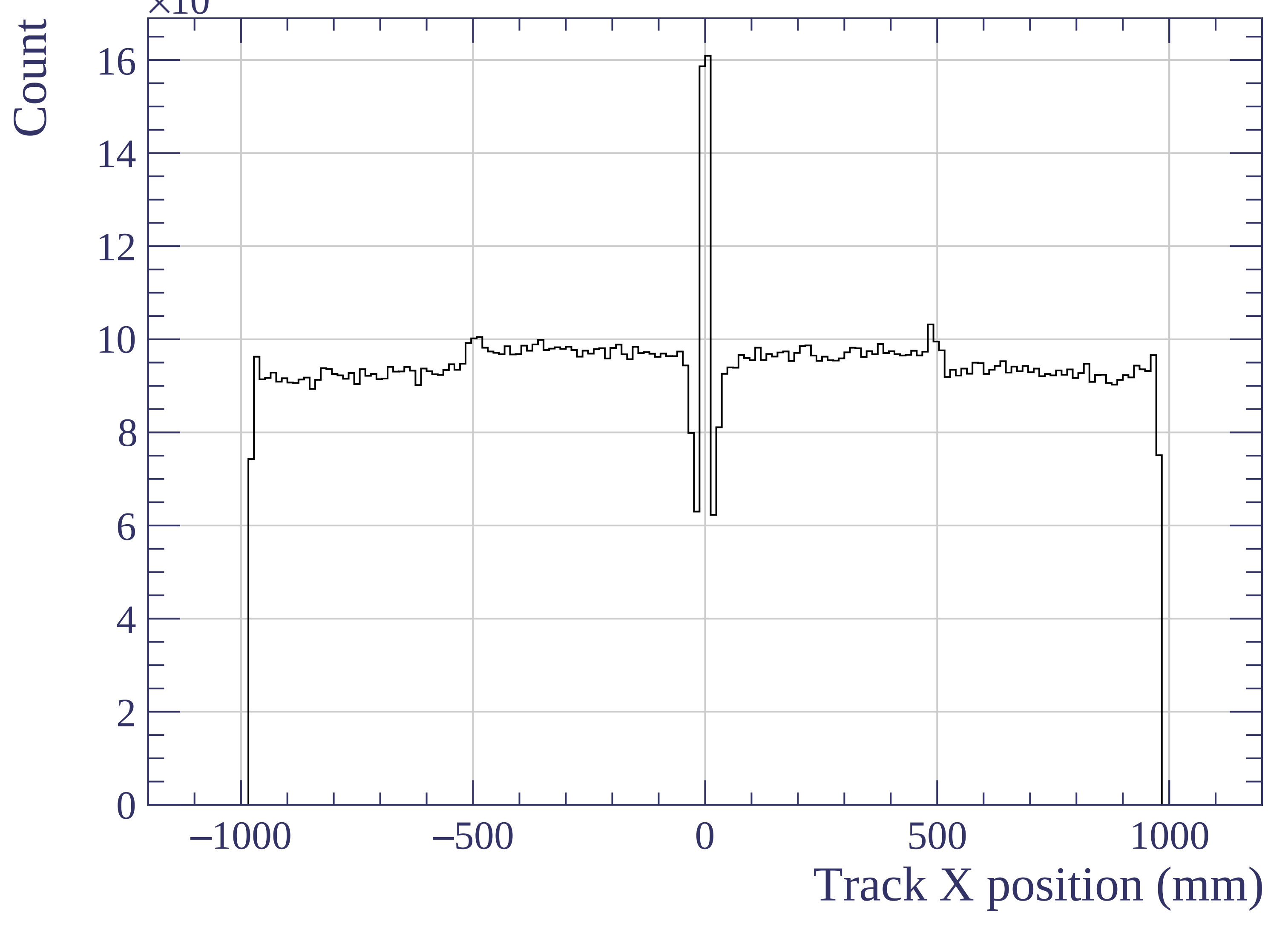


×10^3
Count
250
200
150
100
50
0
−20
−15
−10
−5
0
5
10
15
20
Pull
μ hyp.
e hyp.
p hyp.
S(μ/e) = 1.84 ± 0.00
μ_μ = -0.036 ± 0.001
σ_μ = 0.879 ± 0.001
μ_e = -1.959 ± 0.001
σ_e = 1.189 ± 0.001

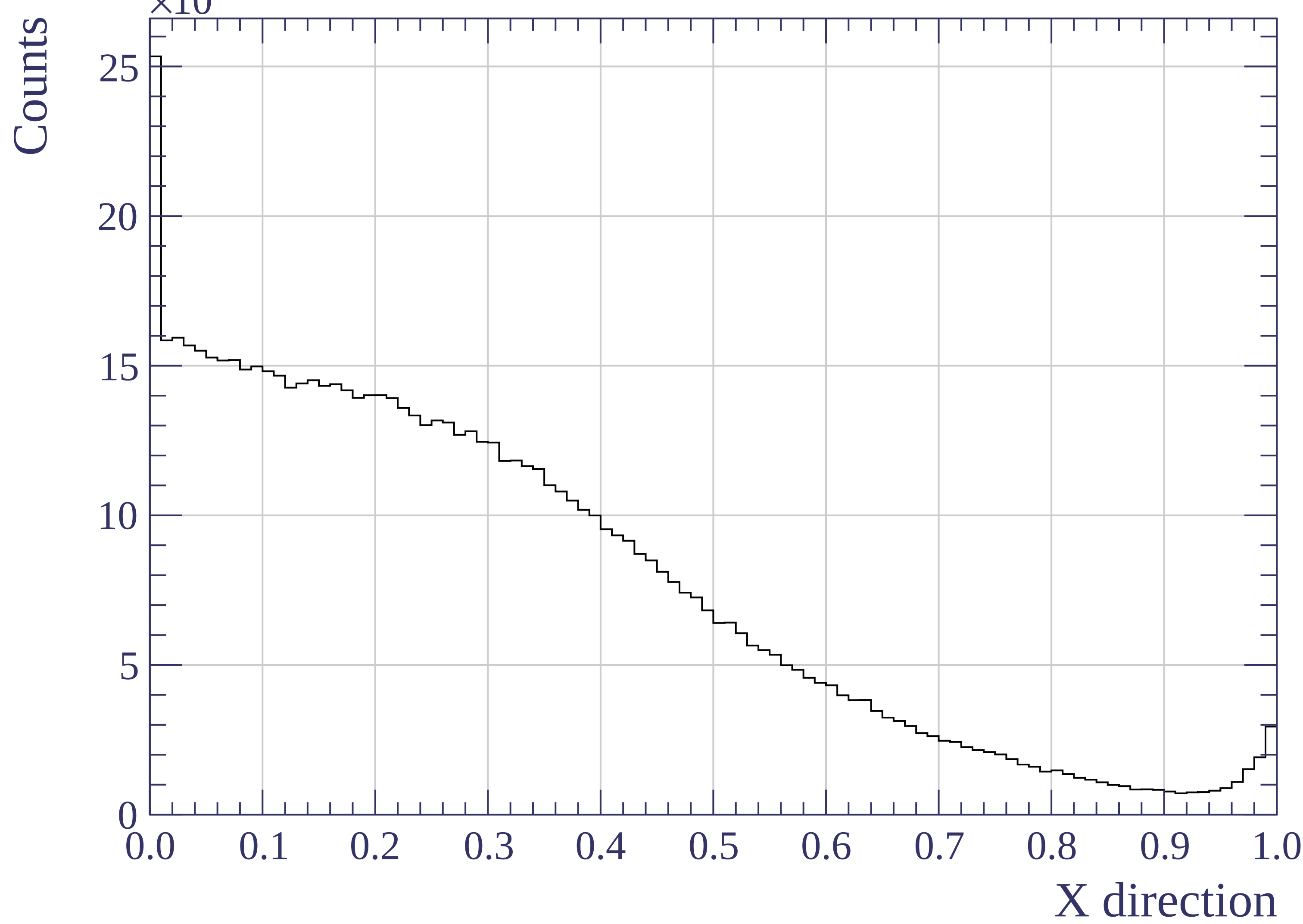


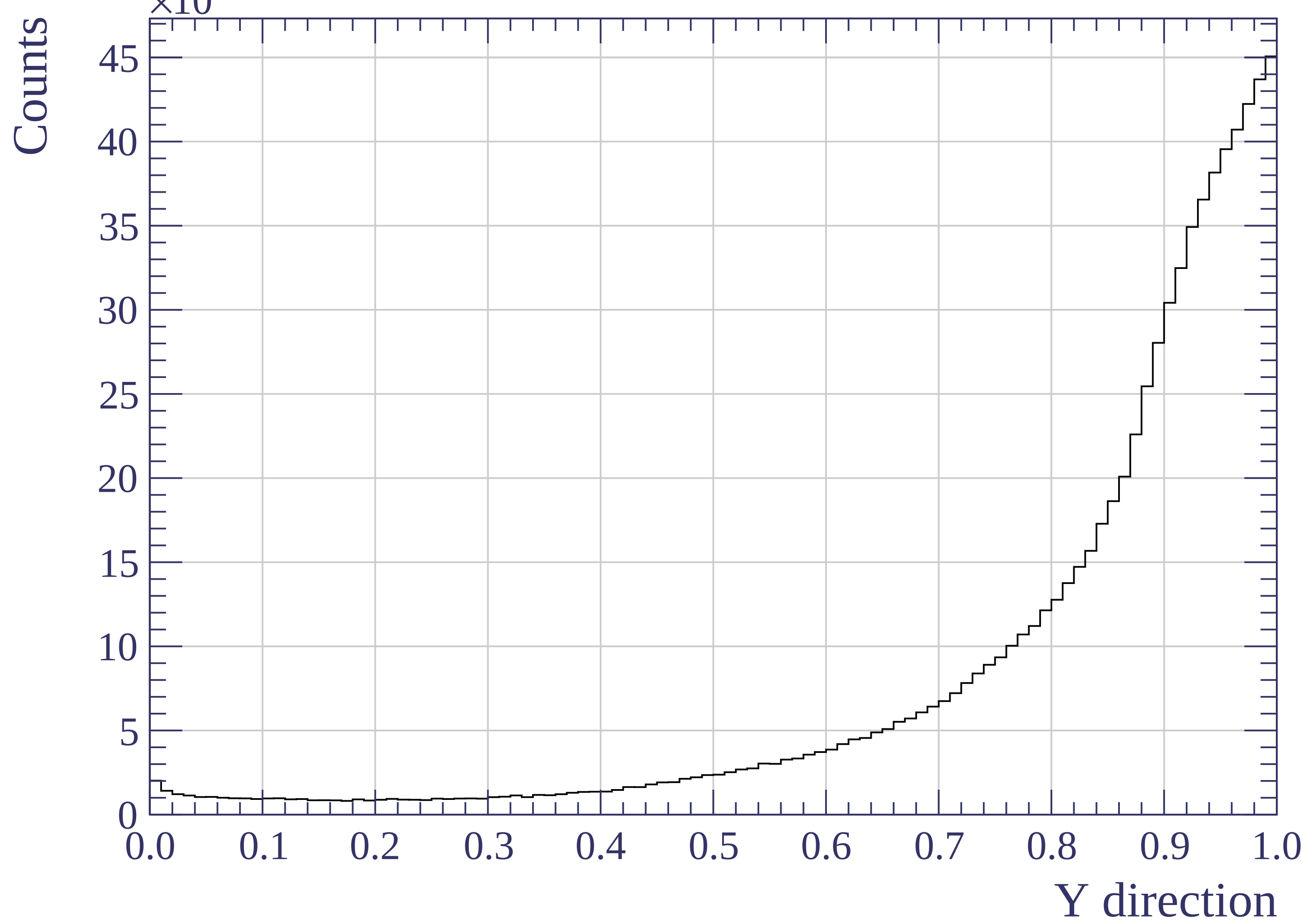
×10
Count
0
10
20
30
40
50
60
70
80
0
20
40
60
80
100
120
140
160
track length (cm)


×10
Count
16
14
12
10
8
6
4
2
0
0
5
10
15
20
25
30
35
40
45
50
χ^2

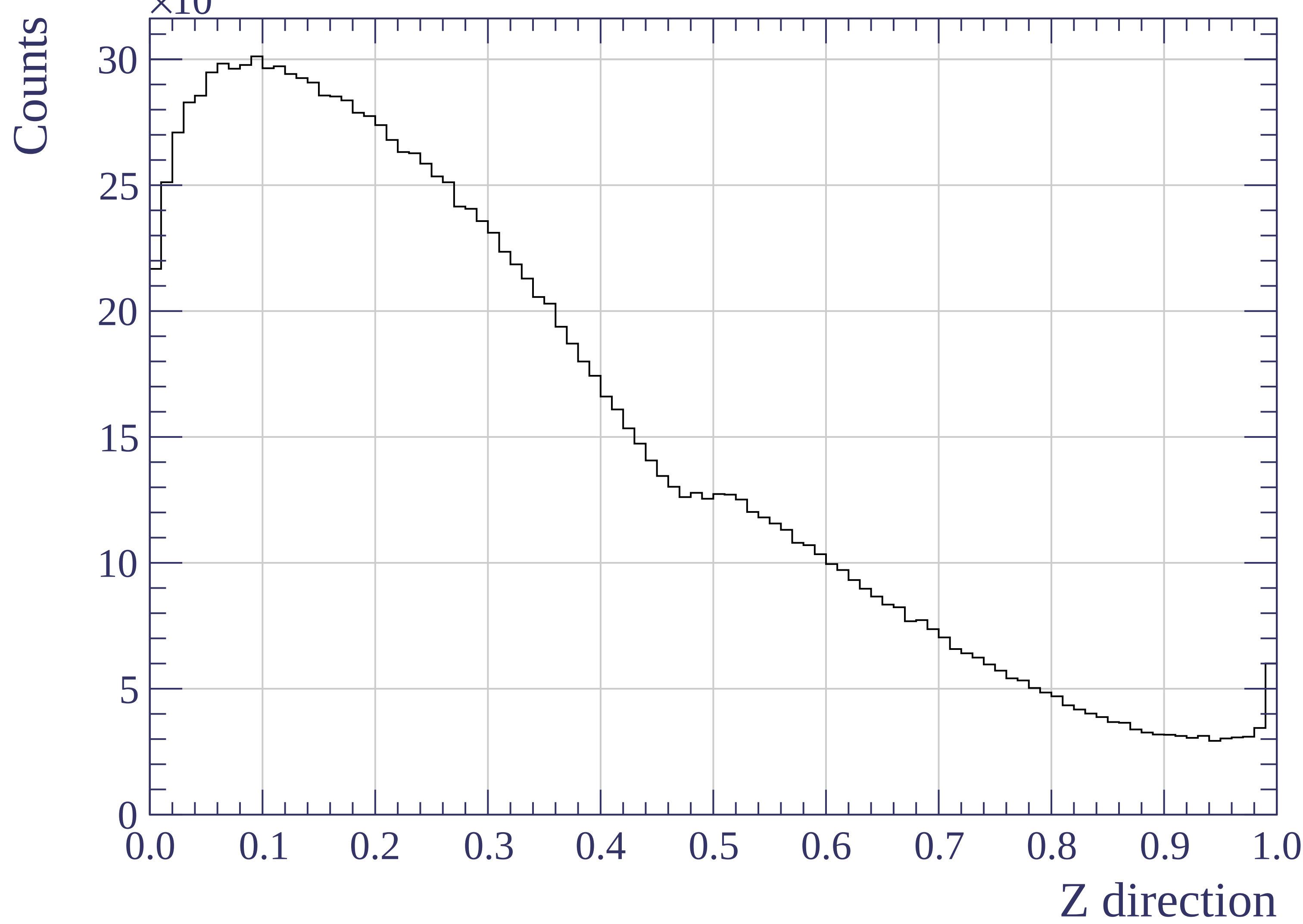


×10
Count
16
14
12
10
8
6
4
2
0
−1000
−500
0
500
1000
Track X position (mm)

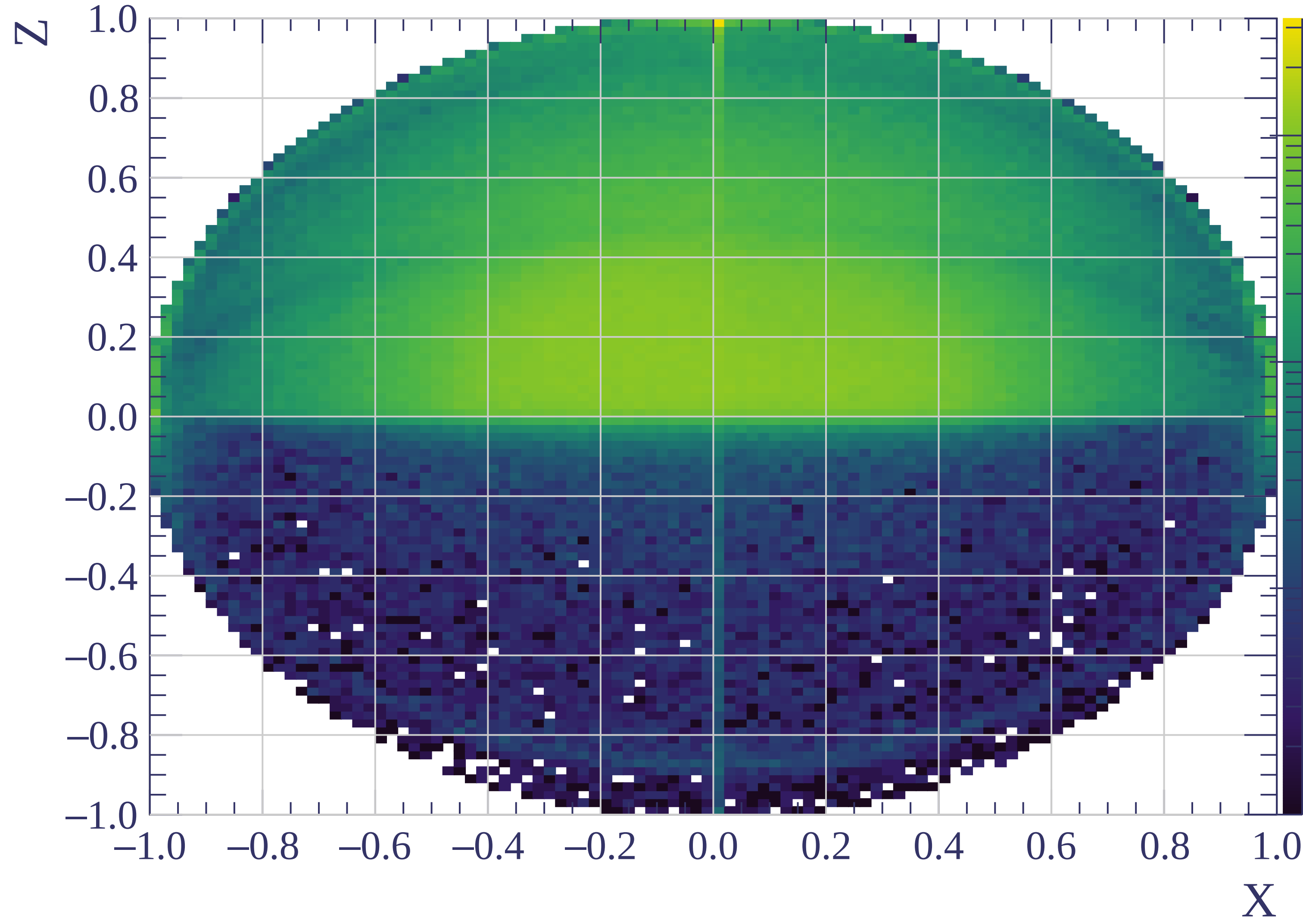

×10
Counts
25
20
15
10
5
0
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
X direction



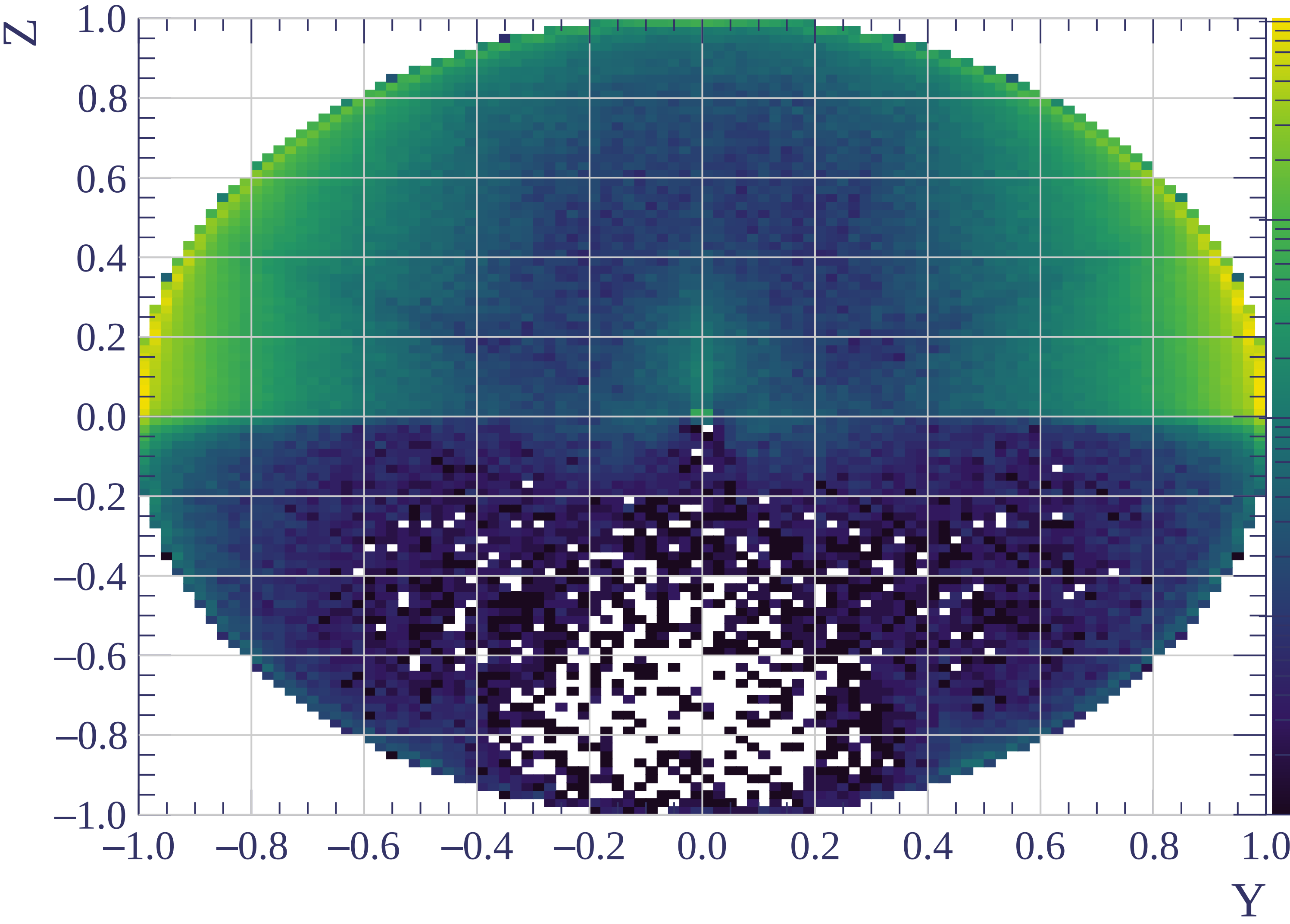
×10
Counts
45
40
35
30
25
20
15
10
5
0
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
Y direction

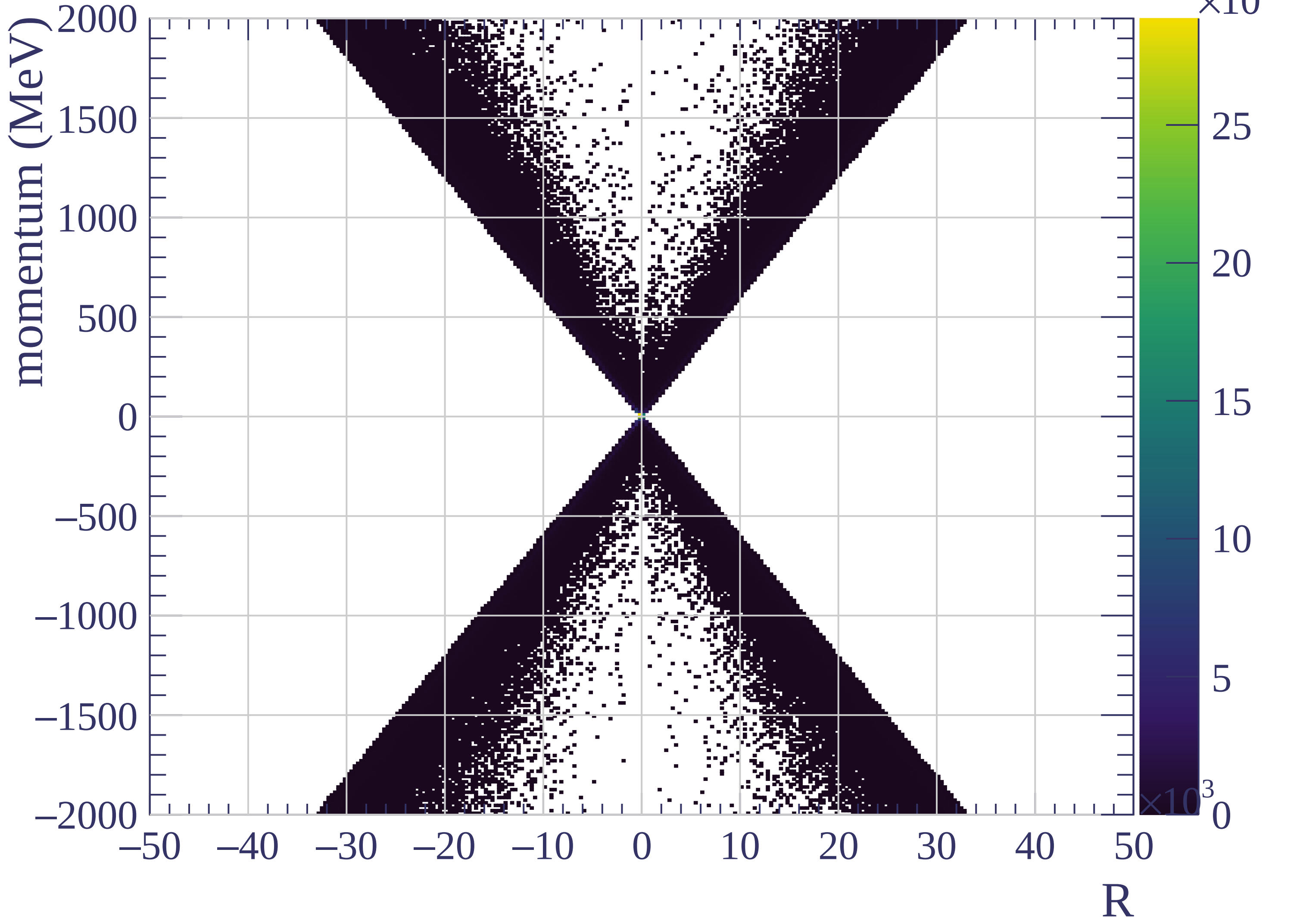


×10
Counts
30
25
20
15
10
5
0
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
Z direction

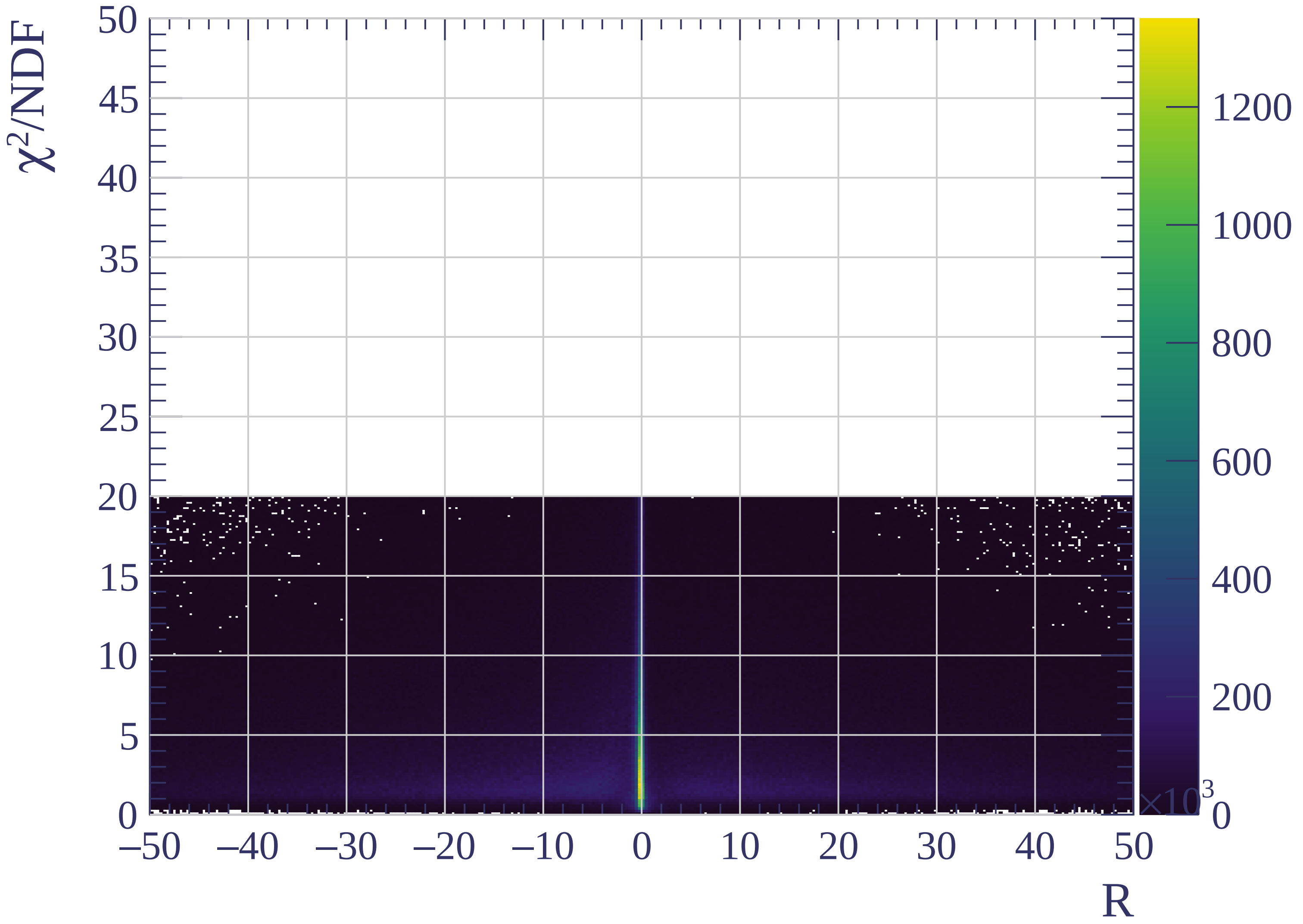


Z
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
−0.8
−1.0
−1.0
−0.8
−0.6
−0.4
−0.2
0.0
0.2
0.4
0.6
0.8
1.0
X

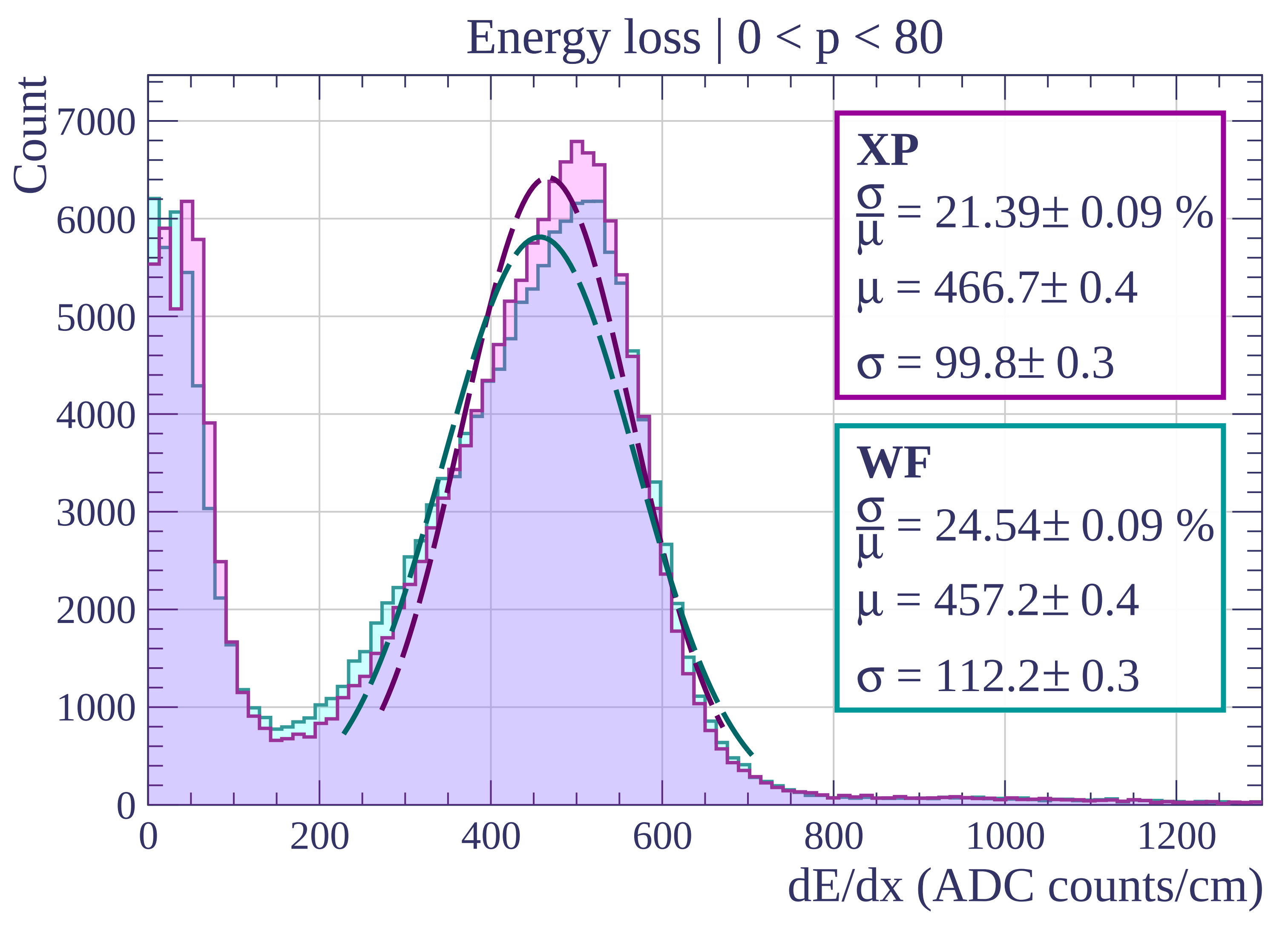

Z
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
−0.8
−1.0
−1.0
−0.8
−0.6
−0.4
−0.2
0.0
0.2
0.4
0.6
0.8
1.0
Y



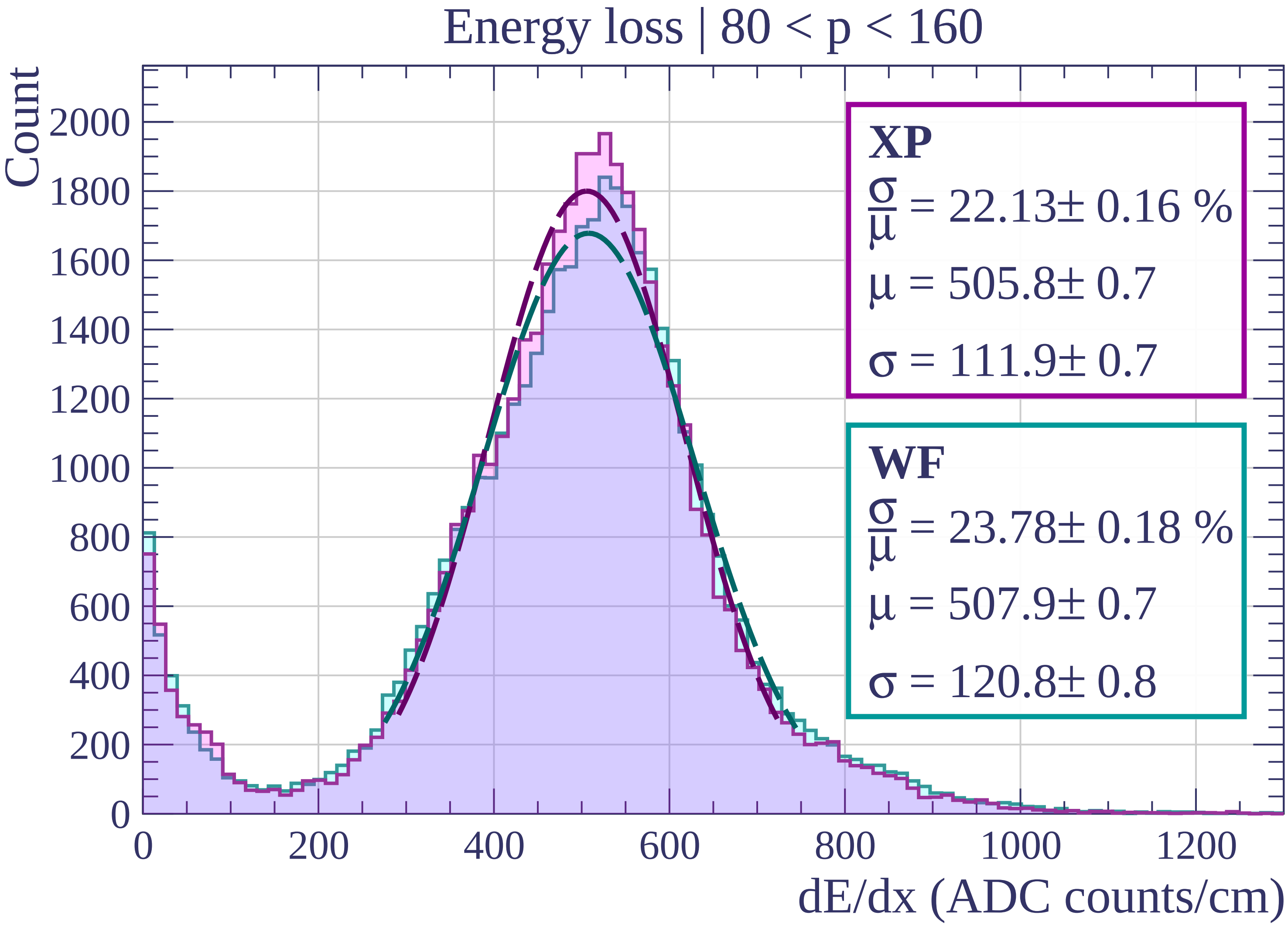
momentum (MeV)
2000
1500
1000
500
0
−500
−1000
−1500
−2000
−50
−40
−30
−20
−10
0
10
20
30
40
50
R
×10^3
25
20
15
10
5
0

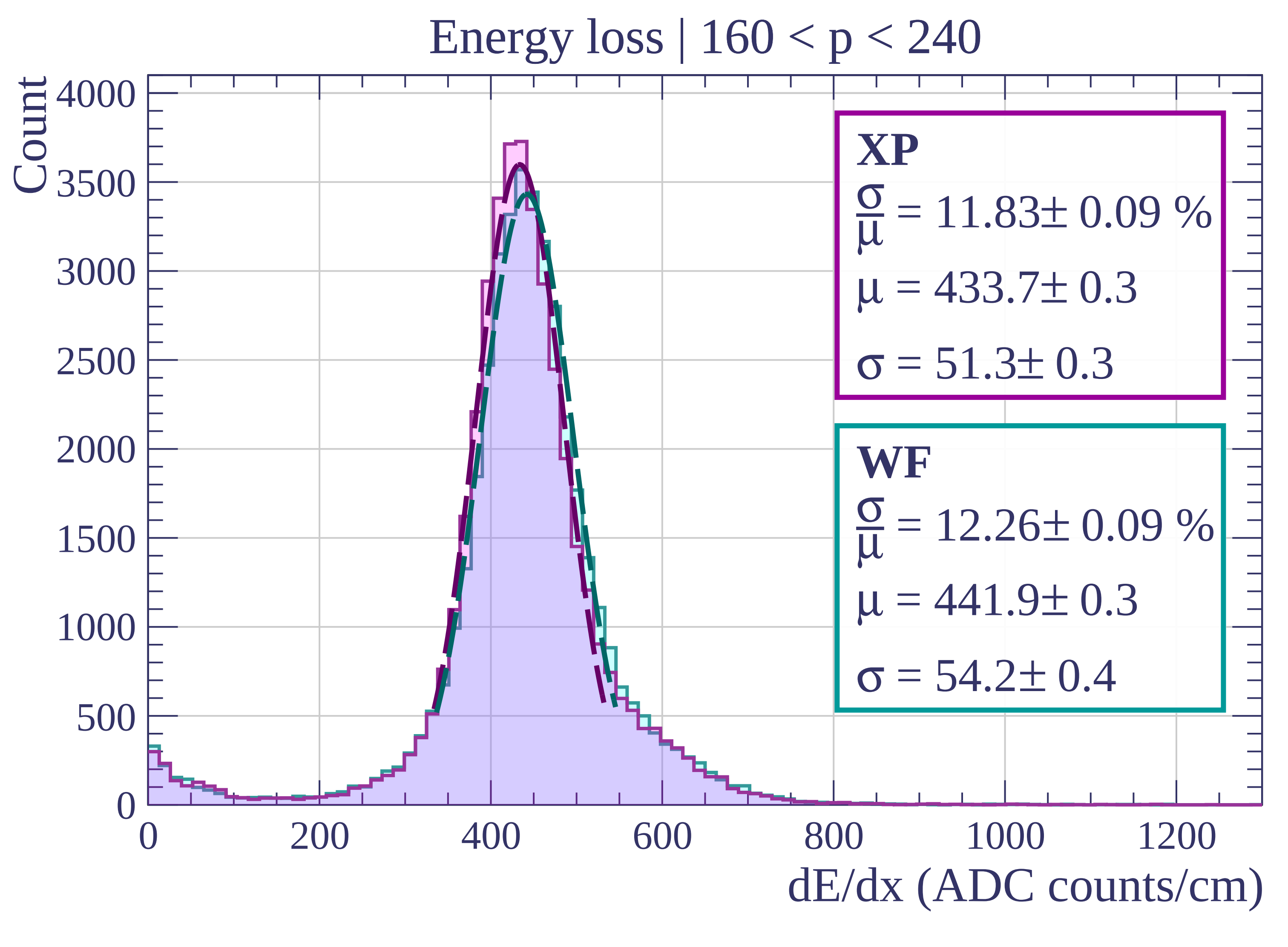


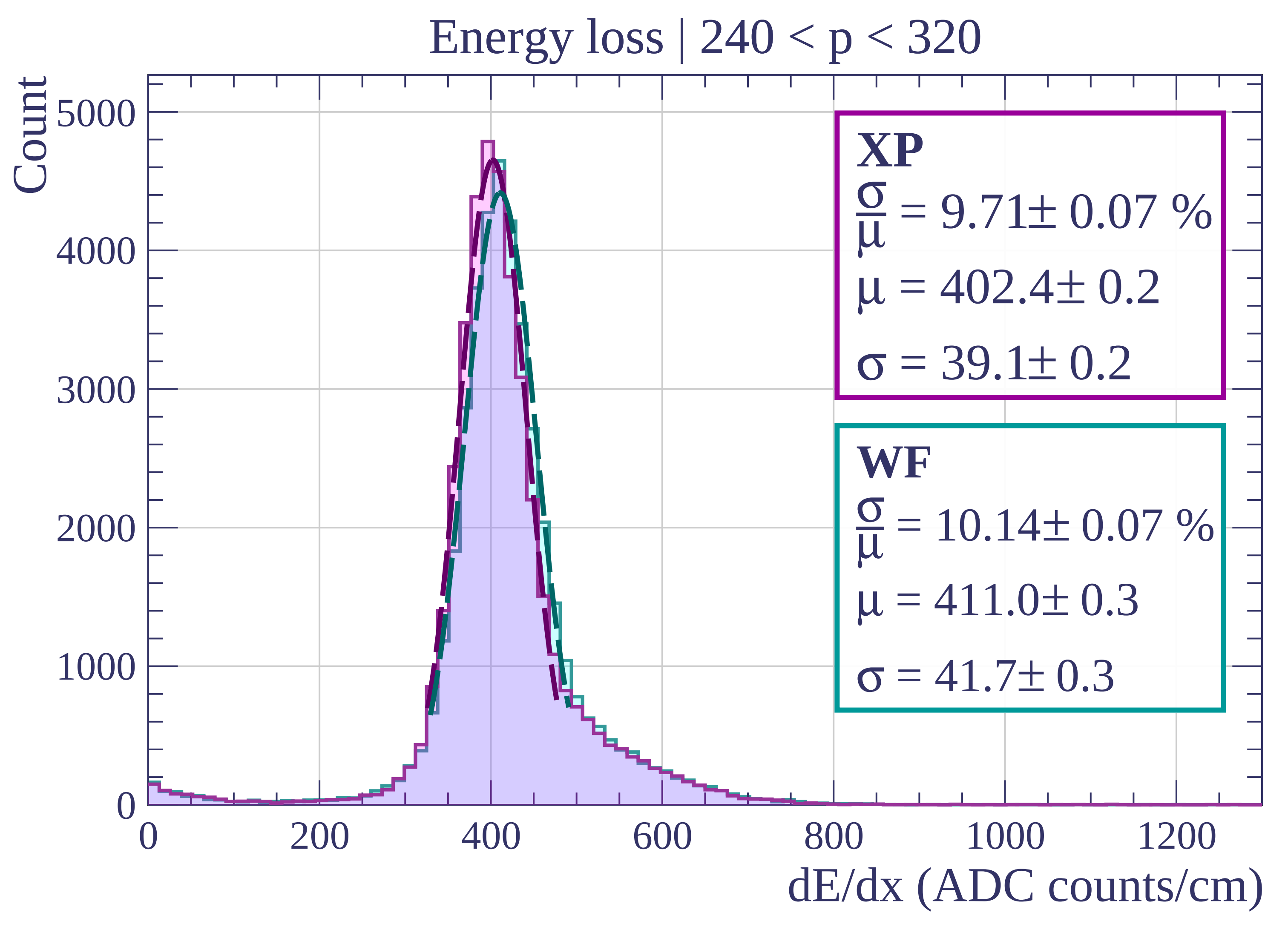
χ^2/NDF
50
45
40
35
30
25
20
15
10
5
0
−50
−40
−30
−20
−10
0
10
20
30
40
50
R
1200
1000
800
600
400
200
0
$\times 10^3$

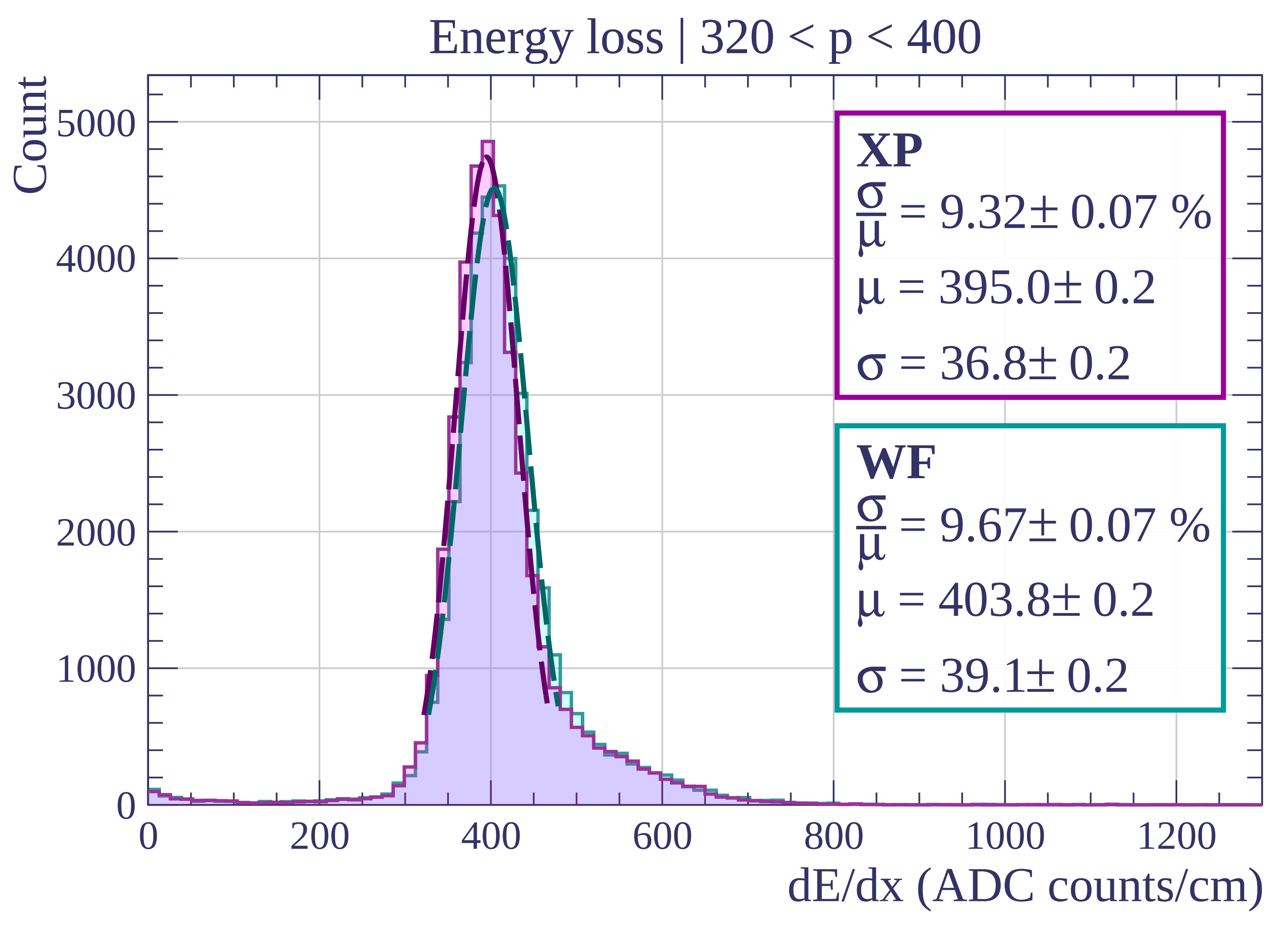


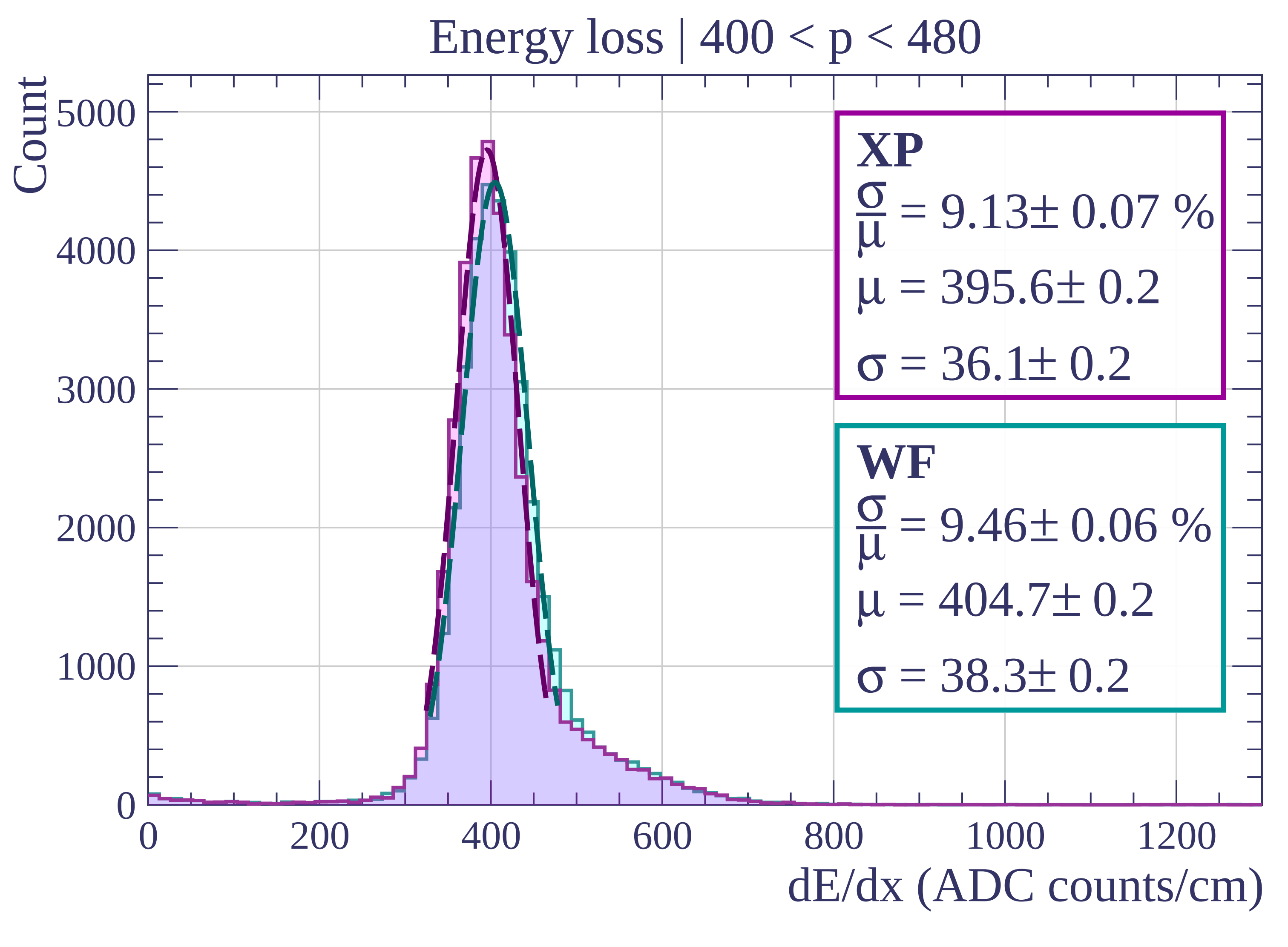
Energy loss | 0 < p < 80
Count
7000
6000
5000
4000
3000
2000
1000
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 21.39± 0.09 %
μ = 466.7± 0.4
σ = 99.8± 0.3
WF
$\frac{\sigma}{\mu}$ = 24.54± 0.09 %
μ = 457.2± 0.4
σ = 112.2± 0.3


Energy loss | 80 < p < 160
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
1400
1600
1800
2000
XP
$\frac{\sigma}{\mu}$ = 22.13± 0.16 %
μ = 505.8± 0.7
σ = 111.9± 0.7
WF
$\frac{\sigma}{\mu}$ = 23.78± 0.18 %
μ = 507.9± 0.7
σ = 120.8± 0.8

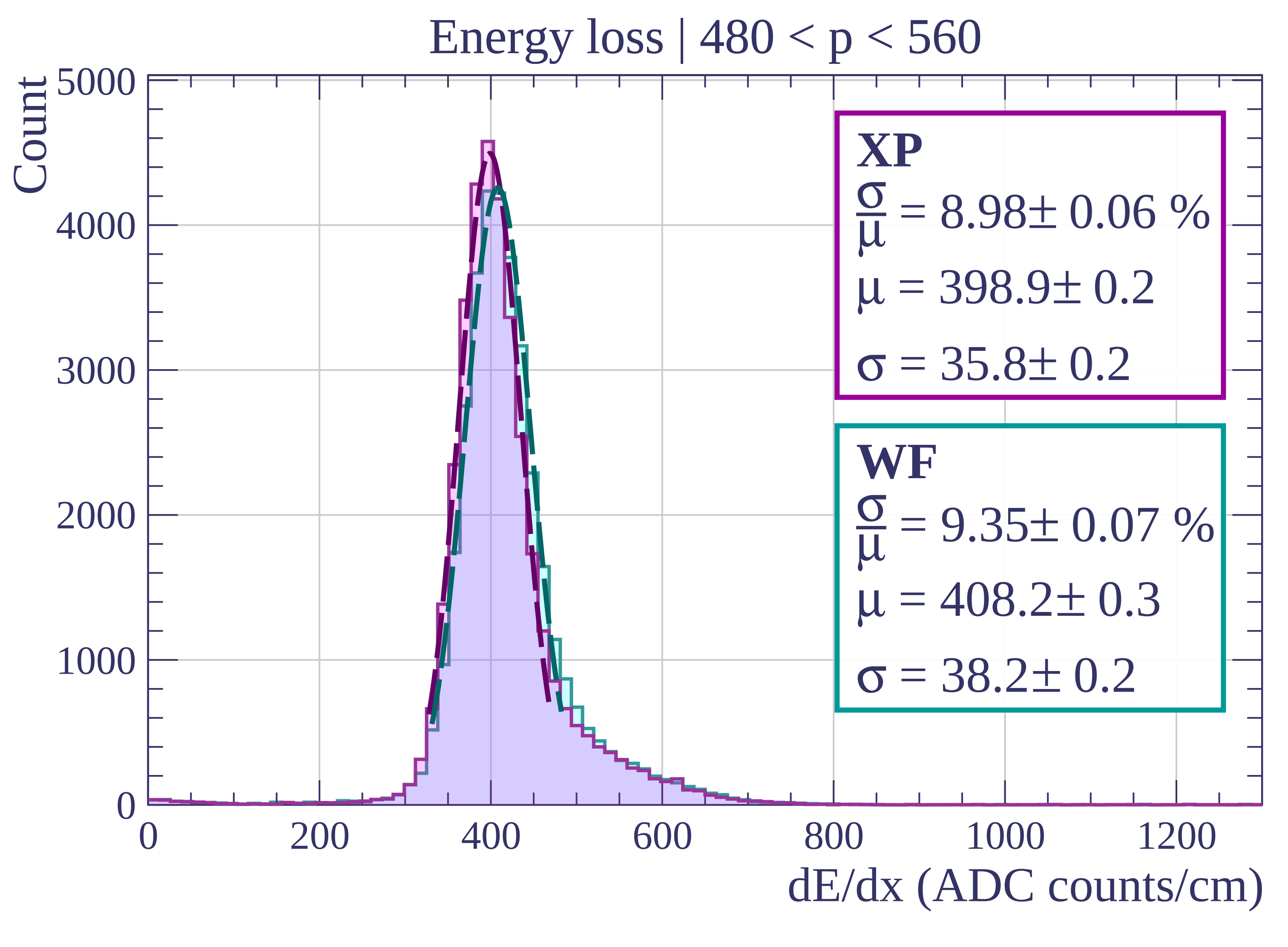

Energy loss | 160 < p < 240
Count
4000
3500
3000
2500
2000
1500
1000
500
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 11.83± 0.09 %
μ = 433.7± 0.3
σ = 51.3± 0.3
WF
σ/μ = 12.26± 0.09 %
μ = 441.9± 0.3
σ = 54.2± 0.4


Energy loss | 240 < p < 320
Count
5000
4000
3000
2000
1000
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu} = 9.71 \pm 0.07$ %
$\mu = 402.4 \pm 0.2$
$\sigma = 39.1 \pm 0.2$
WF
$\frac{\sigma}{\mu} = 10.14 \pm 0.07$ %
$\mu = 411.0 \pm 0.3$
$\sigma = 41.7 \pm 0.3$

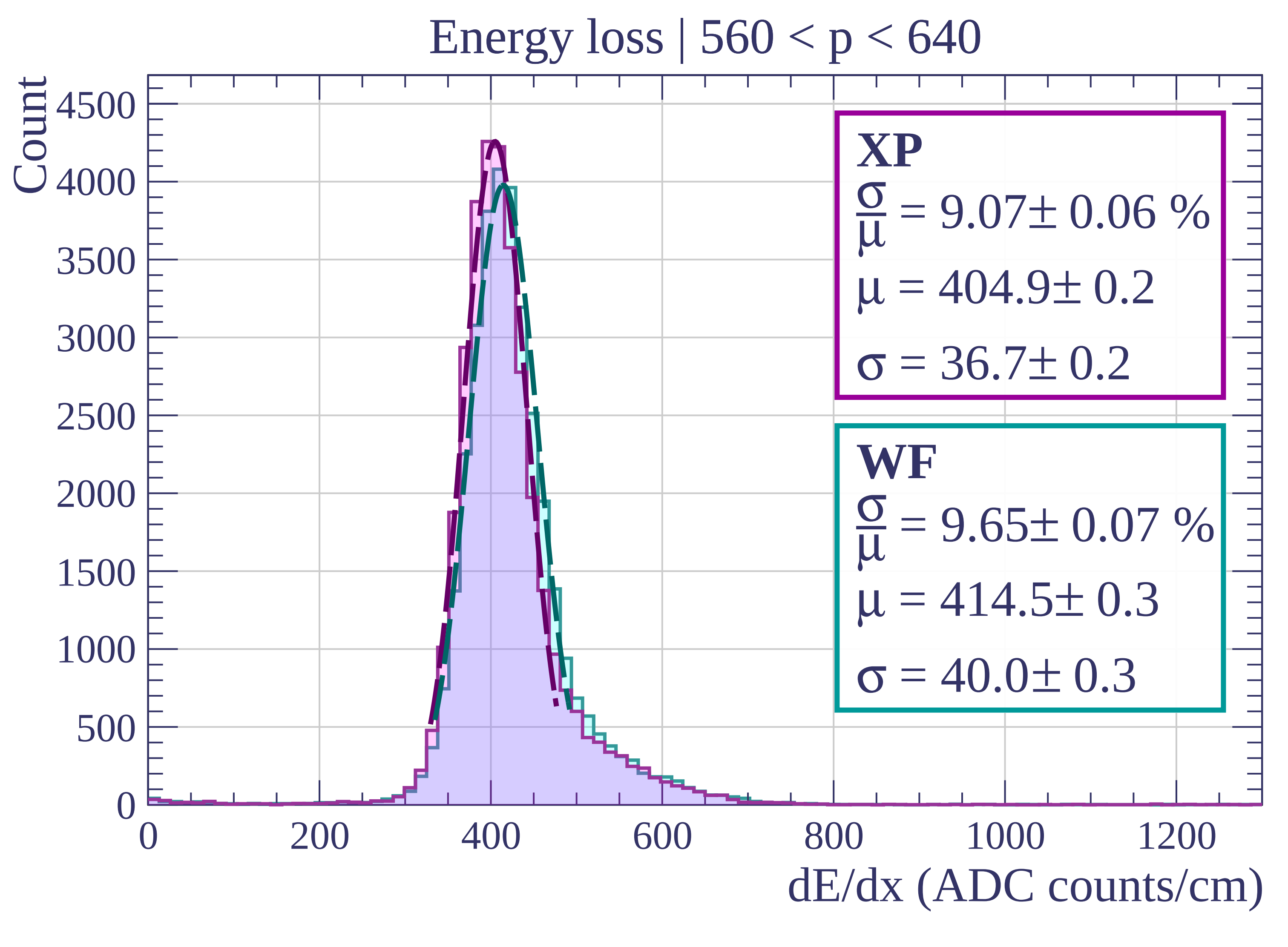

Energy loss | 320 < p < 400
Count
dE/dx (ADC counts/cm)
0
1000
2000
3000
4000
5000
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 9.32± 0.07 %
μ = 395.0± 0.2
σ = 36.8± 0.2
WF
$\frac{\sigma}{\mu}$ = 9.67± 0.07 %
μ = 403.8± 0.2
σ = 39.1± 0.2



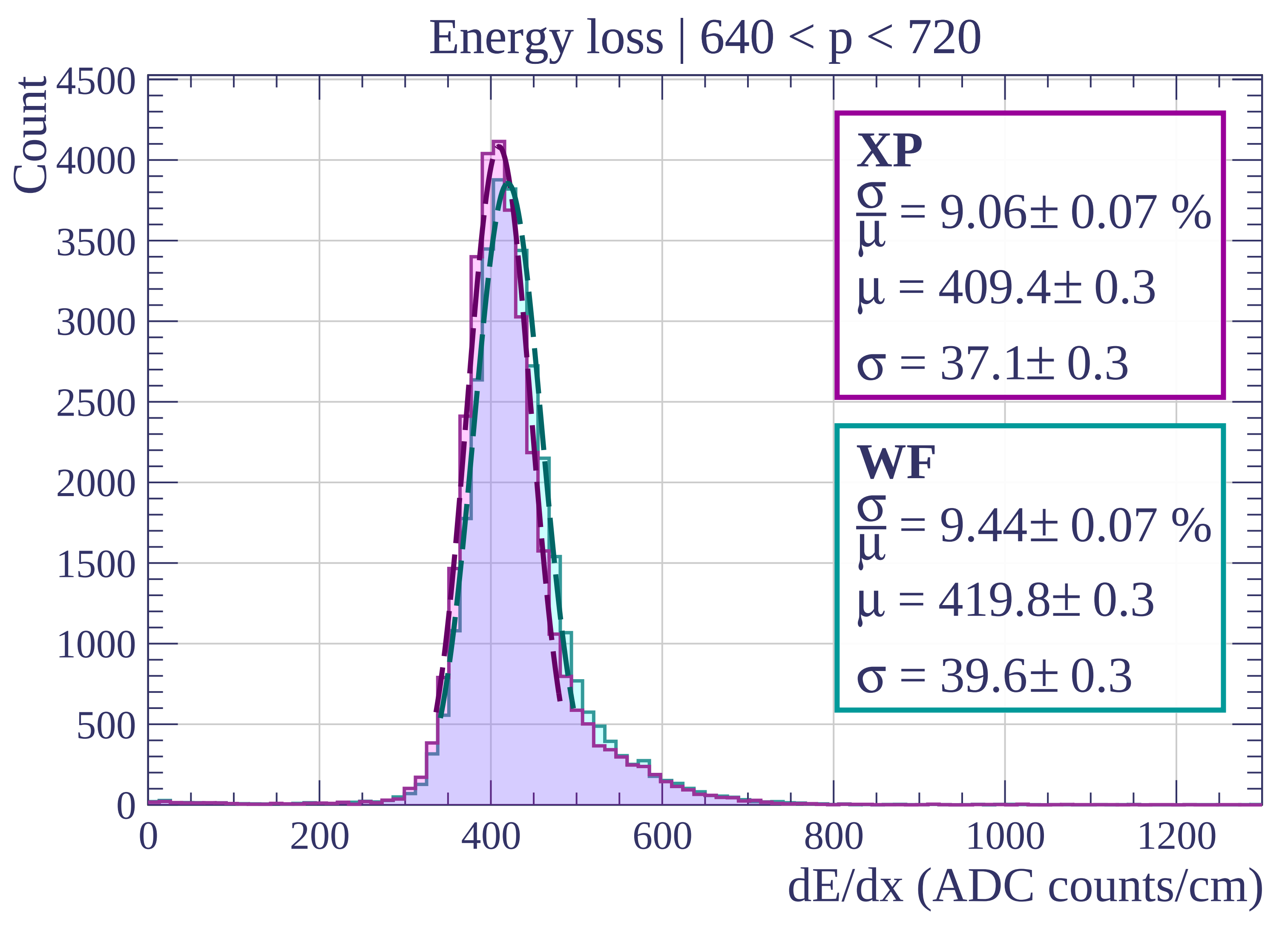
Energy loss | 400 < p < 480
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
1000
2000
3000
4000
5000
XP
$\frac{\sigma}{\mu} = 9.13 \pm 0.07$ %
$\mu = 395.6 \pm 0.2$
$\sigma = 36.1 \pm 0.2$
WF
$\frac{\sigma}{\mu} = 9.46 \pm 0.06$ %
$\mu = 404.7 \pm 0.2$
$\sigma = 38.3 \pm 0.2$



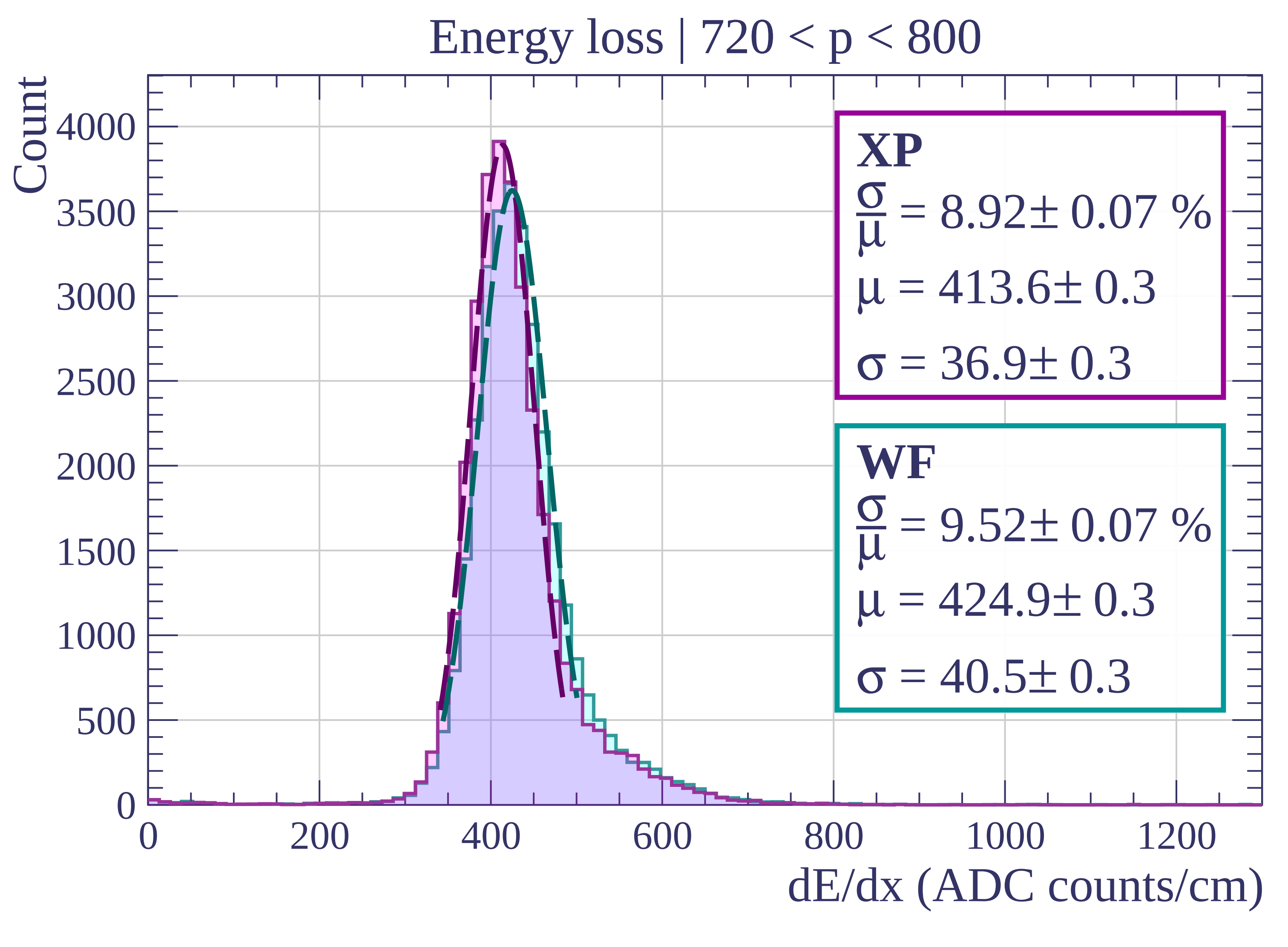
Energy loss | 480 < p < 560
Count
5000
4000
3000
2000
1000
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 8.98± 0.06 %
μ = 398.9± 0.2
σ = 35.8± 0.2
WF
σ/μ = 9.35± 0.07 %
μ = 408.2± 0.3
σ = 38.2± 0.2



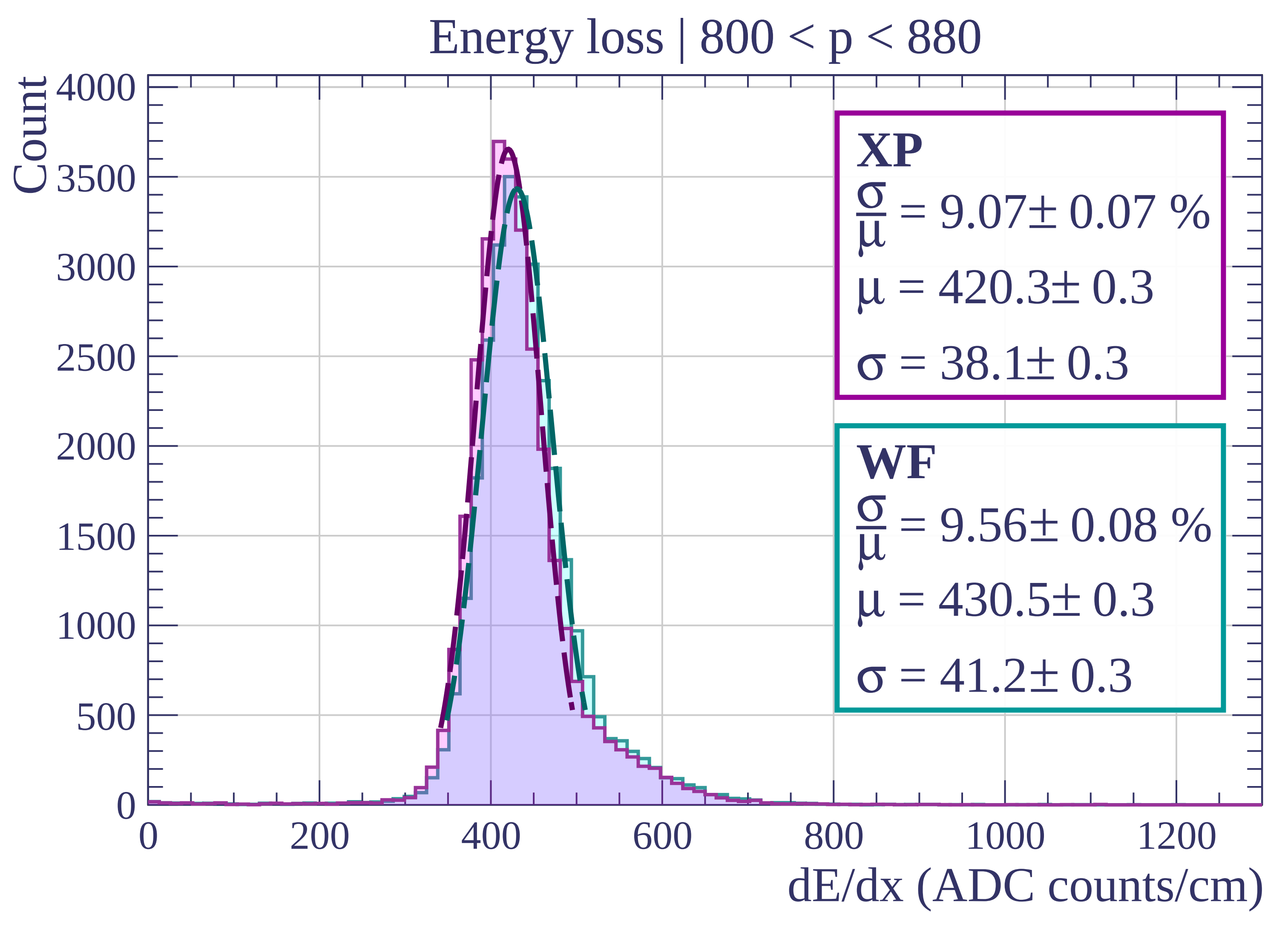
Energy loss | 560 < p < 640
Count
dE/dx (ADC counts/cm)
0
500
1000
1500
2000
2500
3000
3500
4000
4500
0
200
400
600
800
1000
1200
XP
σ/μ = 9.07± 0.06 %
μ = 404.9± 0.2
σ = 36.7± 0.2
WF
σ/μ = 9.65± 0.07 %
μ = 414.5± 0.3
σ = 40.0± 0.3



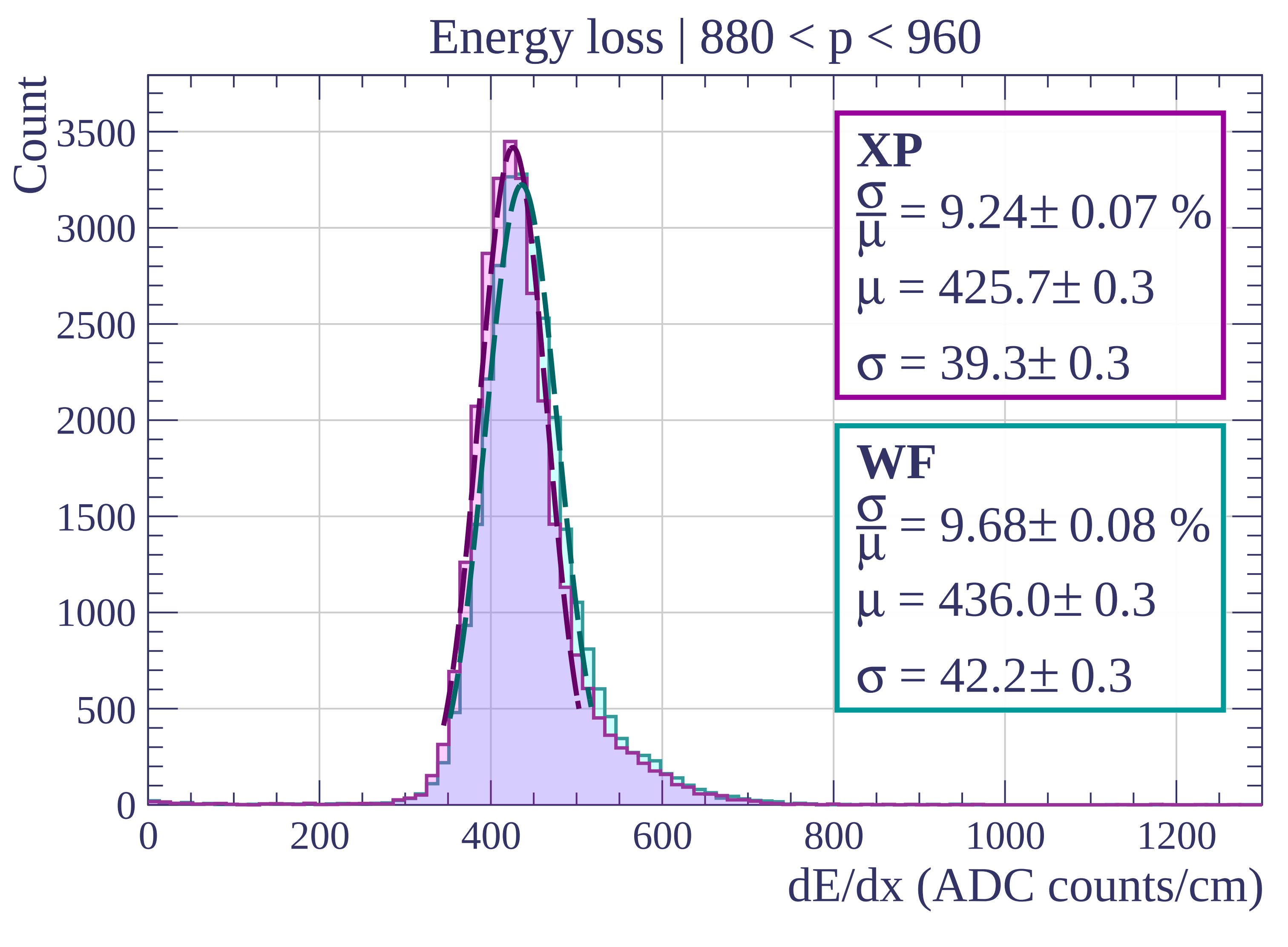
Energy loss | 640 < p < 720
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 9.06± 0.07 %
μ = 409.4± 0.3
σ = 37.1± 0.3
WF
σ/μ = 9.44± 0.07 %
μ = 419.8± 0.3
σ = 39.6± 0.3

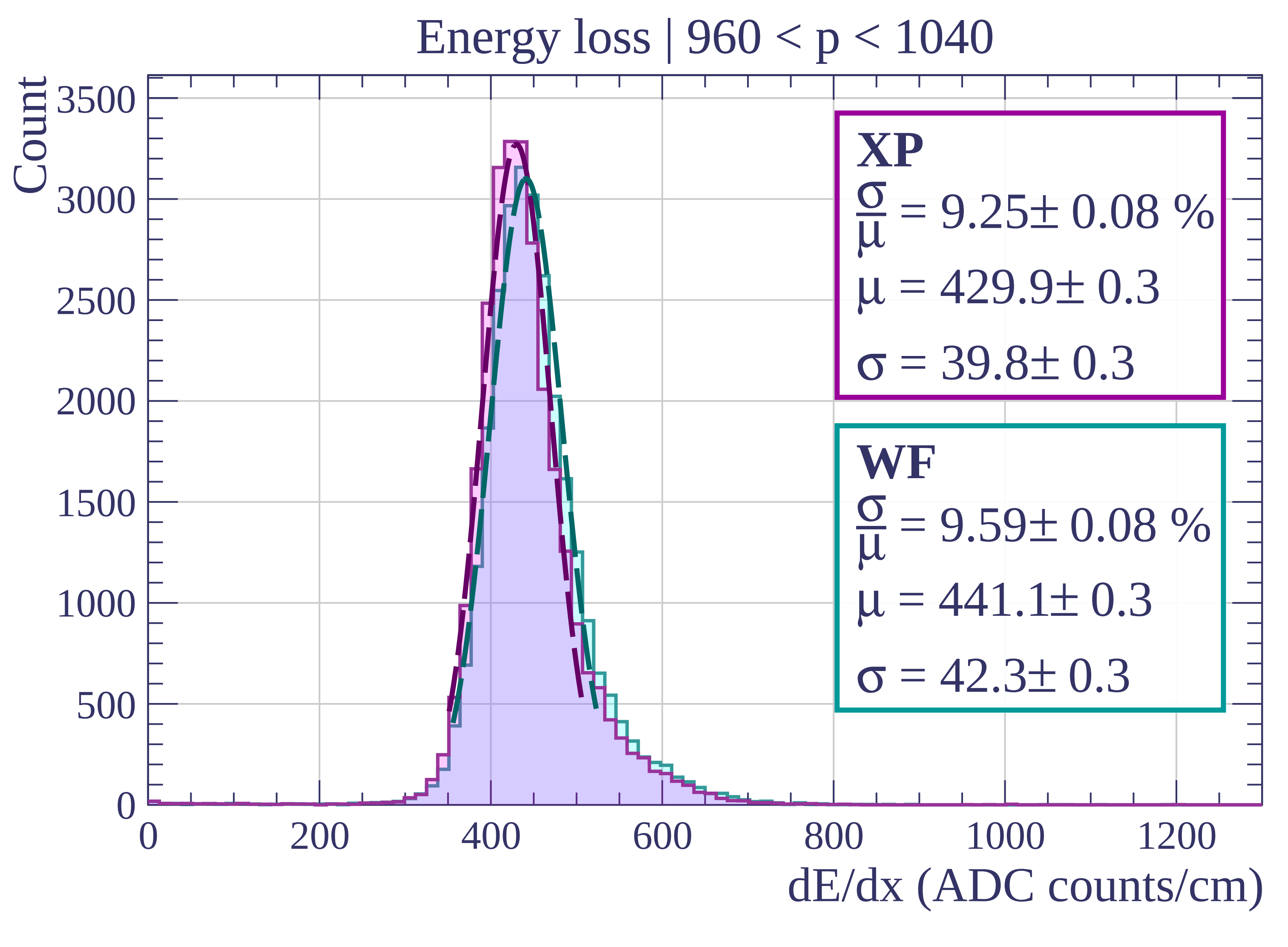


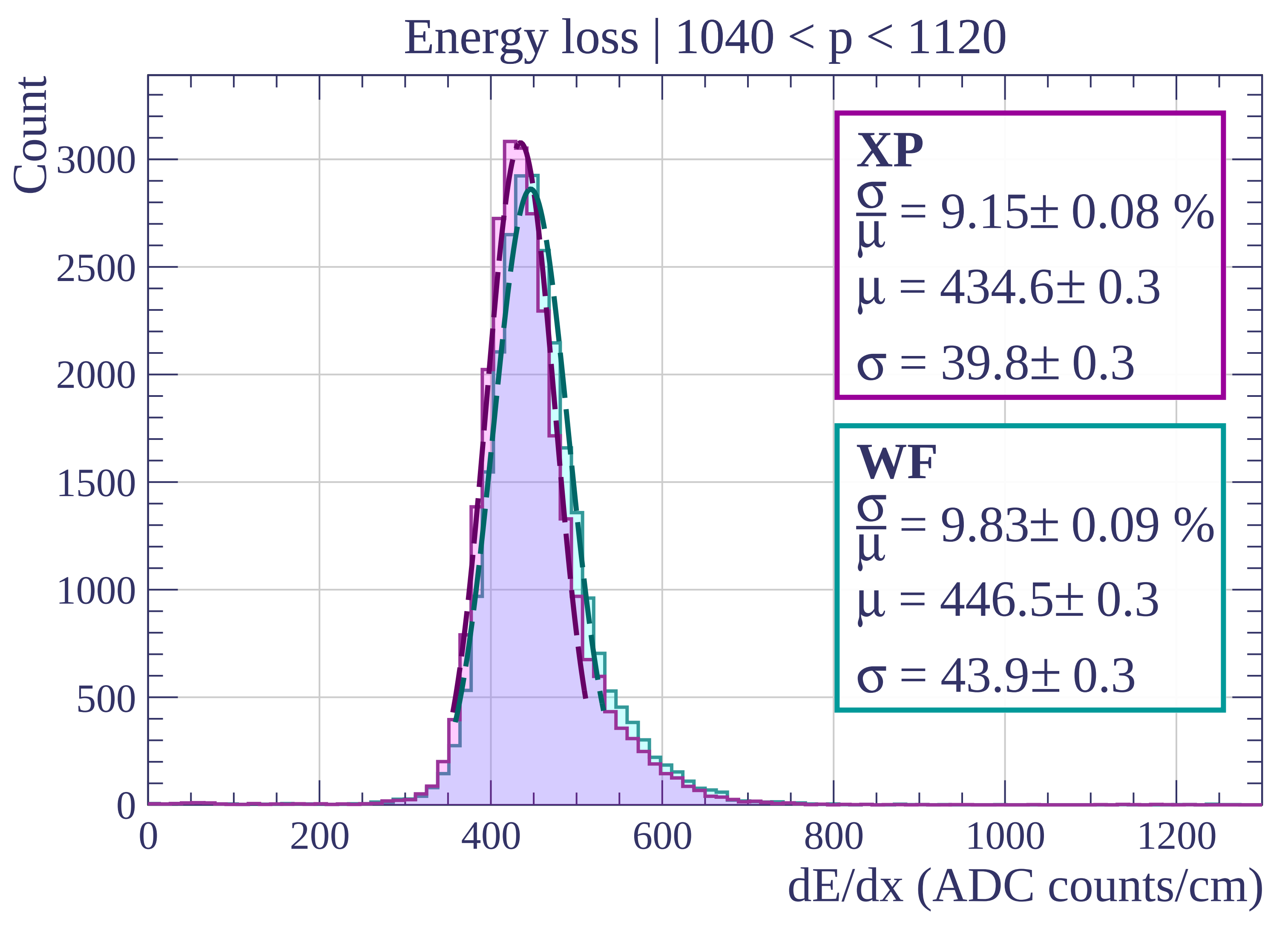
Energy loss | 720 < p < 800
Count
4000
3500
3000
2500
2000
1500
1000
500
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 8.92± 0.07 %
μ = 413.6± 0.3
σ = 36.9± 0.3
WF
σ/μ = 9.52± 0.07 %
μ = 424.9± 0.3
σ = 40.5± 0.3

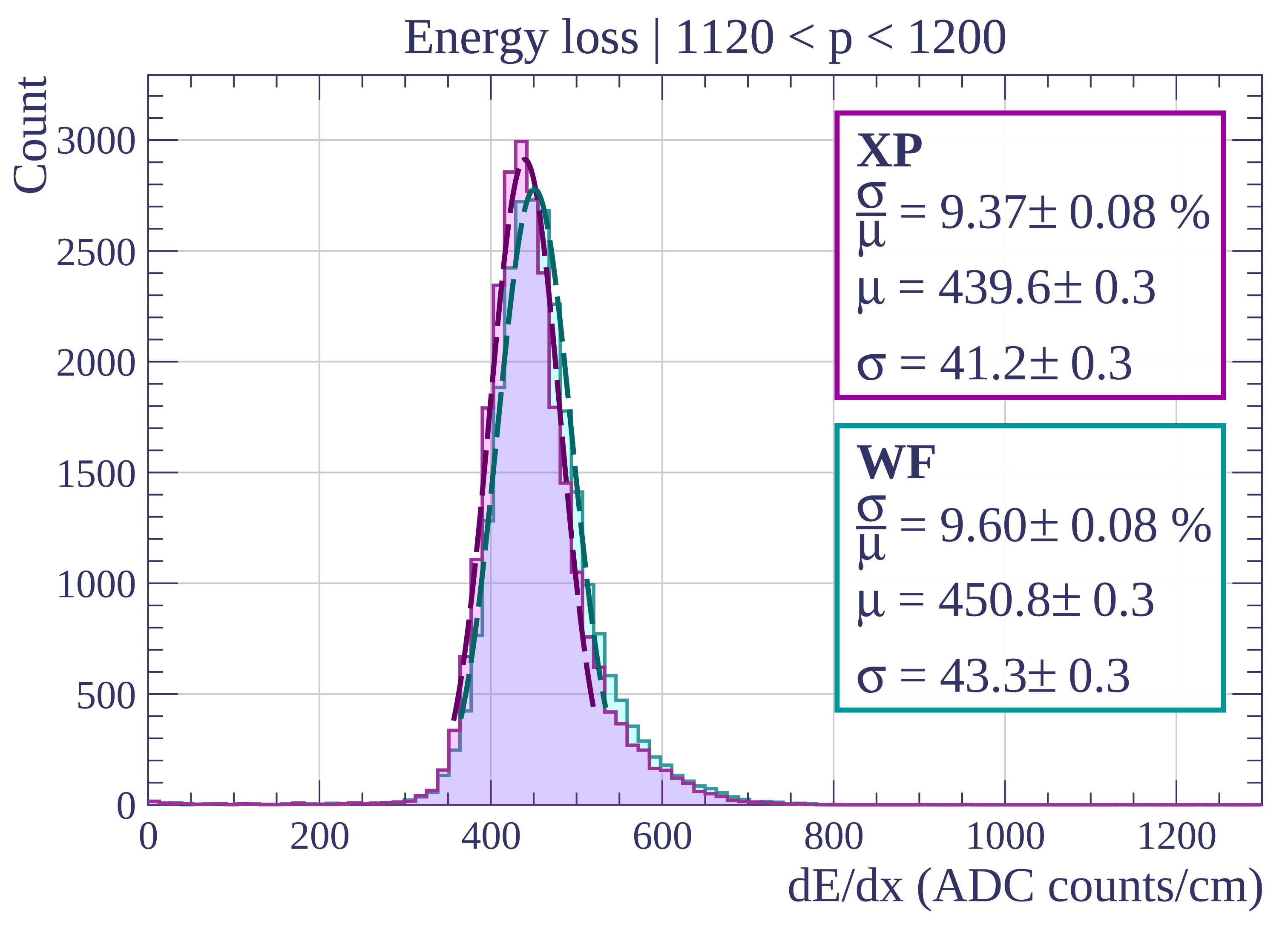


Energy loss | 800 < p < 880
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
500
1000
1500
2000
2500
3000
3500
4000
XP
$\frac{\sigma}{\mu}$ = 9.07± 0.07 %
μ = 420.3± 0.3
σ = 38.1± 0.3
WF
$\frac{\sigma}{\mu}$ = 9.56± 0.08 %
μ = 430.5± 0.3
σ = 41.2± 0.3

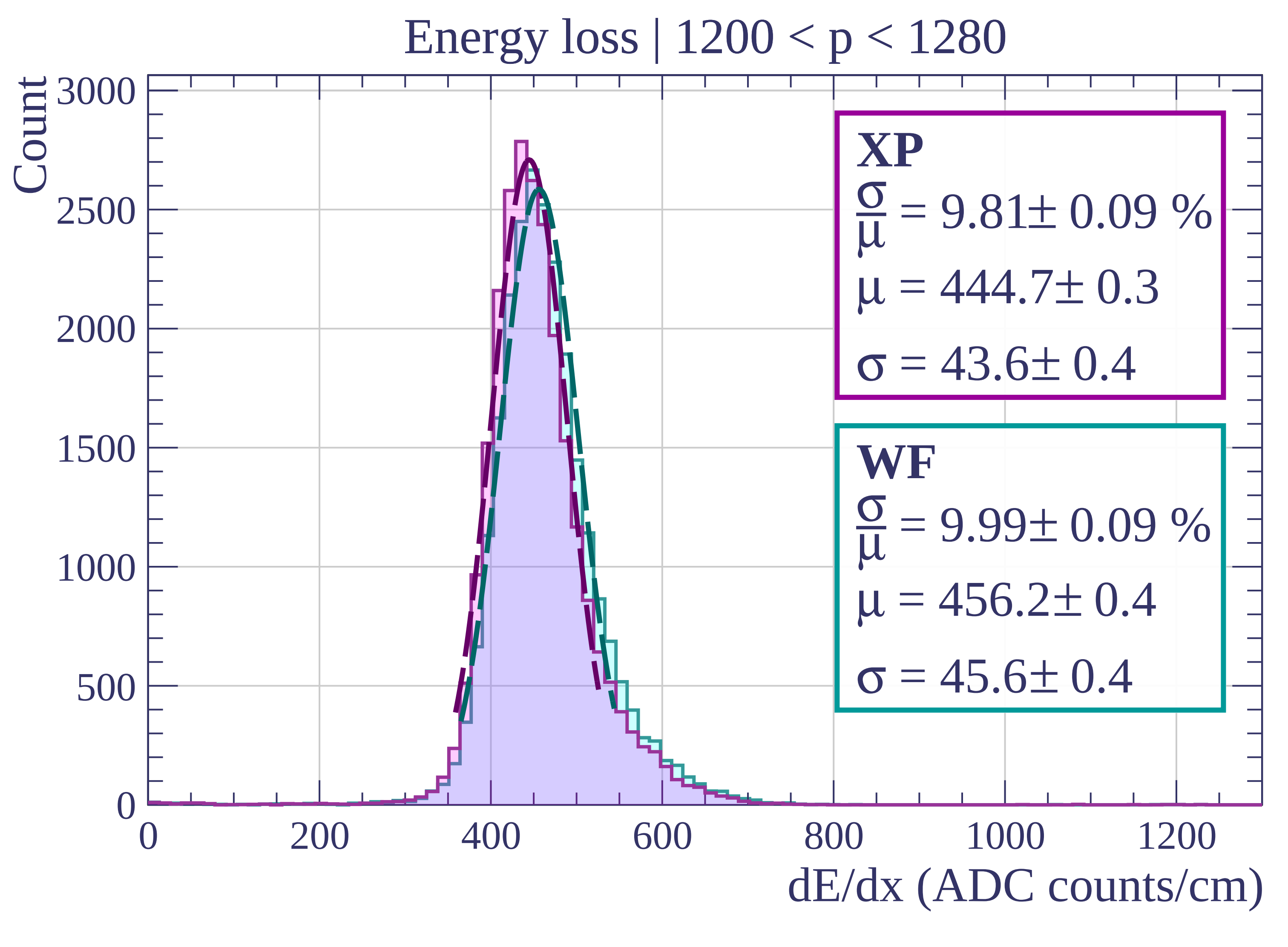

Energy loss | 880 < p < 960
Count
3500
3000
2500
2000
1500
1000
500
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 9.24± 0.07 %
μ = 425.7± 0.3
σ = 39.3± 0.3
WF
$\frac{\sigma}{\mu}$ = 9.68± 0.08 %
μ = 436.0± 0.3
σ = 42.2± 0.3

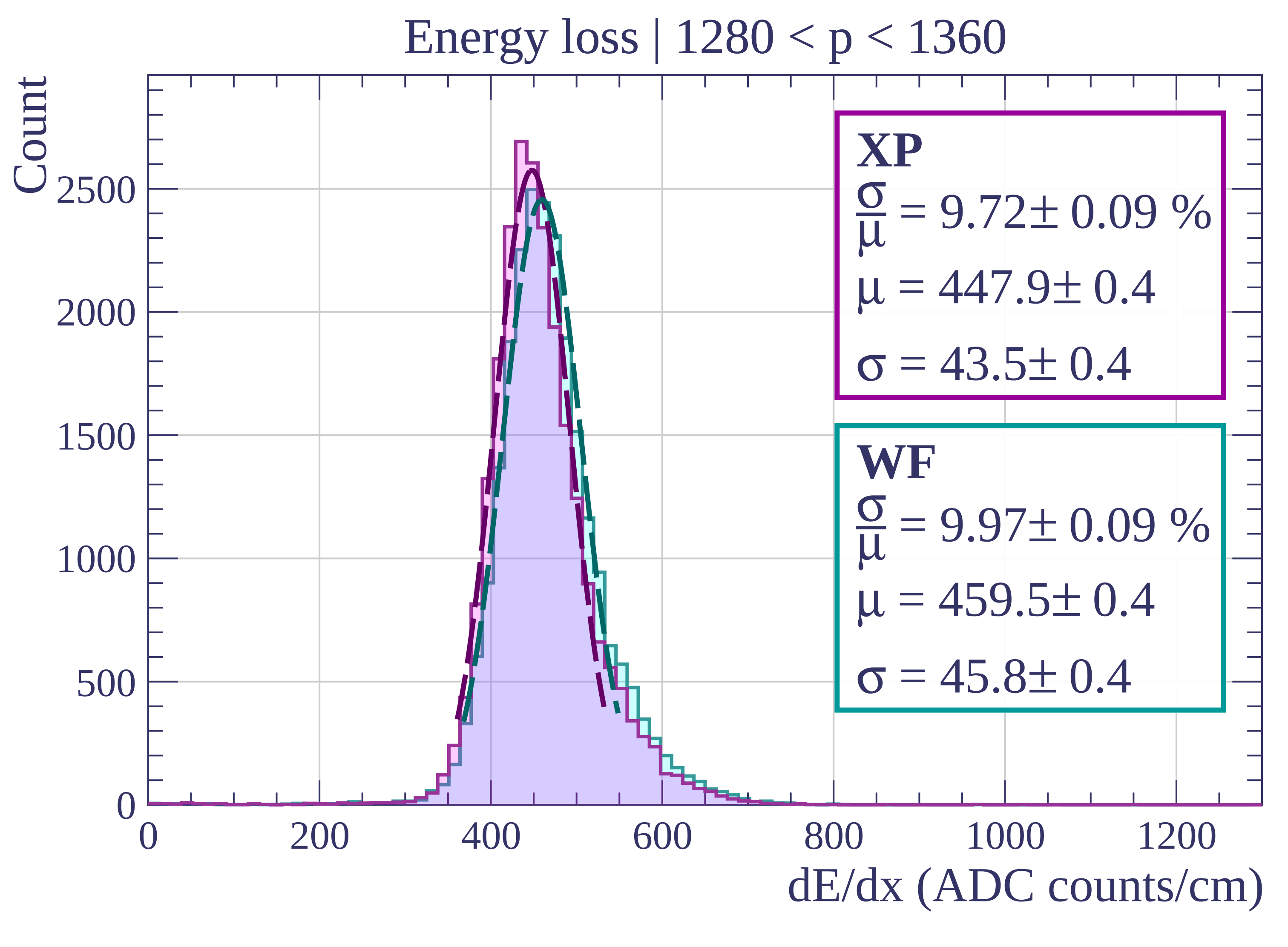

Energy loss | 960 < p < 1040
Count
dE/dx (ADC counts/cm)
0
500
1000
1500
2000
2500
3000
3500
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 9.25± 0.08 %
μ = 429.9± 0.3
σ = 39.8± 0.3
WF
$\frac{\sigma}{\mu}$ = 9.59± 0.08 %
μ = 441.1± 0.3
σ = 42.3± 0.3

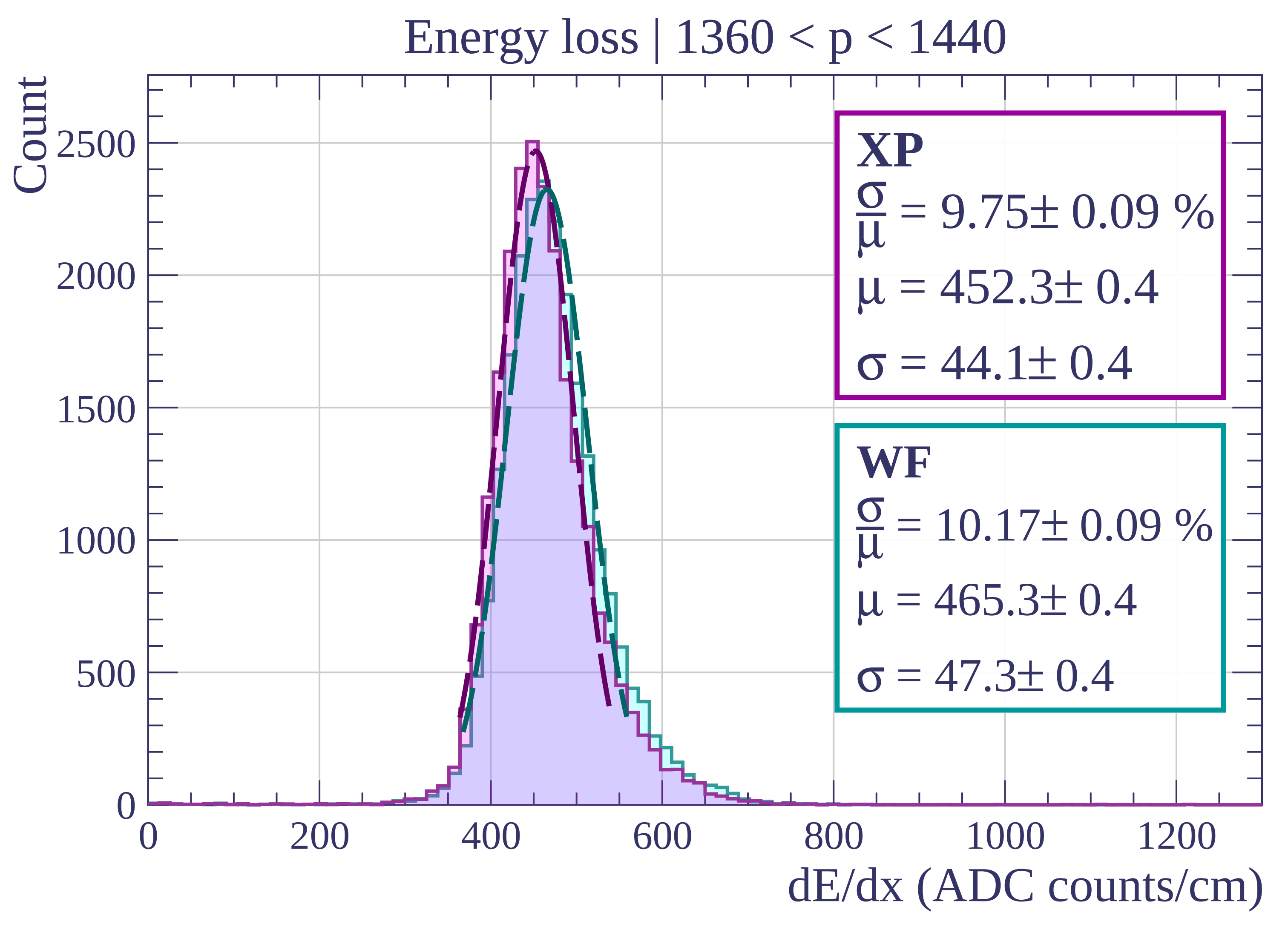

Energy loss | 1040 < p < 1120
Count
dE/dx (ADC counts/cm)
0
500
1000
1500
2000
2500
3000
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 9.15± 0.08 %
μ = 434.6± 0.3
σ = 39.8± 0.3
WF
$\frac{\sigma}{\mu}$ = 9.83± 0.09 %
μ = 446.5± 0.3
σ = 43.9± 0.3



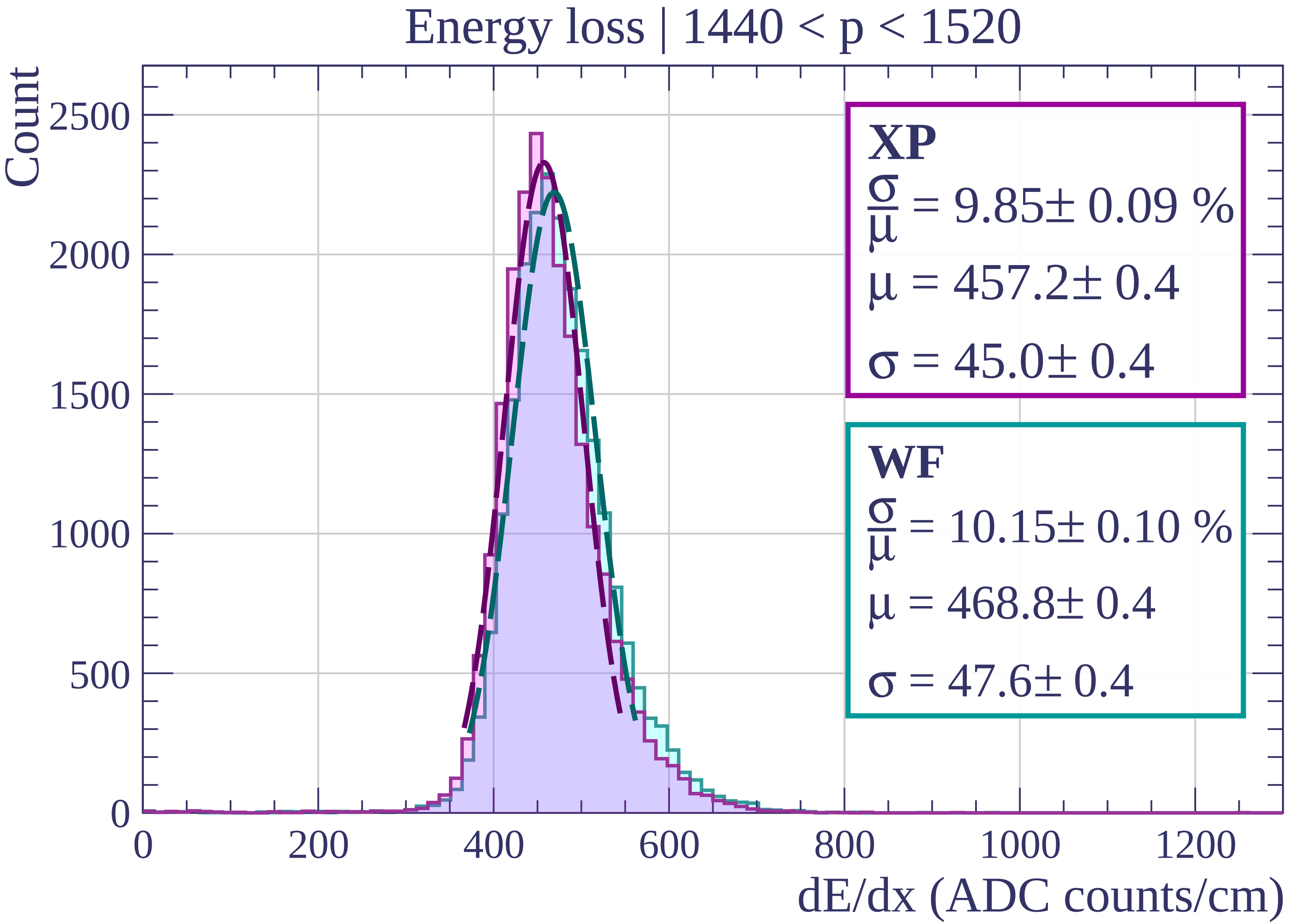
Energy loss | 1120 < p < 1200
Count
dE/dx (ADC counts/cm)
0
500
1000
1500
2000
2500
3000
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 9.37± 0.08 %
μ = 439.6± 0.3
σ = 41.2± 0.3
WF
$\frac{\sigma}{\mu}$ = 9.60± 0.08 %
μ = 450.8± 0.3
σ = 43.3± 0.3


Energy loss | 1200 < p < 1280
Count
dE/dx (ADC counts/cm)
0
500
1000
1500
2000
2500
3000
0
200
400
600
800
1000
1200
XP
σ/μ = 9.81± 0.09 %
μ = 444.7± 0.3
σ = 43.6± 0.4
WF
σ/μ = 9.99± 0.09 %
μ = 456.2± 0.4
σ = 45.6± 0.4

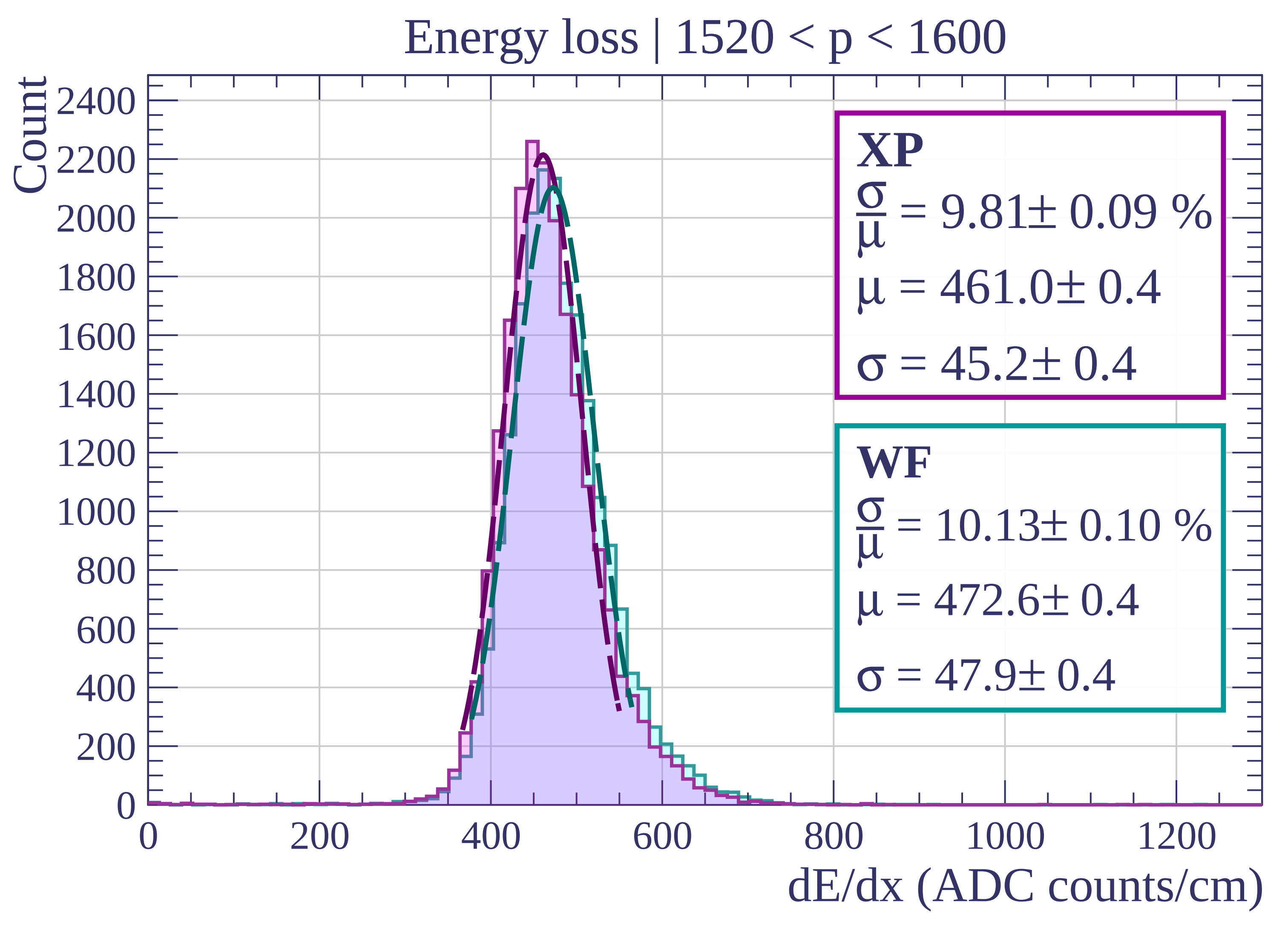

Energy loss | 1280 < p < 1360
Count
dE/dx (ADC counts/cm)
0
500
1000
1500
2000
2500
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 9.72± 0.09 %
μ = 447.9± 0.4
σ = 43.5± 0.4
WF
$\frac{\sigma}{\mu}$ = 9.97± 0.09 %
μ = 459.5± 0.4
σ = 45.8± 0.4

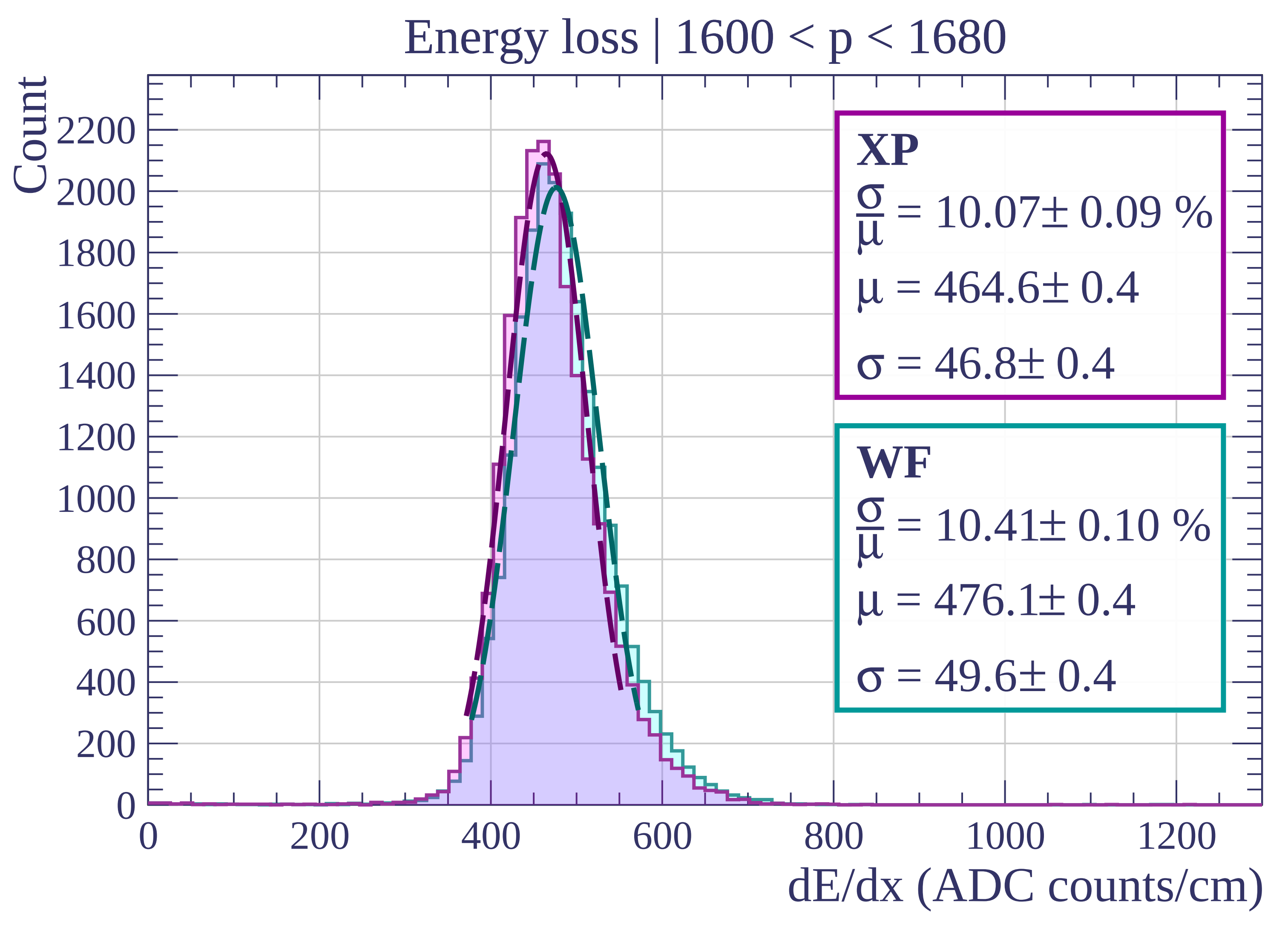


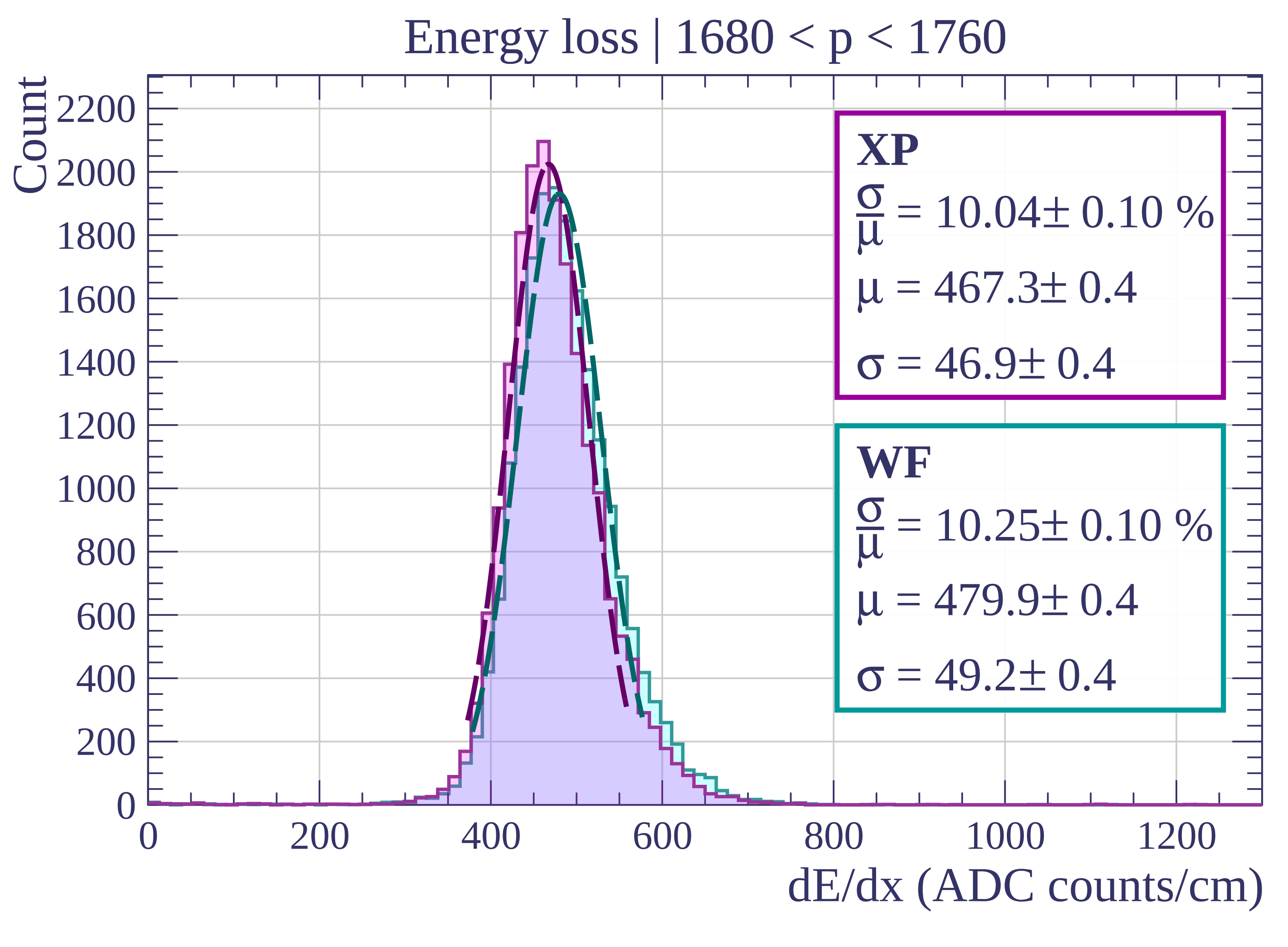
Energy loss | 1360 < p < 1440
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
500
1000
1500
2000
2500
XP
$\frac{\sigma}{\mu}$ = 9.75± 0.09 %
μ = 452.3± 0.4
σ = 44.1± 0.4
WF
$\frac{\sigma}{\mu}$ = 10.17± 0.09 %
μ = 465.3± 0.4
σ = 47.3± 0.4

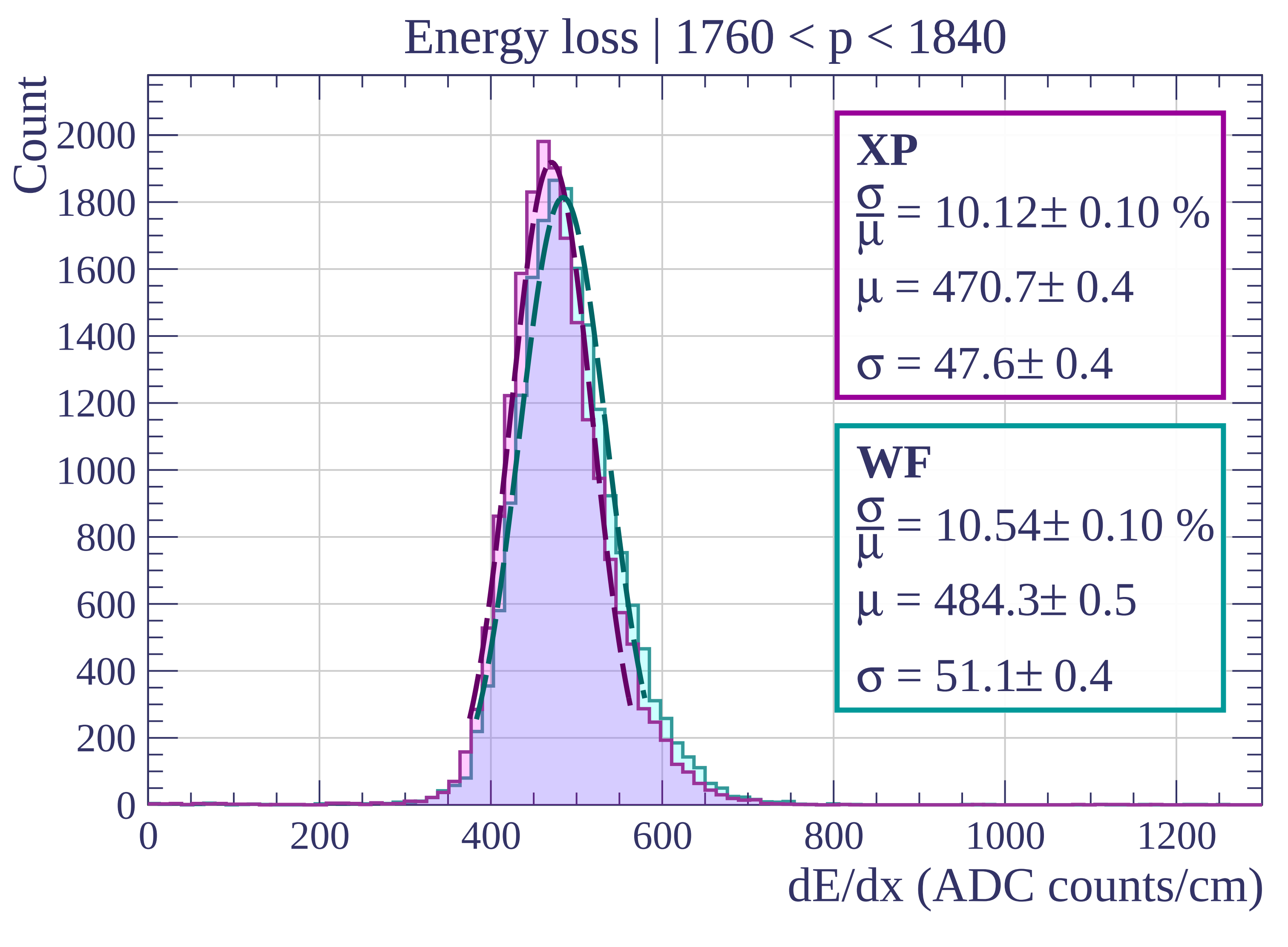


Energy loss | 1440 < p < 1520
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
500
1000
1500
2000
2500
XP
σ/μ = 9.85± 0.09 %
μ = 457.2± 0.4
σ = 45.0± 0.4
WF
σ/μ = 10.15± 0.10 %
μ = 468.8± 0.4
σ = 47.6± 0.4

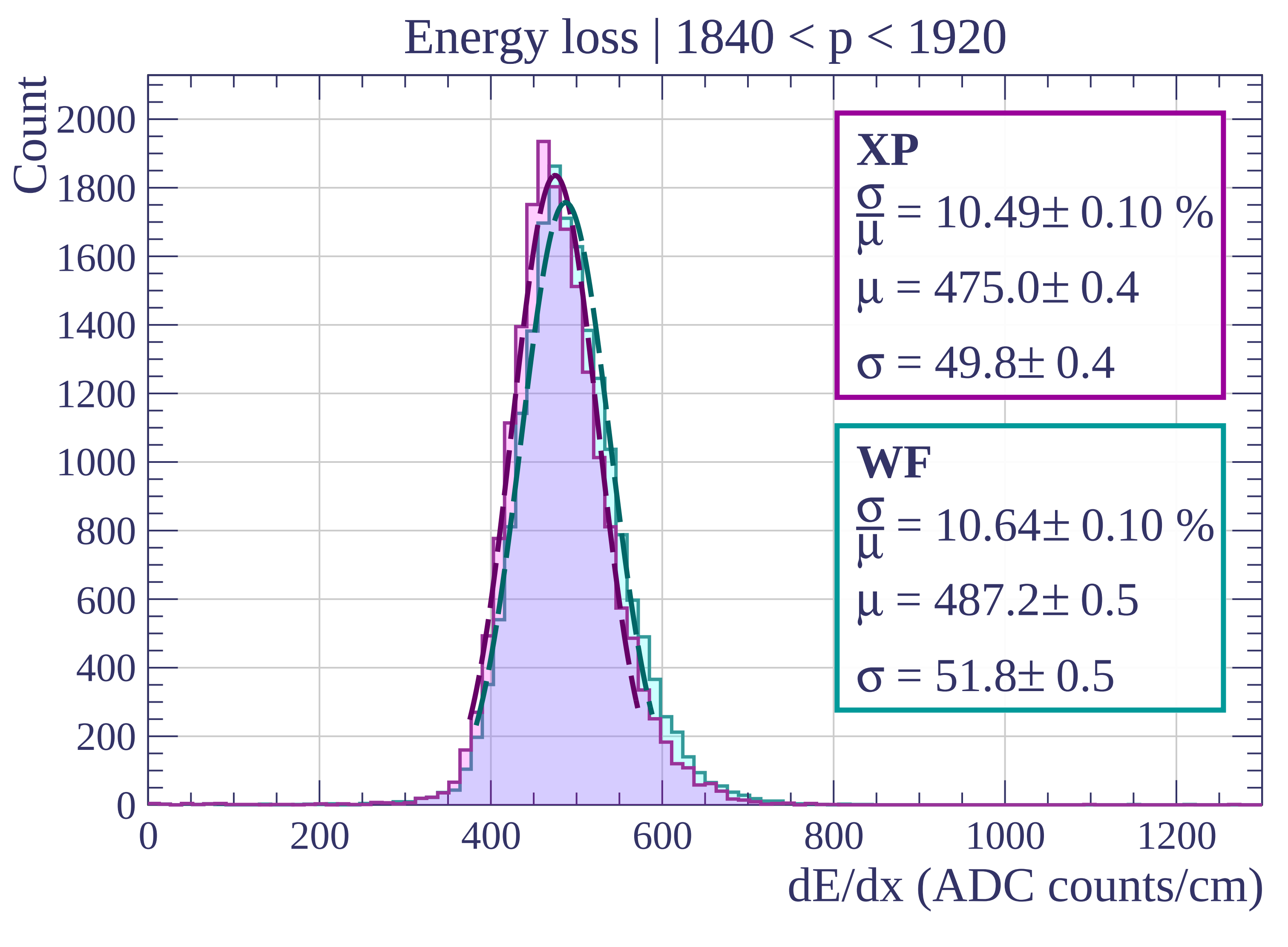

Energy loss | 1520 < p < 1600
Count
2400
2200
2000
1800
1600
1400
1200
1000
800
600
400
200
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 9.81± 0.09 %
μ = 461.0± 0.4
σ = 45.2± 0.4
WF
σ/μ = 10.13± 0.10 %
μ = 472.6± 0.4
σ = 47.9± 0.4

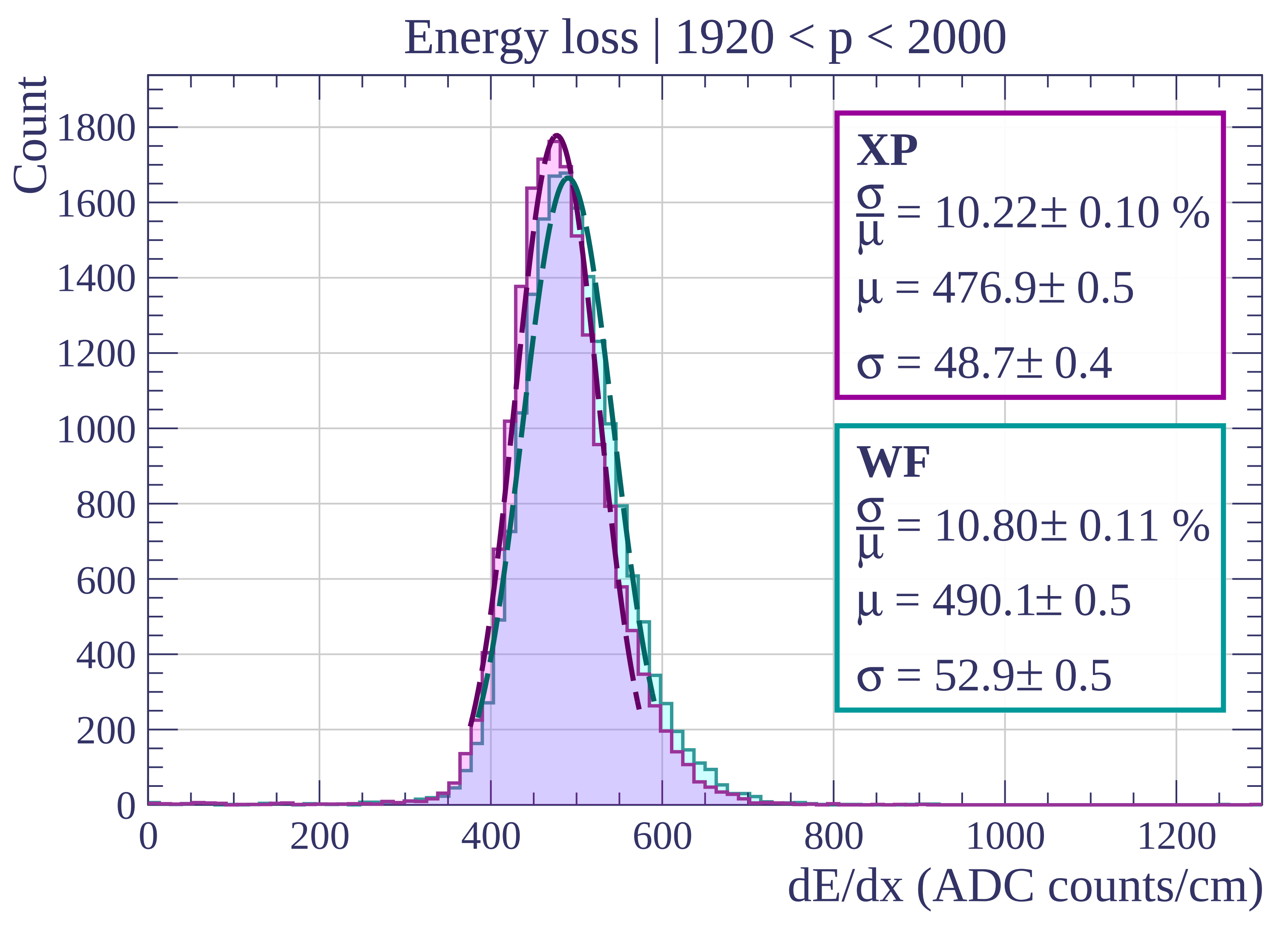

Energy loss | 1600 < p < 1680
Count
0
200
400
600
800
1000
1200
1400
1600
1800
2000
2200
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 10.07± 0.09 %
μ = 464.6± 0.4
σ = 46.8± 0.4
WF
σ/μ = 10.41± 0.10 %
μ = 476.1± 0.4
σ = 49.6± 0.4


Energy loss | 1680 < p < 1760
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 10.04± 0.10 %
μ = 467.3± 0.4
σ = 46.9± 0.4
WF
$\frac{\sigma}{\mu}$ = 10.25± 0.10 %
μ = 479.9± 0.4
σ = 49.2± 0.4

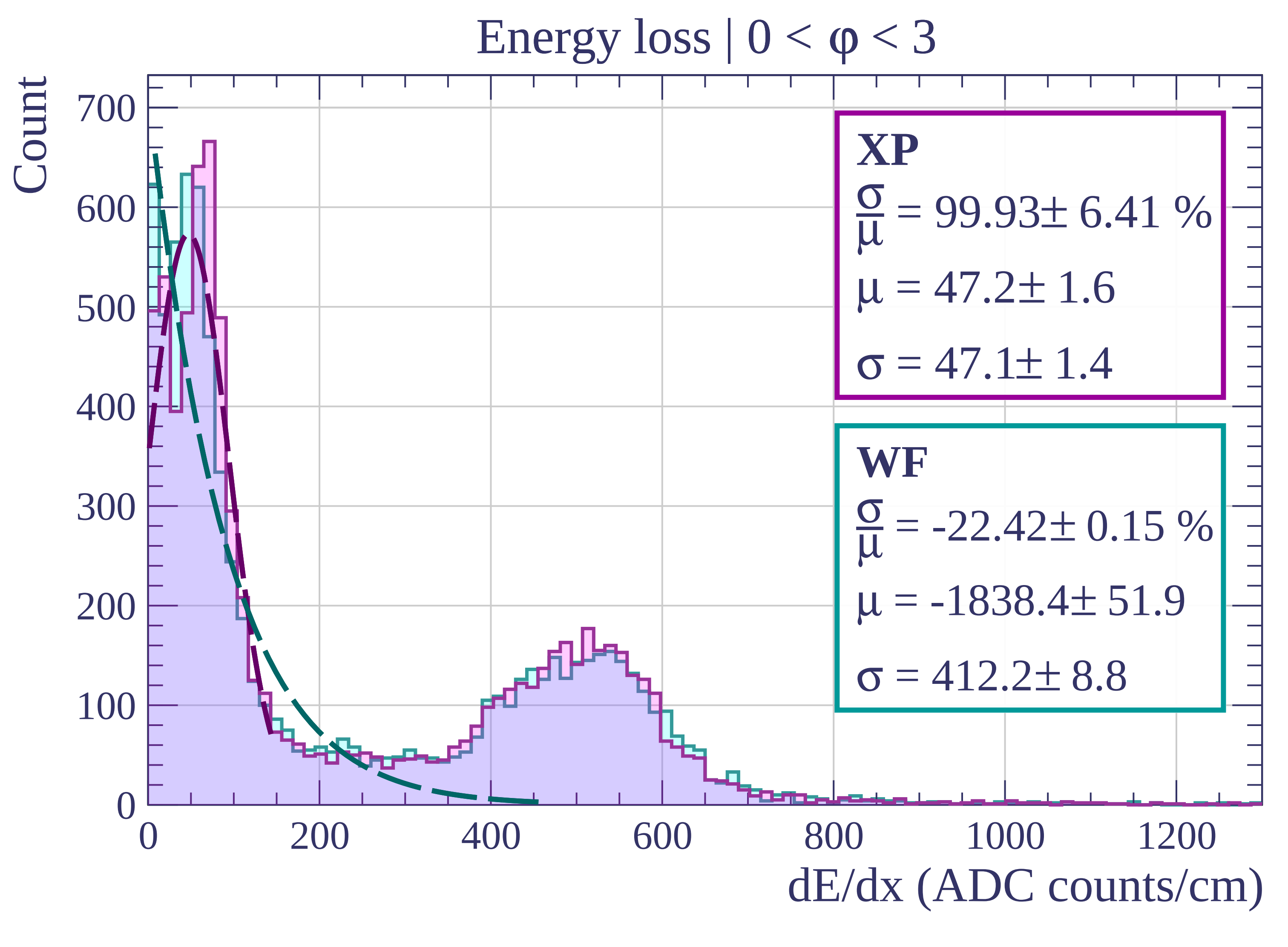


Energy loss | 1760 < p < 1840
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
1400
1600
1800
2000
XP
σ/μ = 10.12± 0.10 %
μ = 470.7± 0.4
σ = 47.6± 0.4
WF
σ/μ = 10.54± 0.10 %
μ = 484.3± 0.5
σ = 51.1± 0.4

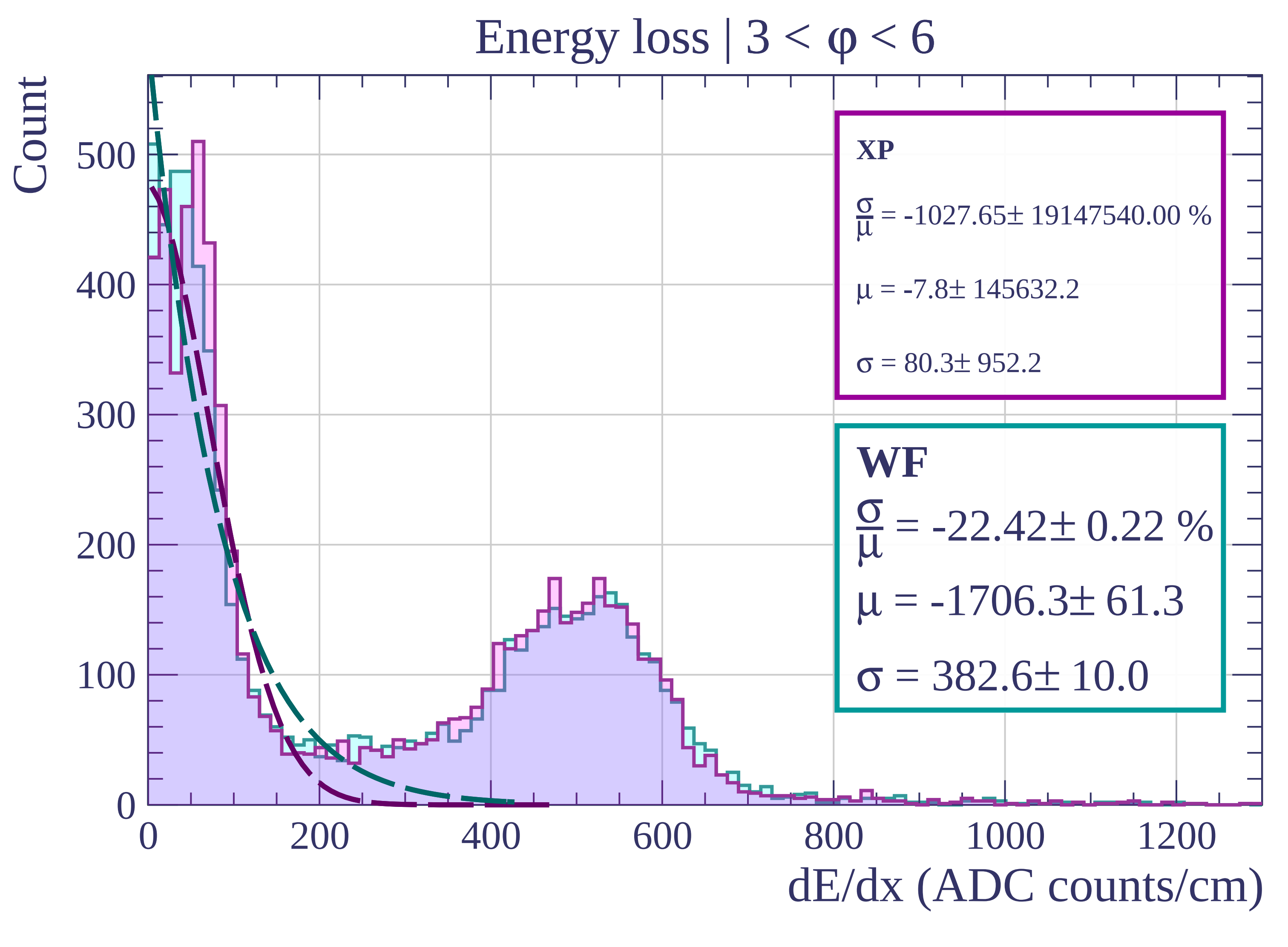

Energy loss | 1840 < p < 1920
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
1400
1600
1800
2000
XP
σ/μ = 10.49± 0.10 %
μ = 475.0± 0.4
σ = 49.8± 0.4
WF
σ/μ = 10.64± 0.10 %
μ = 487.2± 0.5
σ = 51.8± 0.5



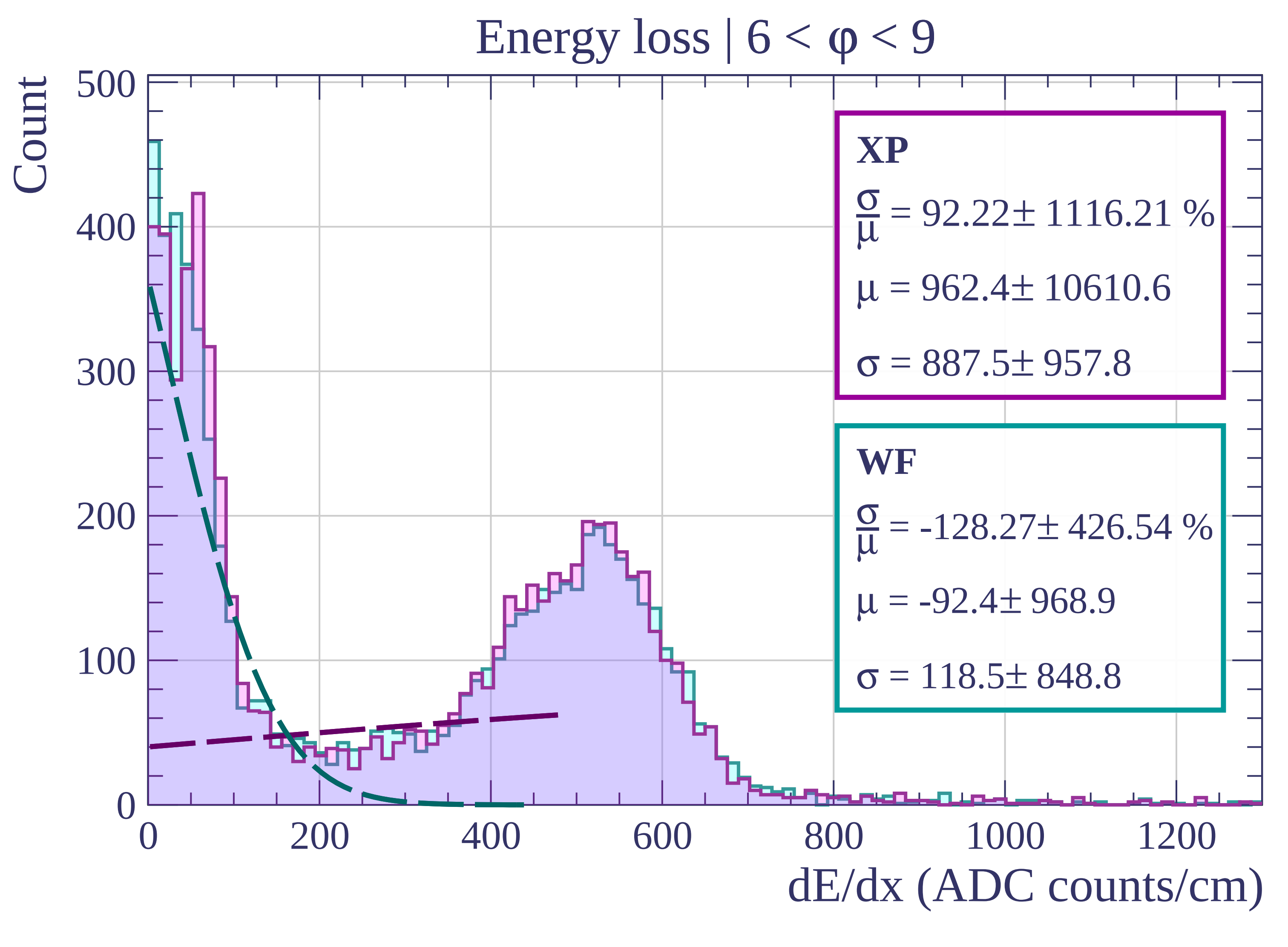
Energy loss | 1920 < p < 2000
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
1400
1600
1800
0
200
400
600
800
1000
1200
XP
σ/μ = 10.22± 0.10 %
μ = 476.9± 0.5
σ = 48.7± 0.4
WF
σ/μ = 10.80± 0.11 %
μ = 490.1± 0.5
σ = 52.9± 0.5

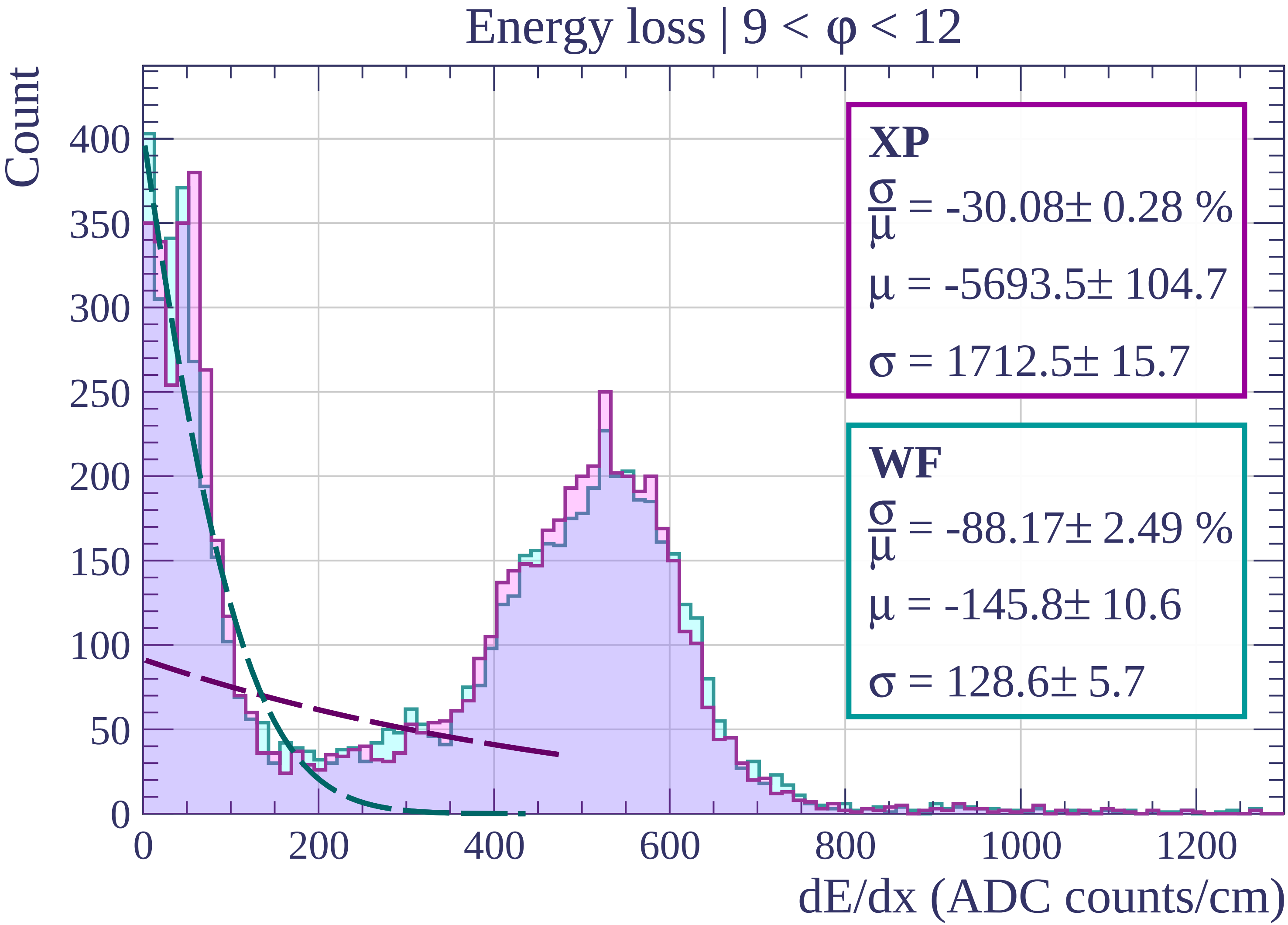


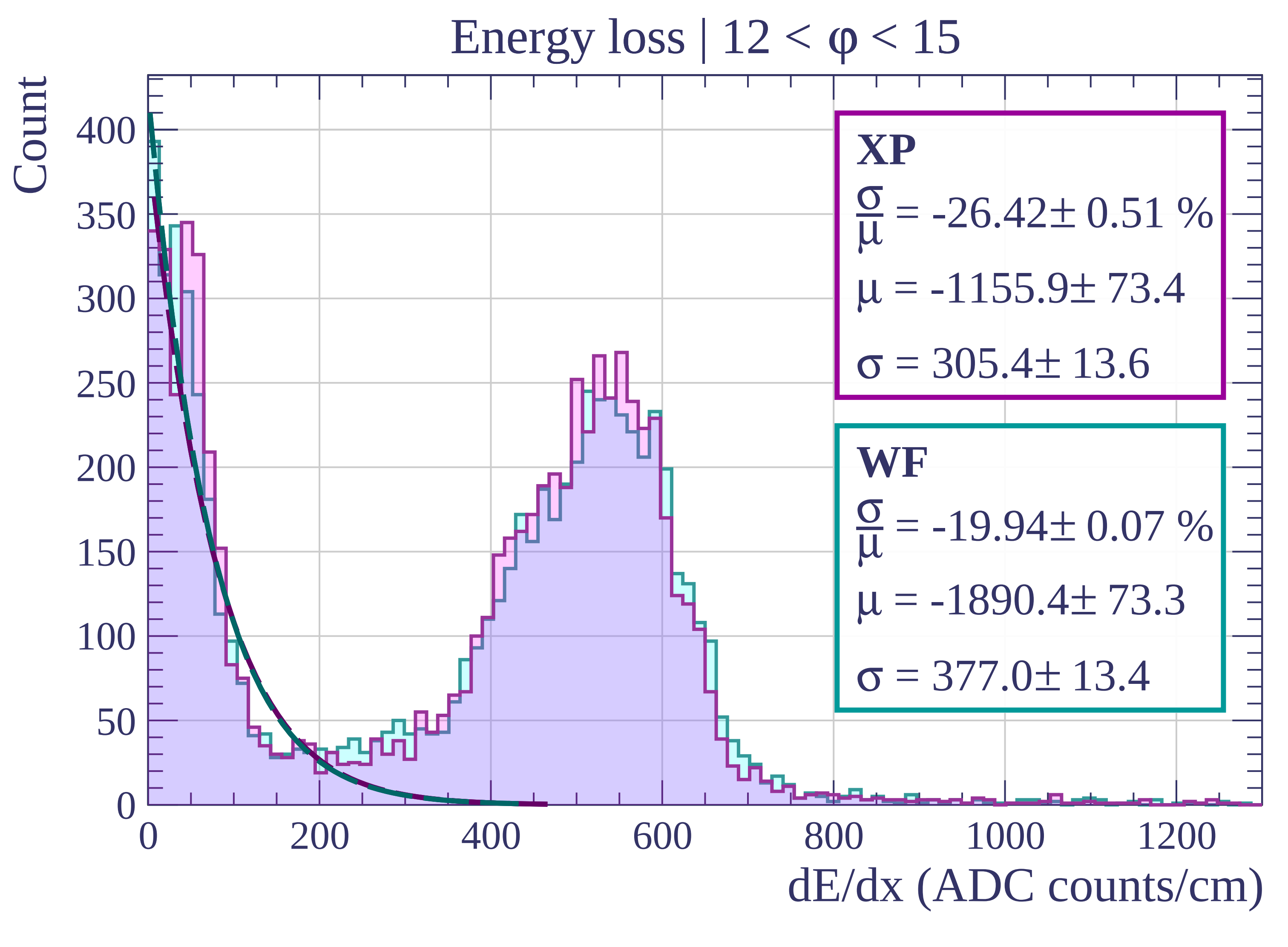
Energy loss | 0 < φ < 3
Count
700
600
500
400
300
200
100
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 99.93± 6.41 %
μ = 47.2± 1.6
σ = 47.1± 1.4
WF
$\frac{\sigma}{\mu}$ = -22.42± 0.15 %
μ = -1838.4± 51.9
σ = 412.2± 8.8



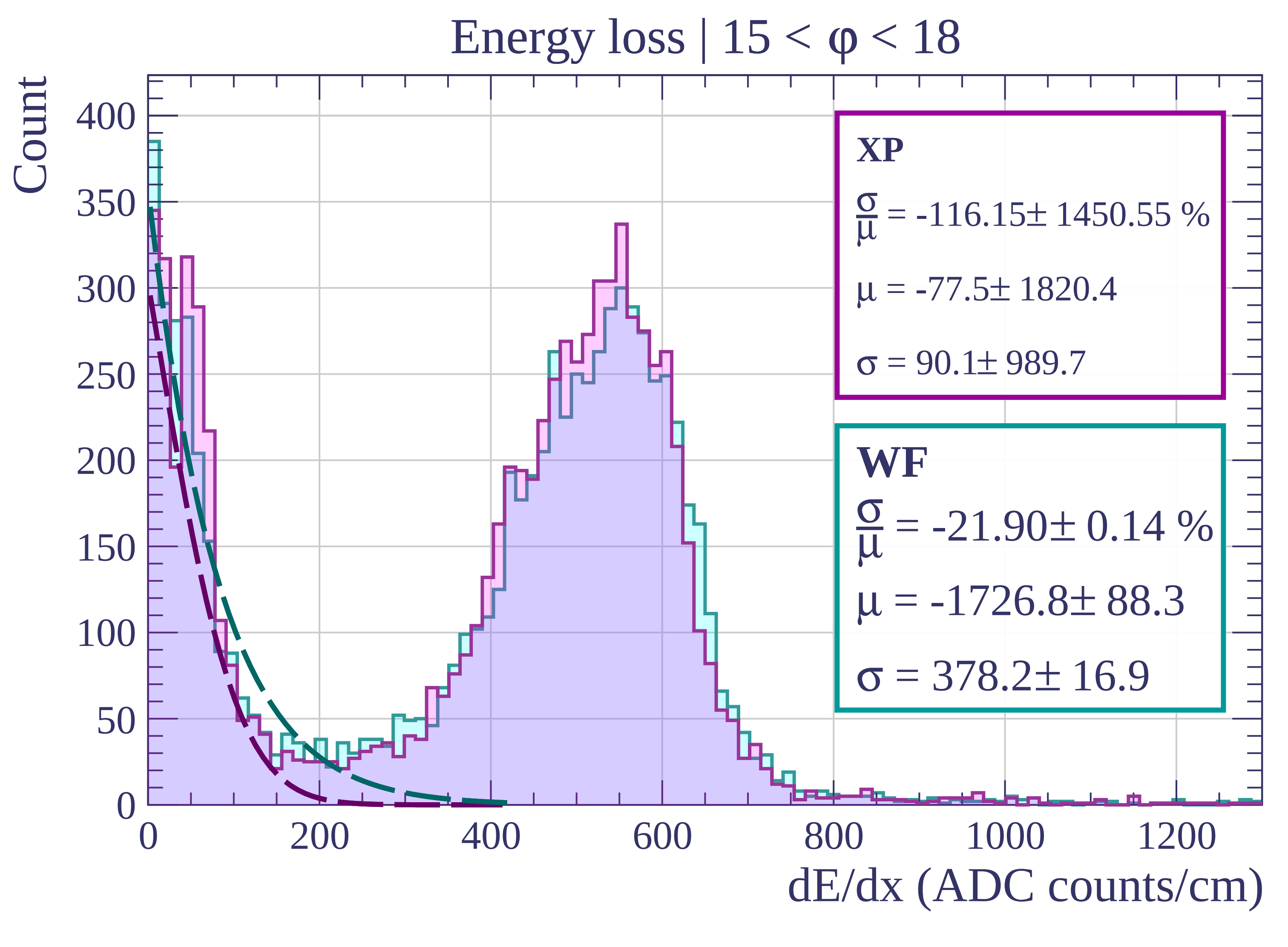
Energy loss | 3 < φ < 6
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
0
200
400
600
800
1000
1200
XP
σ/μ = -1027.65± 19147540.00 %
μ = -7.8± 145632.2
σ = 80.3± 952.2
WF
σ/μ = -22.42± 0.22 %
μ = -1706.3± 61.3
σ = 382.6± 10.0


Energy loss | 6 < φ < 9
Count
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 92.22± 1116.21 %
μ = 962.4± 10610.6
σ = 887.5± 957.8
WF
σ/μ = -128.27± 426.54 %
μ = -92.4± 968.9
σ = 118.5± 848.8

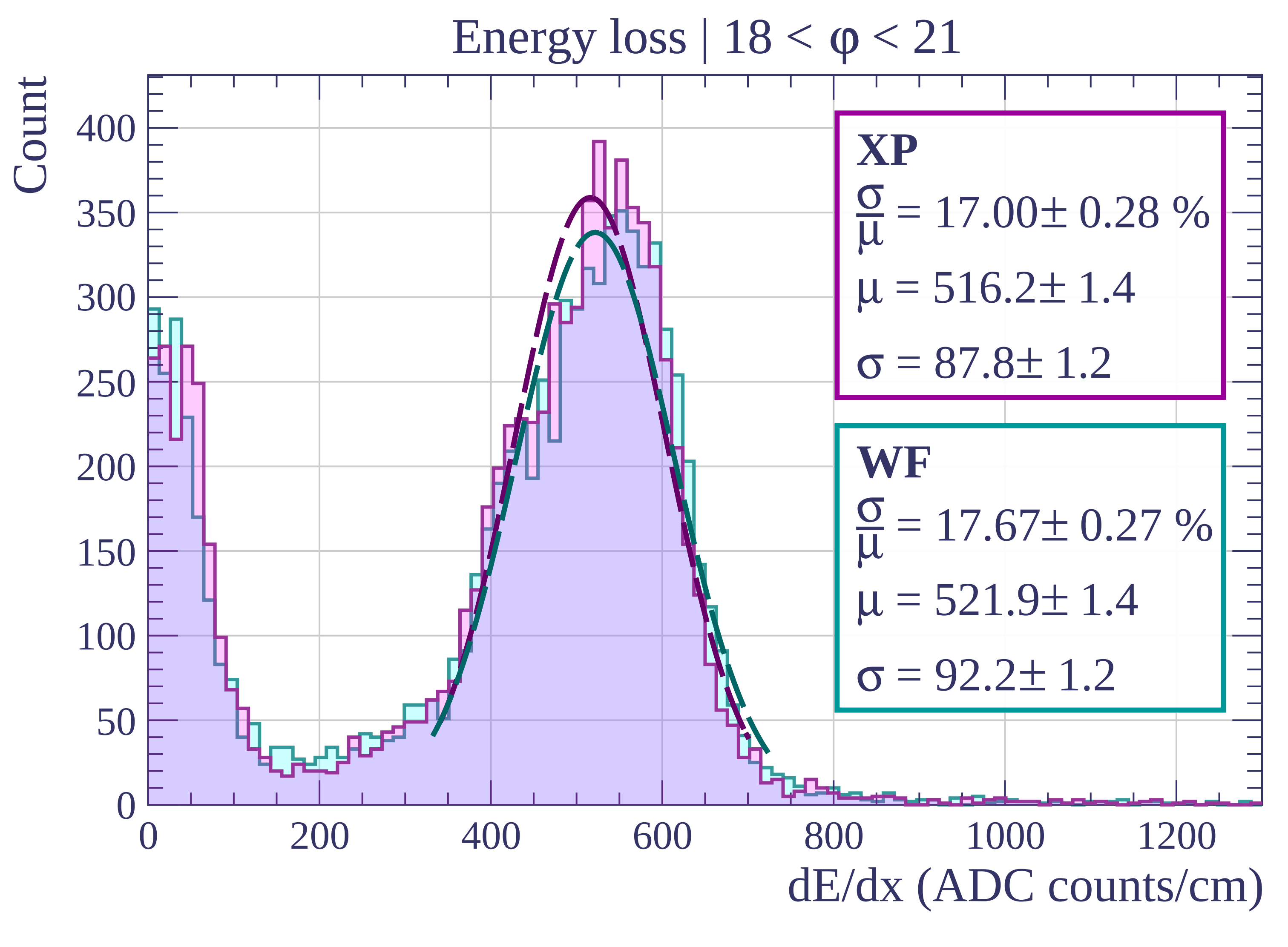

Energy loss | 9 < φ < 12
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = -30.08± 0.28 %
μ = -5693.5± 104.7
σ = 1712.5± 15.7
WF
$\frac{\sigma}{\mu}$ = -88.17± 2.49 %
μ = -145.8± 10.6
σ = 128.6± 5.7
0
50
100
150
200
250
300
350
400
0
200
400
600
800
1000
1200



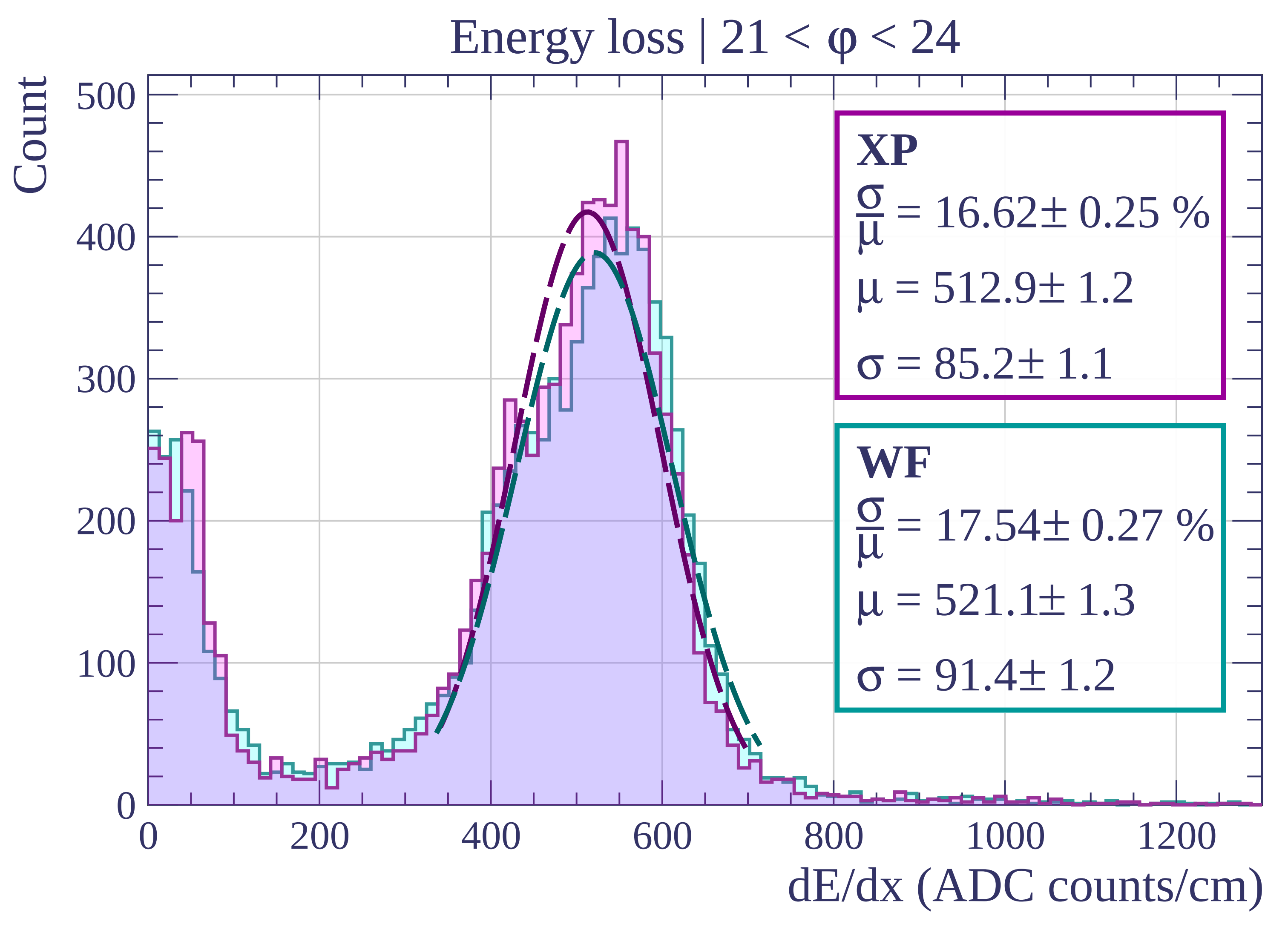
Energy loss | 12 < φ < 15
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
0
200
400
600
800
1000
1200
XP
σ/μ = -26.42± 0.51 %
μ = -1155.9± 73.4
σ = 305.4± 13.6
WF
σ/μ = -19.94± 0.07 %
μ = -1890.4± 73.3
σ = 377.0± 13.4



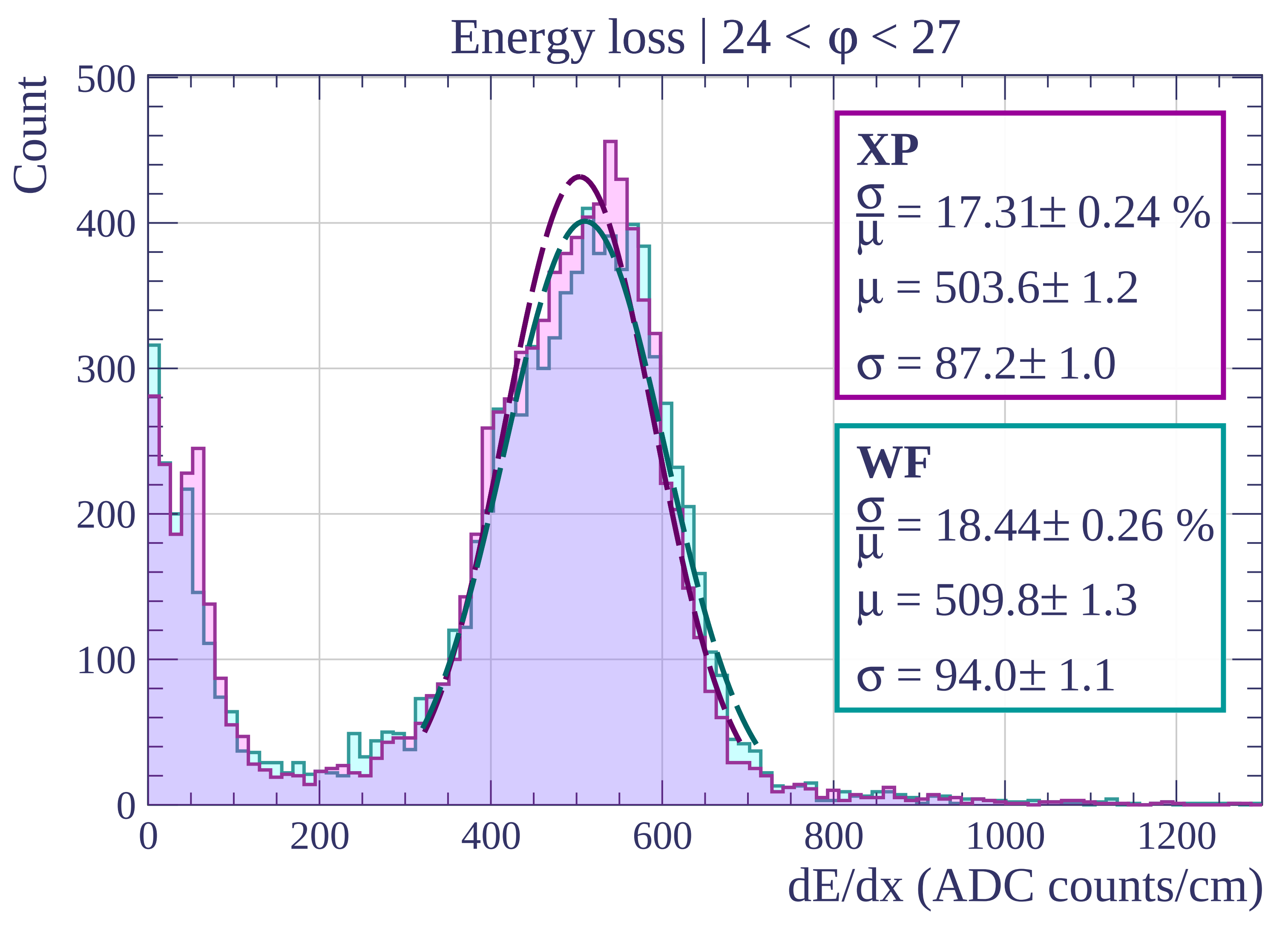
Energy loss | 15 < φ < 18
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
0
200
400
600
800
1000
1200
XP
σ/μ = -116.15± 1450.55 %
μ = -77.5± 1820.4
σ = 90.1± 989.7
WF
σ/μ = -21.90± 0.14 %
μ = -1726.8± 88.3
σ = 378.2± 16.9

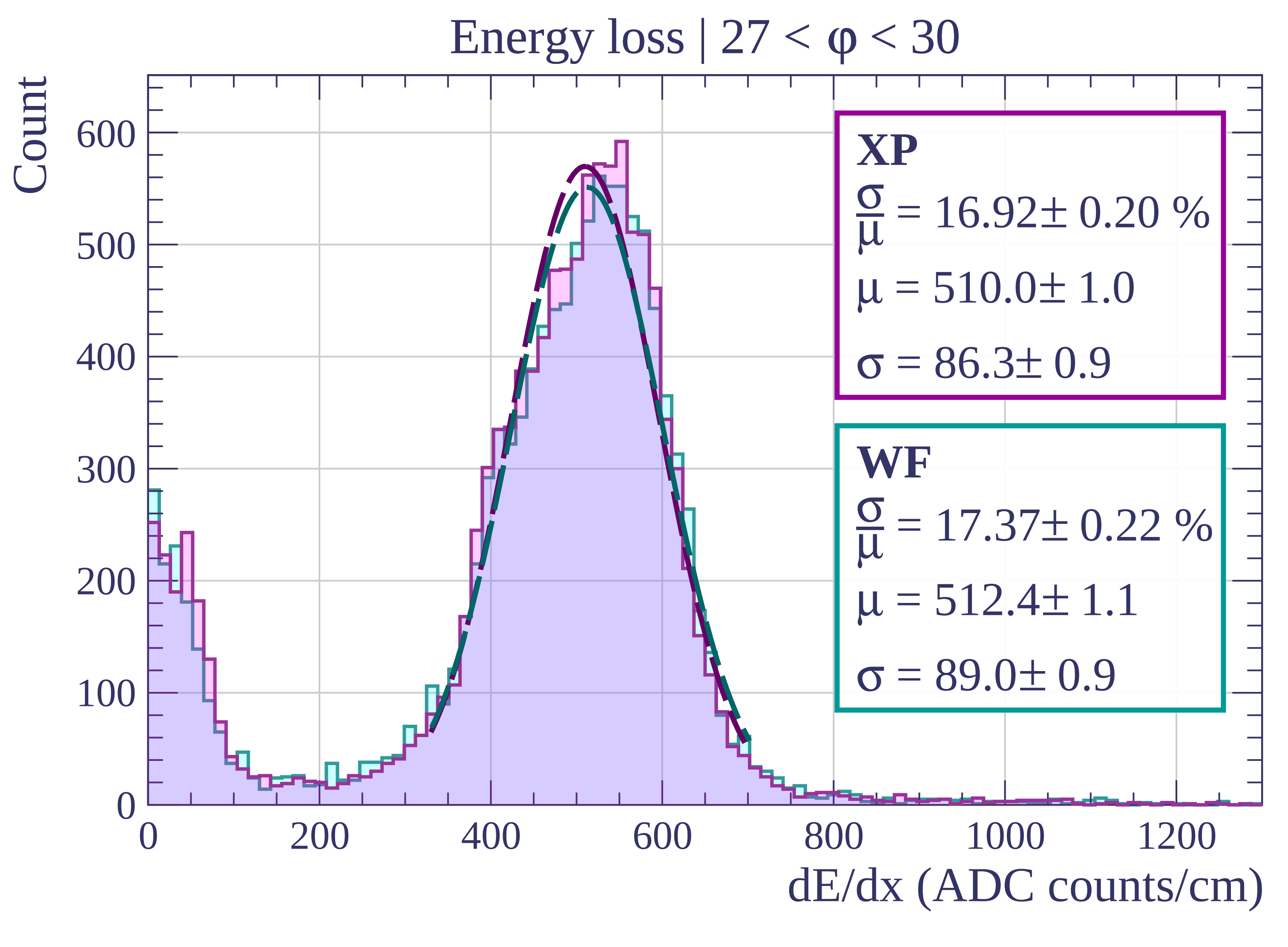


Energy loss | 18 < φ < 21
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
50
100
150
200
250
300
350
400
XP
σ/μ = 17.00± 0.28 %
μ = 516.2± 1.4
σ = 87.8± 1.2
WF
σ/μ = 17.67± 0.27 %
μ = 521.9± 1.4
σ = 92.2± 1.2

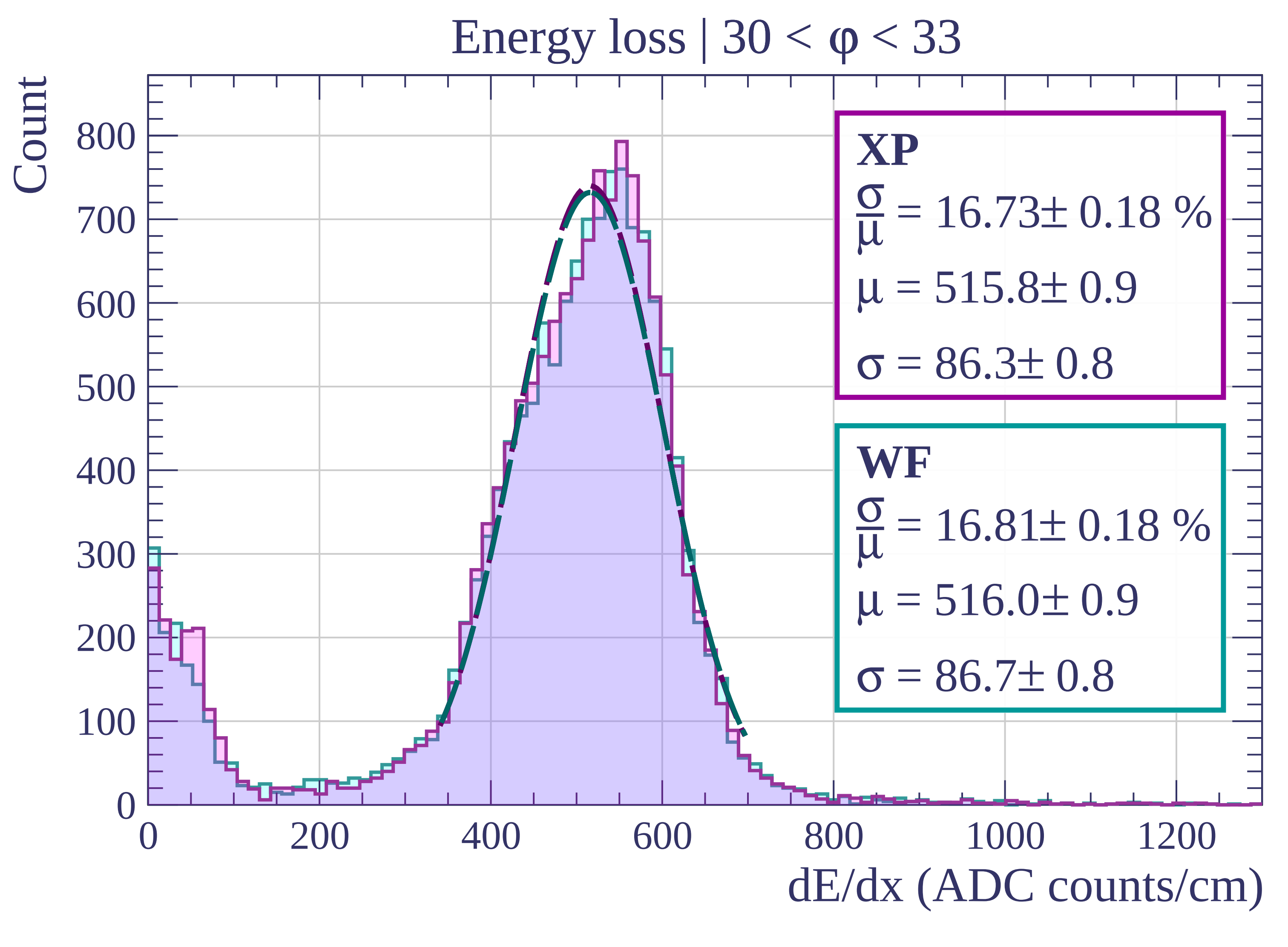


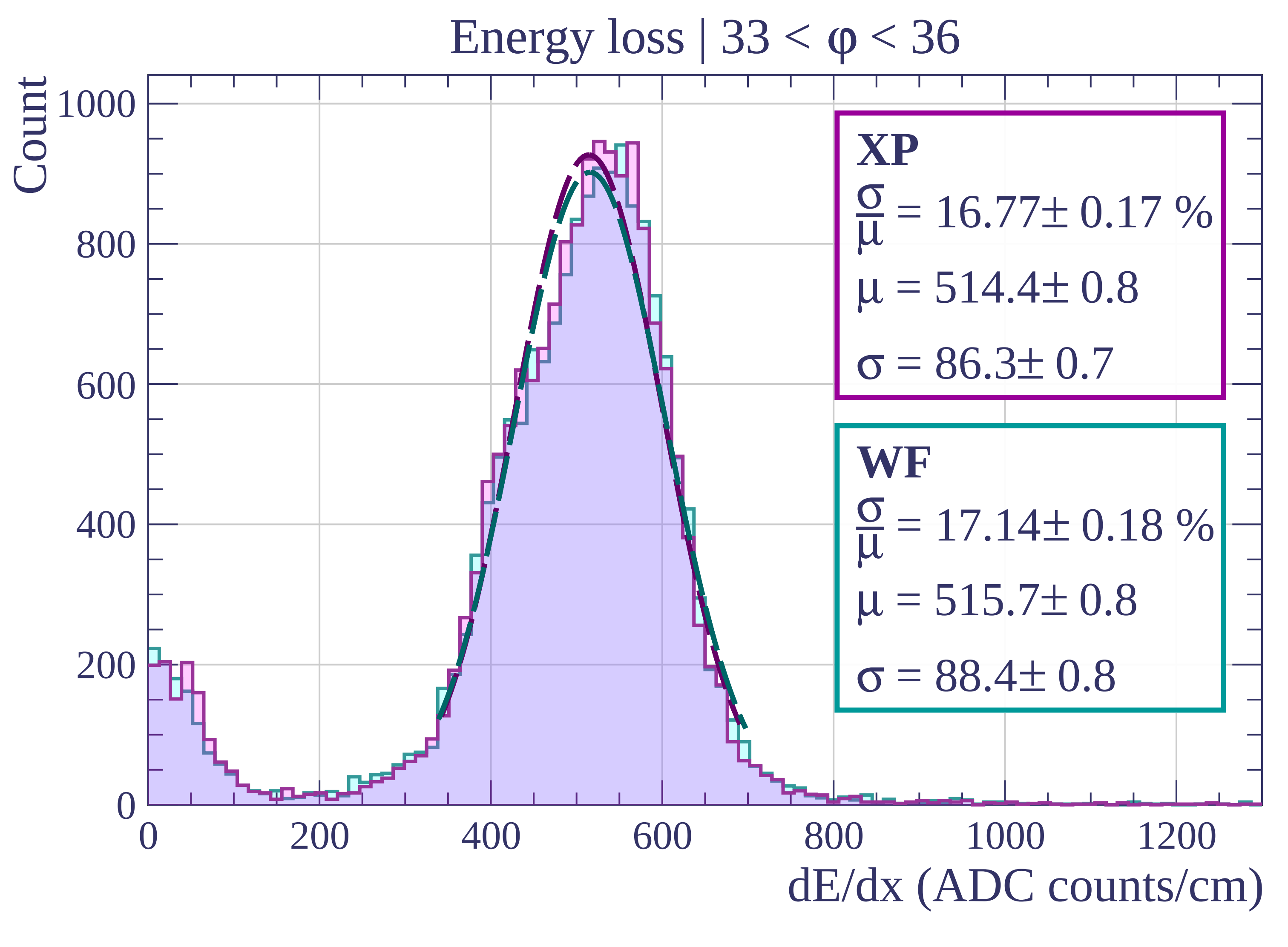
Energy loss | 21 < φ < 24
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
100
300
500
XP
$\frac{\sigma}{\mu}$ = 16.62± 0.25 %
μ = 512.9± 1.2
σ = 85.2± 1.1
WF
$\frac{\sigma}{\mu}$ = 17.54± 0.27 %
μ = 521.1± 1.3
σ = 91.4± 1.2

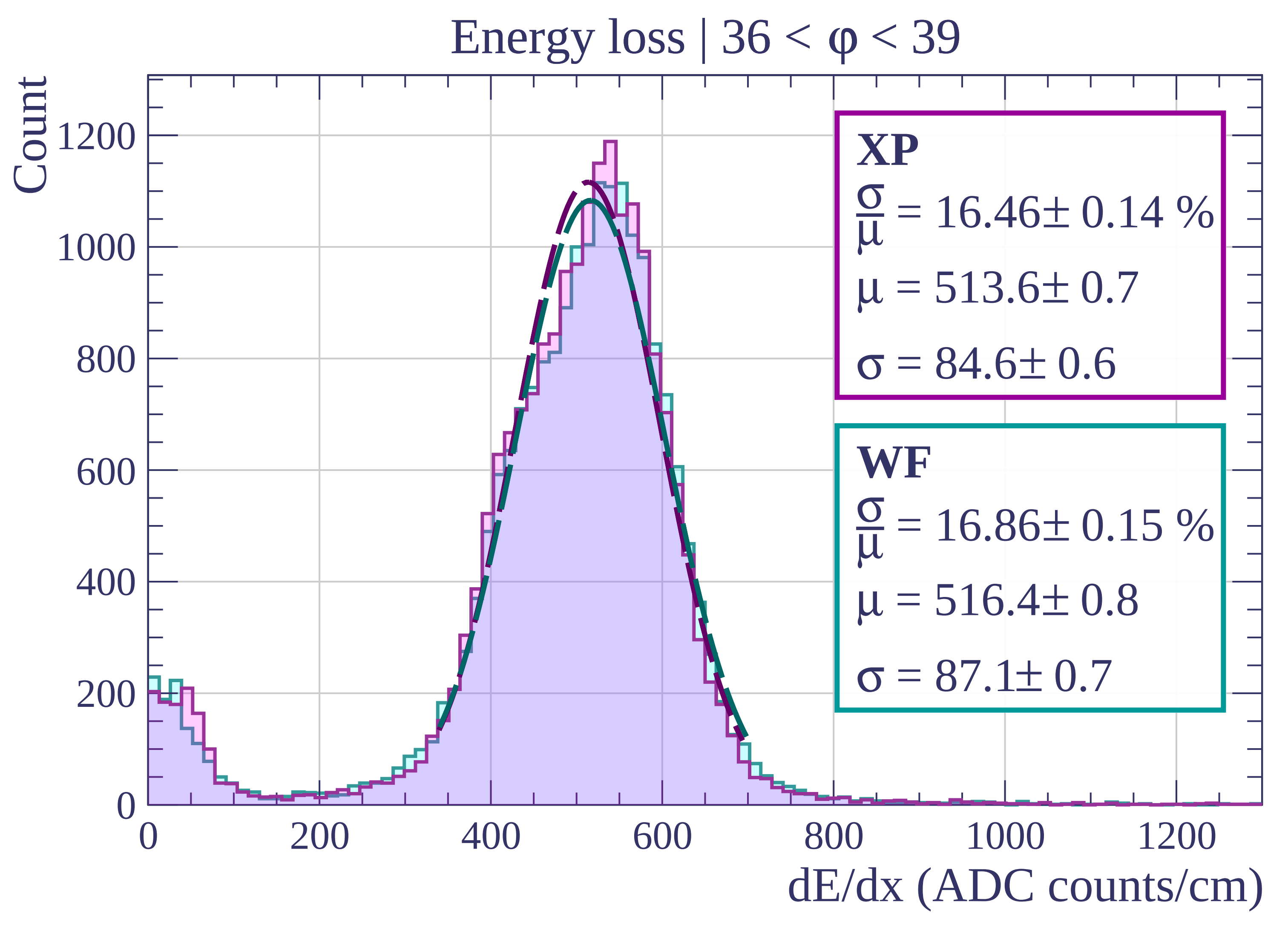

Energy loss | 24 < φ < 27
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
0
200
400
600
800
1000
1200
XP
σ/μ = 17.31± 0.24 %
μ = 503.6± 1.2
σ = 87.2± 1.0
WF
σ/μ = 18.44± 0.26 %
μ = 509.8± 1.3
σ = 94.0± 1.1



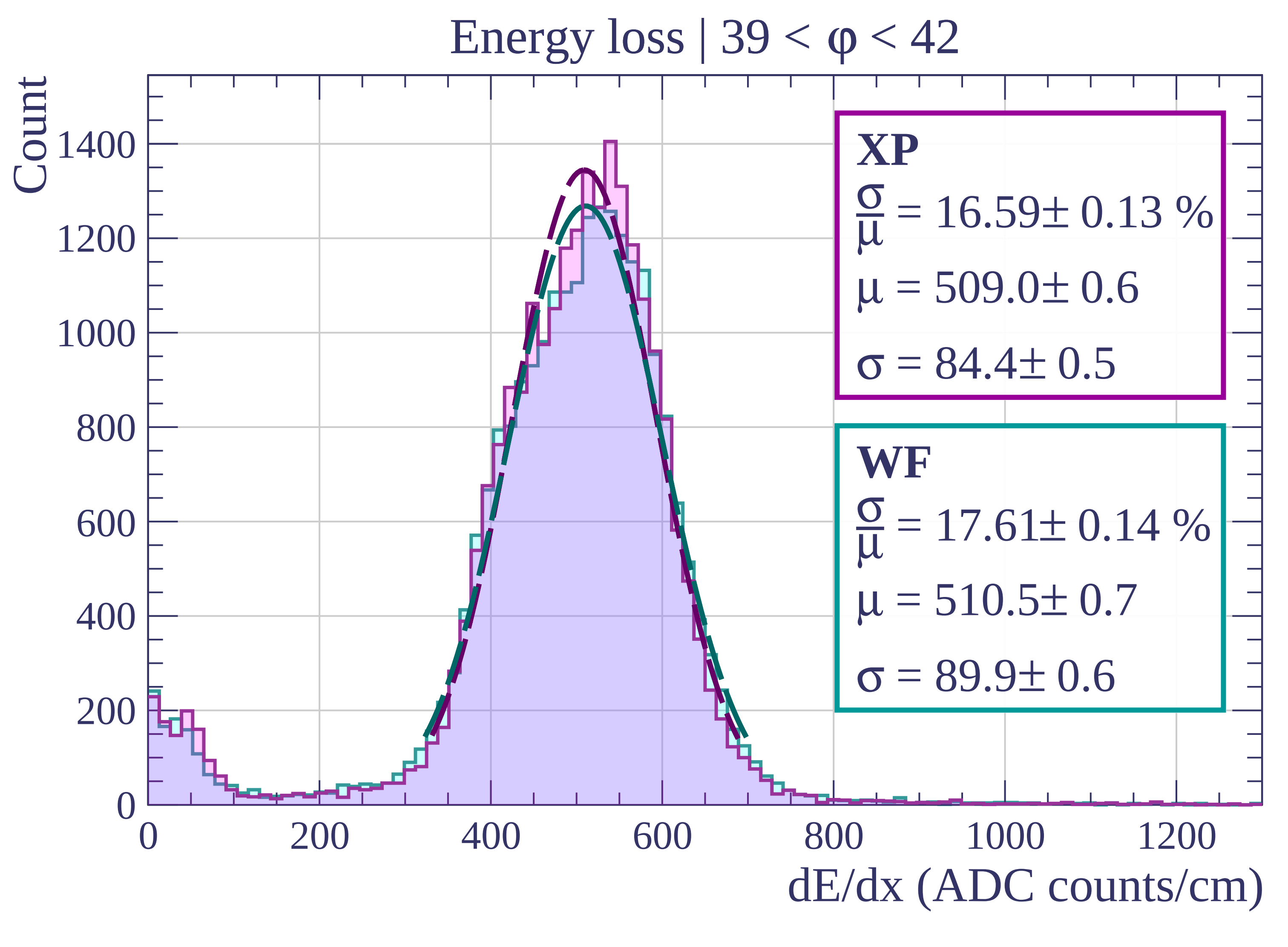
Energy loss | 27 < φ < 30
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
0
200
400
600
800
1000
1200
XP
σ/μ = 16.92± 0.20 %
μ = 510.0± 1.0
σ = 86.3± 0.9
WF
σ/μ = 17.37± 0.22 %
μ = 512.4± 1.1
σ = 89.0± 0.9


Energy loss | 30 < φ < 33
Count
800
700
600
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 16.73± 0.18 %
μ = 515.8± 0.9
σ = 86.3± 0.8
WF
σ/μ = 16.81± 0.18 %
μ = 516.0± 0.9
σ = 86.7± 0.8

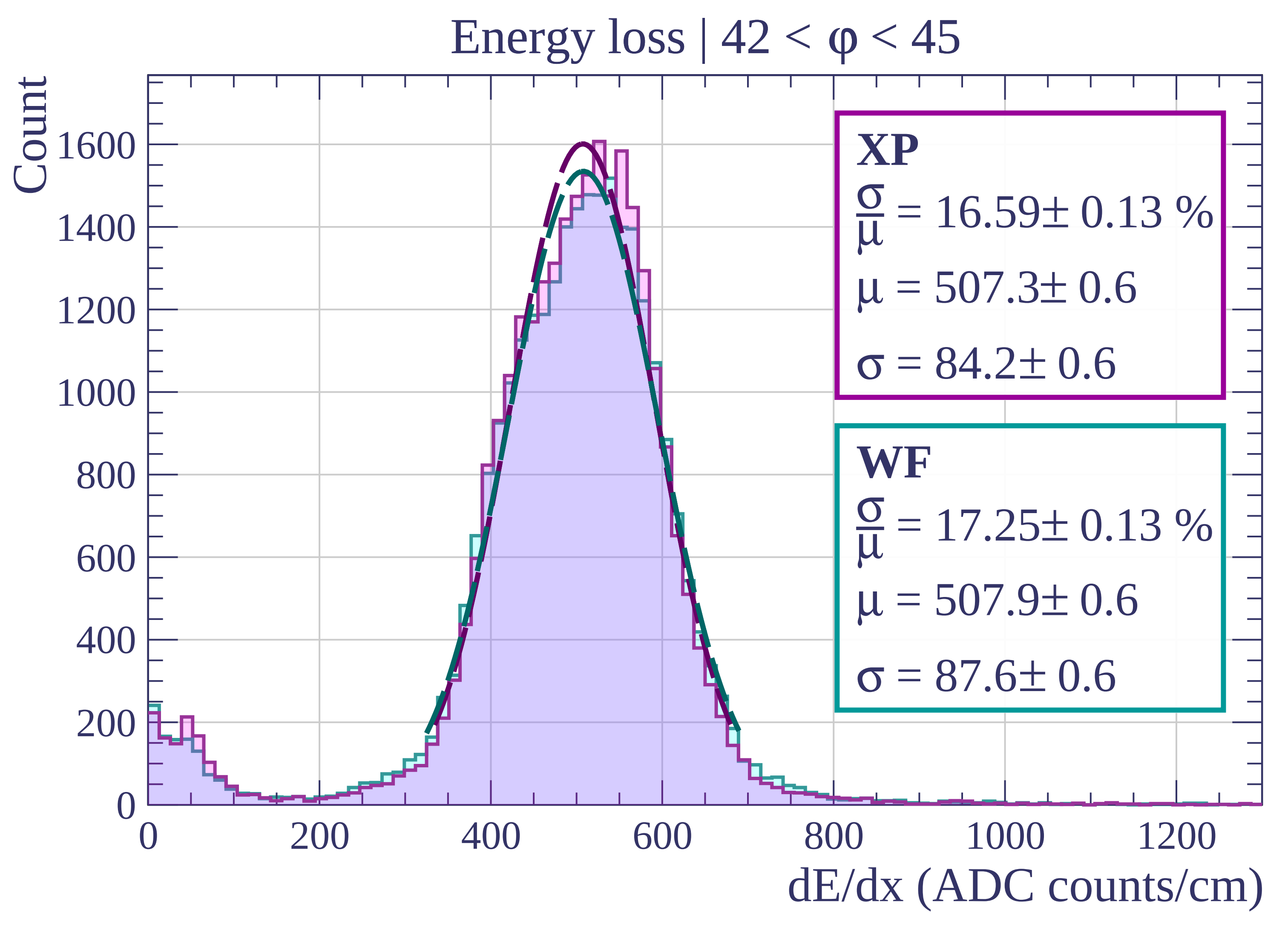

Energy loss | 33 < φ < 36
Count
0
200
400
600
800
1000
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 16.77± 0.17 %
μ = 514.4± 0.8
σ = 86.3± 0.7
WF
σ/μ = 17.14± 0.18 %
μ = 515.7± 0.8
σ = 88.4± 0.8

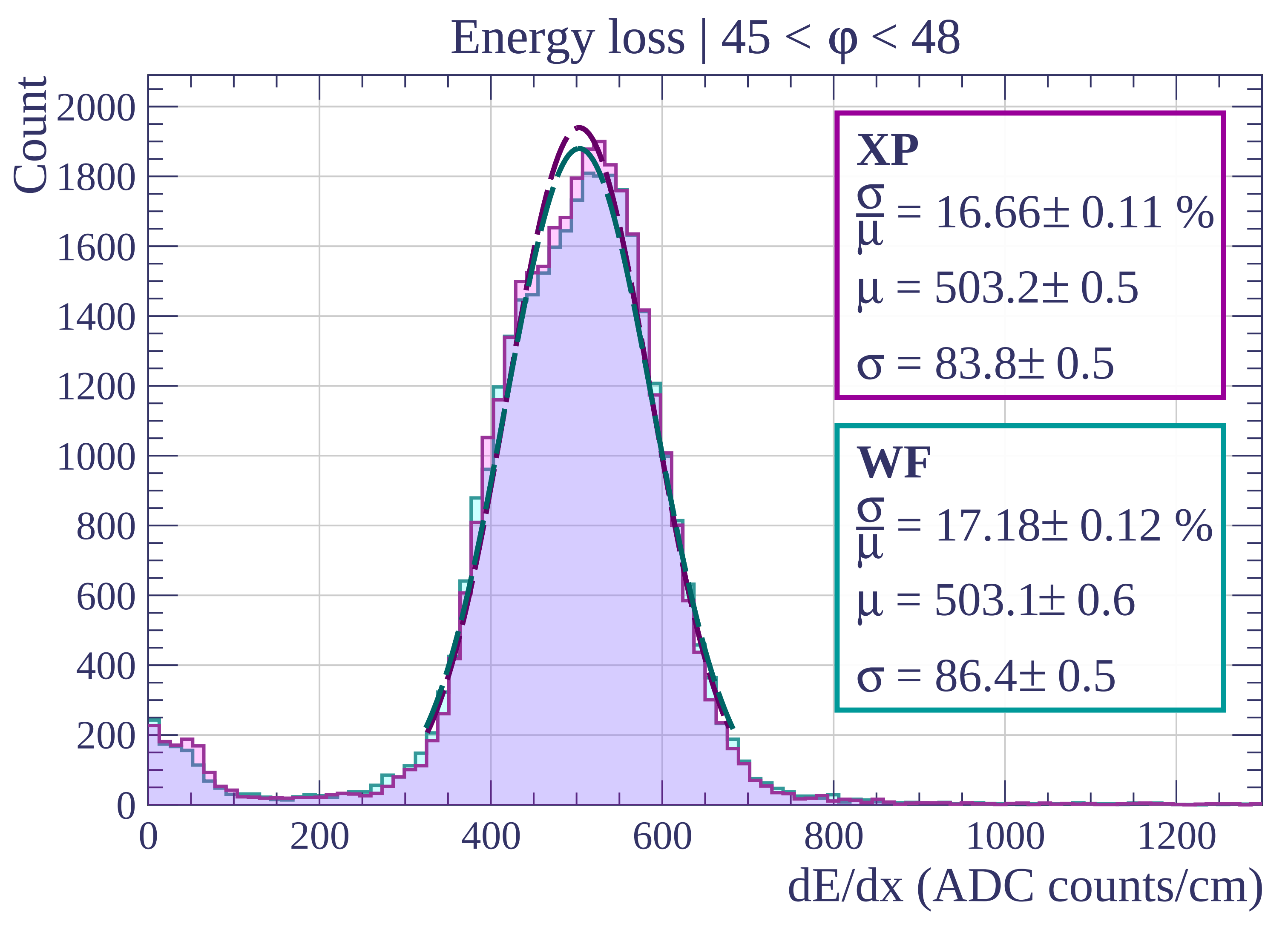


Energy loss | 36 < φ < 39
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 16.46± 0.14 %
μ = 513.6± 0.7
σ = 84.6± 0.6
WF
$\frac{\sigma}{\mu}$ = 16.86± 0.15 %
μ = 516.4± 0.8
σ = 87.1± 0.7

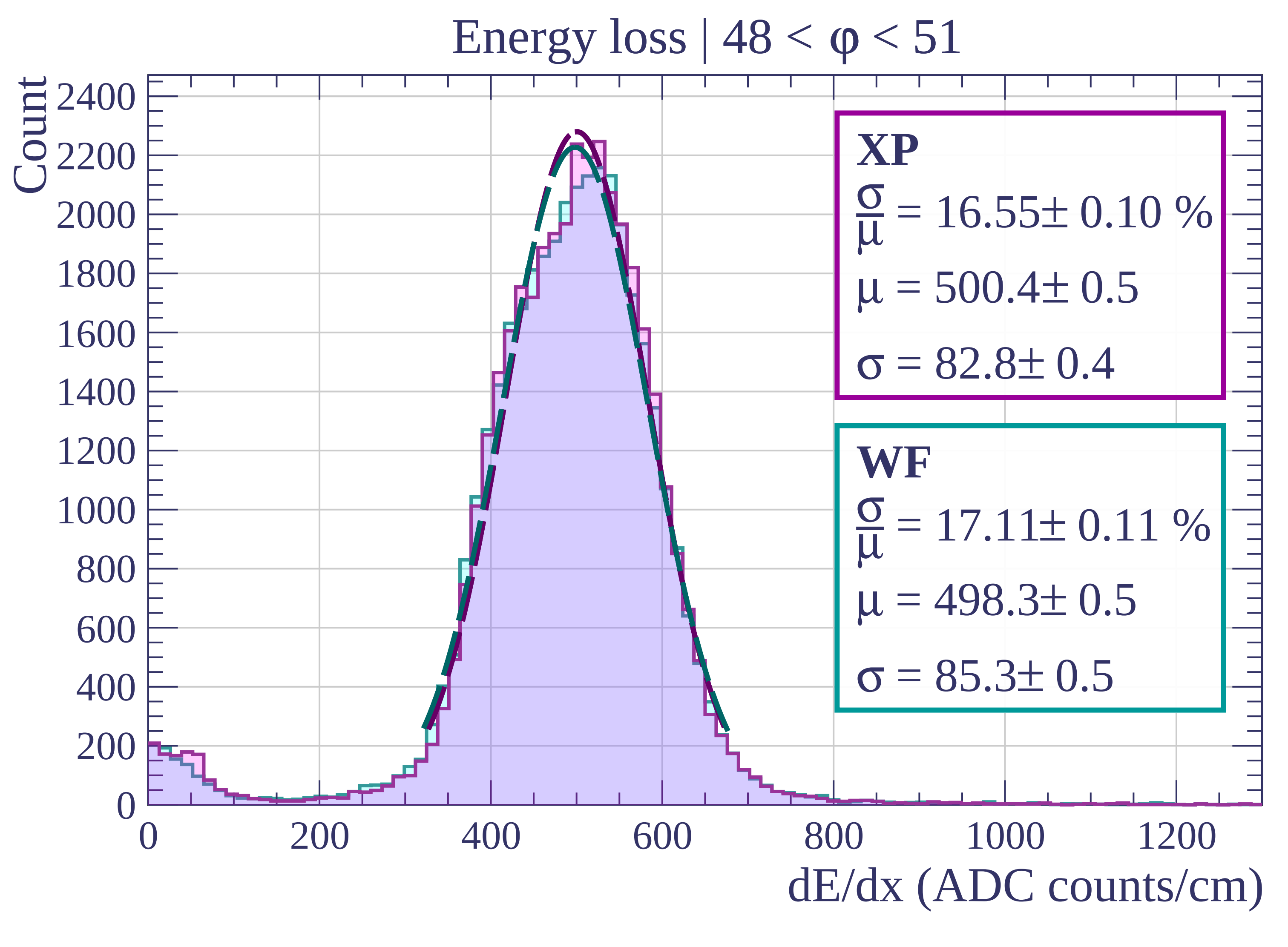


Energy loss | 39 < φ < 42
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 16.59± 0.13 %
μ = 509.0± 0.6
σ = 84.4± 0.5
WF
σ/μ = 17.61± 0.14 %
μ = 510.5± 0.7
σ = 89.9± 0.6
0
200
400
600
800
1000
1200
1400

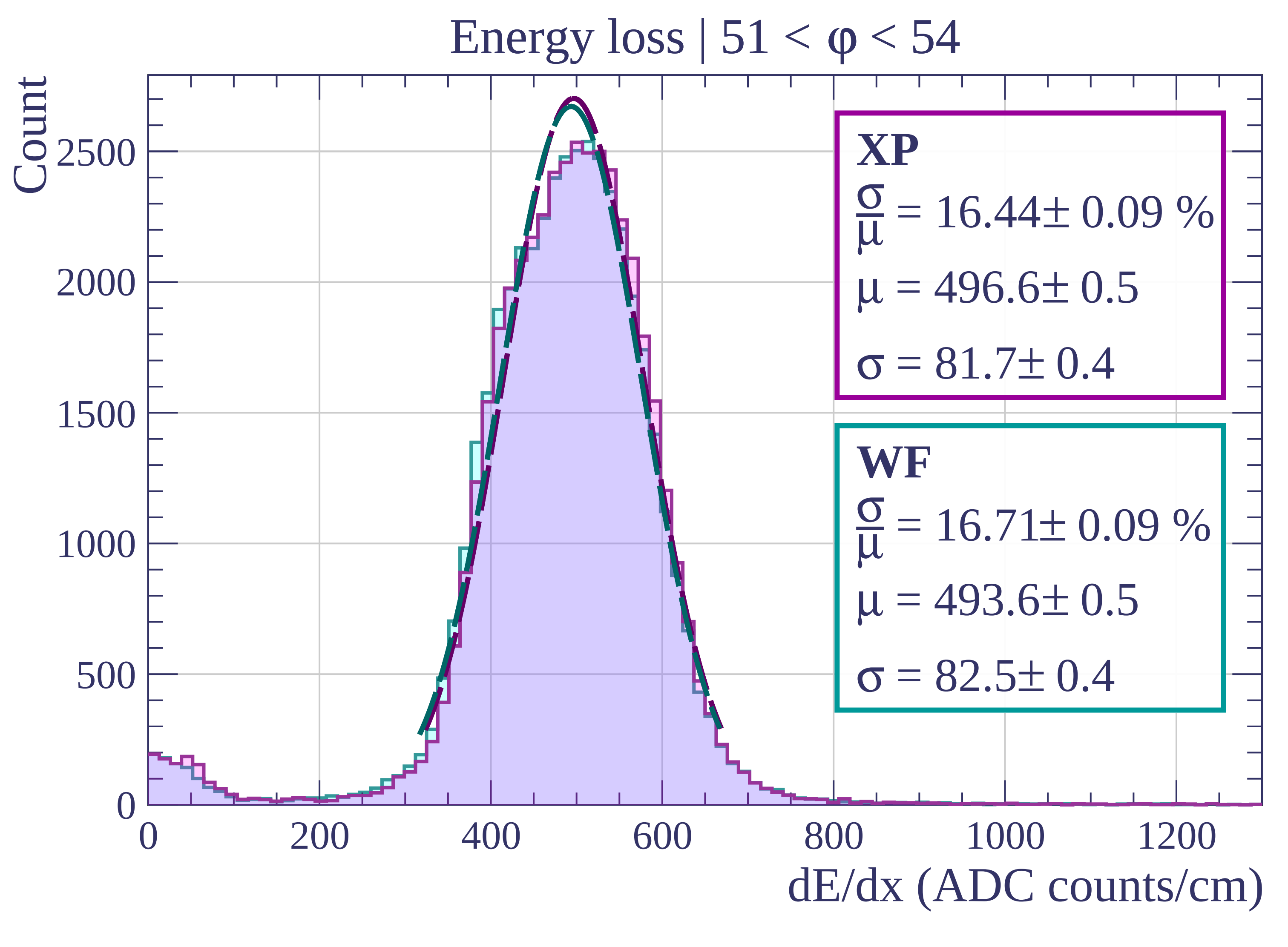

Energy loss | 42 < φ < 45
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 16.59± 0.13 %
μ = 507.3± 0.6
σ = 84.2± 0.6
WF
σ/μ = 17.25± 0.13 %
μ = 507.9± 0.6
σ = 87.6± 0.6
0
200
400
600
800
1000
1200
1400
1600



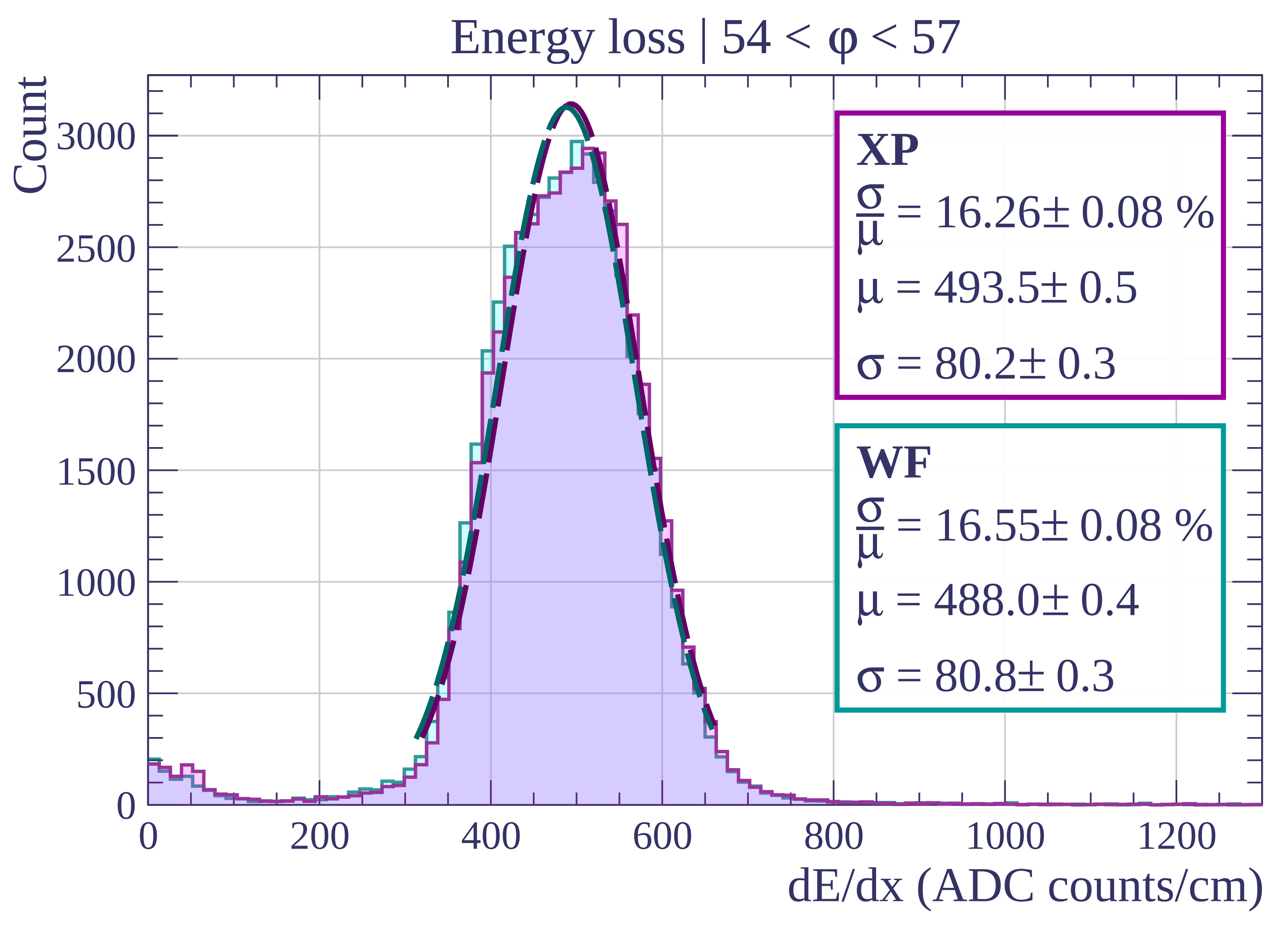
Energy loss | 45 < φ < 48
Count
2000
1800
1600
1400
1200
1000
800
600
400
200
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 16.66± 0.11 %
μ = 503.2± 0.5
σ = 83.8± 0.5
WF
σ/μ = 17.18± 0.12 %
μ = 503.1± 0.6
σ = 86.4± 0.5

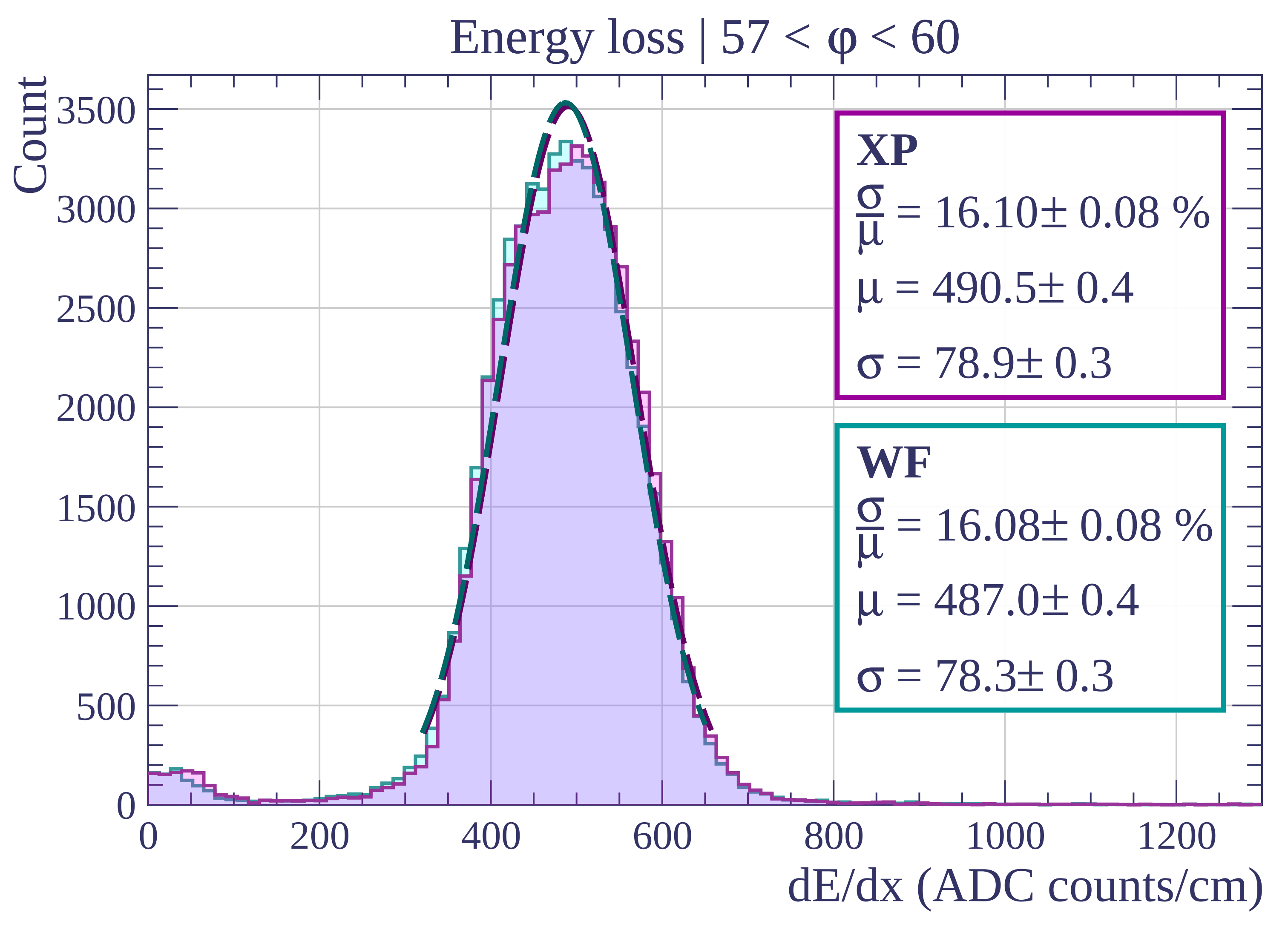


Energy loss | 48 < φ < 51
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
1400
1600
1800
2000
2200
2400
0
200
400
600
800
1000
1200
XP
σ/μ = 16.55± 0.10 %
μ = 500.4± 0.5
σ = 82.8± 0.4
WF
σ/μ = 17.11± 0.11 %
μ = 498.3± 0.5
σ = 85.3± 0.5

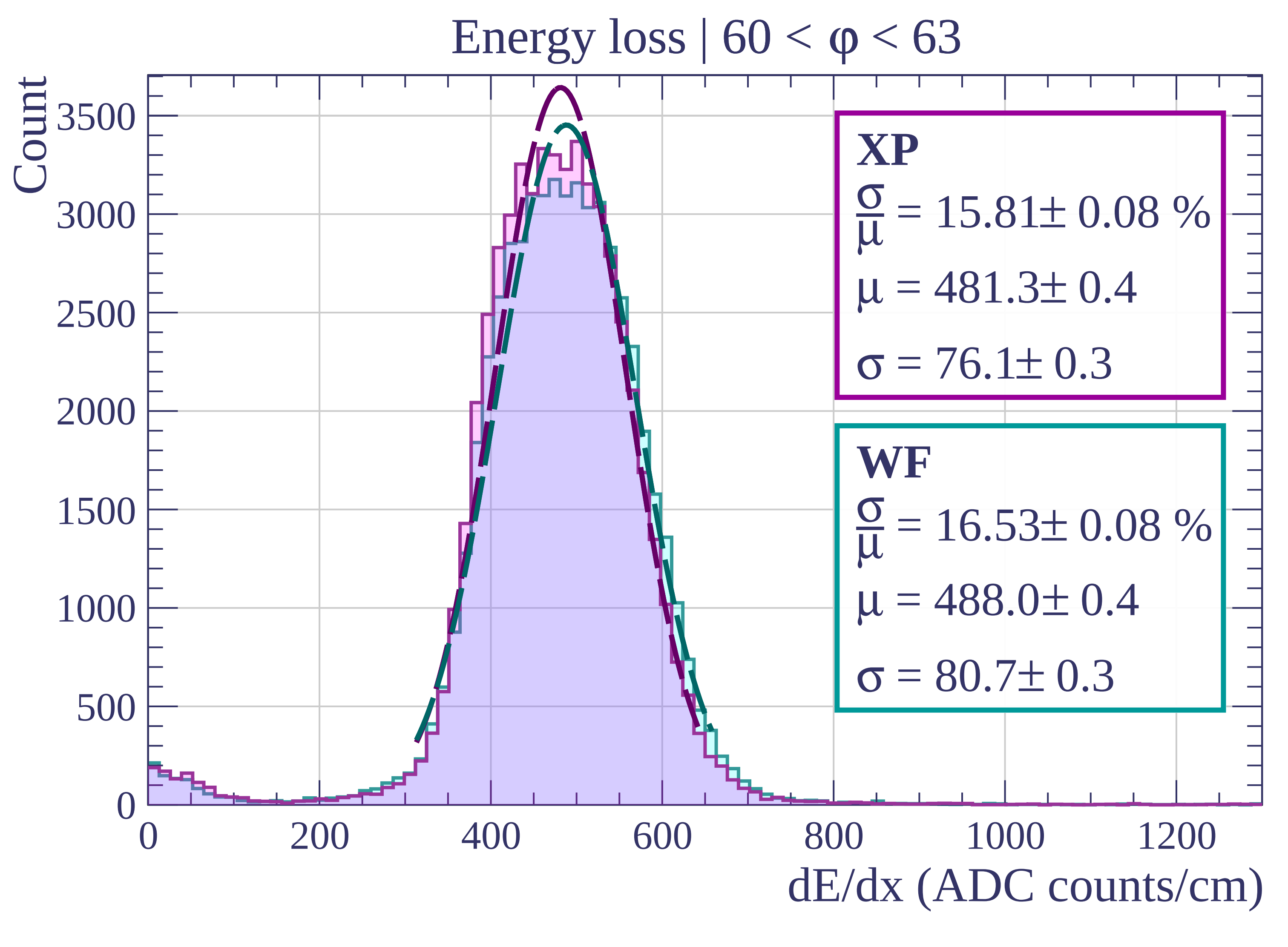


Energy loss | 51 < φ < 54
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
500
1000
1500
2000
2500
XP
σ/μ = 16.44± 0.09 %
μ = 496.6± 0.5
σ = 81.7± 0.4
WF
σ/μ = 16.71± 0.09 %
μ = 493.6± 0.5
σ = 82.5± 0.4

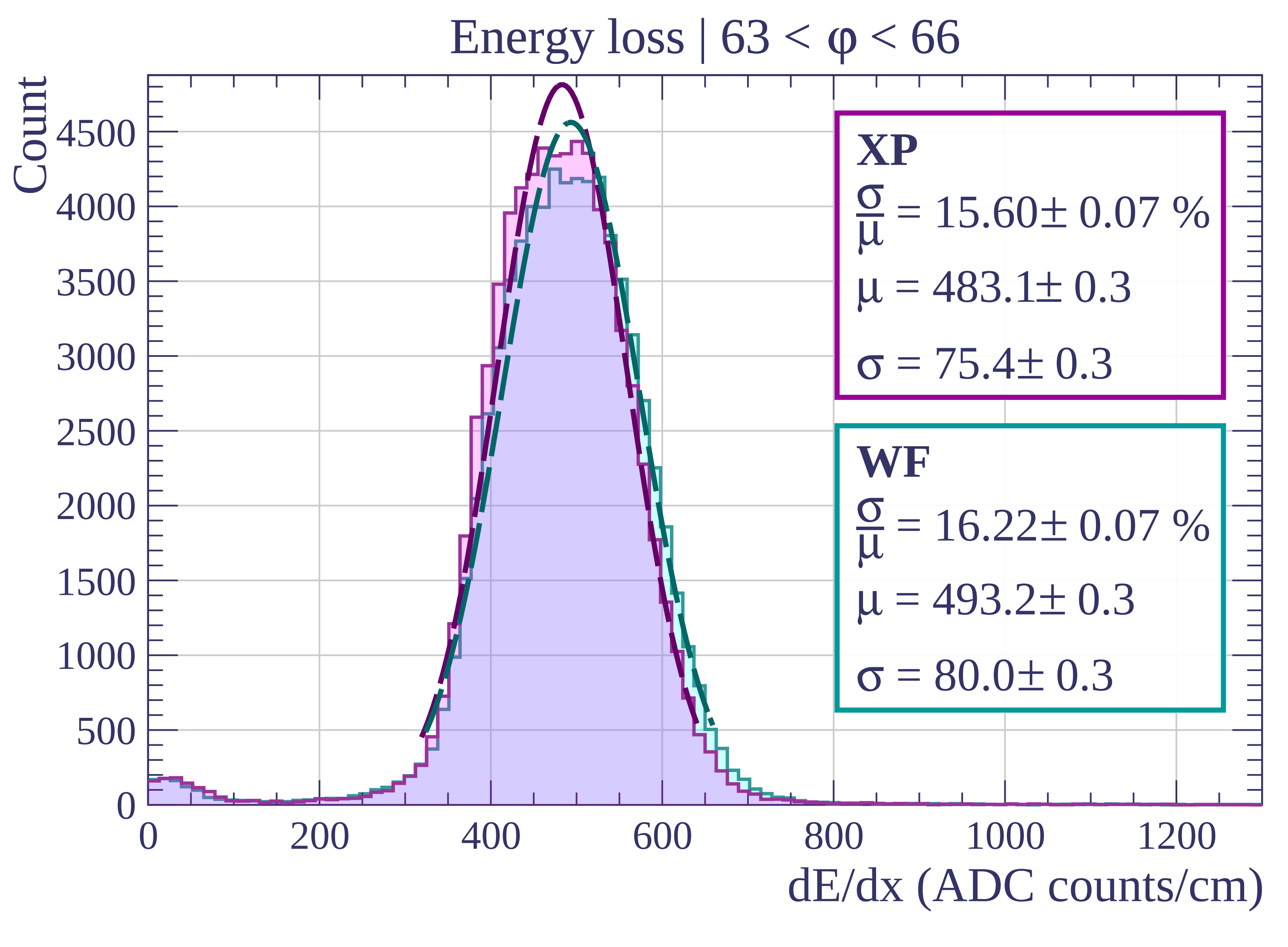

Energy loss | 54 < φ < 57
Count
3000
2500
2000
1500
1000
500
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 16.26± 0.08 %
μ = 493.5± 0.5
σ = 80.2± 0.3
WF
σ/μ = 16.55± 0.08 %
μ = 488.0± 0.4
σ = 80.8± 0.3

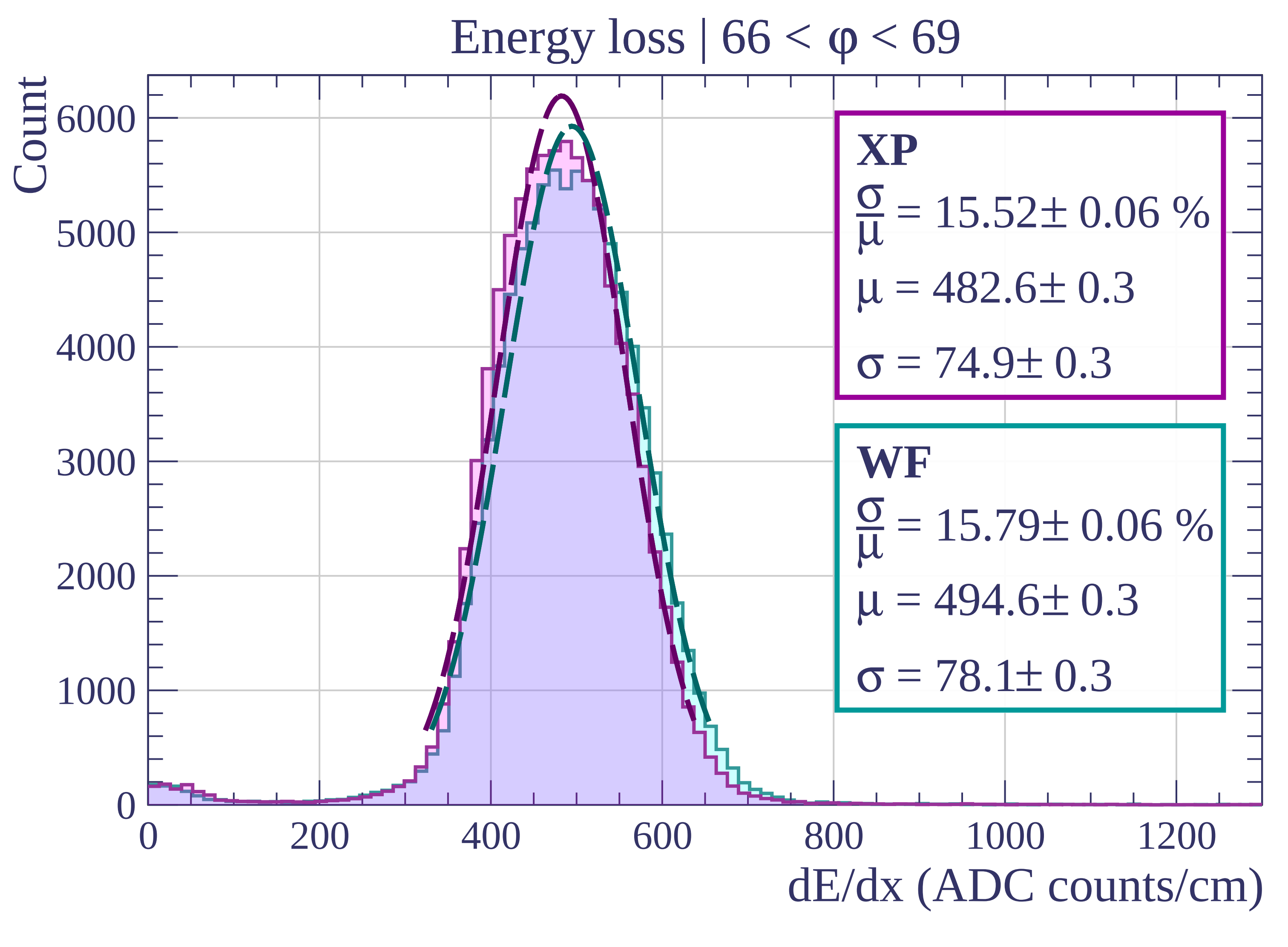


Energy loss | 57 < φ < 60
Count
3500
3000
2500
2000
1500
1000
500
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 16.10± 0.08 %
μ = 490.5± 0.4
σ = 78.9± 0.3
WF
σ/μ = 16.08± 0.08 %
μ = 487.0± 0.4
σ = 78.3± 0.3

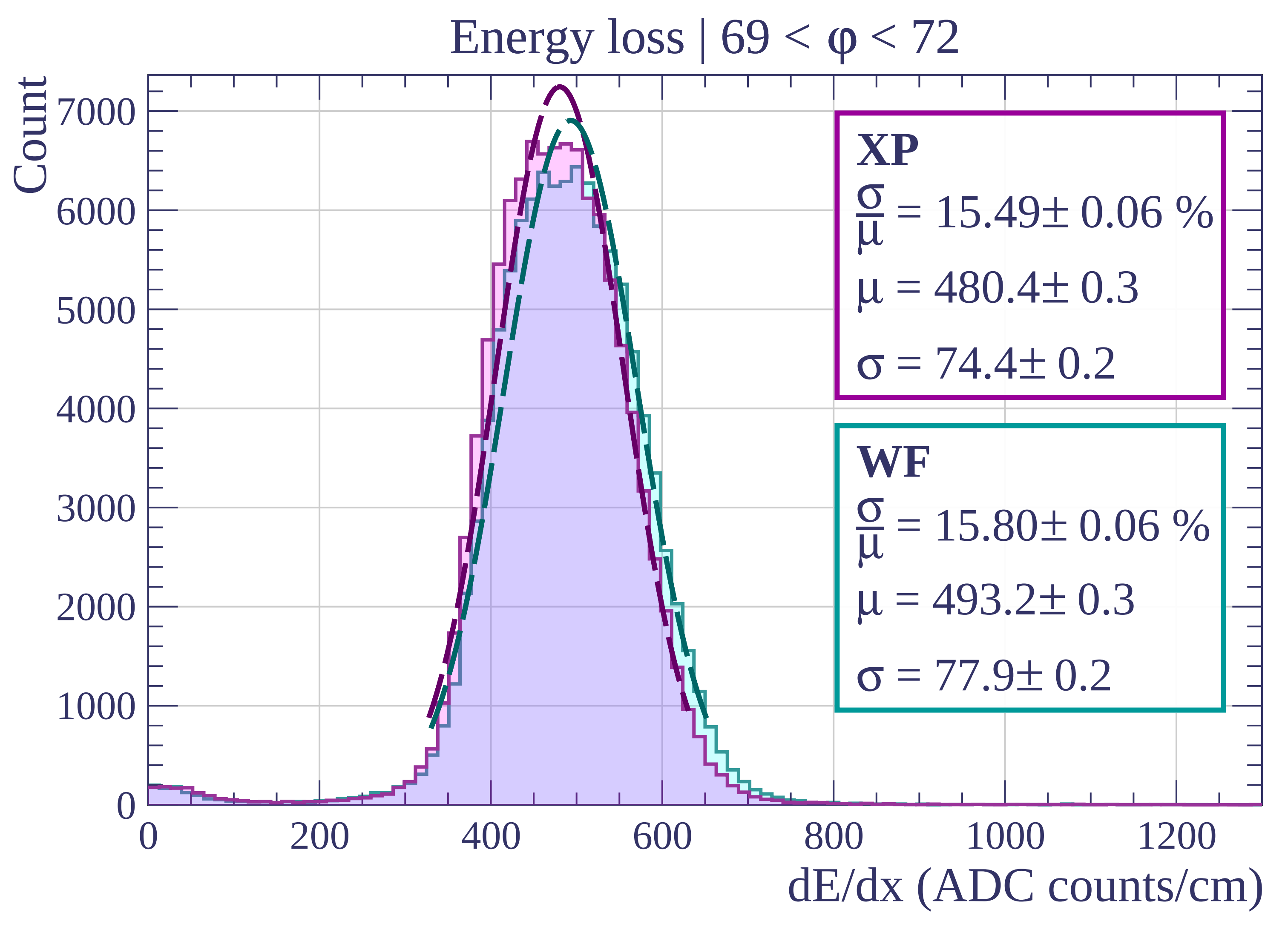


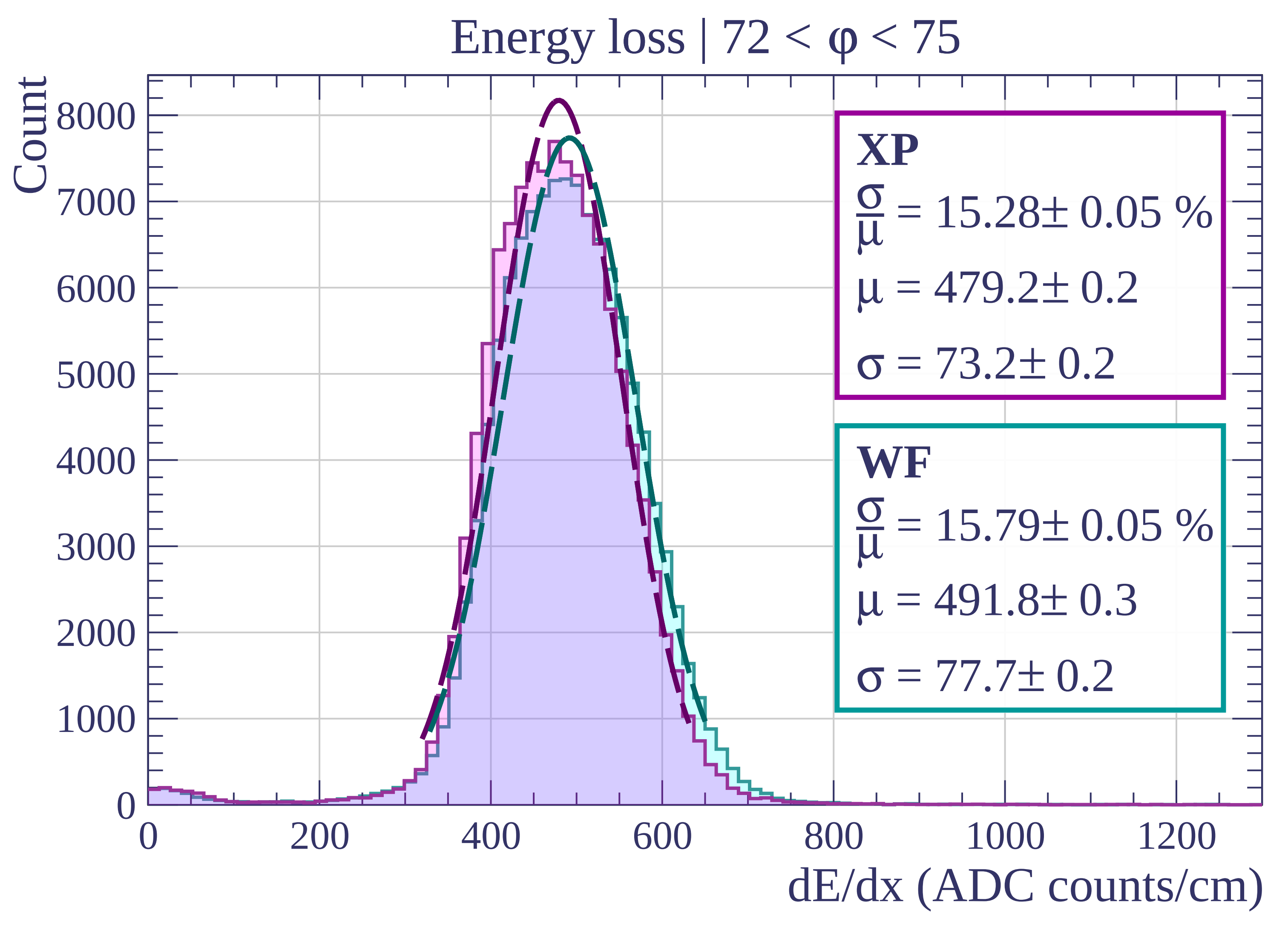
Energy loss | 60 < φ < 63
Count
dE/dx (ADC counts/cm)
0
500
1000
1500
2000
2500
3000
3500
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 15.81± 0.08 %
μ = 481.3± 0.4
σ = 76.1± 0.3
WF
$\frac{\sigma}{\mu}$ = 16.53± 0.08 %
μ = 488.0± 0.4
σ = 80.7± 0.3

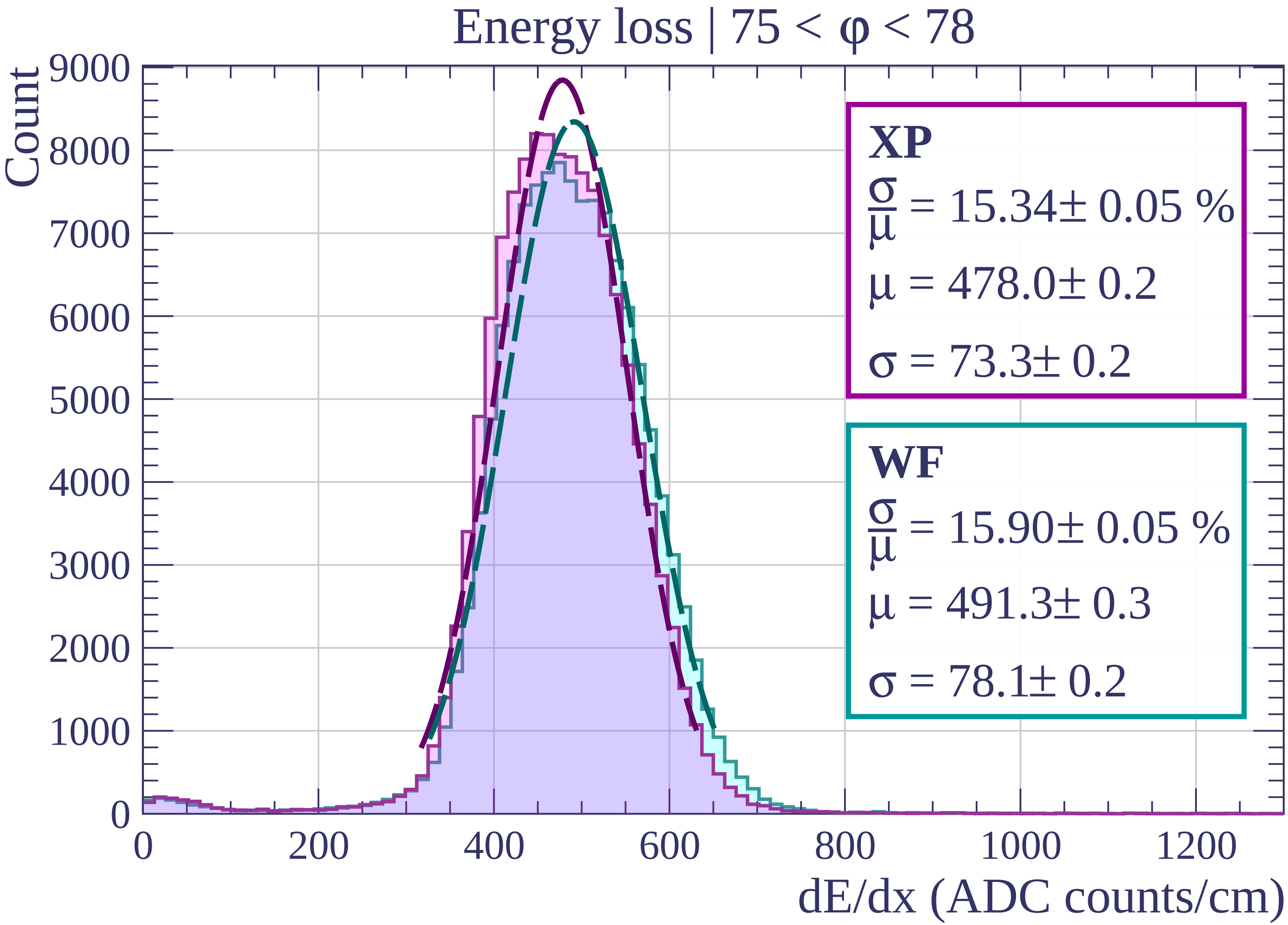

Energy loss | 63 < φ < 66
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 15.60± 0.07 %
μ = 483.1± 0.3
σ = 75.4± 0.3
WF
$\frac{\sigma}{\mu}$ = 16.22± 0.07 %
μ = 493.2± 0.3
σ = 80.0± 0.3
0
500
1000
1500
2000
2500
3000
3500
4000
4500
0
200
400
600
800
1000
1200



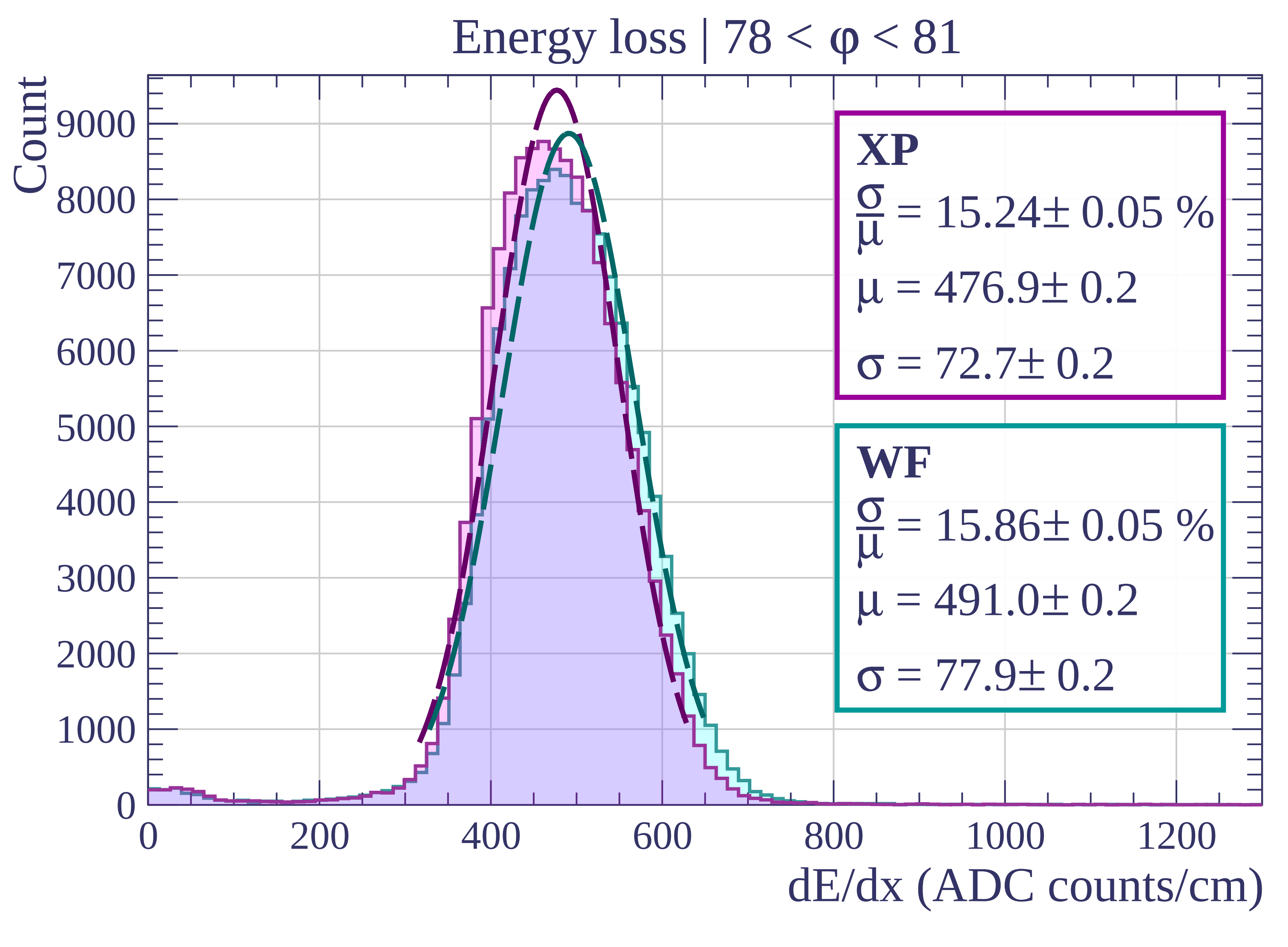
Energy loss | 66 < φ < 69
Count
dE/dx (ADC counts/cm)
0
1000
2000
3000
4000
5000
6000
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 15.52± 0.06 %
μ = 482.6± 0.3
σ = 74.9± 0.3
WF
$\frac{\sigma}{\mu}$ = 15.79± 0.06 %
μ = 494.6± 0.3
σ = 78.1± 0.3


Energy loss | 69 < φ < 72
Count
dE/dx (ADC counts/cm)
0
1000
2000
3000
4000
5000
6000
7000
0
200
400
600
800
1000
1200
XP
σ/μ = 15.49± 0.06 %
μ = 480.4± 0.3
σ = 74.4± 0.2
WF
σ/μ = 15.80± 0.06 %
μ = 493.2± 0.3
σ = 77.9± 0.2

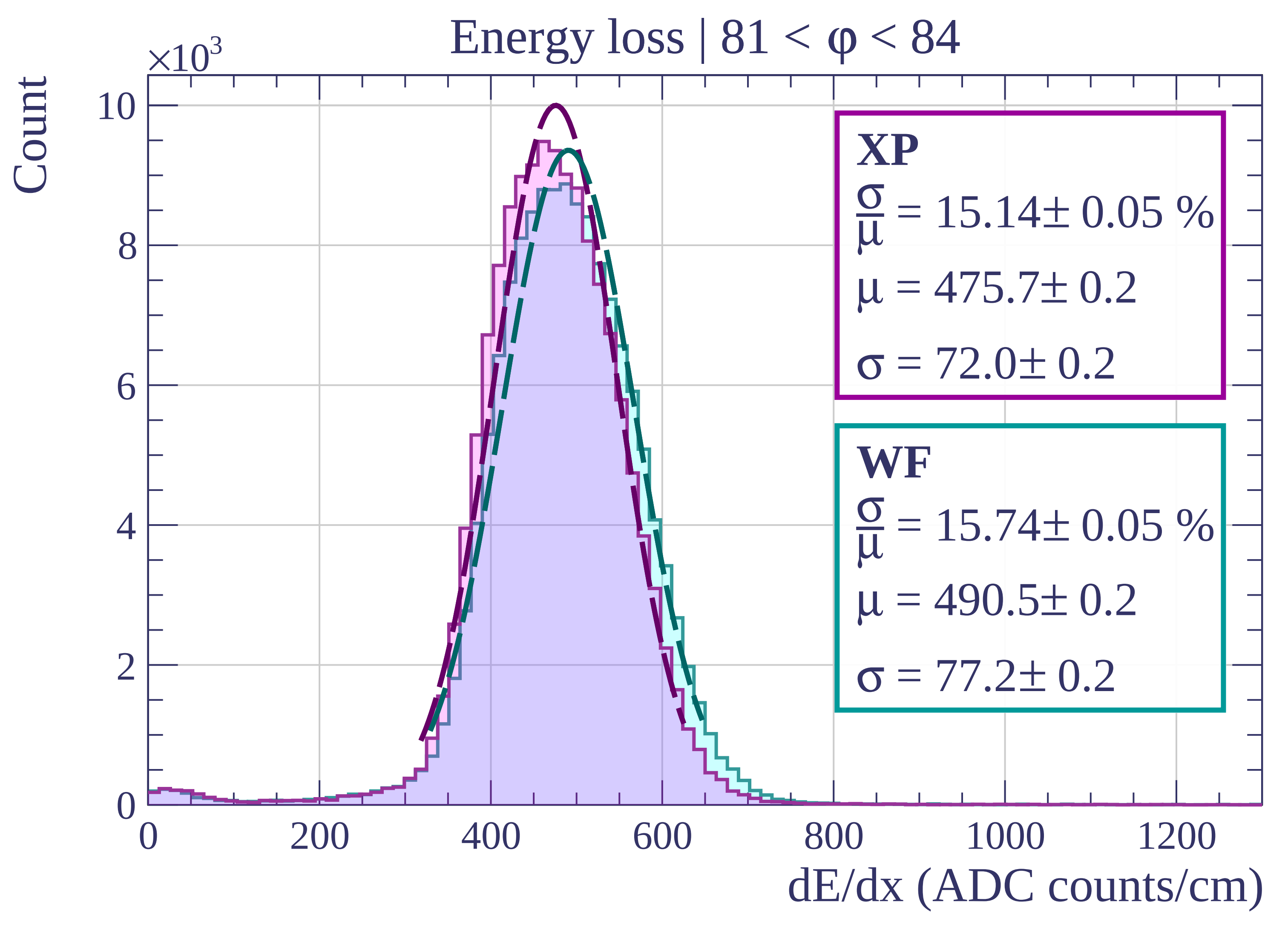

Energy loss | 72 < φ < 75
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 15.28± 0.05 %
μ = 479.2± 0.2
σ = 73.2± 0.2
WF
$\frac{\sigma}{\mu}$ = 15.79± 0.05 %
μ = 491.8± 0.3
σ = 77.7± 0.2
0
1000
2000
3000
4000
5000
6000
7000
8000
0
200
400
600
800
1000
1200

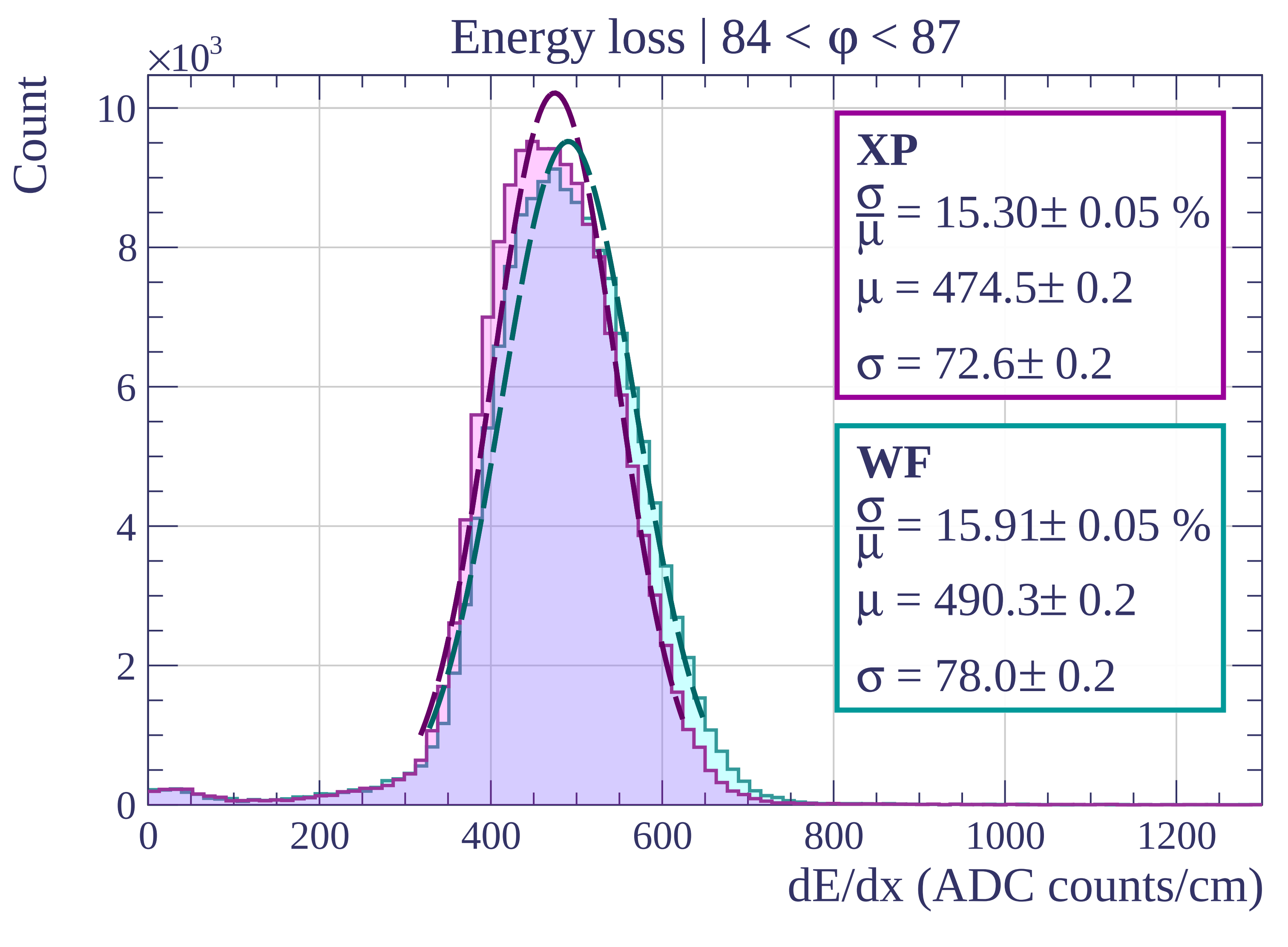


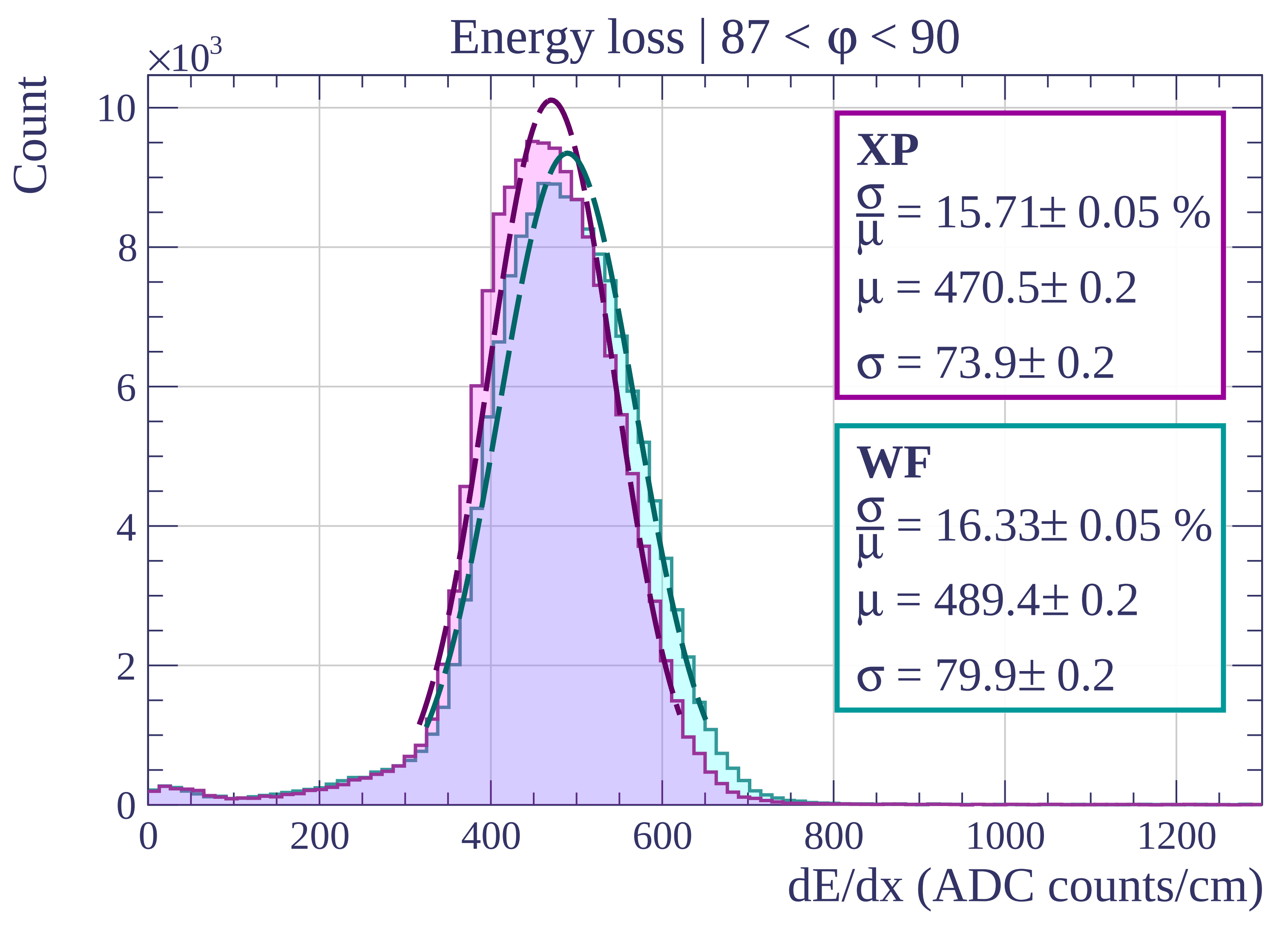
Energy loss | 75 < φ < 78
Count
9000
8000
7000
6000
5000
4000
3000
2000
1000
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 15.34± 0.05 %
μ = 478.0± 0.2
σ = 73.3± 0.2
WF
σ/μ = 15.90± 0.05 %
μ = 491.3± 0.3
σ = 78.1± 0.2

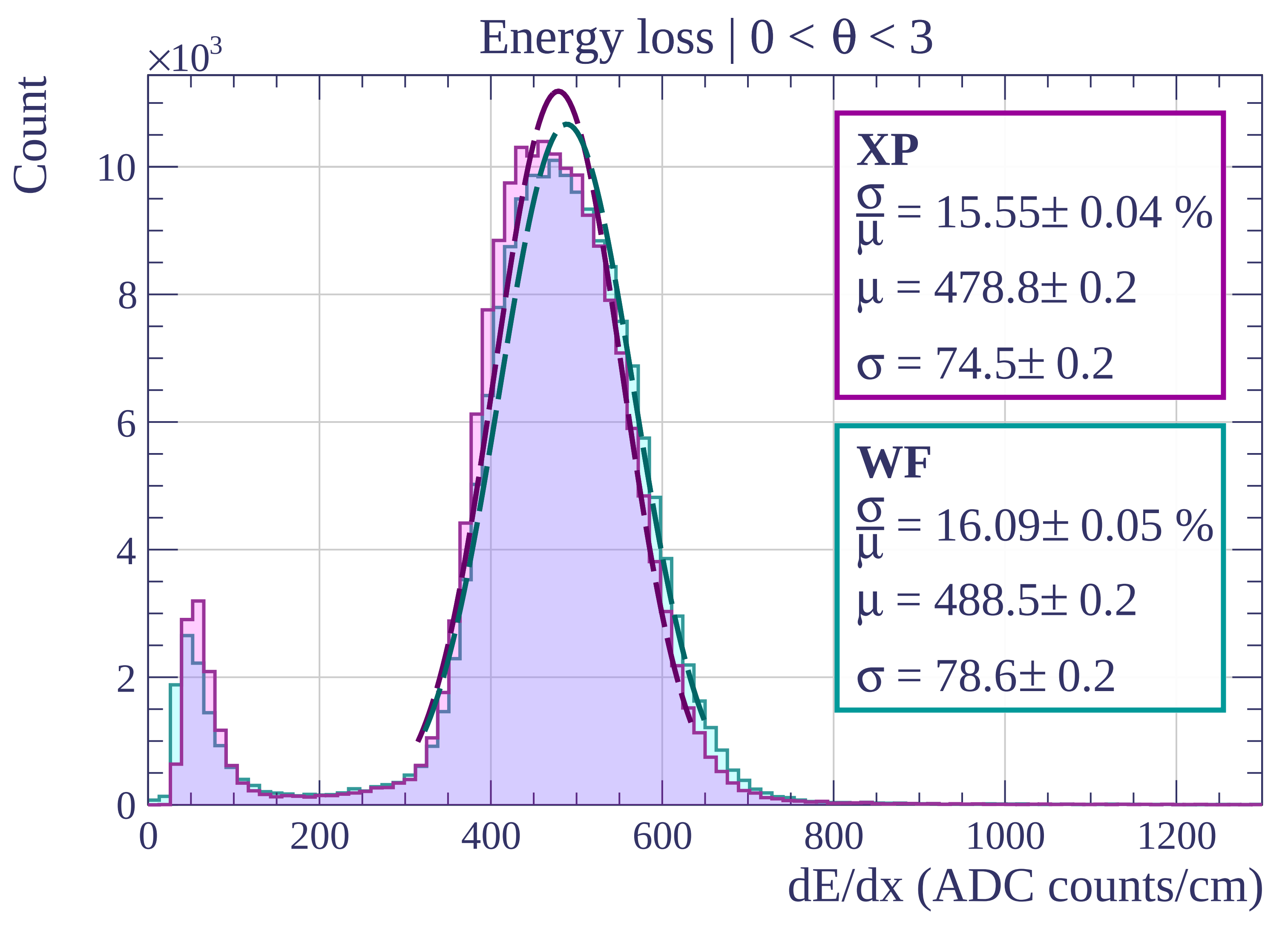


Energy loss | 78 < φ < 81
Count
9000
8000
7000
6000
5000
4000
3000
2000
1000
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 15.24± 0.05 %
μ = 476.9± 0.2
σ = 72.7± 0.2
WF
σ/μ = 15.86± 0.05 %
μ = 491.0± 0.2
σ = 77.9± 0.2

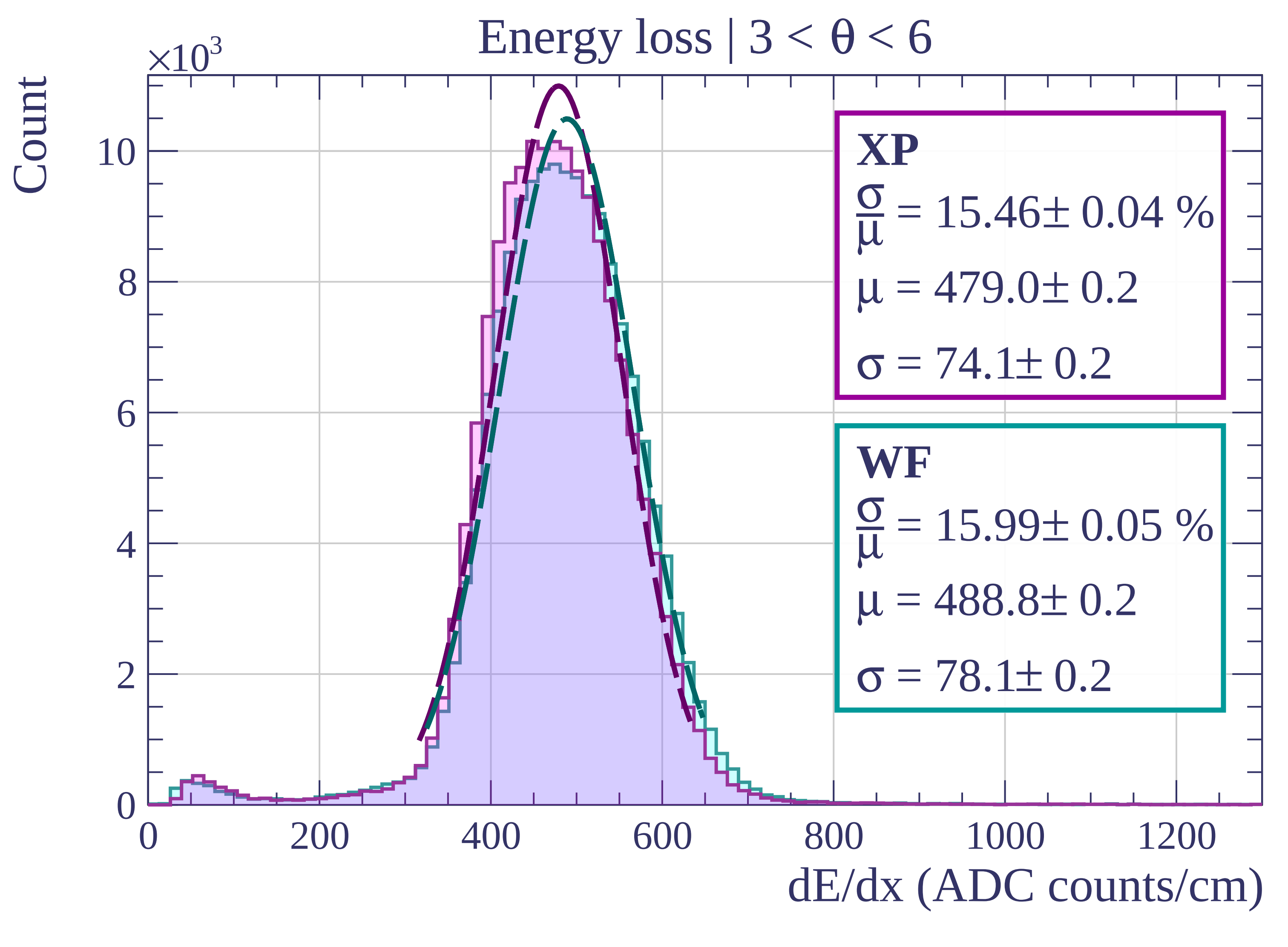

Energy loss | 81 < φ < 84
×10^3
Count
10
8
6
4
2
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 15.14± 0.05 %
μ = 475.7± 0.2
σ = 72.0± 0.2
WF
σ/μ = 15.74± 0.05 %
μ = 490.5± 0.2
σ = 77.2± 0.2


Energy loss | 84 < φ < 87
×10³
Count
10
8
6
4
2
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 15.30± 0.05 %
μ = 474.5± 0.2
σ = 72.6± 0.2
WF
σ/μ = 15.91± 0.05 %
μ = 490.3± 0.2
σ = 78.0± 0.2

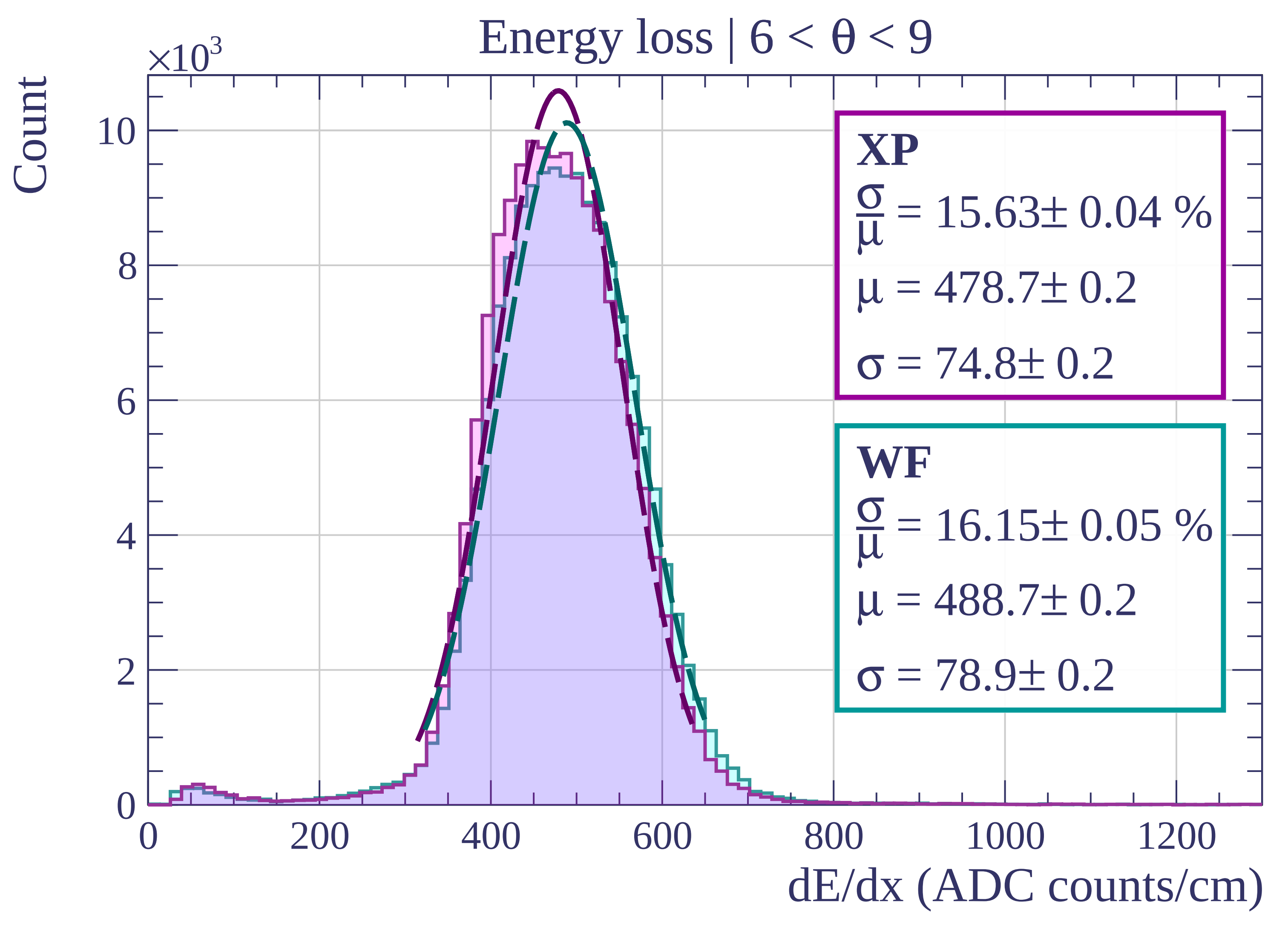

Energy loss | 87 < φ < 90
×10^3
Count
XP
σ/μ = 15.71± 0.05 %
μ = 470.5± 0.2
σ = 73.9± 0.2
WF
σ/μ = 16.33± 0.05 %
μ = 489.4± 0.2
σ = 79.9± 0.2
dE/dx (ADC counts/cm)

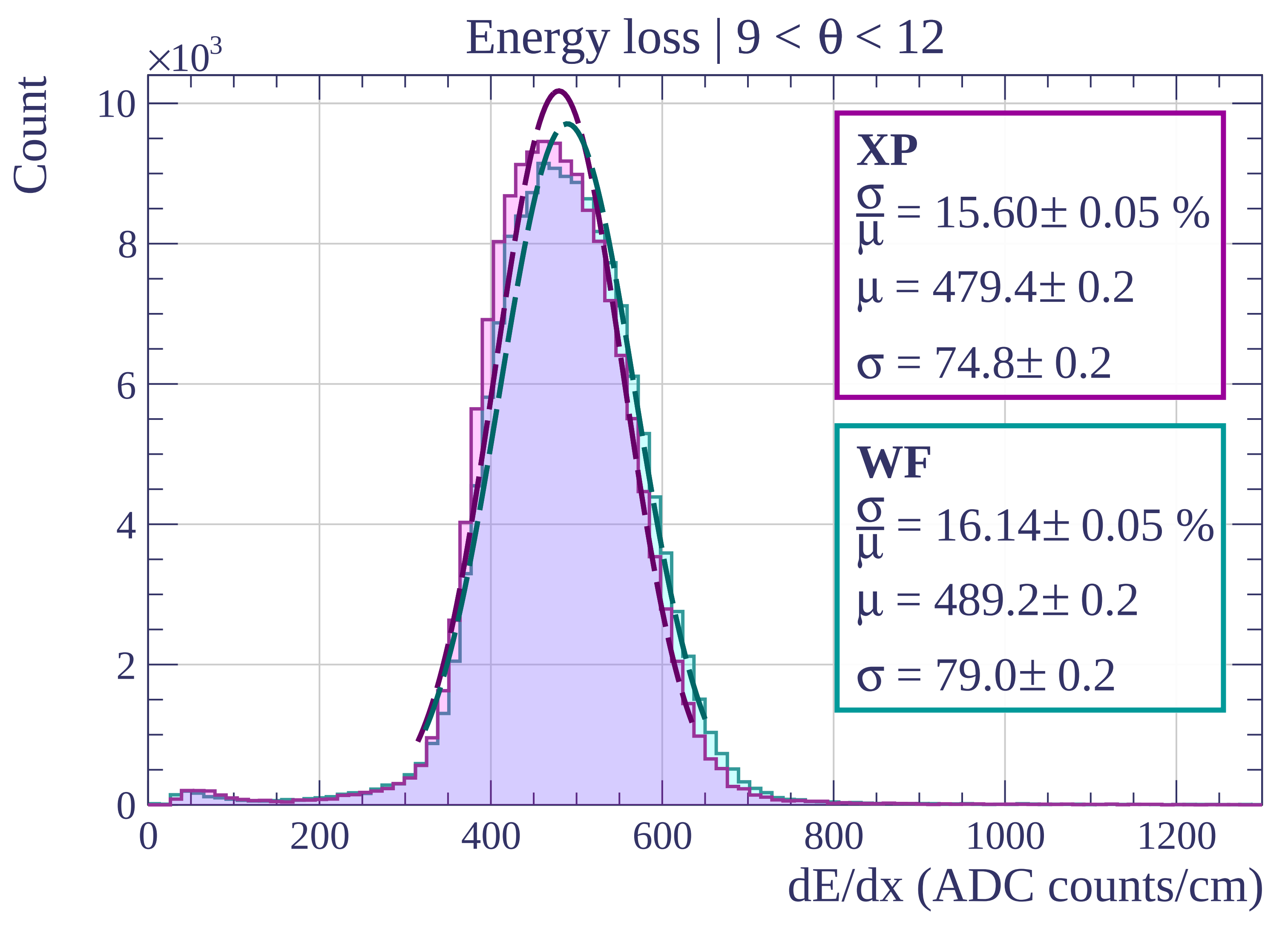


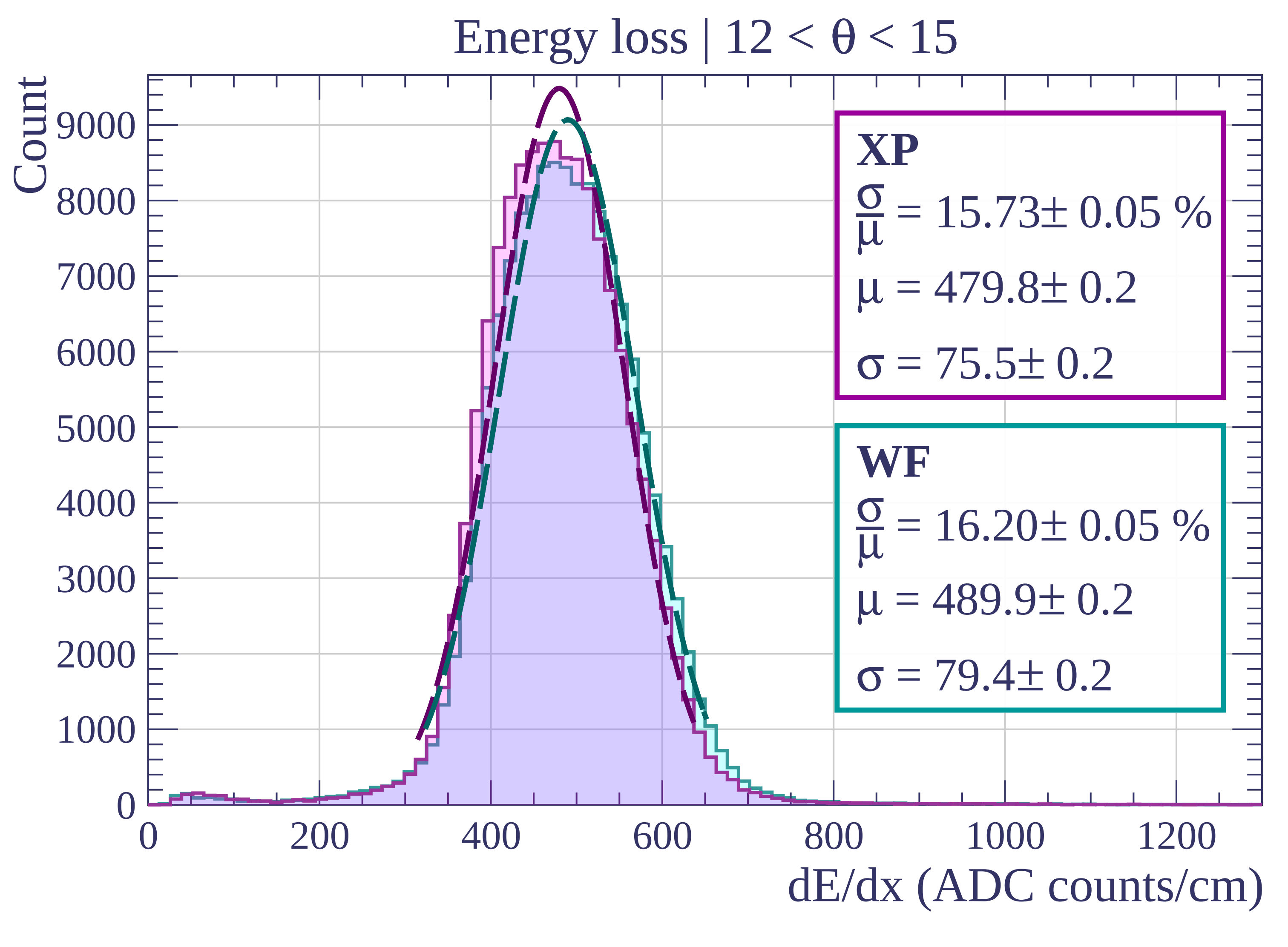
Energy loss | 0 < θ < 3
×10³
Count
0
2
4
6
8
10
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 15.55± 0.04 %
μ = 478.8± 0.2
σ = 74.5± 0.2
WF
σ/μ = 16.09± 0.05 %
μ = 488.5± 0.2
σ = 78.6± 0.2

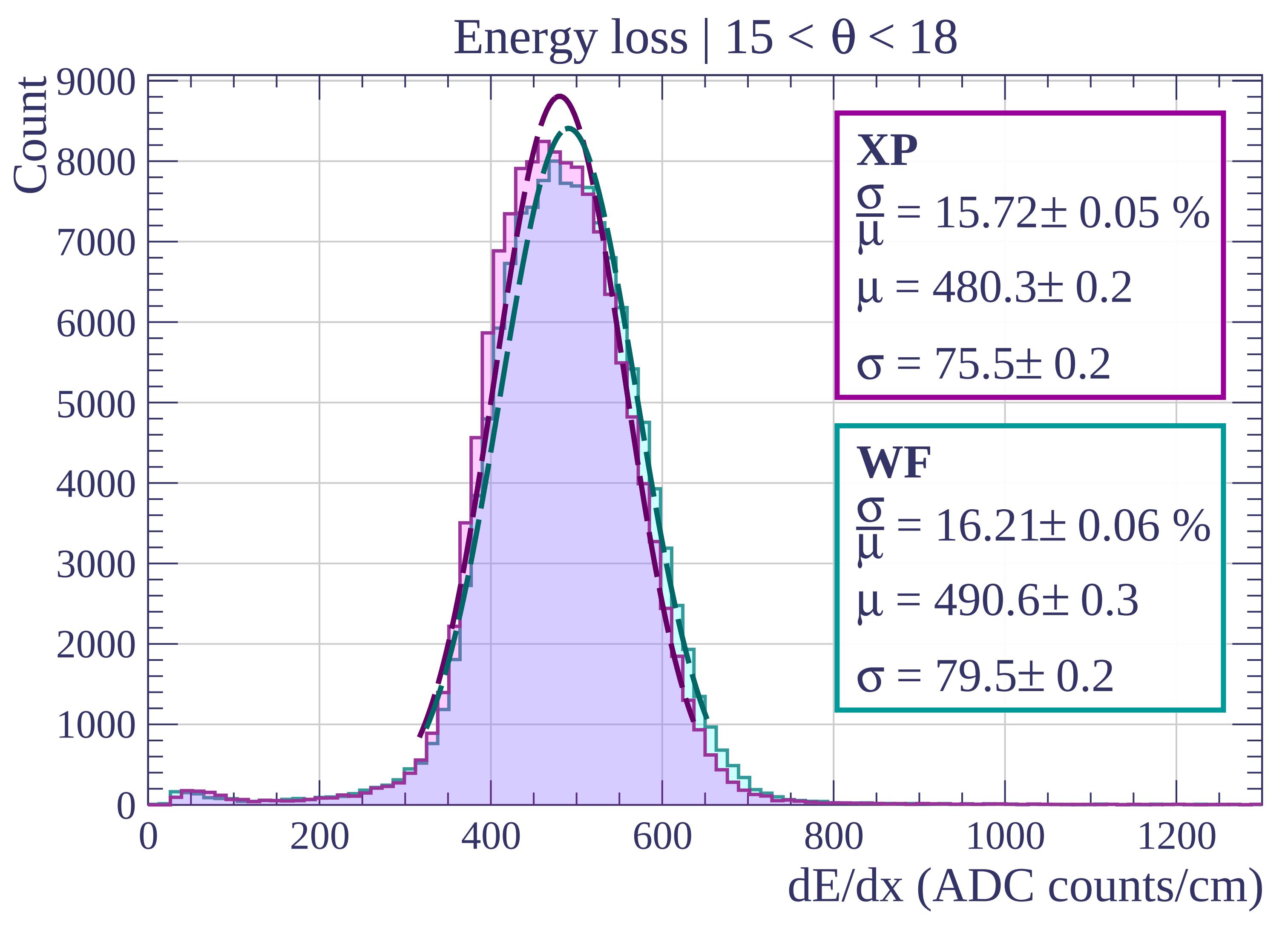


Energy loss | 3 < θ < 6
×10³
Count
10
8
6
4
2
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 15.46± 0.04 %
μ = 479.0± 0.2
σ = 74.1± 0.2
WF
σ/μ = 15.99± 0.05 %
μ = 488.8± 0.2
σ = 78.1± 0.2

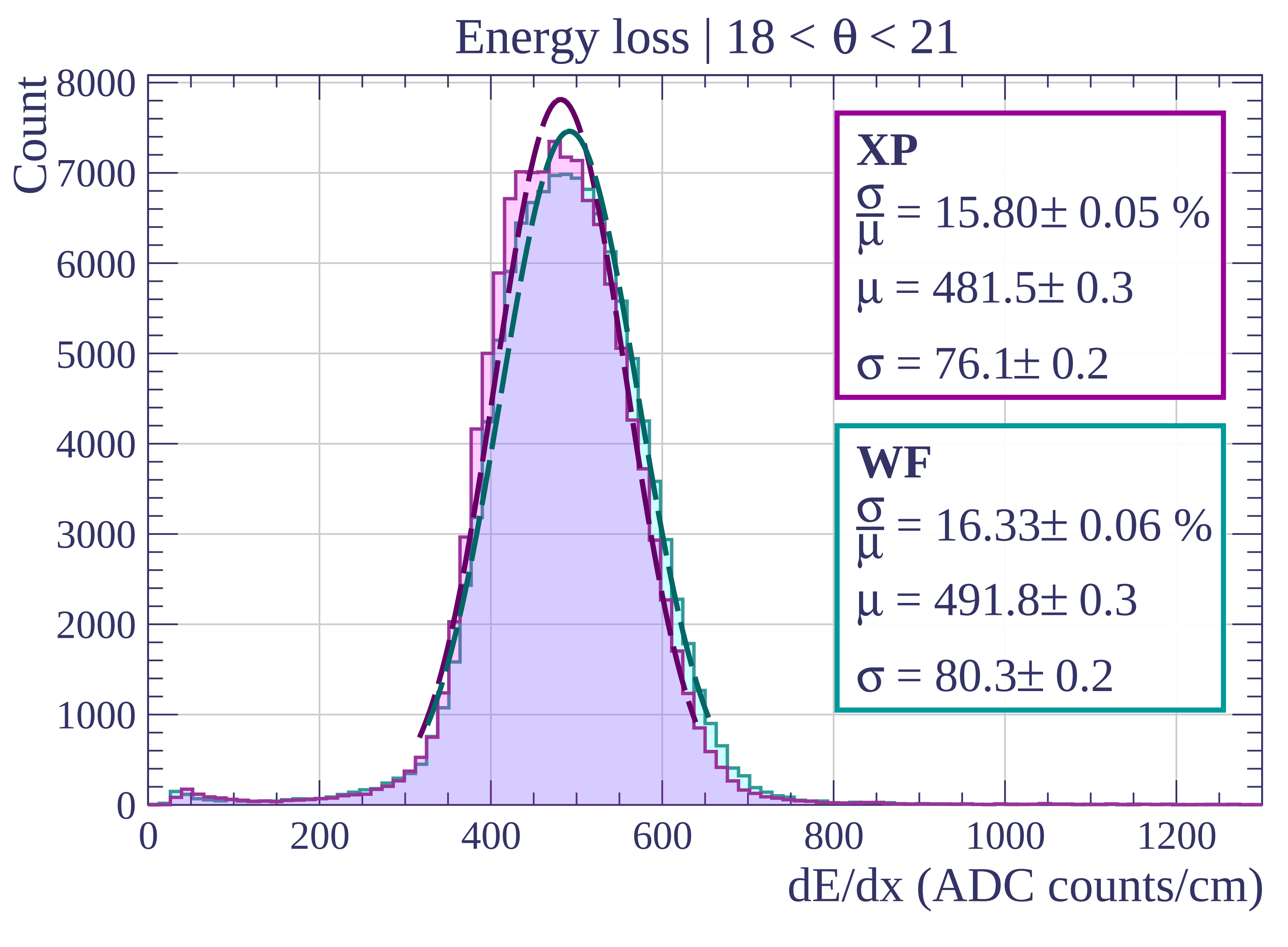

Energy loss | 6 < θ < 9
×10³
Count
10
8
6
4
2
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 15.63± 0.04 %
μ = 478.7± 0.2
σ = 74.8± 0.2
WF
σ/μ = 16.15± 0.05 %
μ = 488.7± 0.2
σ = 78.9± 0.2


Energy loss | 9 < θ < 12
×10^3
Count
XP
σ/μ = 15.60± 0.05 %
μ = 479.4± 0.2
σ = 74.8± 0.2
WF
σ/μ = 16.14± 0.05 %
μ = 489.2± 0.2
σ = 79.0± 0.2
dE/dx (ADC counts/cm)
0
2
4
6
8
10
200
400
600
800
1000
1200

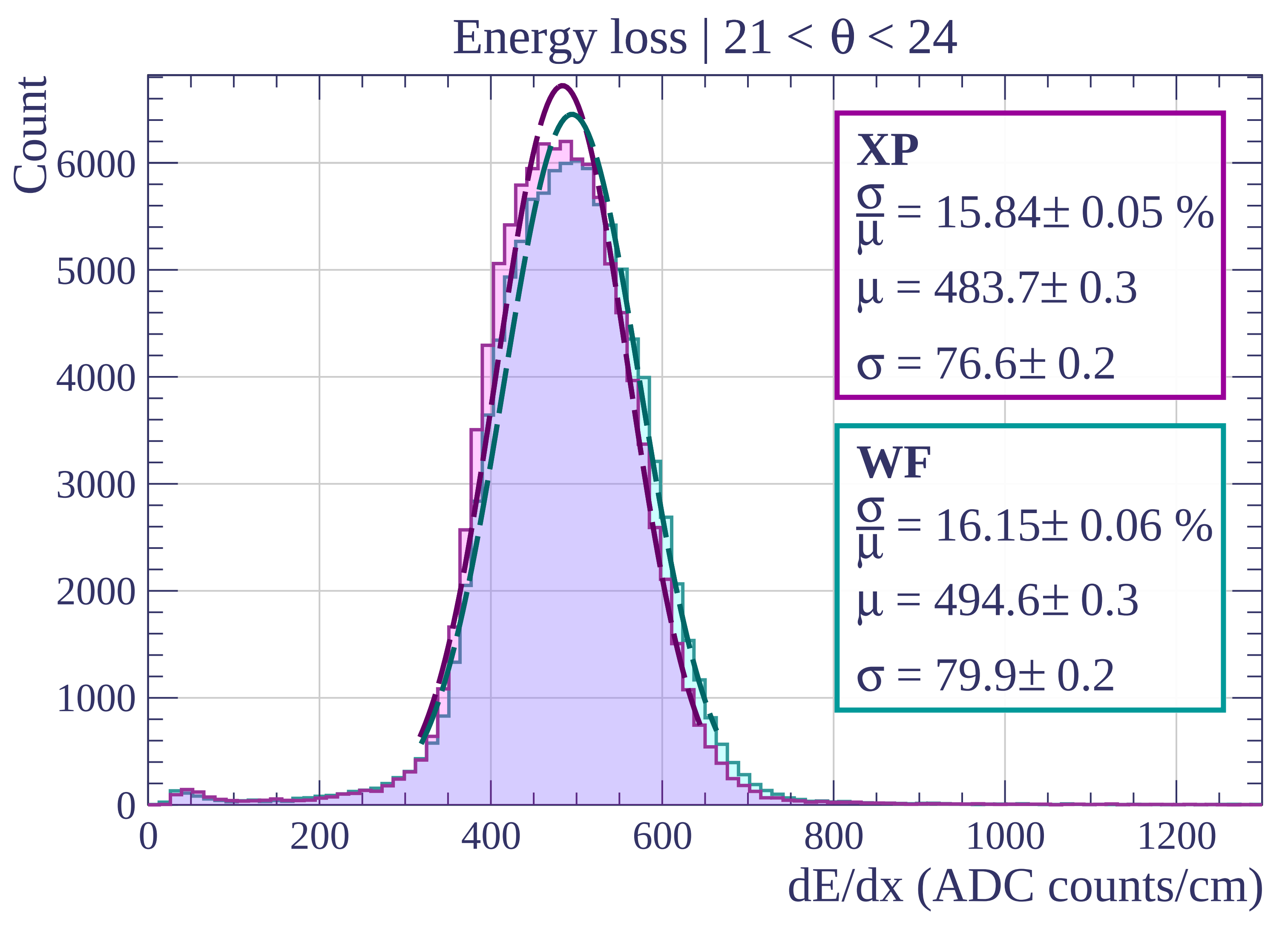

Energy loss | 12 < θ < 15
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
1000
2000
3000
4000
5000
6000
7000
8000
9000
XP
σ/μ = 15.73± 0.05 %
μ = 479.8± 0.2
σ = 75.5± 0.2
WF
σ/μ = 16.20± 0.05 %
μ = 489.9± 0.2
σ = 79.4± 0.2



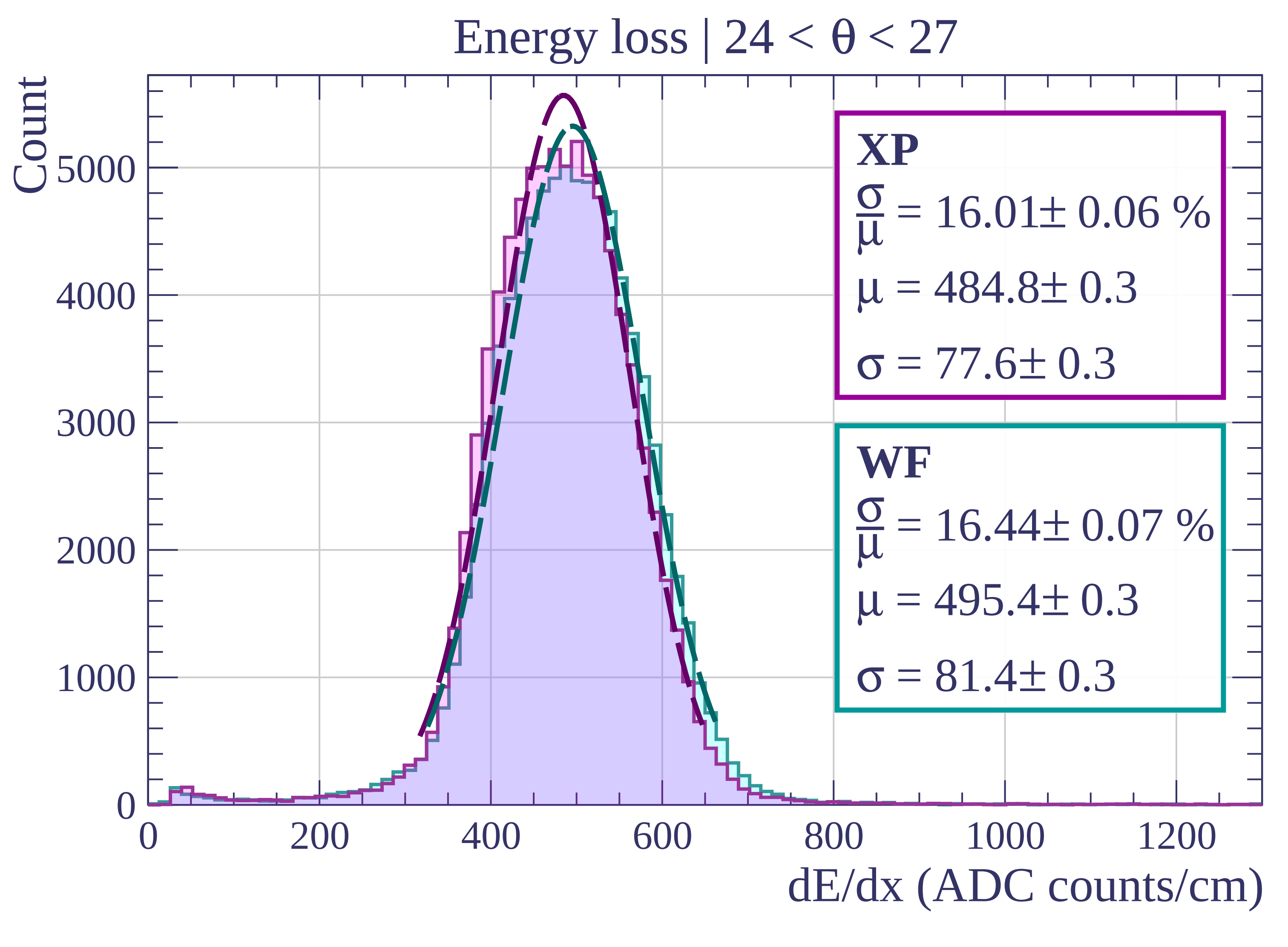
Energy loss | 15 < θ < 18
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
1000
2000
3000
4000
5000
6000
7000
8000
9000
XP
σ/μ = 15.72± 0.05 %
μ = 480.3± 0.2
σ = 75.5± 0.2
WF
σ/μ = 16.21± 0.06 %
μ = 490.6± 0.3
σ = 79.5± 0.2



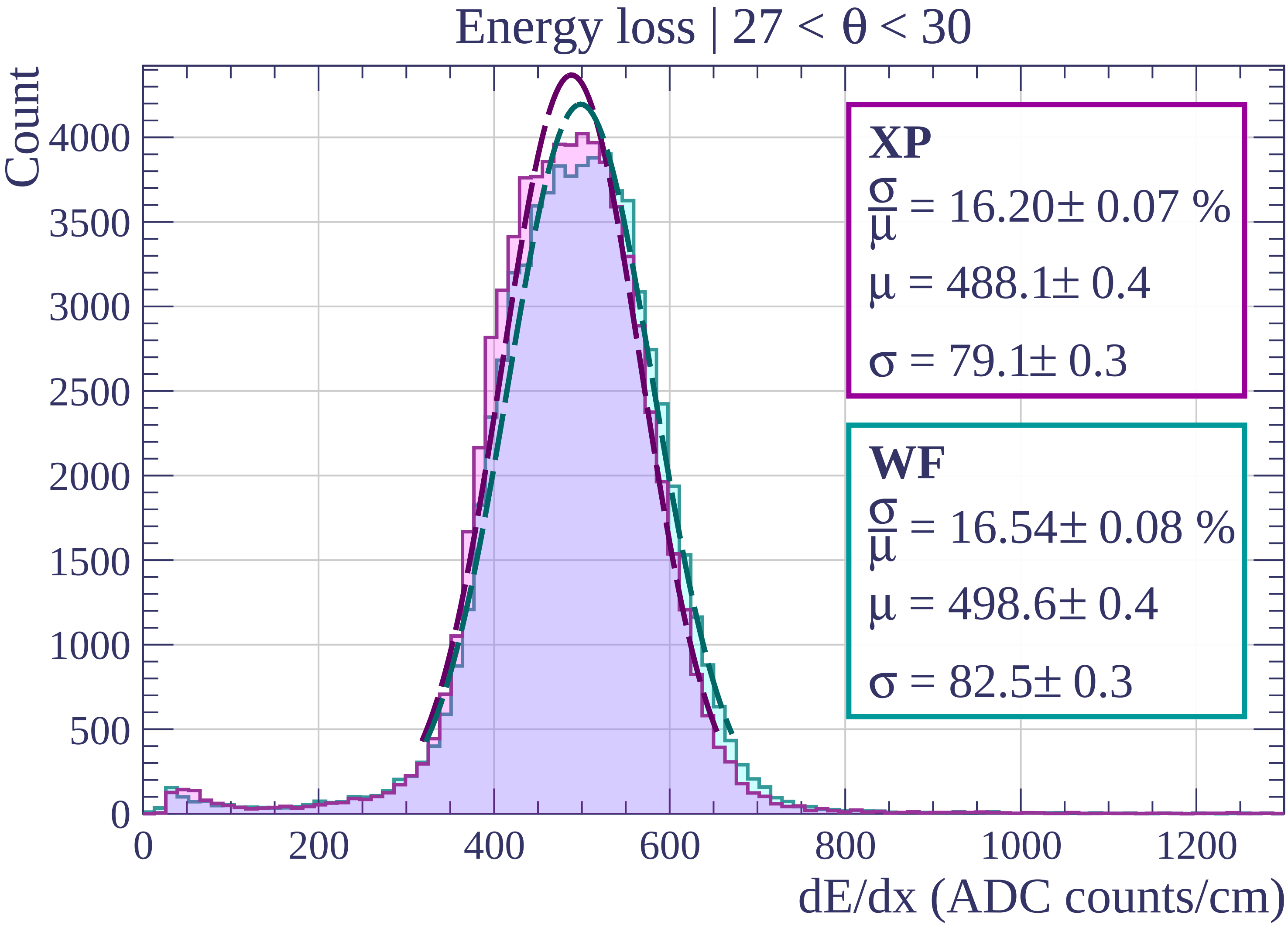
Energy loss | 18 < θ < 21
Count
8000
7000
6000
5000
4000
3000
2000
1000
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 15.80± 0.05 %
μ = 481.5± 0.3
σ = 76.1± 0.2
WF
$\frac{\sigma}{\mu}$ = 16.33± 0.06 %
μ = 491.8± 0.3
σ = 80.3± 0.2

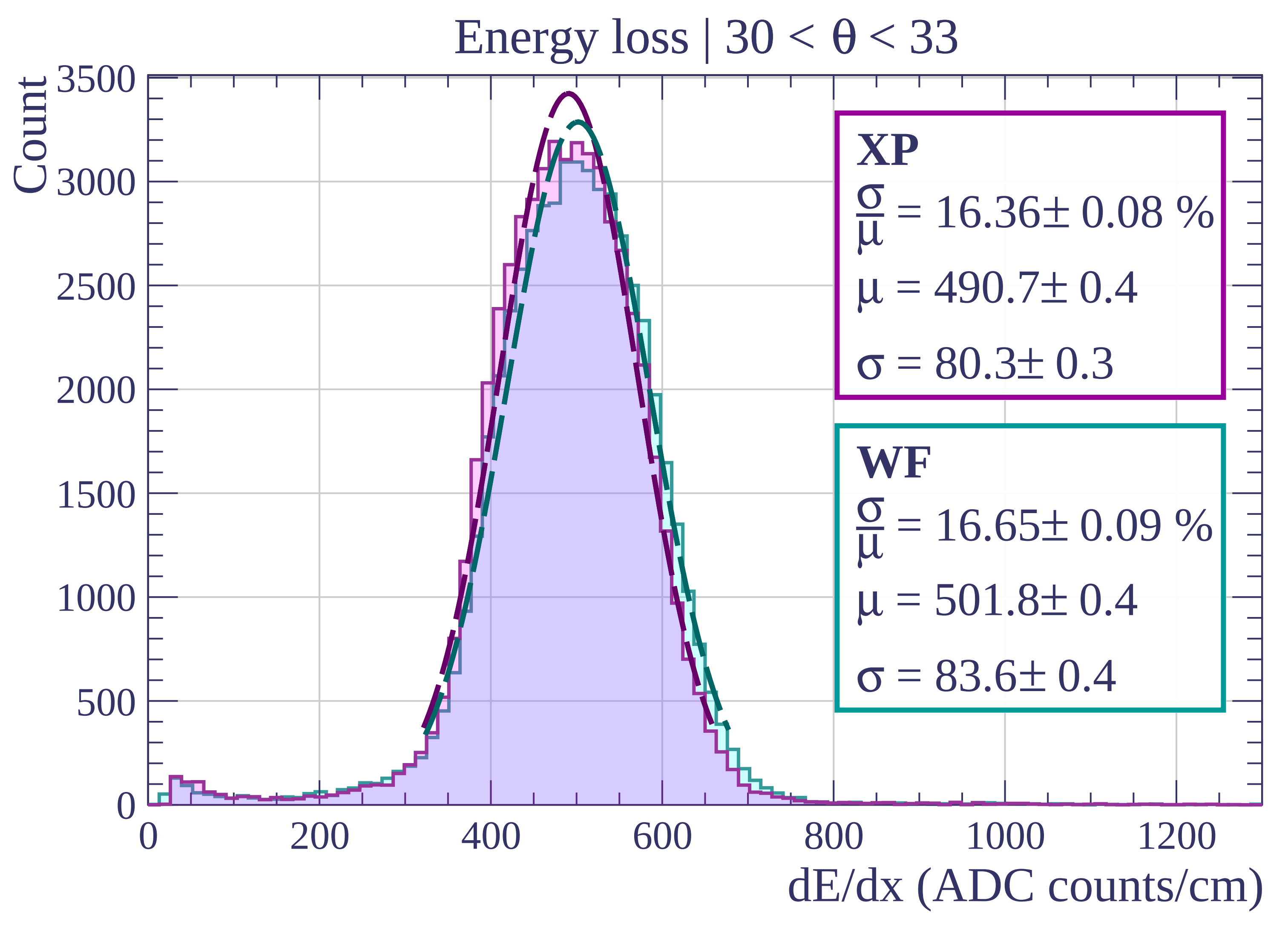


Energy loss | 21 < θ < 24
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
1000
2000
3000
4000
5000
6000
XP
σ/μ = 15.84± 0.05 %
μ = 483.7± 0.3
σ = 76.6± 0.2
WF
σ/μ = 16.15± 0.06 %
μ = 494.6± 0.3
σ = 79.9± 0.2

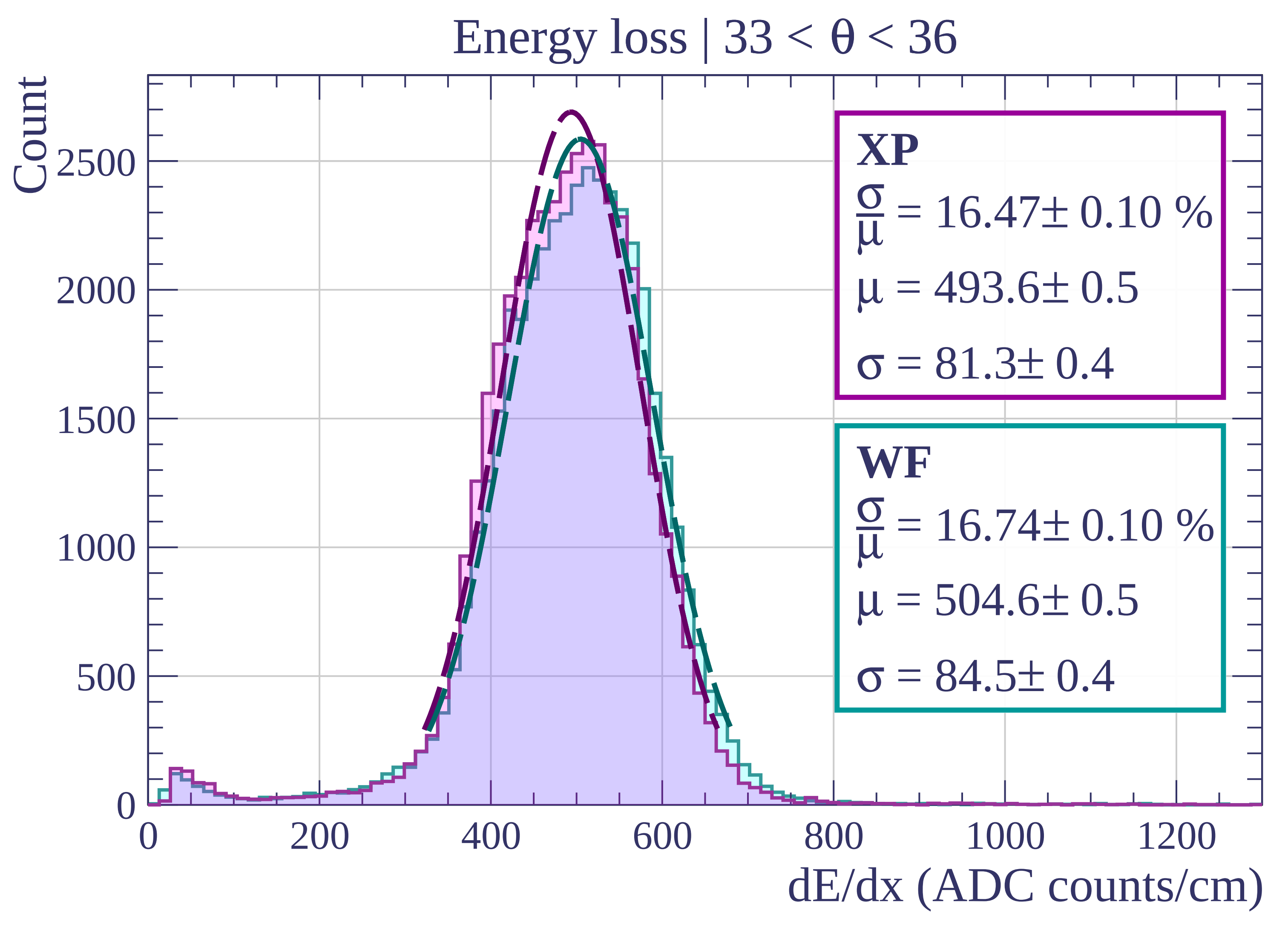


Energy loss | 24 < θ < 27
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
1000
2000
3000
4000
5000
XP
σ/μ = 16.01± 0.06 %
μ = 484.8± 0.3
σ = 77.6± 0.3
WF
σ/μ = 16.44± 0.07 %
μ = 495.4± 0.3
σ = 81.4± 0.3

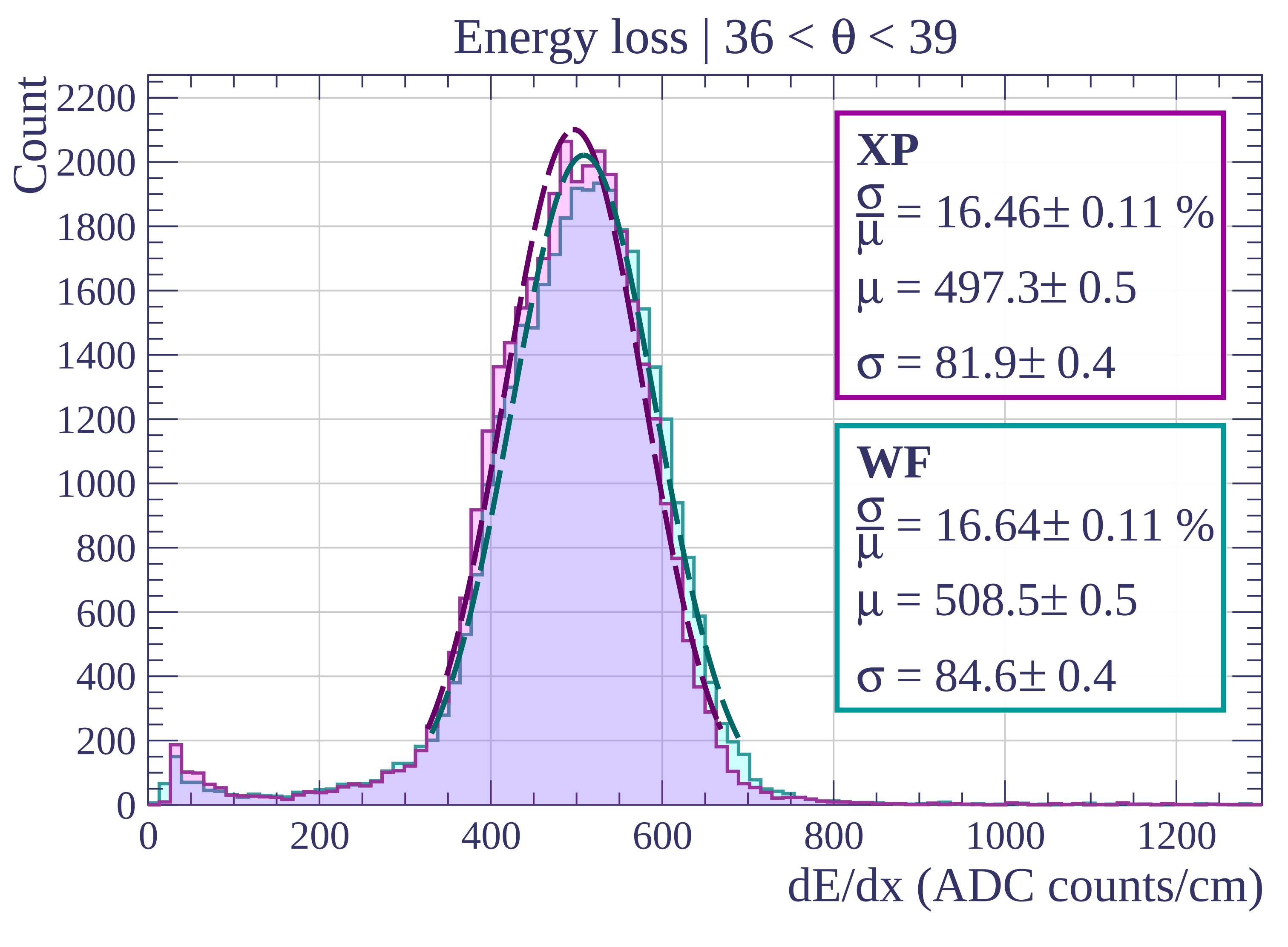

Energy loss | 27 < θ < 30
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
500
1000
1500
2000
2500
3000
3500
4000
XP
σ/μ = 16.20± 0.07 %
μ = 488.1± 0.4
σ = 79.1± 0.3
WF
σ/μ = 16.54± 0.08 %
μ = 498.6± 0.4
σ = 82.5± 0.3



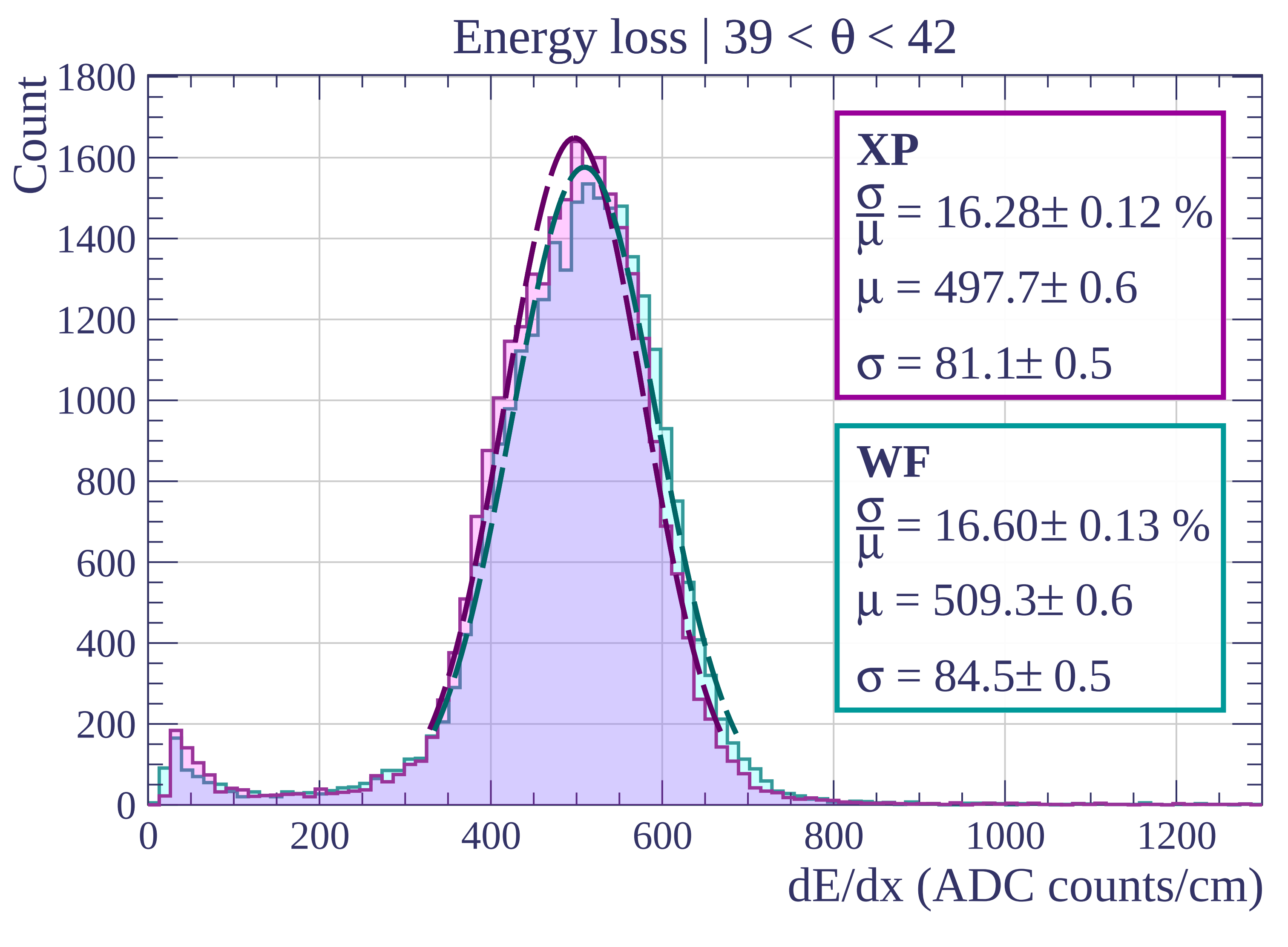
Energy loss | 30 < θ < 33
Count
3500
3000
2500
2000
1500
1000
500
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 16.36± 0.08 %
μ = 490.7± 0.4
σ = 80.3± 0.3
WF
σ/μ = 16.65± 0.09 %
μ = 501.8± 0.4
σ = 83.6± 0.4

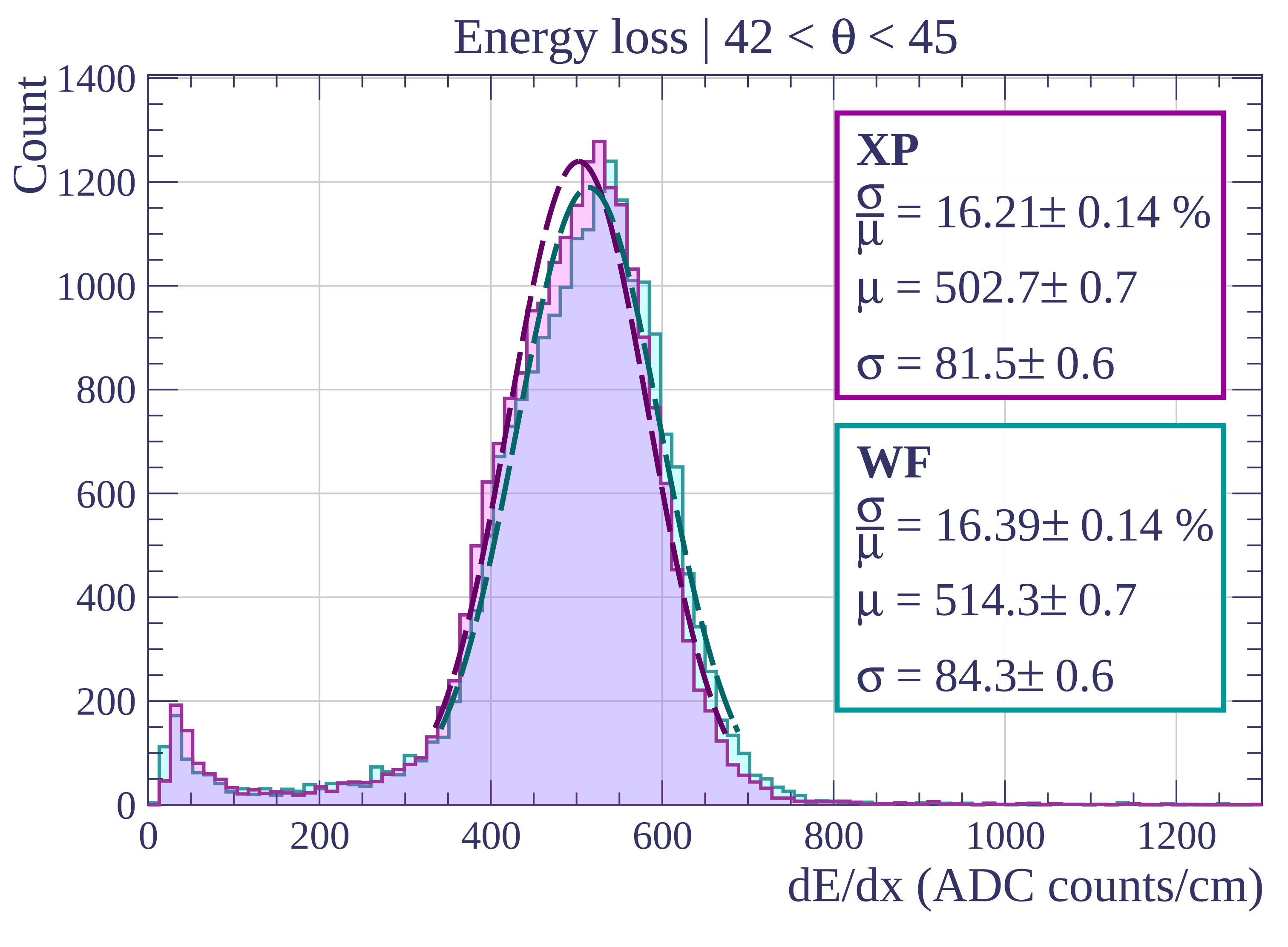


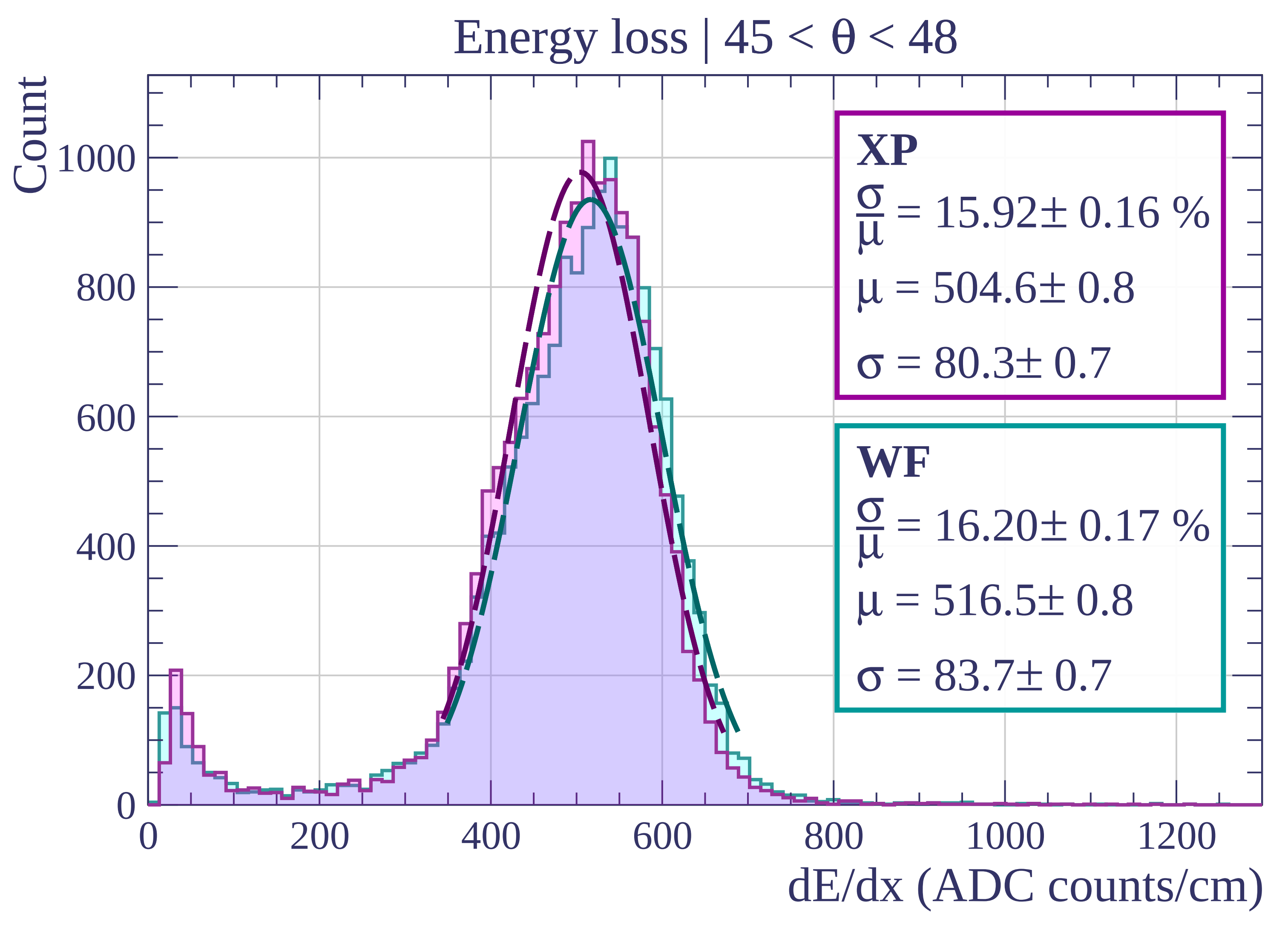
Energy loss | 33 < θ < 36
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 16.47± 0.10 %
μ = 493.6± 0.5
σ = 81.3± 0.4
WF
σ/μ = 16.74± 0.10 %
μ = 504.6± 0.5
σ = 84.5± 0.4
0
500
1000
1500
2000
2500
0
200
400
600
800
1000
1200



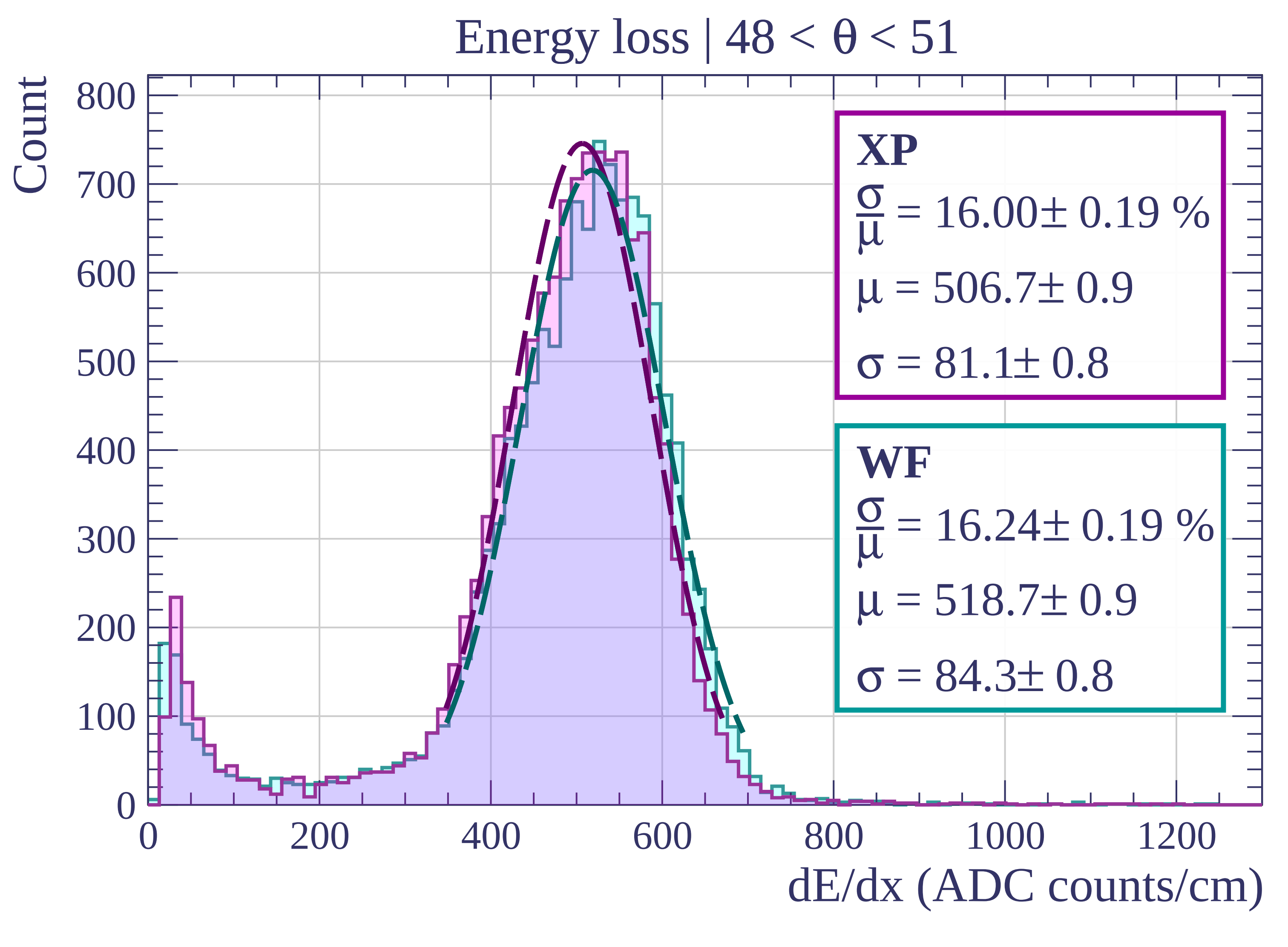
Energy loss | 36 < θ < 39
Count
2200
2000
1800
1600
1400
1200
1000
800
600
400
200
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 16.46± 0.11 %
μ = 497.3± 0.5
σ = 81.9± 0.4
WF
σ/μ = 16.64± 0.11 %
μ = 508.5± 0.5
σ = 84.6± 0.4

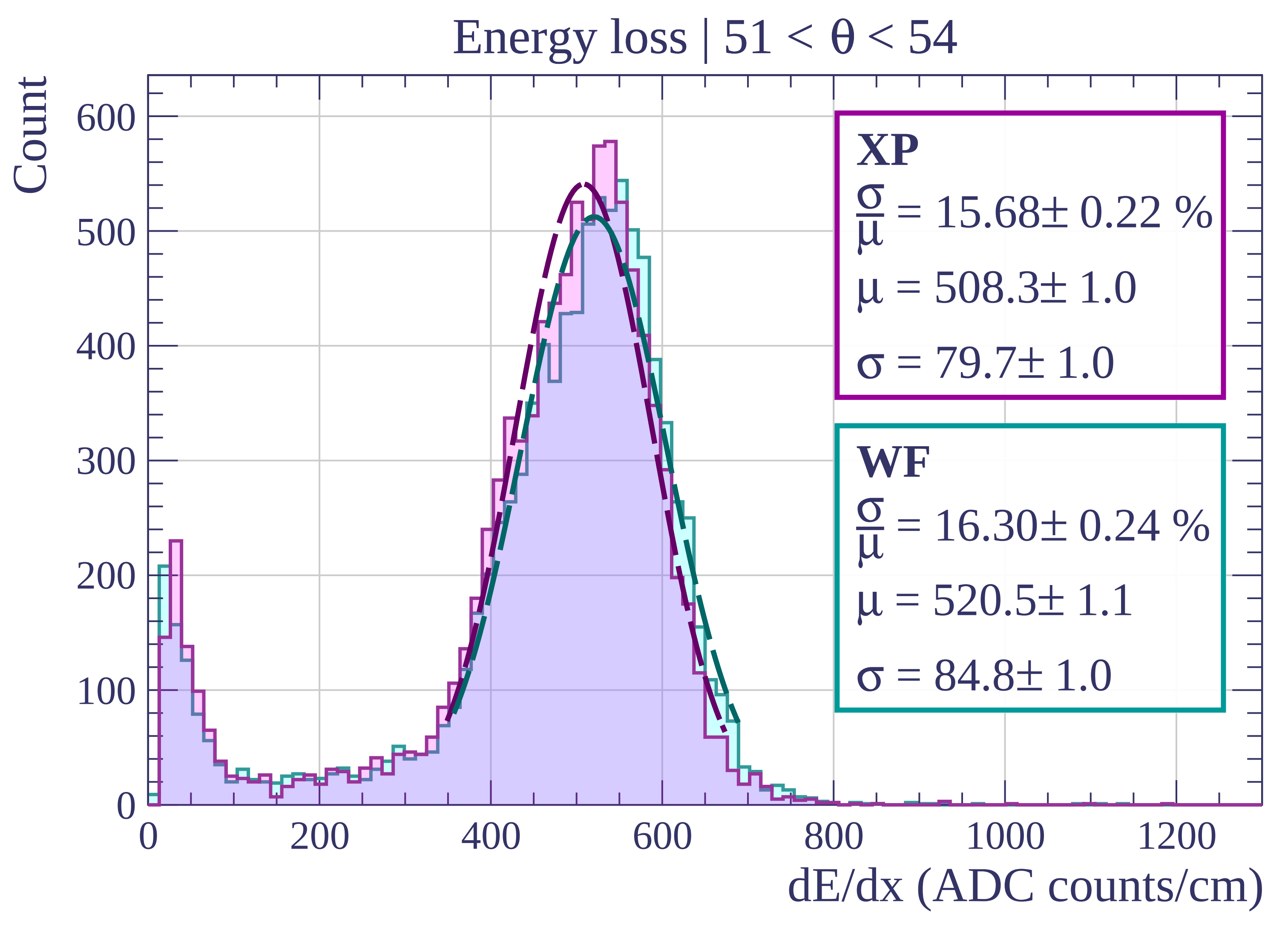

Energy loss | 39 < θ < 42
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 16.28± 0.12 %
μ = 497.7± 0.6
σ = 81.1± 0.5
WF
σ/μ = 16.60± 0.13 %
μ = 509.3± 0.6
σ = 84.5± 0.5
0
200
400
600
800
1000
1200
1400
1600
1800
0
200
400
600
800
1000
1200


Energy loss | 42 < θ < 45
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 16.21± 0.14 %
μ = 502.7± 0.7
σ = 81.5± 0.6
WF
$\frac{\sigma}{\mu}$ = 16.39± 0.14 %
μ = 514.3± 0.7
σ = 84.3± 0.6
0
200
400
600
800
1000
1200
1400

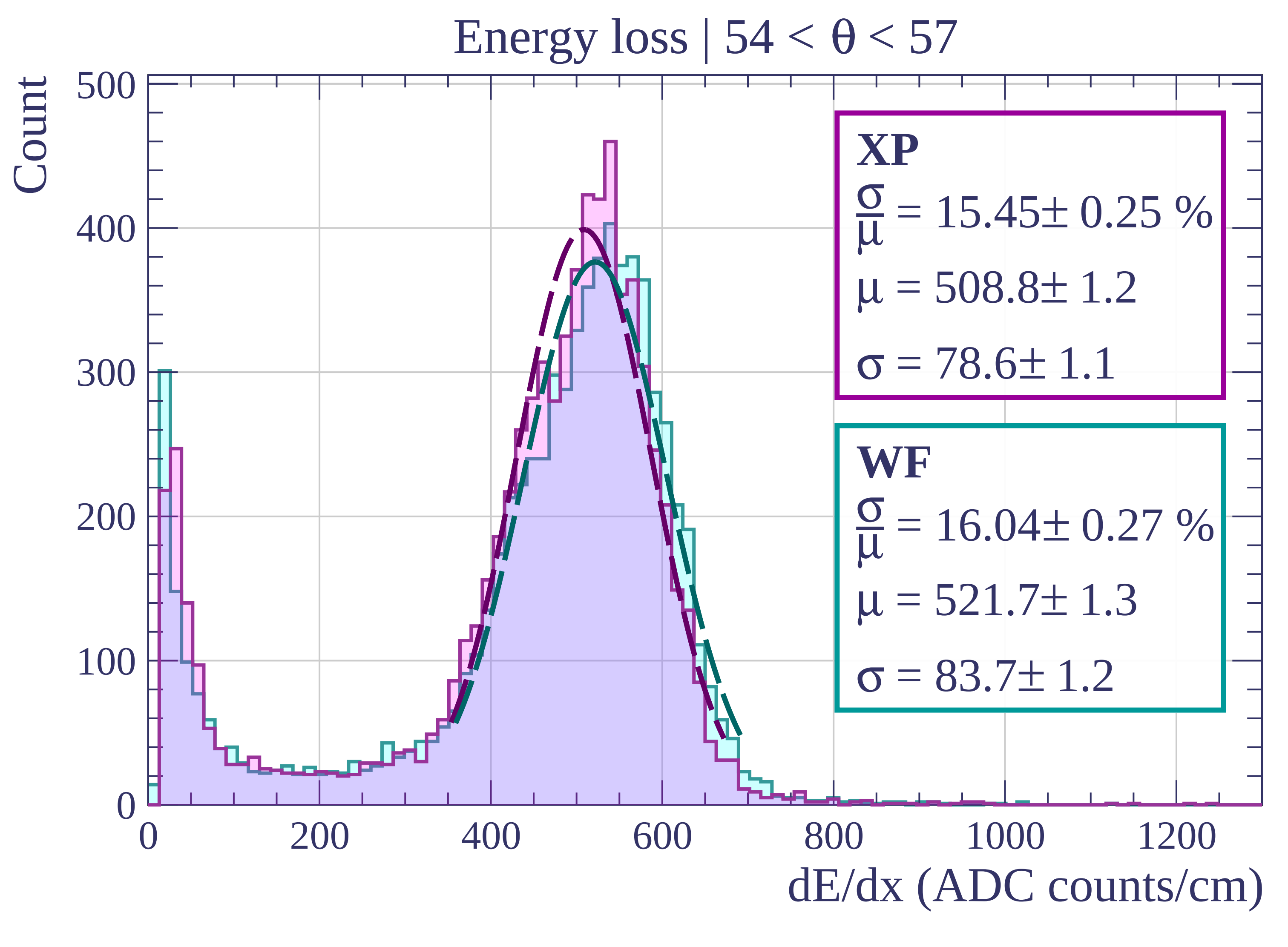


Energy loss | 45 < θ < 48
Count
1000
800
600
400
200
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 15.92± 0.16 %
μ = 504.6± 0.8
σ = 80.3± 0.7
WF
σ/μ = 16.20± 0.17 %
μ = 516.5± 0.8
σ = 83.7± 0.7

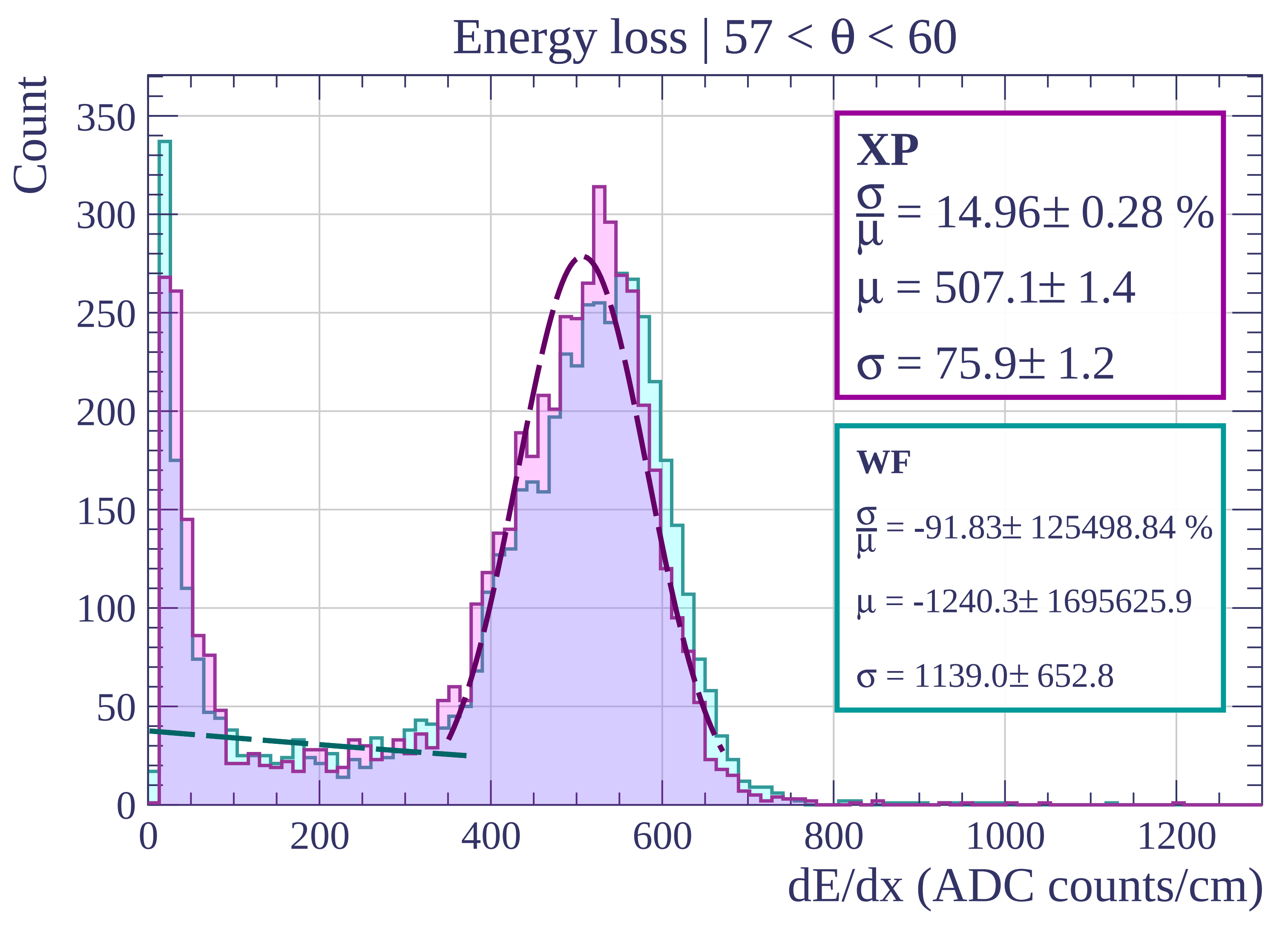

Energy loss | 48 < θ < 51
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 16.00± 0.19 %
μ = 506.7± 0.9
σ = 81.1± 0.8
WF
σ/μ = 16.24± 0.19 %
μ = 518.7± 0.9
σ = 84.3± 0.8
0
100
200
300
400
500
600
700
800
1000
1200

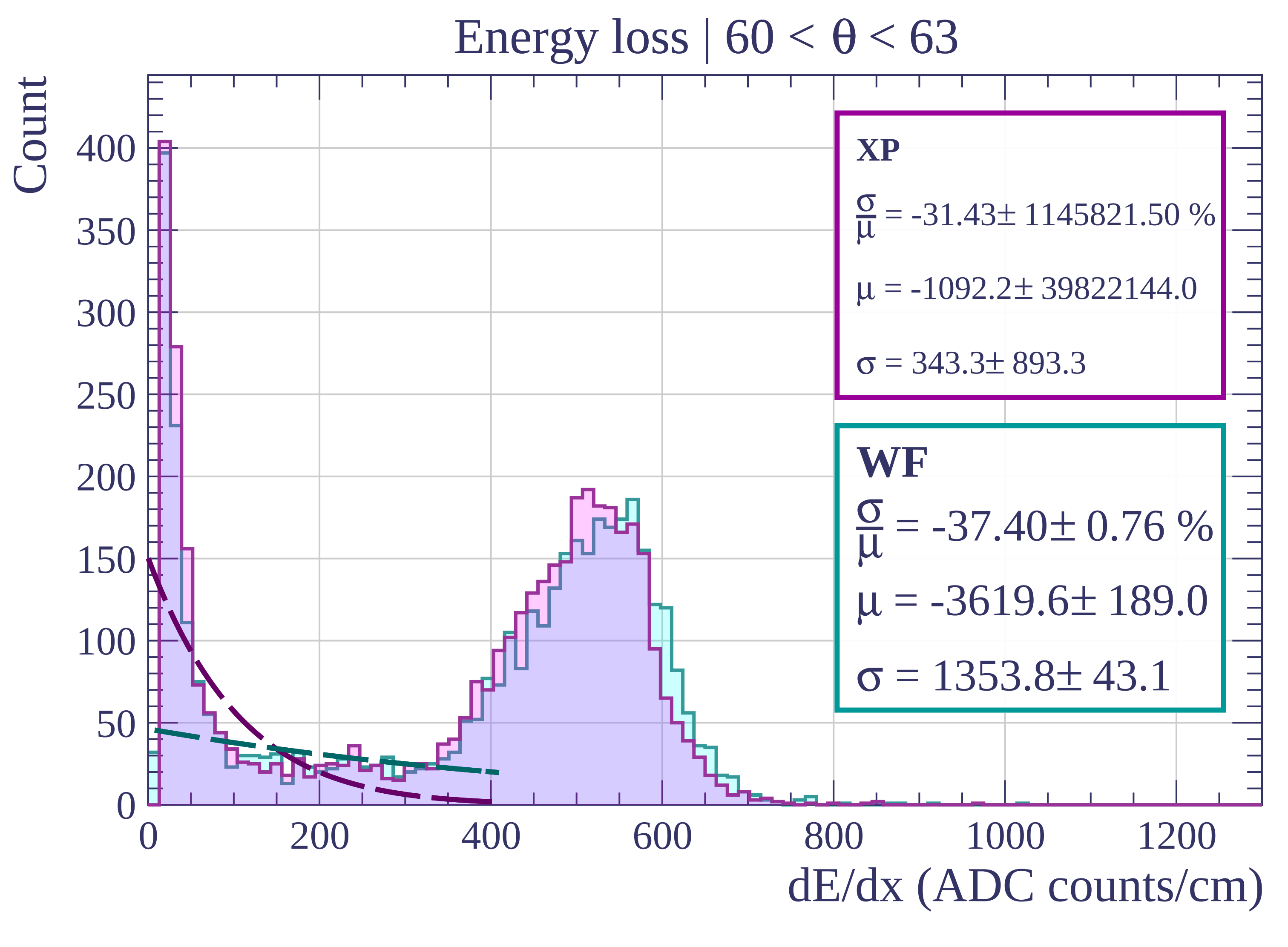


Energy loss | 51 < θ < 54
Count
600
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 15.68± 0.22 %
μ = 508.3± 1.0
σ = 79.7± 1.0
WF
σ/μ = 16.30± 0.24 %
μ = 520.5± 1.1
σ = 84.8± 1.0

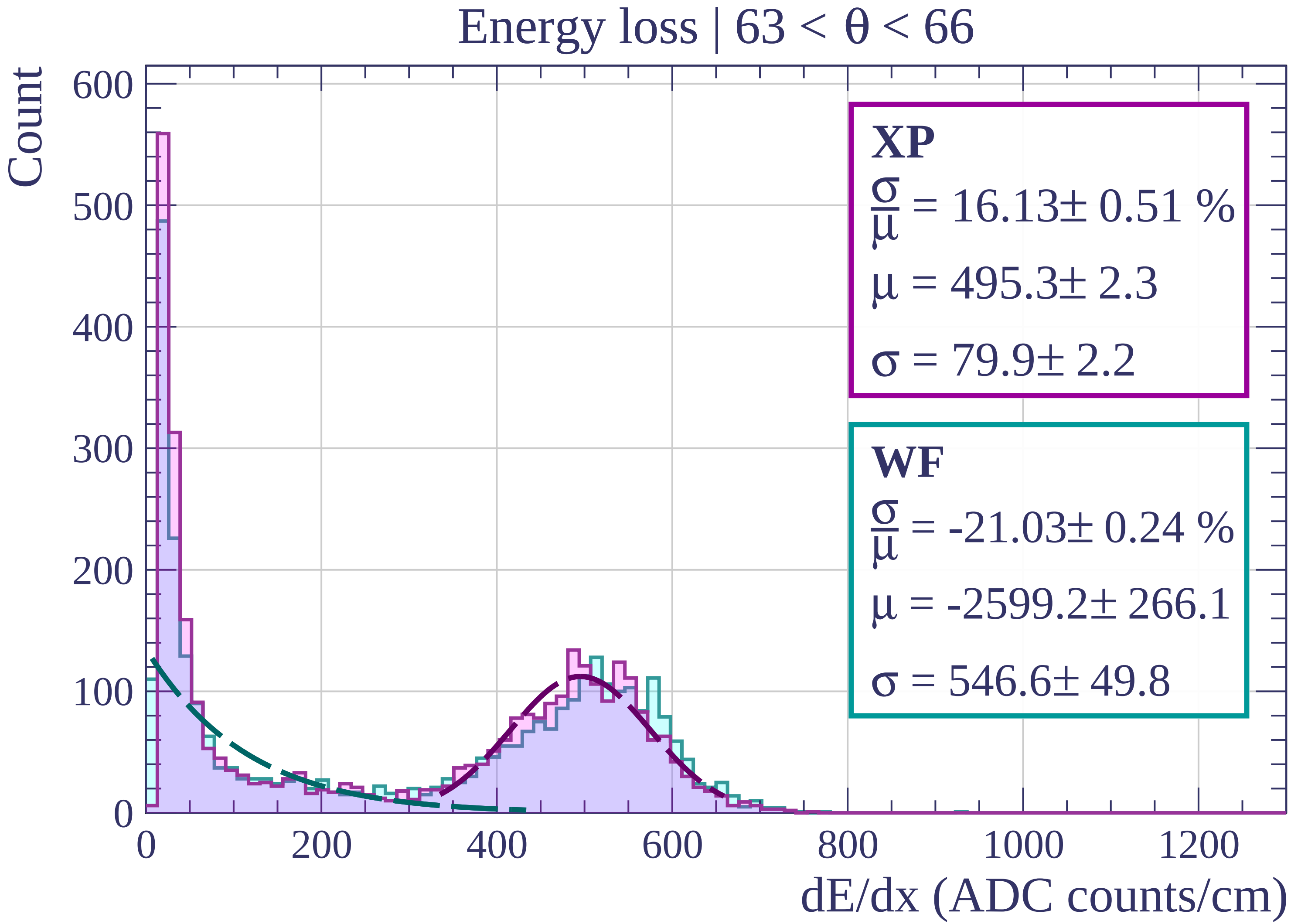


Energy loss | 54 < θ < 57
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 15.45± 0.25 %
μ = 508.8± 1.2
σ = 78.6± 1.1
WF
$\frac{\sigma}{\mu}$ = 16.04± 0.27 %
μ = 521.7± 1.3
σ = 83.7± 1.2

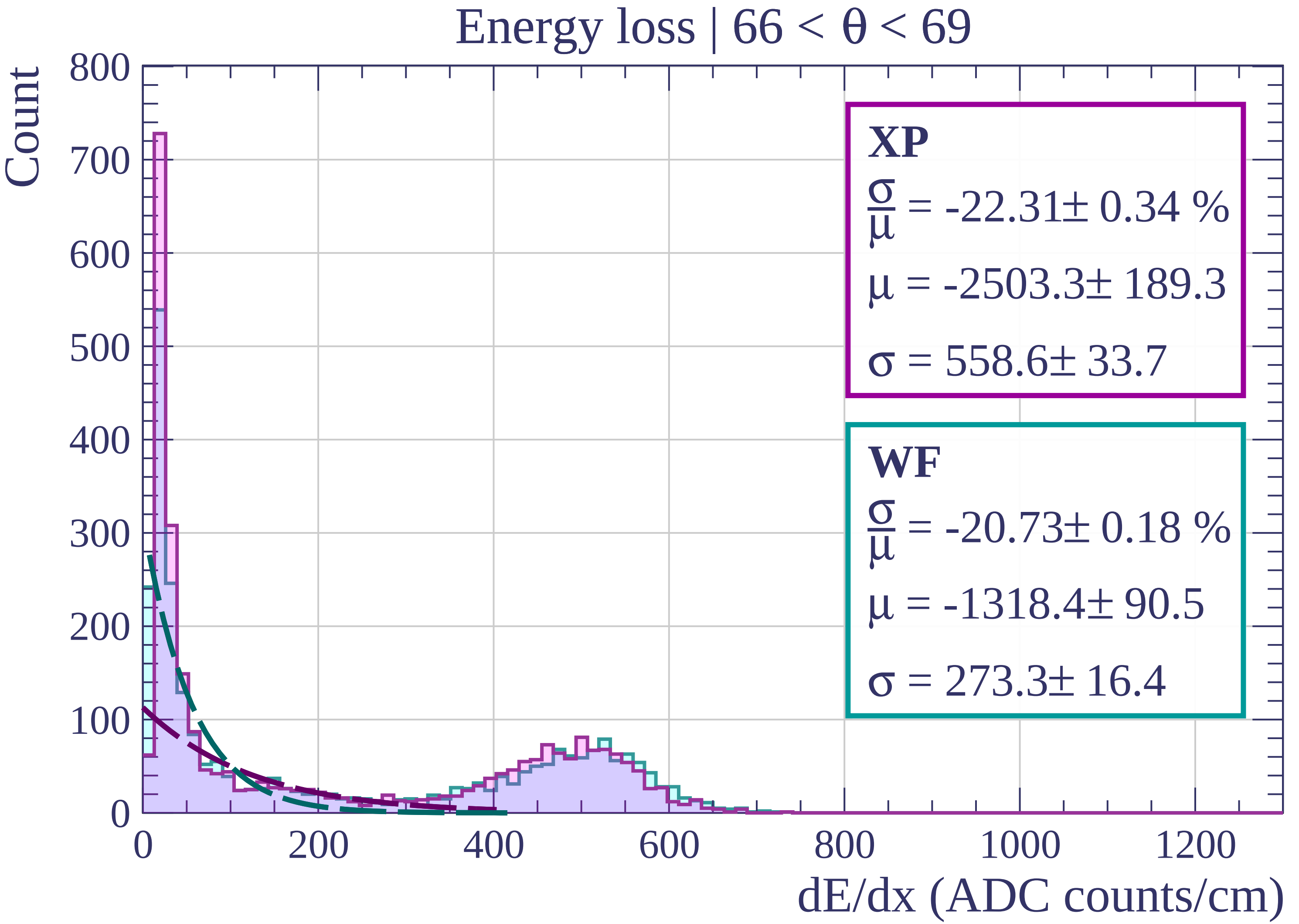

Energy loss | 57 < θ < 60
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200
XP
σ/μ = 14.96± 0.28 %
μ = 507.1± 1.4
σ = 75.9± 1.2
WF
σ/μ = -91.83± 125498.84 %
μ = -1240.3± 1695625.9
σ = 1139.0± 652.8



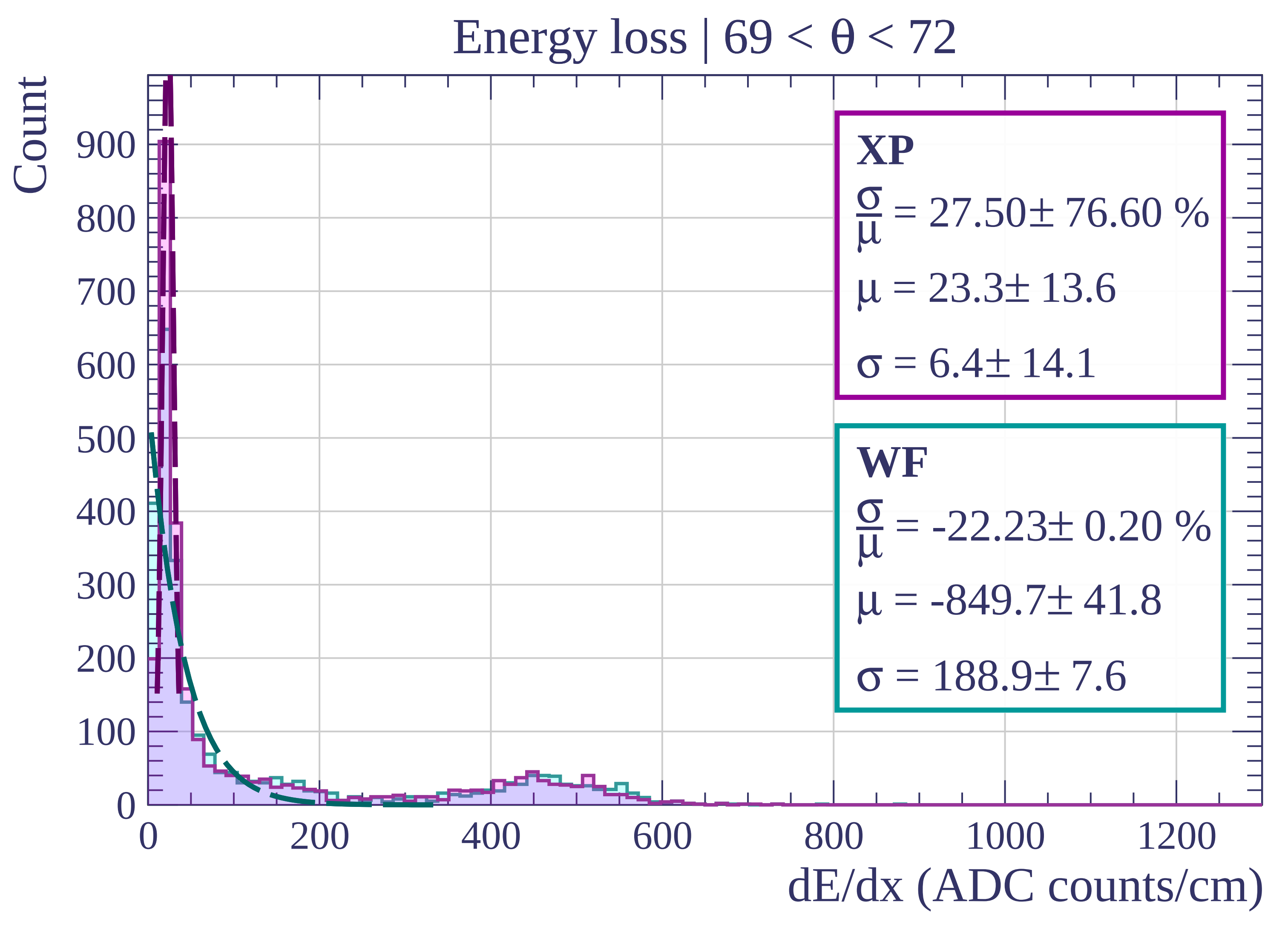
Energy loss | 60 < θ < 63
Count
400
350
300
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -31.43± 1145821.50 %
μ = -1092.2± 39822144.0
σ = 343.3± 893.3
WF
σ/μ = -37.40± 0.76 %
μ = -3619.6± 189.0
σ = 1353.8± 43.1



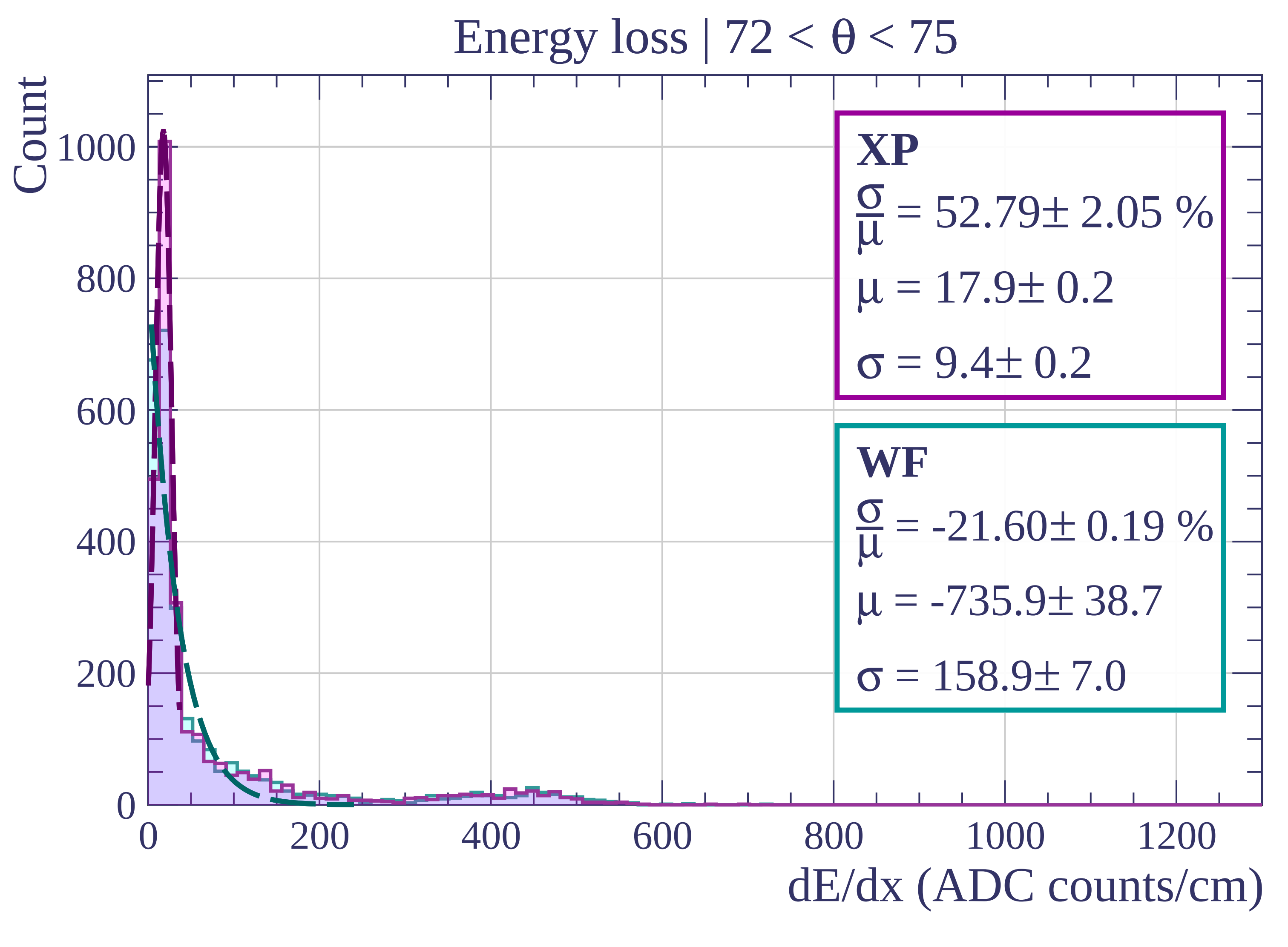
Energy loss | 63 < θ < 66
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 16.13± 0.51 %
μ = 495.3± 2.3
σ = 79.9± 2.2
WF
σ/μ = -21.03± 0.24 %
μ = -2599.2± 266.1
σ = 546.6± 49.8
0
100
200
300
400
500
600
0
200
400
600
800
1000
1200



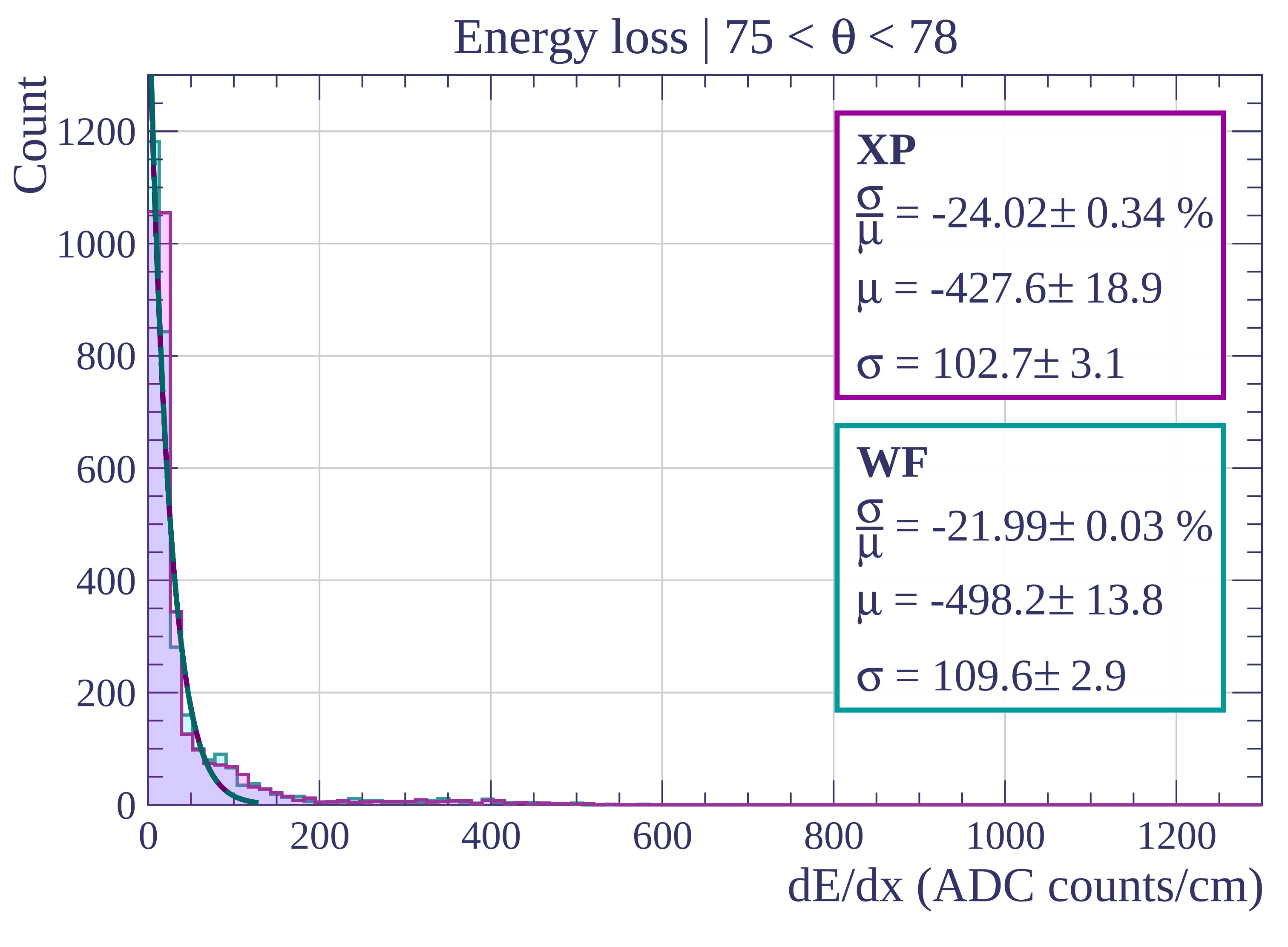
Energy loss | 66 < θ < 69
Count
800
700
600
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -22.31± 0.34 %
μ = -2503.3± 189.3
σ = 558.6± 33.7
WF
σ/μ = -20.73± 0.18 %
μ = -1318.4± 90.5
σ = 273.3± 16.4



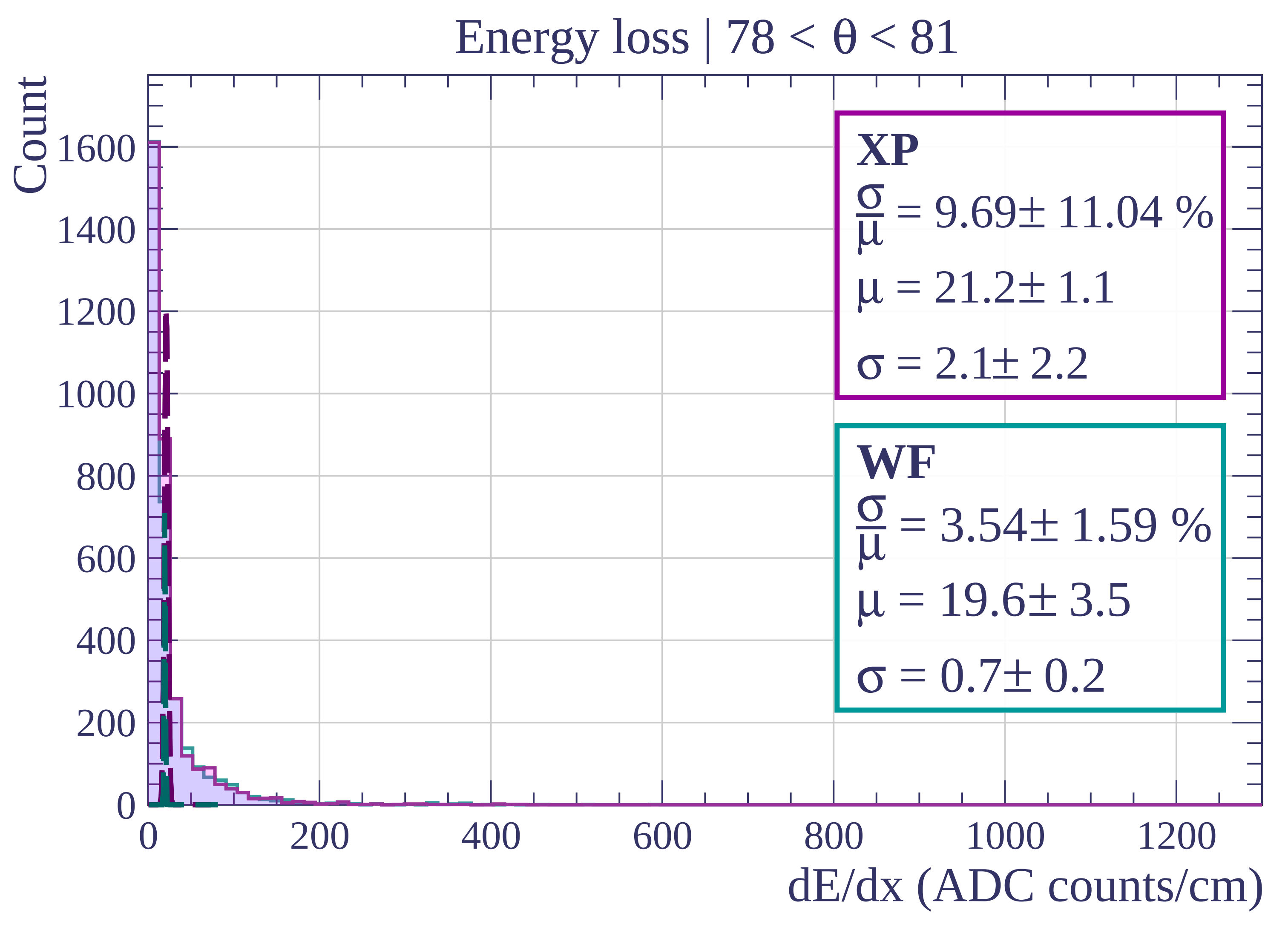
Energy loss | 69 < θ < 72
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
800
900
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 27.50± 76.60 %
μ = 23.3± 13.6
σ = 6.4± 14.1
WF
$\frac{\sigma}{\mu}$ = -22.23± 0.20 %
μ = -849.7± 41.8
σ = 188.9± 7.6

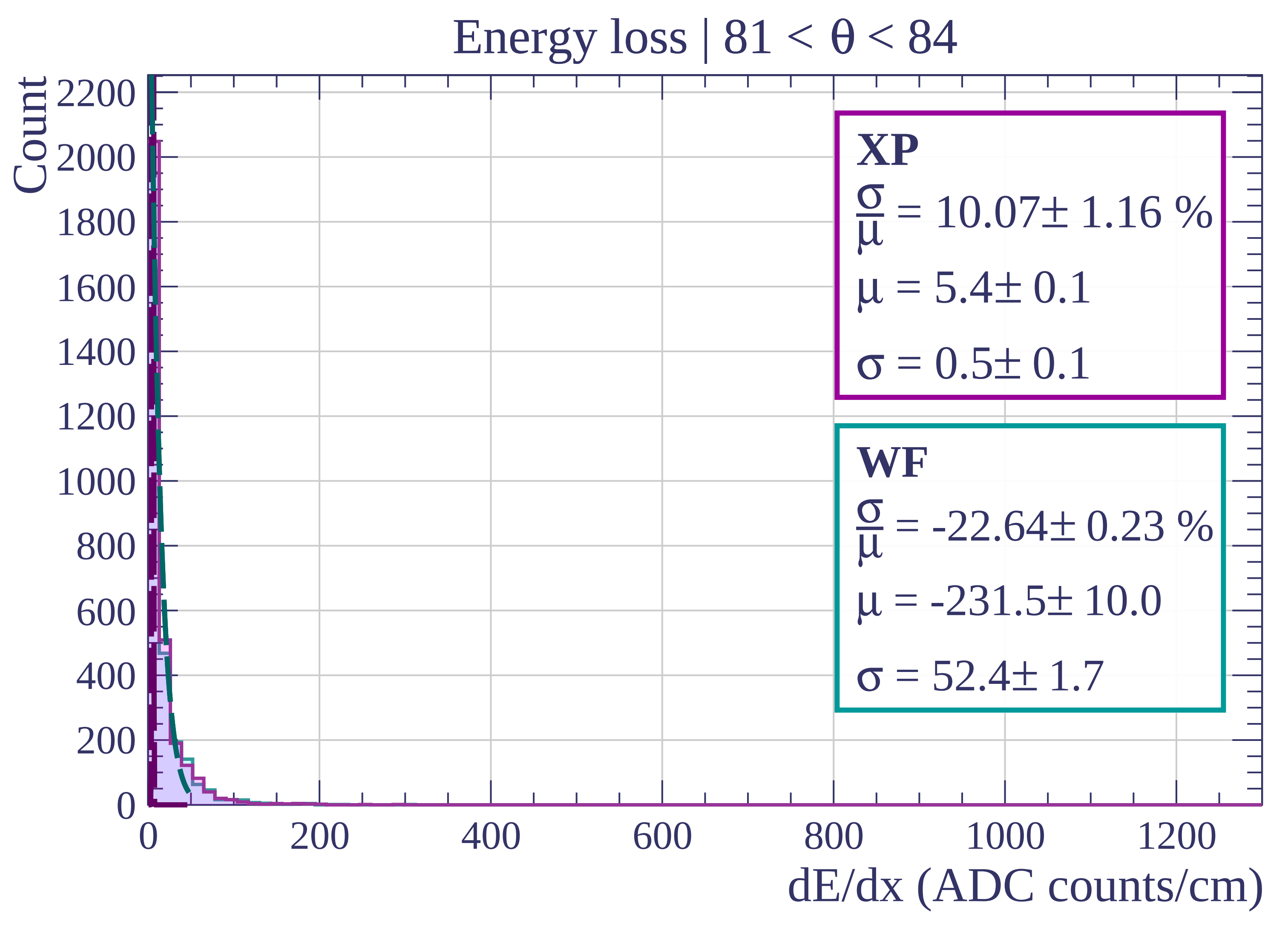


Energy loss | 72 < θ < 75
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
XP
σ/μ = 52.79± 2.05 %
μ = 17.9± 0.2
σ = 9.4± 0.2
WF
σ/μ = -21.60± 0.19 %
μ = -735.9± 38.7
σ = 158.9± 7.0

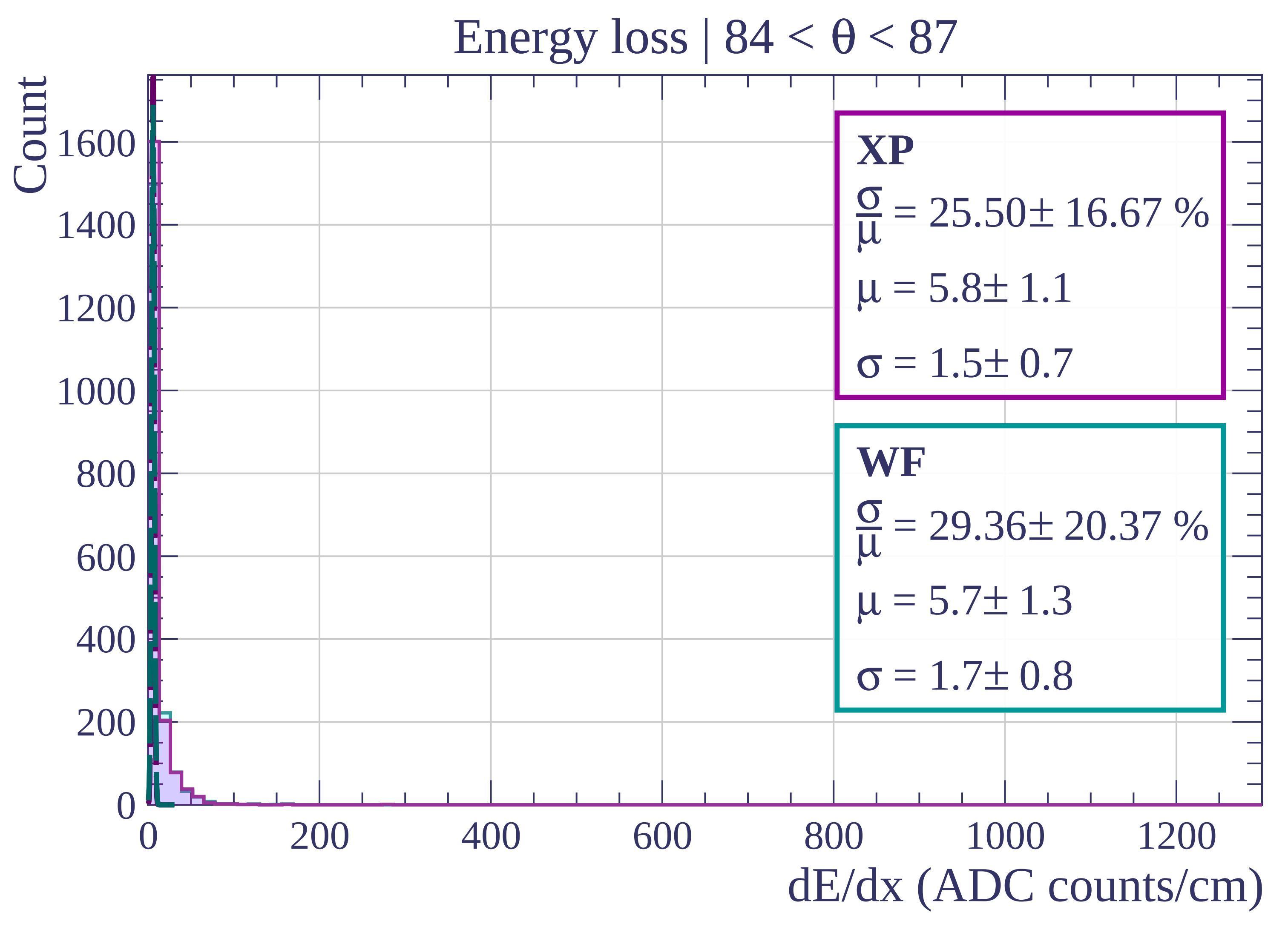


Energy loss | 75 < θ < 78
Count
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -24.02± 0.34 %
μ = -427.6± 18.9
σ = 102.7± 3.1
WF
σ/μ = -21.99± 0.03 %
μ = -498.2± 13.8
σ = 109.6± 2.9

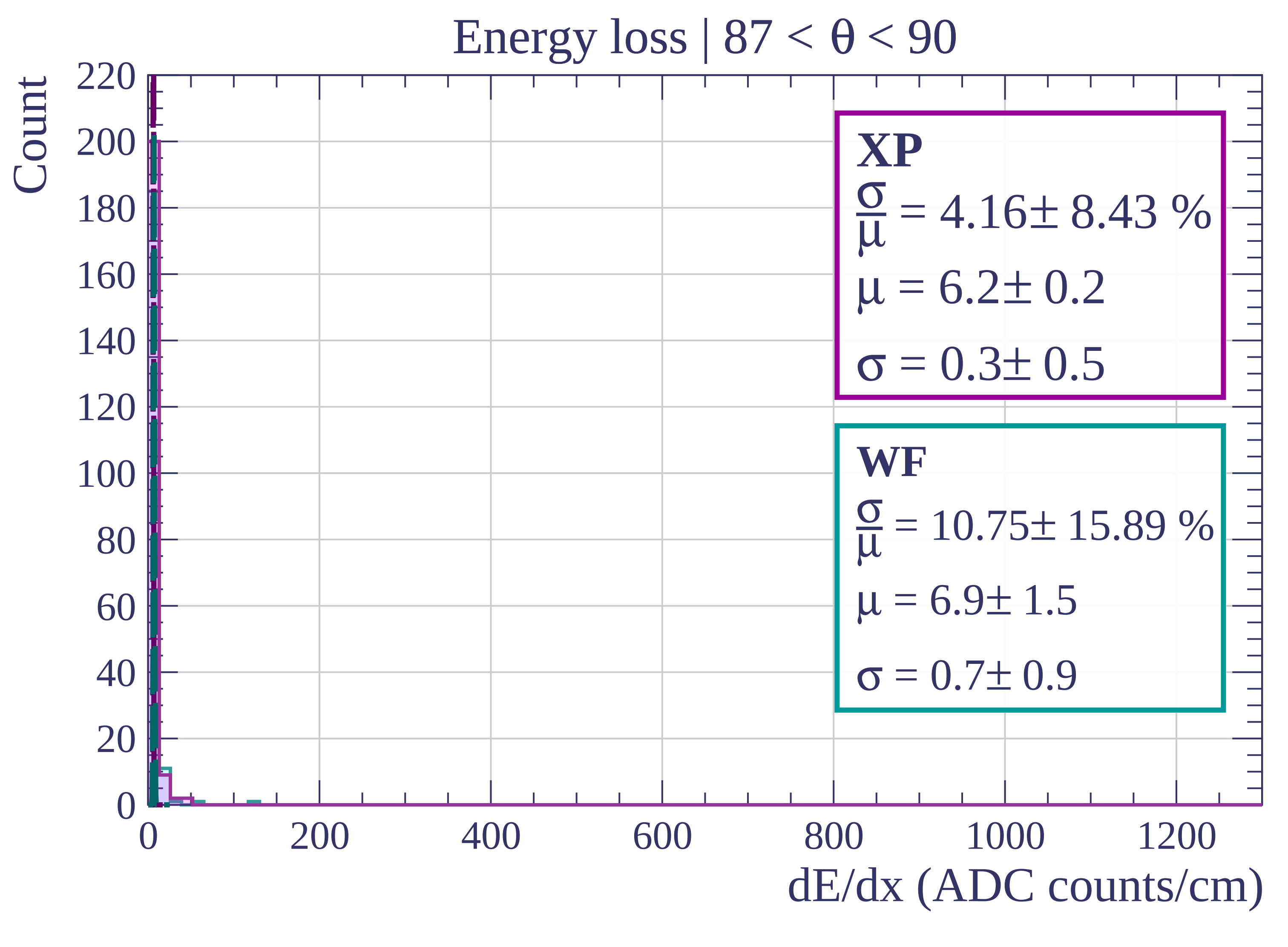

Energy loss | 78 < θ < 81
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
1400
1600
XP
$\frac{\sigma}{\mu}$ = 9.69± 11.04 %
μ = 21.2± 1.1
σ = 2.1± 2.2
WF
$\frac{\sigma}{\mu}$ = 3.54± 1.59 %
μ = 19.6± 3.5
σ = 0.7± 0.2

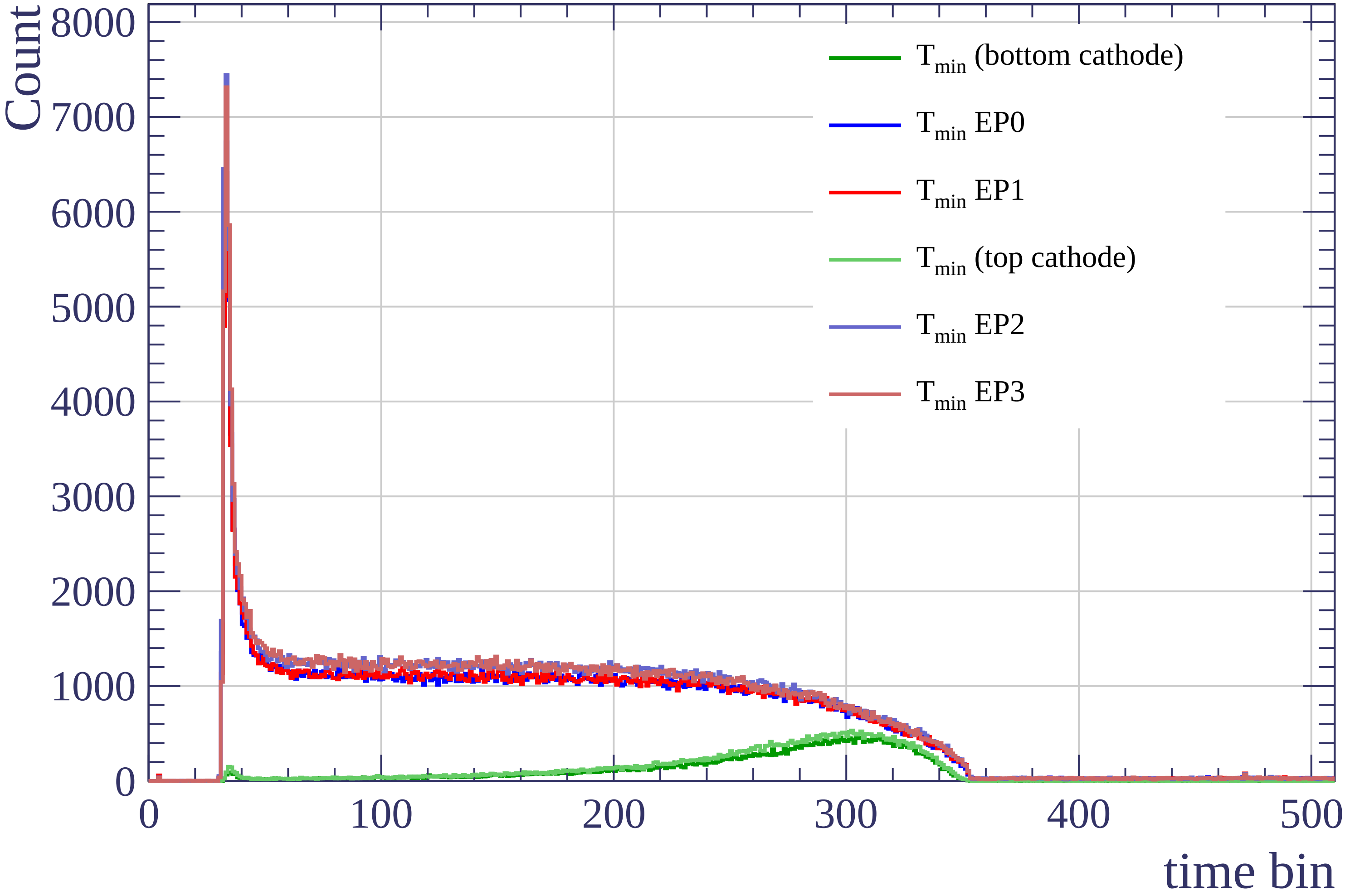


Energy loss | 81 < θ < 84
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 10.07± 1.16 %
μ = 5.4± 0.1
σ = 0.5± 0.1
WF
$\frac{\sigma}{\mu}$ = -22.64± 0.23 %
μ = -231.5± 10.0
σ = 52.4± 1.7

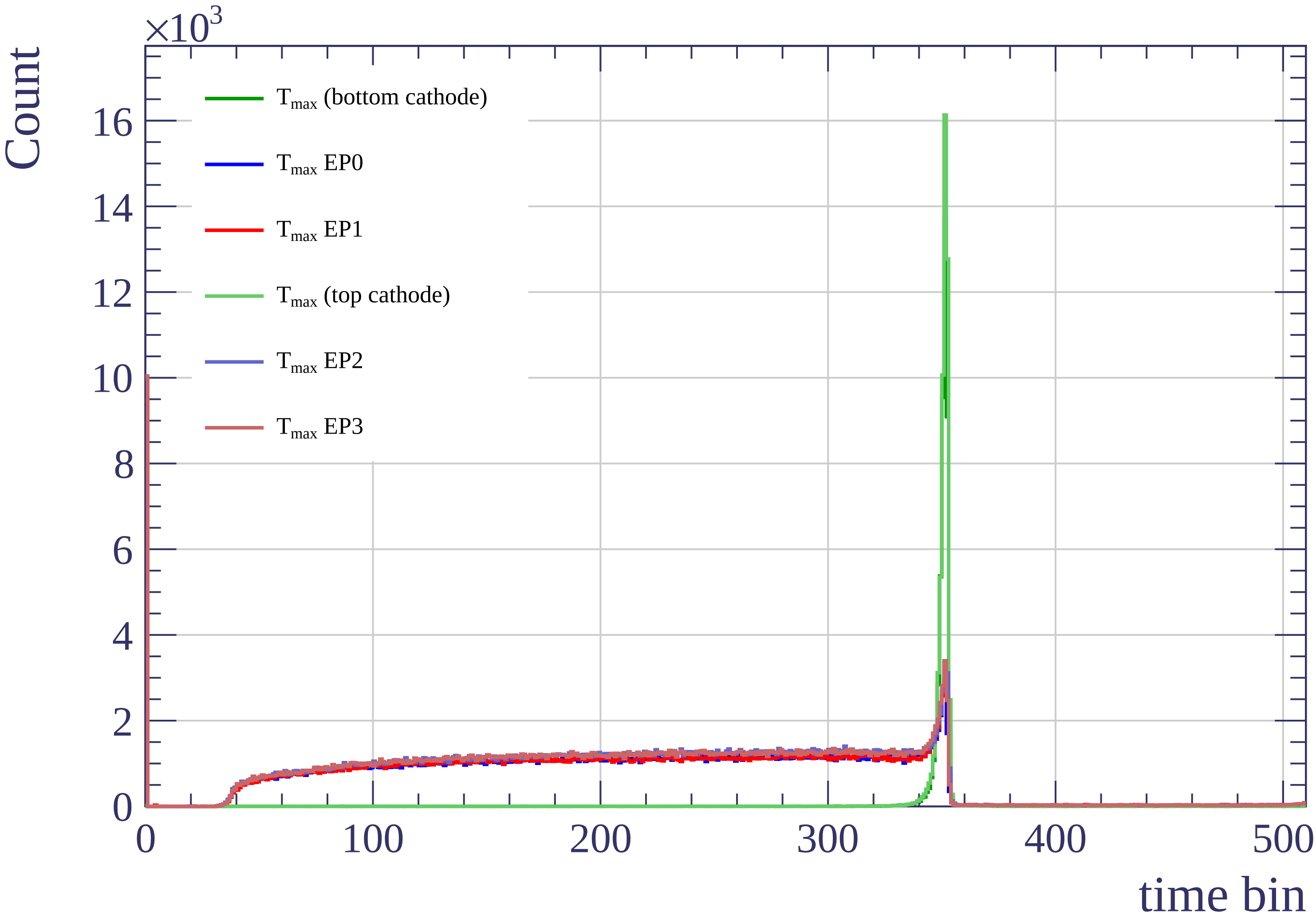


Energy loss | 84 < θ < 87
Count
0
200
400
600
800
1000
1200
1400
1600
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 25.50± 16.67 %
μ = 5.8± 1.1
σ = 1.5± 0.7
WF
$\frac{\sigma}{\mu}$ = 29.36± 20.37 %
μ = 5.7± 1.3
σ = 1.7± 0.8


Energy loss | 87 < θ < 90
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
180
200
220
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 4.16± 8.43 %
μ = 6.2± 0.2
σ = 0.3± 0.5
WF
$\frac{\sigma}{\mu}$ = 10.75± 15.89 %
μ = 6.9± 1.5
σ = 0.7± 0.9



Count
8000
7000
6000
5000
4000
3000
2000
1000
0
0
100
200
300
400
500
time bin
T_{min} (bottom cathode)
T_{min} EP0
T_{min} EP1
T_{min} (top cathode)
T_{min} EP2
T_{min} EP3



Count
$\times10^3$
16
14
12
10
8
6
4
2
0
0
100
200
300
400
500
time bin
T_{max} (bottom cathode)
T_{max} EP0
T_{max} EP1
T_{max} (top cathode)
T_{max} EP2
T_{max} EP3